October 27, 2015

operator's guide

BombardierChallenger 605/650
Pro Line 21™
Advanced Avionics System

FOR
ENTERTAINMENT
ONLY



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INTRODUCTION

This operator's guide describes the components and operation of the Bombardier Challenger 605/650 Pro Line 21 Advanced Avionics System. The general operational information in this guide must be supplemented with information contained in the Aircraft Flight Manual for the Challenger 605/650.

An overview of each of the major Rockwell Collins avionics systems that make up the Bombardier Challenger 605/650 Pro Line 21 Advanced Avionics System follows this introduction. Information about the operation of the flight displays (Multifunction Display (MFD) and Primary Flight Display (PFD)) is contained in the appropriate operation chapters that follow the overviews. The Menus and Displays chapter covers the control panels, flight displays, and menus used to operate the on-board systems. Information regarding messages, annunciations, and flags that require action from the pilot, or indicate a malfunction or failure of a system, is contained in the Messages and Annunciators chapter.

Refer to the following documents for further information:

- Rockwell Collins publication Bombardier Challenger 605/650 Pro Line 21™ Flight Management System, (CPN 523-0822377) for more information on operations involving the FMS.
- Rockwell Collins publication CMU-900, CMU-4000, RIU-40X0 ATN and FANS Interlocked Controller Pilot Datalink Communications (CPDLC) Application Operator's Guide, (CPN 523-0522435) for more information about CPDLC operations.
- Rockwell Collins publication Regional Datalink System CMU-4XXX/RIU-40X0 Operator's Guide, (CPN 523-0818402) for more information about regional AOC operations.
- Rockwell Collins publication Corporate Datalink System CMU-4000/RIU-40X0 Operator's Guide, (CPN 523-0790499) for more information about corporate AOC operations.
- Rockwell Collins publication DBU-5000/5010/5010E Data Base Unit Operator's Guide, (CPN 523-0808703) for more information about loading databases through the DBU.
- Rockwell Collins publication HUD-6605 Head-Up Display Operator's Guide, (CPN 523-0816693) for information about the HUD system.

- Rockwell Collins RTA-4100 Series MultiScan™ Automatic Weather Radar Operator's Guide, (CPN 523-0816797) for more information about weather radar operation.

At the end of this Operator's Guide are five appendices that address the areas of:

- APPENDIX A: FANS and LINK 2000+
- APPENDIX B: TAWS COLORS
- APPENDIX C: IFIS MENU NAVIGATION
- APPENDIX D: TCAS Aurals.
- APPENDIX E: Weather Radar Techniques

NOTE

The illustrations used for CDU screens, PFD, and MFD displays are representative of typical operation and are illustrative only. The illustrations for IFIS displays are representative and for illustrative purposes only.

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SAFETY SUMMARY

CAUTION

Some aircraft operators can have special procedures that are different from those given in this operator's guide. Refer to the applicable aircraft flight manual for instructions specified for your aircraft.

CAUTION

Be careful if you wear sunglasses with lenses that cause you not to see some colors. These types of lenses can have an unwanted effect on how some colors show on the EFIS displays. You cannot see some items on the display when you wear these types of sunglasses. Also, the color of some items can change. For example, some blue lenses can cause a magenta display item to show as red. If the displays are read incorrectly, possible damage to the equipment could occur.

CAUTION

Monitor all instruments to identify if an Flight Management System (FMS) malfunction occurs. The FMS-6200 and related components operate as a system and usually give the pilot an indication when a malfunction occurs. But the pilot must also know that it is not possible to monitor the system for all possible malfunctions. Also, incorrect operation could occur without an indication of a malfunction.

NOTICES

NOTE

The Pro Line 21 Advanced Avionics System can hold a large database of navigation aids, waypoints, and airport data to help the crew with navigation. But the crew must have and use the applicable charts, terminal procedures, and facility directories necessary for the flight. Applicable federal regulations give instructions about the requirement to have the applicable documents available.

NOTE

Defects that you can see on the display surface of the Adaptive Flight Display (AFD) are permitted. But these defects must not cause a distraction or make the crew read the display incorrectly. Also, defects you cannot see in an operational format from a minimum view distance are permitted.

Defective ON or defective OFF rows or columns can make servicing of the AFD necessary. Groups (three or more adjacent elements) of defective ON elements are not permitted. Defective OFF elements are permitted if they do not make the crew read the display incorrectly or cause distraction.

The number of defective ON elements that are permitted include no more than: 6 red, 6 green, 10 blue, or 5 sets of two adjacent defects. The number of defective OFF elements that are permitted include no more than: 5 sets of two adjacent defects, 1 set of 3 adjacent defects, or 30 defective OFF elements.

LIST OF ACRONYMS AND ABBREVIATIONS

ACP	Audio Control Panel
ADC	Air Data Computer
ADF	Automatic Direction Finder
ADS	Air Data System
ADS-B	Automatic Dependent Surveillance – Broadcast
AFCS	Automatic Flight Control System
AFD	Adaptive Flight Display
AFM	Aircraft Flight Manual
ALT	Radio Altimeter
AME	Amplitude Modulation Equivalent
AOC	Air Operations Center
AP	Autopilot
APPR	Approach
ARP	Airport Reference Point
ATC	Air Traffic Control
C&I	Controls and Indicators
CAT2	Category 2
CCP	Cursor Control Panel
CDU	Control Display Unit
CMU	Communication Management Unit
COM	VHF Communications Transceiver
CPDLC	Controller/Pilot Datalink Communication
CW	Continuous Wave
DBU	Data Base Unit
DCP	Display Control Panel
DCU	Data Concentrator Unit
DME	Distance Measuring Equipment
ECU	External Compensation Unit
EFIS	Electronic Flight Instrument System
EICAS	Engine Indication and Crew Alerting System
EVS	Enhanced Vision System
FAF	Final Approach Fix
FANS	Future Air Navigation System
FCC	Flight Control Computer
FCP	Flight Control Panel
FD	Flight Director
FGC	Flight Guidance Computer
FGP	Flight Guidance Panel
FGS	Flight Guidance System
FLIR	Forward Looking Infrared

FM	Frequency Modulation
FMC	Flight Management Computer
FMS	Flight Management System
FPV	Flight Path Vector
FSU	File Server Unit
G/S	Glideslope
GA	Go-Around
GNSS	Global Navigation Satellite System
GPS	Global Positioning System
GPWS	Ground Proximity Warning System
HF	High Frequency
HGS	Head-Up Guidance System
HUD	Head-Up Display
IAPS	Integrated Avionics Processor System
IFIS	Integrated Flight Information System
IRS	Inertial Reference System
IRU	Inertial Reference Unit
ISU	Infrared Sensor Unit
ITU	International Telecommunication Union
LDS	Lightening Detection System
LSB	Lower Sideband
MDC	Maintenance Diagnostics Computer
MDS	Maintenance Diagnostic System
MSG	Message
NAV	Navigation
PFD	Primary Flight Display
RAD ALT	Radio Altitude System
RDC	Remote Data Concentrator
RNP	Required Navigation Performance
RNP AR	Required Navigation Performance Authorization Required
RSP	Reversion Switch Panel
RSS	Radio Sensor System
RTA	Receiver Transmitter Antenna
SATCOM	Satellite Communication
SBAS	Satellite Based Augmentation Systems
SCIT	Storm Cell Identification and Tracking
SVC	Synthetic Vision Computer
SVS	Synthetic Vision System
TAWS	Terrain Awareness Warning System
TCAS	Traffic Alert and Collision Avoidance System

TDR	Air Traffic Control Transponder
TSS	Traffic Surveillance System
TTR	TCAS Transceiver
USB	Universal Serial Bus
USB	Upper Sideband
VDL	VHF Data Link
VSI	Vertical Speed Indicator
WXR	Weather Radar System
XFR	Transfer
YD	Yaw Damper

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OVERVIEW – SYSTEM

INTRODUCTION

This operator's guide describes the system components, operation, and operating procedures for the Rockwell Collins Pro Line 21 Advanced Avionics System with Integrated Flight Information System (IFIS) as installed in the Challenger 605/650. It is not intended to be a training manual. It is a guide to understanding the system as designed by Rockwell Collins.

NOTE

The Rockwell Collins avionics units described in this operator's guide show typical operation and capabilities. The non-Rockwell Collins avionics units may or may not cause typical operation. Refer to the appropriate vendor documentation for specific details on the non-Rockwell Collins avionics units.

SYSTEM DESCRIPTION

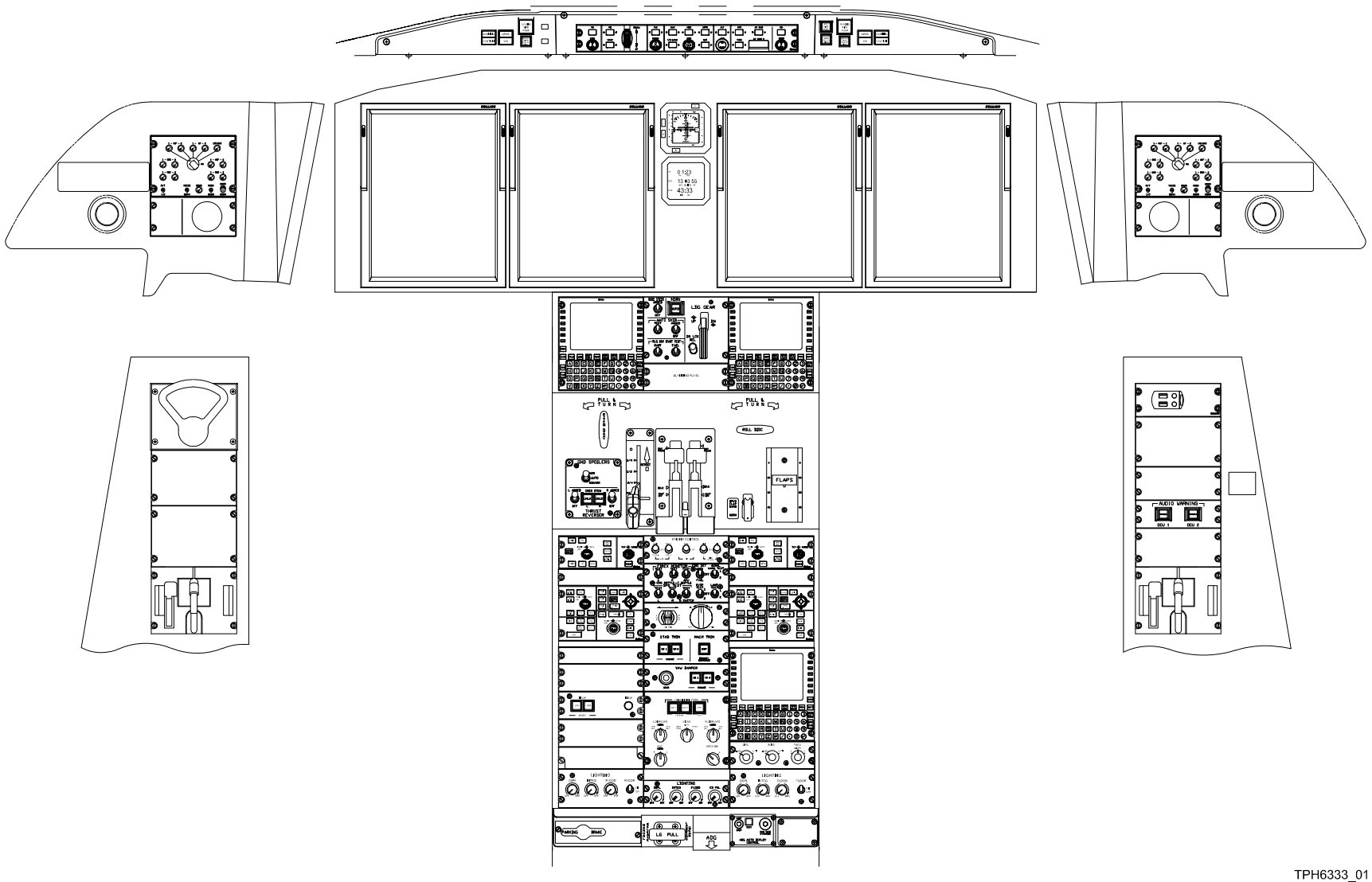
The Rockwell Collins Pro Line 21 Advanced Avionics System for the Challenger 605/650 is a completely integrated flight instrument, Autopilot, and navigation system. All functions have been combined into a compact, highly reliable system designed for ease of operation and reduced flight deck workload.

KEY OPERATING FEATURES

- Large, uncluttered LCD flight displays
- Control and display integration
- Intuitive, easy to interpret menu formats
- Reduced number of control panels.

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Figure 2-1 Challenger 605/650 Flight Deck Layout



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COMPONENTS

Air Data System

The Air Data System (ADS) supplies processed air data to the Automatic Flight Control System, Electronic Flight Instrument System, and navigation systems. The ADS is made up of two Air Data Computers (ADC), pilot and copilot-side pitot and static probes, and pilot and copilot-side temperature sensors. The ADC is functionally isolated, and each acts as a separate ADS. The ADC receives pitot and static air pressure and air temperature inputs from the aircraft sensors. The ADC processes the raw data, then sends digital air data to various aircraft systems that use air data inputs. Processed air data provided by the ADC includes: uncorrected pressure altitude, total and static pressure, vertical speed, airspeed, Mach, maximum airspeed/Mach, true airspeed, total air temperature, static air temperature, and International Standard Atmosphere (ISA) delta temperature.

Automatic Flight Control System

The Automatic Flight Control System (AFCS) supplies a dual Flight Guidance System (FGS), a three-axis Autopilot (AP), including Yaw Damper (YD), and pitch trim control. The pilot selects the Flight Guidance Computer (FGC) in control with the Flight Guidance Panel (FGP) Transfer (XFR) switch. Each Primary Flight Display (PFD) shows the Flight Director (FD) commands from the FGC that is selected with the XFR switch, except for Go-Around (GA), Approach (APPR) and Takeoff modes. A XFR arrow in each FGS annunciation field indicates which side of the aircraft is the controlling side. Lateral and vertical modes selected by the pilot are used by the FGC to generate steering commands. The steering commands are supplied to the flight directors and the AP. The YD supplies control signals to drive the rudder actuator for dutch roll damping and turn coordination. The automatic pitch trim control interfaces with the aircraft pitch trim system to automatically off-load the forces on the elevator and the control column. The AFCS is made up of two dual-channel FGC, a Flight Control Panel (FCP), a YD panel, two primary servos and two rudder actuators. The PFD and Display Control Panel (DCP) are considered to be part of the EFIS but have AFCS functions. Remote-mounted installer-supplied switches on the control wheels and thrust levers have AFCS functions. Other inputs to the AFCS include those from the ADS, Inertial Reference System (IRS)/Attitude Heading Reference System (AHRS), navigation receivers, Data Concentrator Unit (DCU), and the aircraft pitch trim system.

Terrain Awareness Warning System

The non-Rockwell Collins Terrain Awareness Warning System (TAWS) is effectively three independent systems: Ground Proximity, Terrain, and Windshear Systems. The system contains the traditional Ground Proximity Warning System (GPWS) functionality and interface that produces reactive warnings (warnings that require action based on real time input to the GPWS). These warnings are categorized into five modes (GPWS Mode 1 through 5). Each mode has a distinct aural alert, and some modes have associated visual annunciations. The GPWS is also capable of providing a sixth mode that can include altitude related callouts and/or an excessive bank angle callout. The TAWS uses worldwide database information, specific aircraft information from the Data Concentrator Unit, position data from the FMS, GNSS and Navigation receivers, altitude data from the Radio Altimeter System, attitude and heading data from the Inertial Reference System, angle-of-attack data from the AOA transmitter, air data information from the Air Data System to generate system messages, alerts, and terrain symbology. Specified warning and caution alerts show on the PFD. Terrain data is available for display on the MFD and PFD. Specified TAWS mode and fault flags show on the MFD and PFD. System controls are located on the CCP and DCP.

Electronic Flight Instrument System

The Electronic Flight Instrument System (EFIS) is a multicolor, flight instrument display and control system made up of four Adaptive Flight Displays (AFD), two Display Control Panels (DCP), two Cursor Control Panels (CCP), and one Reversion Switch Panel (RSP). The EFIS provides display and control functions for all of the installed communications, navigation, hazard avoidance, and engine indication and crew alerting systems. The EFIS is made up of a pilot-side and copilot-side system. Each side is functionally and physically isolated from each other and is capable of functioning as a complete, independent system. The display portion of the system is made up of four Adaptive Flight Displays. In normal operation, one display is configured as a Primary Flight Display (PFD) and one is configured as a Multifunction Display (MFD) for each side. The DCP provides configuration and selection controls for the on-side PFD. Two CCPs provide for MFD display control. The RSP provides the means for selecting a compressed PFD format on either display, control of the EICAS presentation location, as well as selecting reversion to cross-side attitude/heading data, and air data. The integrated avionics processor system and data concentrator units are used to gather and concentrate the data that shows on the EFIS.

Engine Indicating and Crew Alerting System

The Engine Indication and Crew Alerting System (EICAS) collects data from the many different aircraft systems and converts that data to supply the necessary visual cues and aural tones or voice commands to the crew. Engine conditions, aircraft system conditions, and system messages are generated. The EICAS displays are divided into the engine indicating system area, the primary flight controls area, and the crew alerting system area. The EICAS is made up of two dual-channel Data Concentrator Units (DCU) and one dual-channel Remote Data Concentrator (RDC). The DCU provides Crew Alerting System logic processing and overall system data concentration. The RDC concentrates data from other aircraft systems and routes the data to the DCU for processing. The DCU converts the data to supply the Engine Indication System and aircraft system displays and the appropriate aural alerts or voice commands for the Crew Alerting System portion of the EICAS. The display and control portion of the EICAS is made up of the four Adaptive Flight Displays (AFD) and two Cursor Control Panels (CCP). The CCP provides control of the display pages on either Multifunction Display (MFD).

Flight Management System

The Flight Management System (FMS) provides lateral and vertical flight plan point-to-point navigation using multiple navigation sensors. The system generates lateral and vertical steering commands for use by the AFCS and EFIS. The Control Display Unit (CDU) provides flight deck management functions that include navigation sensor control, radio tuning, and Multifunction Display (MFD) control menus. The FMS is made up of two Flight Management Computers (FMC), two CDUs, two Global Positioning Global Navigation Satellite System (GNSS), and one Data Base Unit (DBU). The Air Data System (ADS), Inertial Reference System (IRS), NAV receiver, and DME sensor supply data to the FMS. The FMS uses position data from the GNSS, VOR, DME, and sensor data from the IRS and ADS, along with the active flight plan and its own database information to generate lateral and vertical flight plan-based navigation solutions. An optional third FMC and CDU are supported by the FMS. The third FMC will act as a standby FMC which can be reverted in if one FMC fails.

Head-Up Guidance System

The optional Rockwell Collins Head-Up Guidance System (HGS), also referred to as the Head-Up Display (HUD), installed in the Bombardier CL605/650 aircraft may also be equipped with the Enhanced Vision System (EVS). The HUD is an electronic and optical system that displays information in the left-side pilot's forward field-of-view. The HUD focuses the display at optical infinity and presents flight and navigational data conformal to the real world. The system is certified for all phases of flight. The HUD optimizes symbology for full-flight regime use and includes the application of inertial flight path and flight path acceleration. HUD systems integration and symbology allow for extremely precise aircraft control while enhancing situational awareness and energy management. EVS is an electronic and optical system that shows infrared video on the Combiner display. The Combiner display shows a conformal display of the video with the applicable symbology. Refer to the Rockwell Collins HGS Model 6605 for Bombardier CL-605 Aircraft Operator's Guide, (CPN 523-0816693) for full instructions in operating this equipment.

Integrated Flight Information System

The Integrated Flight Information System (IFIS) is a Multifunction Display (MFD) upgrade that optionally may add Electronic Charts (E-Charts), Graphical Weather (GWX), and Enhanced Map (E-Map) features to the MFD. The GWX service provider may be Universal Weather GWX or XM GWX. The IFIS is made up of a File Server Unit (FSU) and two Cursor Control Panels (CCP). A third datalink-capable VHF COM and a Communications Management Unit (CMU) are required when the Universal Weather GWX option is installed. An XM Satellite Receiver is required when the XM GWX option is installed. The FSU provides the processing and storage for all IFIS functions including E-Charts, E-Maps, and GWX. The FSU supplies stored graphical symbology to the MFD by way of a dedicated bi-directional ethernet bus. The FSU also provides an interface with the DBU that allows for operational software (FSA) and databases to be uploaded.. The CCP provides a dedicated interface to the MFD for controlling IFIS functions through onscreen menus. The VHF COM datalink radio and CMU provide air and ground data communications with compatible air-ground networks when the Universal Weather GWX option is installed. The Control Display Units (CDU) are used to request Universal Weather GWX from the ground-based service provider. The optional XM Satellite receiver receives a constant broadcast of graphical and textual weather data from the XM Satellite Radio weather service. The XM Satellite receiver provides the weather data to the FSU. Rockwell Collins supplies hardware and software to support both GWX options, but a non-Rockwell Collins subscription is required. The IFIS illustrations in this operator's guide are for illustrative purposes only.

Inertial Reference System

The non-Rockwell Collins Inertial Reference System (IRS) supplies attitude and magnetic or true heading to the AFCS, EFIS, FMS, navigation systems, and the WXR. The system is made up of two or optionally three Inertial Navigation Units (INU) and remote-mounted installer-supplied controls located on the flight deck. The INU generates and supplies attitude and heading data to the EFIS and other subsystems by way of the IAPS and the system bus structure. Attitude and heading data from the cross-side or optional third INU can be selected to drive on-side systems in the event of an on-side INU failure.

Lightning Detection System

The optional L3 Communications Stormscope® Series II Weather Mapping System Lightening Detection System (LDS) maps electrical discharge activity (lightning) within 200 NM of the aircraft. The Weather Mapping System is referred to as the Lightning Detection System (LDS) in this guide. LDS generated lightning symbology is available for display on the MFD and PFD. The LDS is made up of one LDS processor and one antenna. The LDS processor uses inputs from the antenna to calculate the azimuth and range of the electrical discharges in the area surrounding the aircraft. The LDS processor generates lightning symbology and both operating and fault messages for display on the AFD. The LDS graphic displays are selected with controls on the CCP and DCP. Remote-mounted installer-supplied controls are used to operate dedicated LDS functions. The LDS outputs lightning azimuth, range, intensity, and system mode/status data to the AFD by way of the IAPS and system bus structure.

Maintenance Diagnostic System

The Maintenance Diagnostic System (MDS) monitors avionics units to detect failures, isolate faults to a particular unit, and provide historical fault data. This maintenance and fault information is available for display on either Multifunction Display (MFD). Storage space is also provided for engine trend/performance data and maintenance data from other aircraft systems. Maintenance data can also be downloaded to a portable memory device using the Data Base Unit (DBU). The primary system component is the Maintenance Diagnostics Computer (MDC), which is pre-loaded with fault isolation logic that uses collected avionics unit failure data to identify failed or degraded avionics units. The fault logic is updated through the use of the DBU. The system supplies the fault information to the MFD and DBU by way of the IAPS and system bus structure.

Radio Altimeter System

The Radio Altitude System (RAD ALT) provides a direct radio height measurement from 2500 feet to touchdown for use by the AFCS, EFIS, TAWS, and TCAS. RAD ALT supplied data is used to generate Radio Altitude Minimum (RA MIN) alerts that show on the PFD. The RAD ALT is made up of one or optionally two Radio Altitude Receiver/Transmitter (ALT) and the system antennas. Control inputs are provided by the PFD References Menu. The system supplies digital data to the AFCS, EFIS, and TAWS by way of the IAPS and system bus structure.

Radio Sensor System

The Radio Sensor System (RSS) contains the control heads and radios used for voice communication, VOR/ILS navigation, distance measurement, Automatic Direction Finder (ADF) navigation, TCAS/Mode S communication, and operation within the Air Traffic Control (ATC) environment. The RSS is a dual-independent system made up of pilot and copilot-side radios/sensors. Baseline-equipped aircraft come standard with two VHF COMs, two HF COM systems, two VOR/ILS/MKR/ADF receivers, two DME transceivers, two ATC Mode-S Diversity Transponders with Flight ID, and two GNSS receivers. Tuning control is provided by the Control Display Unit (CDU) and Multifunction Displays (MFD). The RSS provides digital radio data to the EFIS, navigation systems, and hazard avoidance systems by way of the IAPS and system bus structure. Each side RSS (pilot and copilot) is functionally isolated and acts as a stand-alone system. Each side RSS can control the cross-side radios/sensors in the event of a control or display failure.

Synthetic Vision System

The non-Rockwell Collins Synthetic Vision System (SVS) contains a Synthetic Vision Computer (SVC) and two databases (terrain and airport/runway). The SVC provides the crew with a synthetic view of terrain around the aircraft. It is used to estimate the general location and characteristics of the surrounding terrain along with the TAWS. The SVS gives a general sense of the terrain contours, significant terrain features, and whether the terrain is near, above, or below the aircraft altitude. The SVS is not to be used for navigation, and is not a replacement for TAWS.

Traffic Alert and Collision Avoidance System II

The Traffic Alert and Collision Avoidance System (TCAS) is designed to protect a volume of airspace around the TCAS-equipped aircraft by warning the pilots of the threat of other transponder-equipped aircraft penetrating that airspace. The system interrogates Mode C and Mode S transponders in nearby aircraft and analyzes the replies to identify potential and predicted collision threats. The system advises the pilot when to climb, descend, or maintain altitude to avoid passing too close to or colliding with threat aircraft. When an intruder aircraft is equipped with TCAS, the system coordinates avoidance maneuvers with this aircraft using the datalink capability of the Mode S transponders. The TCAS is made up of a TCAS transceiver (TTR for Challenger 605 or TSS for Challenger 650), two antennas, and two Mode S transponder functions (TSS provides left transponder function in Challenger 650). The system receives altitude/vertical speed information from the air data system, radio altitude information from the radio altitude system, and heading information from the IRS. The TCAS system generates vertical guidance commands that show on the Vertical Speed Indicator (VSI) and aural alerts that sound over the flight deck audio system. The controls for operating the TCAS are located on the Display Control Panel (DCP) and Cursor Control Panel (CCP). An ATC/TCAS control menu that shows on the MFD and CDU supplies TCAS mode selection.

Weather Radar System

The Weather Radar System (WXR) locates and depicts the bearing, range, and precipitation rate of all detectable precipitation within the scan area and display range. The Challenger 605 is equipped with a standard RTA-854 Turbulence Weather Radar System (TWR) or an optional RTA-4114 MultiScan™ Weather Radar. The RTA-4114 is standard on the Challenger 650. Both systems locate precipitation-related weather targets, can be used for ground mapping, and show precipitation-related turbulence targets. The primary difference is that the MultiScan system is capable of faster scan rates, database/GPS location comparison for more accurate severity depictions, and contains more accurate depictions during vertical and horizontal attitude changes.

KEY TROUBLESHOOTING FEATURES

- The cross-side air data source can be selected as the source of on-side air data in the event of an on-side failure.
- The cross-side attitude/heading source can be selected as the source of on-side attitude/heading data in the event of an on-side failure or alternate in the case of the optional triple IRS installation.
- Display reversion can be selected to show a composite PFD/MFD format on the remaining display in the event on an on-side PFD or MFD failure.
- Automatic EICAS reversion transfers the display of EICAS data from a failed or powered-down MFD to the cross-side MFD.
- Manual EICAS reversion can be selected to show a compressed EICAS format on both the pilot-side and copilot-side PFD.
- Dedicated self-test modes available for individual subsystems.
- Continuous self-monitoring that activates a system flag when a fault is detected.
- Fault data is recorded by the Maintenance Diagnostic System.

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OVERVIEW – SYSTEM COMPONENTS

INTRODUCTION

This chapter provides a simplified block diagram of the baseline configuration system architecture and descriptions of each of the system components necessary for the operation of the Pro Line 21 Advanced Avionics system.

SYSTEM DESCRIPTION

The Bombardier Challenger 605/650 Pro Line 21 Advanced Avionics System is a fully integrated avionics system. Four Adaptive Flight Displays (AFD) provide display and control functions for the Automatic Flight Control System (AFCS), flight instruments, navigation subsystems, hazard avoidance subsystems, Radio Sensor System (RSS), and Maintenance Diagnostics System (MDS). Two or optionally three Control and Display Units (CDU) provide display and control functions for the Flight Management System (FMS) and RSS. Additional control functions are provided by the Flight Control Panel (FCP), pilot and copilot-side Cursor Control Panels (CCP) and Display Control Panels (DCP), and the remote mounted installed supplied controls. The central feature of the avionics system is the Integrated Avionics Processor System (IAPS). The IAPS provides a central data collection and distribution point for the avionics systems. The AFD and CDU receive data directly from primary sensors as well as concentrated data from the IAPS.

KEY OPERATING FEATURES

- The system is primarily interconnected using ARINC 429 buses.
- The central feature of the avionics system is the Integrated Avionics Processor System (IAPS). The IAPS provides a housing for a number of modules providing lightning and HIRF Protection, data concentration and distribution, Autopilot and flight control functions, Flight Management functions as well as Maintenance Diagnostics and environmental control.
- Data from external systems is provided to the IAPS for use by internal modules and for distribution to other external systems.
- Aircraft systems data is concentrated by dual Data Concentrator Units (DCU), for distribution to the IAPS and the flight deck displays.

- Four Adaptive Flight Displays (AFD) provide for display of primary attitude, heading, altitude, airspeed, navigation, and flight guidance functions. Each display is identical and operates either as a Primary Flight Display (PFD) or Multifunction Display (MFD) based on its location.
- The AFD receives data directly from primary sensors and the DCU as well as concentrated data from the IAPS.
- The Architecture ensures that after loss of data from the IAPS that primary display data is maintained for both plot and copilot-side systems.

Figure 3-1 Challenger 605 Block Diagram

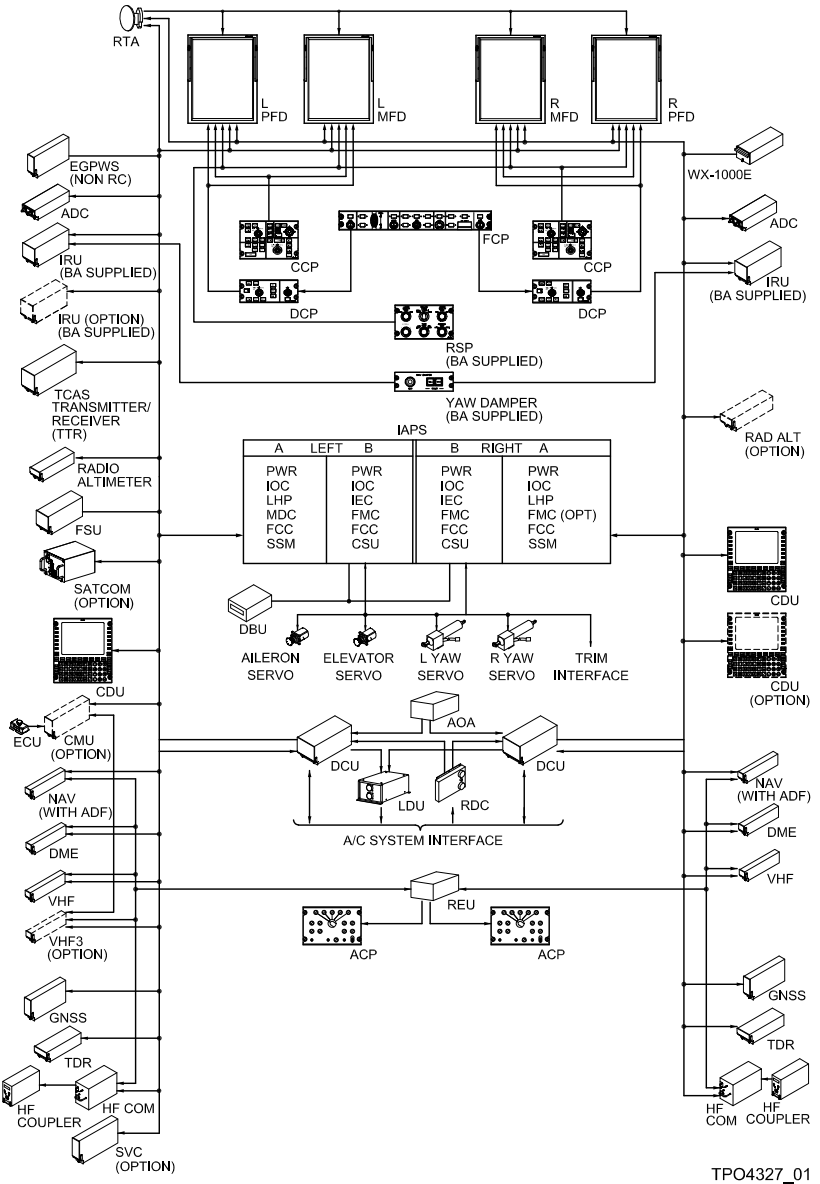
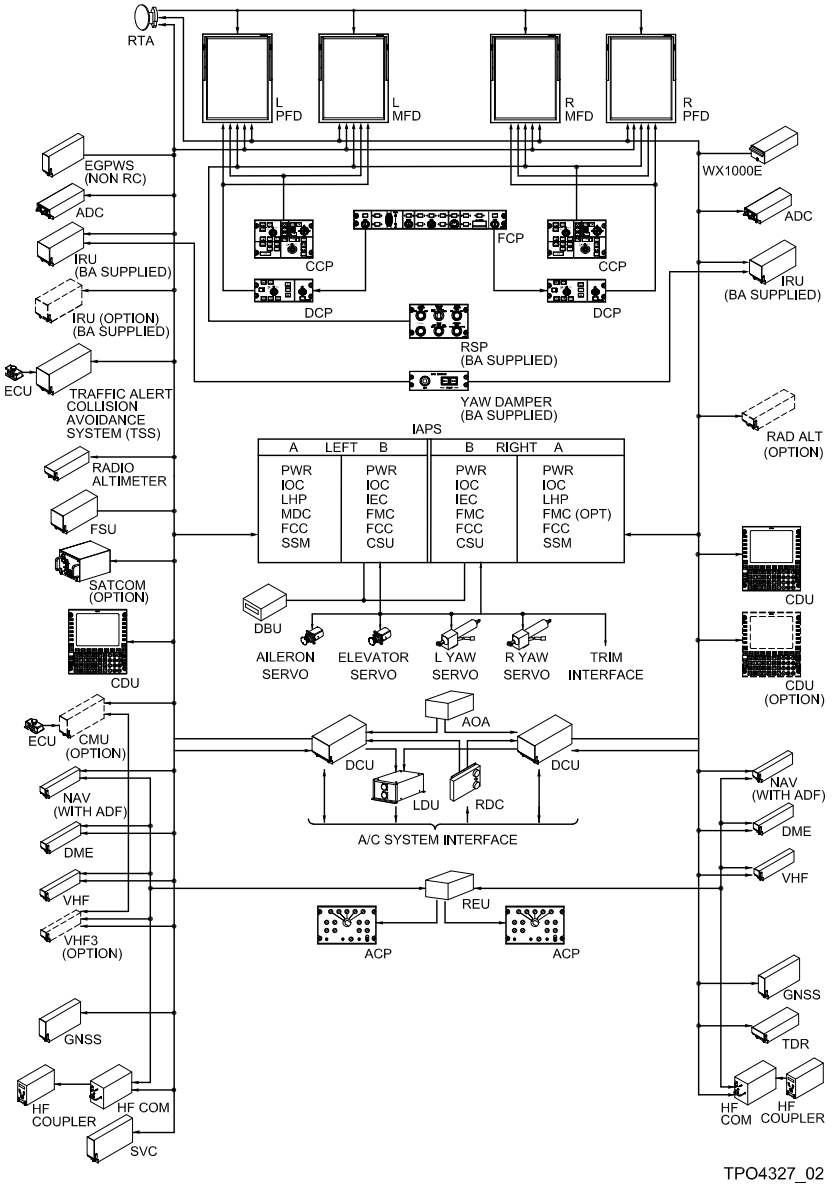
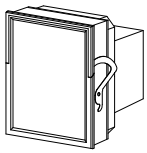


Figure 3-2 Challenger 650 Block Diagram

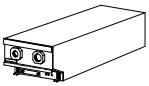


COMPONENTS



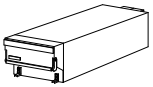
Adaptive Flight Display

The Adaptive Flight Display (AFD) are color, liquid crystal displays that provide display and control functions for the flight instruments, flight guidance, engine instruments, crew alerting system, navigation, hazard avoidance, and communications systems. Four AFD are installed, two at each pilot station. In normal operation, the outboard AFD are configured as Primary Flight Displays (PFD) and the inboard AFD are configured as a Multifunction Displays (MFD). AFD reversion is available in case of an on-side AFD failure.



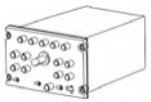
Air Data Computer

The Air Data Computer (ADC) receives pitot/static pneumatic (pressure) and air temperature inputs from the aircraft sensors. The ADC processes the raw data and provides digital air data for display on the AFD and for use by other aircraft subsystems. Two ADCs are installed: one for the pilot-side and one for the copilot-side. ADC reversion is available in case of an on-side ADC failure.



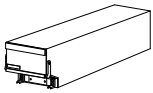
Air Traffic Control Transponder

The Air Traffic Control Transponder (TDR) function supplies identification and altitude reporting for the air traffic control system. The transponder features the familiar 4096 possible Mode A code selections and selectable Mode C altitude reporting. Mode S datalink and diversity reception features that are required for TCAS II operation and Automatic Dependent Surveillance – Broadcast (ADS-B) are also provided. The TDR is also equipped for Flight Identification (Flight ID) elementary and enhanced surveillance. The transponder function is provided by a TSS and right TDR or by dual TDRs.



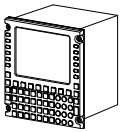
Audio Control Panel

The Audio Control Panel (ACP) provides controls to select the desired receiver(s) source, voice/identifier source, microphone source, transmitter, and emergency backup for the on-side audio system. Two ACPs are installed, one for the pilot-side and one for the copilot-side.



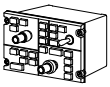
Communication Management Unit

The Communication Management Unit (CMU) supplies information for the AOC and CPDLC (Link2000 and FANS) options or the Universal Weather Graphical Weather (GWX) option. Universal Weather GWX images are requested from the ground station through the use of the control pages on the CDU. The requested GWX images are uplinked through the third COM and supplied to the FSU by way of the CMU. Optionally, the GWX images can be selected to show on the CDU.



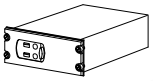
Control Display Unit

The Control Display Unit (CDU) provides centralized control and display functions for the FMS, the optional Datalink system, the RSS, and the optional SATCOM system. Controls on the CDU are used to select FMS Map background symbology for display on the AFD and to request the optional IFIS graphical weather images for uplink. Optionally, the GWX images CDU can be selected to show on the CDU. Two or optionally three CDUs are installed.



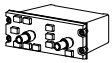
Cursor Control Panel

The Cursor Control Panel (CCP) provides control functions for the on-side MFD. The CCP is used to show MFD menus and sub-menus that are used to control the MFD format and overlay state. Dedicated controls for the MFD navigation format, overlay state, optional E-Charts, CAS, and synoptic displays are resident on the CCP. Two CCPs are installed, one for the pilot-side and one for the copilot-side.



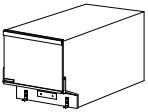
Data Base Unit

The DBU-5010E provides a data loader interface for the LRUs on the aircraft that require database and configuration file updates. Additionally, it provides a means to write information from certain avionics systems to removable media. The Data Base Unit (DBU) is mounted in the aircraft cockpit on the copilot outboard side panel. The DBU transfers files between aircraft LRU and solid state removable flash media (for example, Universal Serial Bus (USB) memory stick or thumbdrive). Refer to Rockwell Collins publication DBU-5000, 5010 and 5010E Data Base Unit Operator's Guide (CPN 523-0808703) for more information.



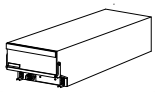
Display Control Panel

The Display Control Panel (DCP) provides control functions for the on-side PFD and MFD. The DCP is used to show PFD menus and sub-menus that are used to control the PFD navigation format, overlay state, NAV source selection and bearing source selection. Dedicated controls for the WXR and for the PFD and MFD display range are resident on the DCP. Two DCPs are installed, one for the pilot-side system and one at the copilot-side system.



Data Concentrator Unit

The Data Concentrator Unit (DCU) collects and organizes data from the on-board systems that interface with the Rockwell Collins avionics systems. The DCU generates visual and aural information for use by the AFD and the flight deck audio system. The DCU performs a master warning/master caution function that controls the setting and clearing of the flight deck master warning and master caution lamps, based on the setting and clearing of warning and caution CAS messages. The DCU contains two independent, segregated channels. One channel provides conversion for each engine and the other provides conversion for aircraft systems data (for example, flap position). Two dual-channel DCUs are installed. Processed DCU data is supplied to the cross-side DCU channel full-time.



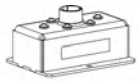
Distance Measuring Equipment

The Distance Measuring Equipment (DME) transceiver provides distance, time-to-station, groundspeed, and station identification information for use by the navigation systems and hazard avoidance subsystems. The DME provides three channels and can track up to three DME stations at a time. Normally, channel one is tuned manually with the on-side Nav receiver and the distance data can be displayed on the AFD. Distance information from channels 2 and 3 is used by the FMS to aid in the calculation of the navigation solution. The selectable FMS autotune feature allows the FMS to automatically tune channel one of each DME. Two DME transceivers are installed.



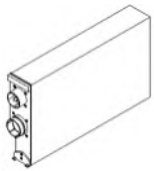
Enhanced Vision System

The optional Enhanced Vision System (EVS) produces a forward-looking image of the natural scene, including the approach lights of the runway, on the MFD. The EVS gives the pilot increased visibility in low-light and obscured environments. The EVS consists of a Forward Looking Infrared (FLIR) camera assembly, an EVS processor assembly, and a window assembly.



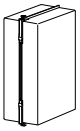
External Compensation Unit

The CMU, FSU, TSS, and FSU each use an External Compensation Unit (ECU). The CMU uses it to store datalink configuration and license storage. The FSU uses it to store license key and configuration storage. The TSS uses it to store configuration data.



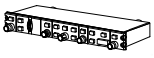
File Server Unit

The File Server Unit (FSU) provides the processing and storage for all Integrated Flight Information System (IFIS) functions including Electronic Charts, Enhanced Maps, and Graphical Weather. FSU supplied IFIS symbology as well as operating and fault messages is available for display on the MFD.



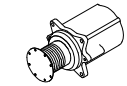
Flight Control Computer

The Flight Control Computer (FCC) provide flight guidance, autopilot, yaw damper, and automatic pitch trim functions. The FCC contains two identical flight guidance channels. Both channels work together to generate lateral and vertical steering commands. The steering commands are supplied to the flight directors and to the autopilot servos when the autopilot is engaged. Yaw damping and turn coordination commands are supplied to the yaw damper actuator when the yaw damper is engaged. Automatic pitch trim arm and command logic is supplied to the onboard pitch trim system when the autopilot is engaged. Four FCCs are installed. One pair is selected for use at a time.

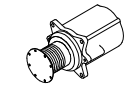


Flight Control Panel

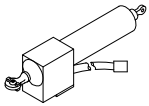
The Flight Control Panel (FCP) provides most of the controls for operation of the AFCS. The FCP provides controls to select the active lateral and vertical mode of operation, to engage and disengage the autopilot, transfer flight guidance, and remove power from the servos. Controls located on the FCP are also used to set the altitude, heading, airspeed, vertical speed, and pitch references. The FCP is a single unit that supplies control inputs to both channels of the selected FCC.



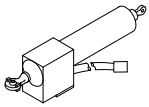
ROLL SERVO



PITCH SERVO



L YAW ACTUATOR



R YAW ACTUATOR

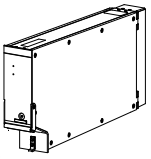
Flight Control System Servos

The pitch servo, roll servo, and yaw actuators physically position the control surfaces in response to commands by the autopilot and yaw damper. Servo position feedback is supplied to the FCCs.



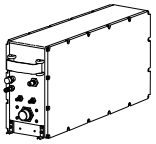
Flight Management Computer

The Flight Management Computer (FMC) generates flight plan-based lateral and vertical point-to-point navigation using an internal Navigation database and multiple navigation sensors. The FMC stores user input flight plans for use in providing guidance used by the FCS and for display on the AFD. Two or three FMC are installed.



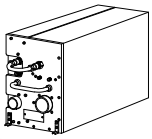
Global Navigation Satellite System Receiver

The Global Navigation Satellite System (GNSS) receiver receives navigation data from a constellation of Global Positioning System (GPS) satellites and Satellite Based Augmentation Systems (SBAS). The GNSS processor uses these inputs to generate precise position, velocity, and time data for use by the navigation and hazard avoidance subsystems. Two GNSS receivers are installed.



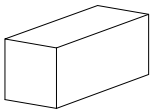
HF Coupler

The High Frequency (HF) antenna coupler quickly matches the power amplifier in the HF COM receiver-transmitter to the antenna to maximize the radiated power. The HF antenna coupler has the ability to learn and store tuning data for 99 preset channels, as well as storing the last 50 manually tuned frequencies. One HF coupler is installed for each HF transceiver that is installed.



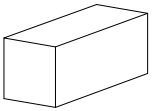
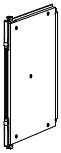
HF Transceiver

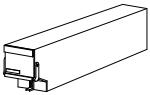
The HF transceiver is a multichannel, high frequency transceiver that provides voice communications. Communication is possible in both simplex and half-duplex in Upper Sideband (USB), Lower Sideband (LSB), Amplitude Modulation Equivalent (AME), Continuous Wave (CW), and Frequency Modulation (FM) mode. The HF transceiver has the capability to store 99 user-programmed preset channels. In addition, 6 emergency channel frequencies and all 176 half-duplex International Telecommunication Union (ITU) maritime radiotelephone channels (public correspondence network) are preprogrammed into nonvolatile memory. Two HF transceivers are installed.



Inertial Reference Unit

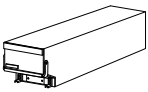
The non-Rockwell Collins Inertial Reference Unit (IRU) generates attitude and heading data, acceleration, and lat/long for display on the AFD and for use by other avionics subsystems. Two or optionally three IRU are installed. IRU reversion is available in case of an on-side IRU failure.

	Integrated Avionics Processor System	The Integrated Avionics Processor System (IAPS) is a part of the on-board wiring and physically houses some avionics units. The IAPS provides a central data collection and distribution point for the avionics systems. The IAPS receives data inputs from on-board systems, sorts the data, then transmits only the data of interest to each receiving avionics unit.
	LDS Processor	The non-Rockwell Collins LDS processor calculates the azimuth and range of the electrical discharges within approximately 200 NM of the present position. The LDS processor generates lightning symbology as well as operating and fault messages for display on the AFD.
	Maintenance Diagnostics Computer	The Maintenance Diagnostics Computer (MDC) uses internal fault isolation logic to identify failed or degraded avionics units. The MDC supplies fault information to the multifunction displays and Data Base Unit (DBU) by way of the IAPS and system bus structure. The MFD fault logic software is periodically updated by way of the DBU. The MDC is physically located in the IAPS card cage.



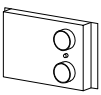
Navigation Receiver

The Navigation (NAV) receiver provides VOR/LOC, glideslope, ADF and marker beacon outputs for display on the AFD and for use by the navigation and hazard avoidance subsystems. The NAV receiver also receives identification and voice audio.



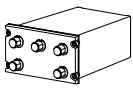
Radio Altimeter

The Radio Altimeter (ALT) measures height Above Ground Level (AGL) from 0 (zero) to 2500 feet. The ALT uses this input to generate digital altitude data for display on the PFD and for use by the hazard avoidance subsystems. One or optionally two ALTs are installed.



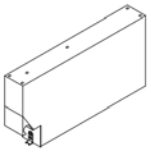
Remote Data Concentrator

The Remote Data Concentrator (RDC) expands the data input to the Data Concentrator Unit (DCU). The RDC digitizes discrete data from on-board subsystems for use by the DCU.



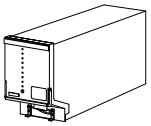
Reversion Switch Panel

The non-Rockwell Collins Reversion Switch Panel (RSP) provides the means for selecting ADC reversion and AFD reversion. The AFCS 1 or 2 selection is also controlled by way of the RSP. The RSP is also used to control the location of the master EICAS presentation.



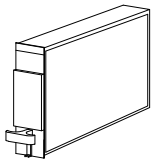
Synthetic Vision Computer

The Synthetic Vision Computer (SVC) takes inputs from the IRS, GNSS, AFD, FMC, and IOC to determine the exact location of the aircraft, direction of travel, altitude above the terrain, and attitude before placing the sky/ground depiction on the AFD. The SVC also puts depictions of the physical terrain in place of the standard brown depiction for ground, based on a terrain database.



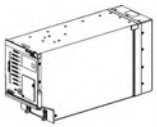
TCAS Transceiver (Challenger 605 Aircraft Only)

The TCAS Transceiver (TTR) interrogates nearby Mode C and Mode S transponders and analyzes the replies to identify potential collision threats. The TTR advises the pilot when to climb, descend, or maintain altitude to avoid passing too close to or colliding with threat aircraft. When an intruder is equipped with TCAS, the TTR coordinates avoidance maneuvers by way of the datalink capability of the Mode S Transponders. TTR generated vertical speed cues, as well as operating and fault messages, show on the PFD. TTR generated intruder symbology is available for display on the AFD.



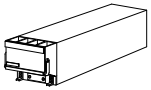
Terrain Awareness Warning System Computer

The Terrain Awareness Warning System (TAWS) computer uses an internal worldwide database, aircraft information from the DCU, position data from the FMS, GNSS and NAV receivers, altitude data from the ALT, attitude and heading data from the IRS, processed air data information from the ADC, and AOA data from the AOA transmitter to generate terrain symbology for display on the AFD.



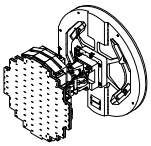
Traffic Surveillance System (Challenger 650 Aircraft Only)

The Traffic Surveillance System (TSS) interrogates nearby Mode C and Mode S transponders and analyzes their replies to identify potential collision threats. The TSS advises the pilot when to climb, descend, or maintain altitude to avoid passing too close to or colliding with threat aircraft. When an intruder aircraft is also equipped with TCAS, the TSS coordinates avoidance maneuvers through the datalink capability of the Mode S Transponders. TSS generated collision avoidance visual fly-to directives are provided by path cues on the AFD attitude displays. TCAS operating and fault messages are also shown on the AFD. TSS generated intruder symbology can be shown on any of the flight displays. One TSS is installed.



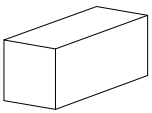
VHF Communications Transceiver

The VHF Communications Transceiver (COM) is a multichannel VHF transceiver that provides AM voice communications. Two or optionally three COMs are installed. The standard frequency range COMs may optionally be replaced with extended-range COMs. The third COM is a datalink capable COM when the optional IFIS Universal Weather GWX option is installed.



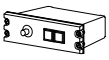
Weather Radar System Receiver Transmitter Antenna

The Receiver Transmitter Antenna (RTA) is an integrated Receiver/Transmitter Antenna unit located in the radome. The RTA is made up of an antenna, antenna tilt and scan motors, a Receiver/Transmitter and signal processing circuits. RTA generated precipitation related weather targets as well as system operation and fault messages are available for display on the AFD.



XM Satellite Receiver

The XM Satellite Receiver is required for installations that include the XM Satellite Weather Graphical Weather (GWX) option. The optional non-Rockwell Collins XM Satellite datalink receiver receives a constant broadcast of graphical and textual weather data from the XM Satellite Radio weather service. The digital weather data as well as operating and fault data is supplied to the MFD by way of the File Server Unit.



Yaw Damper Panel

The non-Rockwell Collins Yaw Damper Panel provides independent engage controls for yaw damper 1 and yaw damper 2 and a disconnect control for both of the dual-independent yaw dampers.

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CDU OPERATION

INTRODUCTION

The CDU(s) provide the normal means of radio tuning. Each CDU provides single point of control of both on-side and cross-side radios from the pilot or copilot position.

A top level tune page shows to give an at-a-glance view of all of the installed radios. When optional equipment is installed, it may be necessary to have two tuning pages. Each radio has its own control page. The control page provides control of all radio functions, including the primary functions available on the top level tune page, as well as the list of numbered presets where applicable. A control page is available for each of the radios that follow:

- VHF COM Radio Tuning and Control
- NAV Radio Tuning and Control
- ADF Radio Tuning and Control
- DME Radio Tuning and Control
- ATC Transponder selection, Ident Beacon Code, and Mode
- TCAS II Control
- HF Radio Tuning and Control.



NOTE

When the FMS AUTO TUNE mode is active, the FMC may automatically tune the NAV and DME radios.

Data entry into the CDU is made by way of the scratchpad line. Alphanumeric data is entered into the scratchpad line from the keyboard and then transferred to the appropriate location with the line select keys. If the contents of the scratchpad (for example, frequency, transponder ID, etc.) are not appropriate for the select key that is selected, the data is not transferred and a warning message shows in the scratchpad. There are three different methods available for radio tuning from the CDU:

- Direct Tuning — the desired frequency, preset number, or (for NAV radios only) station identifier is entered in the scratchpad. Push the appropriate line select key to transfer the contents of the scratchpad to the appropriate radio.

- Recall Tuning — (applicable to COM radios only) push the Recall number line select key for the appropriate radio to tune the radio to the Recall frequency.
- Numbered Preset — on a radio control page and with the scratchpad empty, push the appropriate line select key to tune the radio to the already-entered preset frequency.

Controls located on the CCP are also used for radio control as follows:

- The DME-H (hold) button is used to select the DME hold feature for the selected DME when the active NAV source is VOR or LOC (push-on/push-off).
- The IDENT button is used to command the ATC IDENT feature.
- The ATC button may be used to set the ATC Transponder mode to STBY if the current mode is not STBY and to TA/RA if the current mode is STBY.



TECH DETAIL

Altitude Reporting Source Selection

The ATC transponder receives altitude data from its commanding CDU. During normal operation, the CDU passes through to the transponder altitude data received from whichever ADC the Autopilot or Flight Director is, or will be, coupled to. If altitude data from the coupled ADC is invalid or missing and the altitude data from the cross-side ADC is valid, the cross-side data is passed through to the transponder. If an ADC has been deselected by way of the ADC reversion switch, its altitude data is not passed through.

Each CDU also supports full reversionary tuning of the cross-side radios (except for FMS AUTO TUNE) if the cross-side CDU fails. Refer to the Remote Switches section of this guide for details on tuning reversion.

The Audio Control Panel (ACP) installed at the pilot and copilot positions provides control of the microphone selection, volume controls, and voice/identifier selection at the installed position. Refer to the Audio Control Panel (ACP) section of this guide for details on operation of the ACP.

GENERAL CDU TUNING

RATIONALE:

The CDU(s) provide primary tuning control for all installed radios. This includes all COM, NAV, and ADF radio frequencies, ATC transponder selection, beacon ident codes, TCAS modes of operation, and DME Hold.

SUMMARY:

The CDU TUNE pages are the primary radio tuning control. The TUNE page is in view when the TUN key on the keyboard is pushed. The TUNE page normally provides an operational summary and primary control of all installed COM, NAV, DME, ADF, and ATC radios and TCAS. The HF and optional third VHF radio tuning and control is located on the TUNE 2/2 page.



PRECONDITIONS:

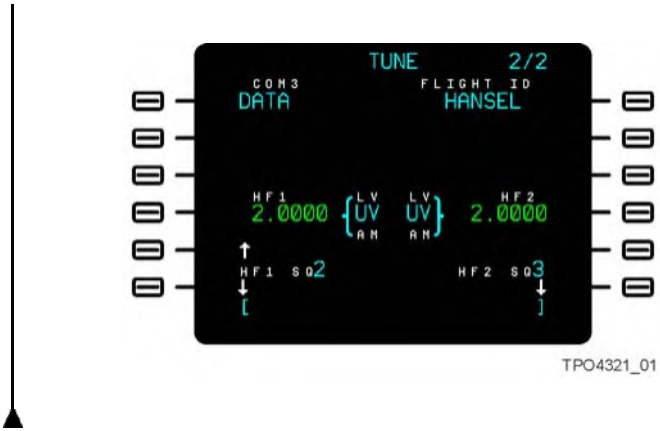
Push TUN key to show the first TUNE page. Push the NEXT or PREV keys to toggle the CDU display between the first (TUNE 1/2) and second (TUNE 2/2) tuning pages. The first radio TUNE page also shows when the radio line select key L1 or R1 on any control page is pushed with no value entered in the scratchpad.

The first radio control page is in view when the radio line select key L1 or R1 on the TUNE page is pushed with no value entered in the scratchpad. Subsequent pushes of the NEXT or PREV move the CDU

display either forward or backward through each of the radio control pages.

▼ CHECKLIST:

1	Radios are directly tuned by changing the active frequency or code. The value of the desired change is entered by way of the keyboard into the scratchpad, then the appropriate line select key is pushed to insert the desired change. Active values show in green.
2	COM radios are recall tuned by entering a recall frequency or preset number into the scratchpad, then selecting the RECALL line select key. The recall frequency becomes the active frequency and the previously active frequency becomes the recall frequency. Recall frequencies show in white.
3	Preset tuning is accomplished by selecting the line select key next to the desired preset frequency. This action automatically tunes the associated radio to the preset value. Preset frequency values show in white and their corresponding descriptions show in cyan.
4	The FLIGHT ID field on the TUNE 2/2 page is used to enter the Flight Identification code. Up to eight alphanumeric characters may be entered into the FLIGHT ID field. The FLIGHT ID code may also be entered or changed on the ATC control page. **NOTE** Once it has been entered, the Flight ID setting shows in green if a transponder is active, otherwise it shows in white.



POST CONDITIONS:

Each installed radio is also provided with a CDU control page that provides all radio functions that show on the TUNE page plus mode control where appropriate and the list of numbered presets.

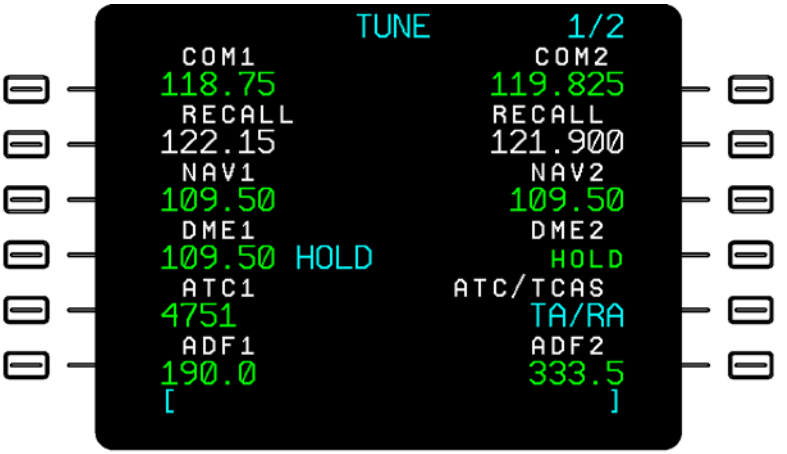
COM RADIOS – TUNING AND CONTROL

RATIONALE:

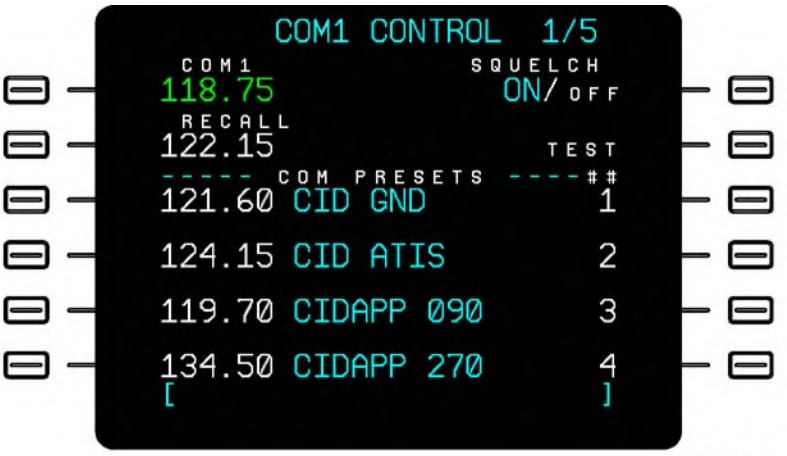
The COM1 and COM2 radio frequencies may be tuned from either the first TUNE page or from the COM-specific control pages. When the optional COM3 datalink transceiver is installed, the COM3 active and recall fields are located on the TUNE 2/2 page. COM radio control and frequency presets are controlled from the COM-specific control pages. If the optional COM3 transceiver is in DATA mode, DATA is shown as the active frequency and the RECALL field is not displayed.

SUMMARY:

The active and recall frequencies are tuned from the COM1 and COM2 active frequency lines on the first TUNE page or from the COM-specific control pages. COM radio squelch, self test, and preset tuning assignments are all controlled from the COM-specific control pages.



TPO4324_01



TPO4618_04

PRECONDITIONS:

If the value entered in the scratchpad is an invalid frequency or preset number when the line select key is pushed, the scratchpad displays either INVALID FREQUENCY (for 25-kHz tuning) or INVALID CHANNEL (for 8.33-kHz tuning) for approximately 1 second and then displays the original scratchpad contents. The active radio frequency remains unchanged.

RULES:

The active COM frequency shows in green for approximately 3 seconds after entry and whenever the commanded CDU frequency matches the echo frequency from the radio. If the echo frequency from the radio is flagged as failed, missing, or does not match the commanded frequency, the active frequency value shows in yellow. The numbered preset data shows in green when it is selected and its active frequency is valid. The numbered preset data shows in yellow when it is selected and its active frequency is invalid. The conditions that follow define a valid frequency or preset number entry.

- For 25-kHz spacing:
  - Frequency tuning increments are 25 kHz with a display precision to tens of kHz.
  - Three to five digits have been entered (for example, 1185 = 118.50).
  - The digits entered are between 118 and 13697 for the baseline COM radio frequency range (118.00 to 136.97 MHz) or between 118 and 15197 (118.00 to 151.97 MHz) for the optional expanded COM frequency range.
- For 8.33-kHz spacing:
  - Tuning entries are made by channel name (decimal points and trailing zeros are not required).
  - Three to six digits have been entered (for example, 1185 = 118.500).
  - The digits entered are between 118 and 136990 for the baseline COM radio frequency range (118.00 to 136.97 MHz) or between 118 and 151975 (118.000 to 151.975 MHz) for the optional expanded COM frequency range.
  - Only 25-kHz spacing is available for the 137.000 to 151.975 MHz frequency range.
- Valid preset numbers are from 1 to 20.

▼ CHECKLIST:

1	To show the first TUNE page, either push the TUN key on the CDU keyboard, or while on the COM-specific control page, push the line select key next to the COM radio active frequency line with the scratchpad empty.
2	To show the first COM-specific control page, while on the TUNE page push the line select key next to the COM1 or COM2 active frequency line with the scratchpad empty.
3	Push the PREV or NEXT keys on the CDU keyboard to advance forward or backward through pages.

COM RADIOS – DIRECT TUNING

SUMMARY:

COM radios are manually tuned by entering the desired frequency into the scratchpad and then using a line select key to command the new frequency.

▼ CHECKLIST:

1	Enter the desired frequency or preset number in the scratchpad.
2	From the TUNE page, push the appropriate line select key (L1 for COM1 or R1 for COM2) to transfer the contents of the scratchpad to the radio active frequency field.
3	From the COM-specific control 1/X page, push line select key L1 to transfer the contents of the scratchpad to the radio active frequency field.

4 The frequency that was active is transferred to the RECALL field and shows in white.

COM RADIOS – RECALL TUNING

SUMMARY:

COM radios are tuned to a previously active frequency by the RECALL line select key.

▼ CHECKLIST:

1 From the TUNE page, push the appropriate line select key (for example, L2 for COM1 or R2 for COM2) to transfer the frequency in the RECALL field to the active frequency field. The previously active frequency is transferred to the RECALL field (the RECALL and active frequencies swap).

2 From the COM-specific control 1/X page, push line select key L2 to transfer the frequency in the RECALL field to the active frequency field. The previously active frequency is transferred to the RECALL field (the recall and active frequencies swap).

3 To change the RECALL frequency value:

1. Enter the desired frequency or preset number in the scratchpad.
2. Push the appropriate line select key (for example, L2 for COM1 or R2 for COM2) to transfer the scratchpad frequency value or the frequency associated with the preset number to the RECALL field.

COM RADIOS – PRESET TUNING

SUMMARY:

COM radios are tuned to a preset frequency by using line select keys on the COM-specific control pages. The frequency value and/or identifier assigned to a preset can also be changed.

▼ CHECKLIST:

1 With the scratchpad empty, push the line select key next to the desired numbered preset to transfer the numbered preset frequency to the active frequency field. The previously active frequency is transferred to the RECALL frequency field and changes color to white.

NOTE

The numbered preset data line changes color to green if the frequency is valid. If the preset frequency is invalid, the preset data line changes color to yellow.

2 To set or change a preset frequency value and/or identifier:

1. Enter a valid frequency value and/or identifier in the scratchpad. A valid identifier is from 1 to 13 characters long, starts with an alpha character, and may contain either alpha or numeric characters.
2. Push the appropriate line select key to transfer the preset frequency and/or identifier from the scratchpad to the numbered preset frequency field. The frequency value shows in white and its identifier (if present) shows in cyan.

NOTE

If an invalid frequency value is entered in the scratchpad, when the line select key is pushed, an INVALID FREQUENCY message shows in the scratchpad and then the original scratchpad contents show.

COM RADIOS – SQUELCH MODE CONTROL

PRECONDITIONS:

The squelch mode (ON or OFF) is controlled from the COM-specific control 1/X page.

CHECKLIST:

1 From the COM-specific control 1/X page, push line select key R1 to change the squelch state. The selected state (ON or OFF) is enlarged and shows in cyan.

NOTE

If squelch is turned off, an SQ advisory shows in cyan in the COM1/COM2 line on the TUNE page.

COM RADIOS – SELF TEST CONTROL

PRECONDITIONS:

The COM radio self-test mode is controlled from the COM-specific control 1/X page.

▼ CHECKLIST:

1 From the COM-specific control 1/X page, push the TEST line select key R2 to initiate a self test of the COM radio. During the self test, the TEST legend is enlarged and shows in cyan.

NOTE

When the self test is complete (after approximately 10 seconds), the TEST legend shrinks and shows in white letters.

- The TEST legend enlarges when active and changes color to cyan.
- A single aural beep at the conclusion of the test indicates pass. A double beep indicates fail.
- Faults determined during self-test are stored in archival fault storage memory
- The self-test period is approximately 10 seconds.
- The TEST legend is reduced in size and shows in white when the self-test inactive.

COM RADIOS – MESSAGES

▼ CHECKLIST:

1 A cyan TX advisory shows to the right of the COM 1 legend or to the left of the COM 2 legend on the TUNE page to indicate that radio is transmitting.

2 A cyan SQ advisory shows next to the radio legend (and next to the TX advisory if it is present) to indicate that squelch for that radio has been turned off.

COM RADIOS – EMERGENCY RADIO TUNING

RATIONALE:

In the event of a failure of all radio tuners (CDUs and MFDs), the procedure that follows can be used to tune COM1 to the emergency frequency of 121.5 MHz.

▼ CHECKLIST:

1 Activate all three TUNE INHIBIT switches, CDU1 INHIB, CDU2 INHIB and MFD INHIB on the tuning reversion panel.

NOTE

Refer to the TUNING REVERSION section for additional information on the operation of the tuning reversion panel.

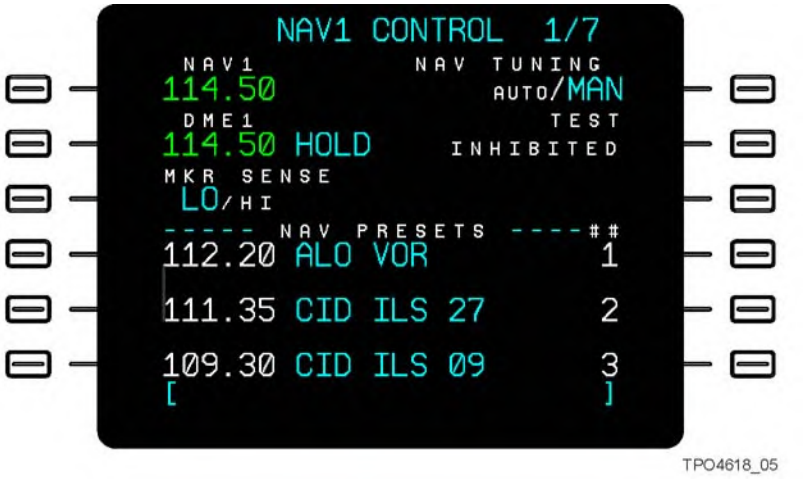
NAV/DME RADIO TUNING AND CONTROL

RATIONALE:

The NAV1/DME1 and NAV2/DME2 radio frequencies may be tuned from either the first TUNE page or from the NAV-specific control pages. NAV and DME radio control and frequency presets are controlled from the NAV-specific control pages. The CDU(s) can also receive VOR and DME tuning commands from the FMS.

SUMMARY:

The NAV radios may be tuned by frequency, preset number, or station identifier. The first TUNE page and the NAV-specific control pages display both frequency and station identifier (if the radio has been manually tuned by station identifier). Active flight plan origin, destination, or alternate airport localizer frequency may be tuned by the localizer identifier. If an identifier has been selected that has multiple associated frequencies the TUNE page may be replaced with a SELECT NAVAID page, refer to the NAV/DME Radios — SELECT NAVAID section for additional information.



PRECONDITIONS:

If the value entered in the scratchpad is an invalid frequency or preset number when the line select key is pushed, the scratchpad displays INVALID FREQUENCY for approximately 1 second and then displays the original scratchpad contents. The active NAV radio frequency remains unchanged.

If an identifier is entered and not found in the database, the scratchpad displays NOT IN DATABASE for approximately 1 second and then displays the original scratchpad contents.

If an invalid station identifier is entered, the scratchpad displays INVALID IDENTIFIER for approximately 1 second and then displays the original scratchpad contents. The active NAV radio frequency remains unchanged.

RULES:

The active NAV frequency, identifier, and DME hold frequency show in green for approximately 3 seconds after entry and whenever the commanded CDU frequency matches the echo frequency from the radio. If the echo frequency from the radio is flagged as failed, missing, or does not match the commanded frequency, the active frequency and identifier show in yellow. The conditions that follow define a valid frequency, preset number entry, and NAVAID identifier:

- Frequency tuning increments are 50 kHz with a display precision to tens of kHz.

- Frequency restrictions:
  - Three to five digits have been entered (for example, 1081 = 108.10).
  - The digits entered are between 108 and 11795 for the baseline NAV and DME radio frequencies (108.00 to 117.95 MHz). If the extended DME frequency range option has been installed, the digits entered are between 108 and 11795 (108.00 to 117.95 MHz) and between 133 and 13595 (133.00 to 135.95 MHz).
- For a preset, valid preset numbers are from 1 to 20.
- For NAVAID identifiers:
  - Two to four characters have been entered.
  - The characters entered are alpha and/or numeric.
  - The character string contains at least one alpha character.
  - The character string may start with a numeric character.

▼ CHECKLIST:

1	To show the first TUNE page, either push the TUN key on the CDU keyboard or while on the NAV-specific control page push the line select key next to the NAV radio active frequency line with the scratchpad empty.
2	To show the NAV-specific control pages, while on the TUNE page push the line select key next to the NAV1 or NAV2 active frequency line with the scratchpad empty.
3	Push the PREV or NEXT keys on the CDU keyboard to advance forward or backward through pages.

NAV/DME RADIOS – FREQUENCY SELECTION

SUMMARY:

Active NAV radio frequencies can be changed from a TUNE page, a NAV-specific control page, or a SELECT NAVAID page. Frequencies

can be changed by direct entry, by the selection of a predefined preset frequency, or by entering a NAVAID station identifier (for example, CID). If the frequency is selected by an identifier, the frequency and identifier show in the active frequency field.

▼ CHECKLIST:

1	To change the active NAV radio frequency from a TUNE page, enter a valid NAV frequency, station identifier, or preset number in the scratchpad and then push the LSK next to the NAV radio legend. The frequency value or the frequency associated with the station identifier or preset number is transferred to the NAV radio active frequency field.
2	To change the active NAV radio frequency from the NAV-specific control page, perform one of the following steps: 1. Enter a valid NAV frequency, station identifier, or preset number in the scratchpad and then push line select key L1 next to the NAV radio legend. 2. With the scratchpad empty, push an associated line select key with a preset number/frequency to transfer the preset frequency to the NAV radio active frequency field.
3	To change the active NAV radio frequency from a SELECT NAVAID page: With the scratchpad empty, push a line select key next to the desired NAVAID to transfer the NAVAID frequency and identifier to the NAV radio active frequency field.

NAV/DME RADIOS – DME HOLD

SUMMARY:

If DME HOLD is not active, the DME radio is tuned to the same frequency as the on-side NAV radio.

▼ CHECKLIST:

1 To place a DME radio in the DME HOLD mode, on the TUNE page push the DME1 line select key L2 or DME2 line select key R2. Operation of the dedicated DME-H button on the CCP can also be used to select DME HOLD mode.

1. The active DME frequency associated with the on-side NAV radio is placed in HOLD.
2. The HOLD legend next to the active DME frequency field shows enlarged and in cyan on both the TUNE and NAV-specific control pages.
3. The NAV tuning mode will change to MAN if AUTO had been selected.

2 To cancel the DME HOLD mode, with the scratchpad empty push the DME1 line select key L2 or DME2 line select key R2. Operation of the dedicated DME-H button on the CCP can also be used to deselect DME HOLD mode. The HOLD legend next to the active DME frequency field shows small and in white on both the TUNE and NAV-specific control pages.

POST CONDITIONS:

The DME frequency shows in yellow on both the TUNE and NAV-specific control pages if DME HOLD is inactive and the echoed frequency is flagged or missing.

NAV/DME RADIOS – OPERATION CONTROL

SUMMARY:

In addition to allowing tuning of the active NAV and DME frequencies, the NAV-specific radio control pages are used to control the NAV and DME radio features that follow:

- NAV test function

- DME Hold
- Independent DME tuning
- NAV AUTO tune selection
- NAV presets
- Marker beacon sensitivity selection.

▼ CHECKLIST:

1

To change the NAV tuning mode, push the NAV TUNING line select key R1. The possible mode selections are AUTO or MAN. The selected mode shows enlarged and in cyan.

1. If the AUTO mode is selected, a cyan AUTO legend shows on the TUNE page next to the NAV active frequency field.
2. If one of the conditions that follow exist, manual (MAN) mode is automatically selected:
   - FMC Mode Word 1 is received indicating NAV radio tuning is in manual mode.
   - FMC Mode Word 3 is received indicating the FMC will not auto tune the NAV radio.
   - FMC Mode Word 1 and/or FCC Mode Word 3 is flagged or missing.
   - The NAV radio is manually tuned by a CDU.
   - The DME is manually tuned or DME HOLD mode is activated.
3. If the FMS indicates the AUTO tune mode will not be accepted and the line select key R1 is pushed, an AUTO NOT AVAILABLE message shows in approximately 1 second and the tuning mode remains in Manual (MAN).

2 To initiate a NAV/DME self test, push the TEST line select key R2. The TEST legend enlarges and shows in cyan when the self test is active. A NAV/DME self-test cycle lasts approximately 10 seconds. A white TEST INHIBITED advisory shows on the TEST line if either the FCC Mode Word 3 is received with inhibit active or if the NAV TEST INHIBIT input is present. When the test is active:

- The TEST legend enlarges when active and changes color to cyan.
- The VOR bearing shows 360 degrees and all Marker Beacons go active when a VOR frequency is tuned.
- A fly up/fly left is indicated and all Marker Beacons go active when a LOC frequency is tuned.
- DME distance shows 100 NM.
- TTG shows 60 minutes.
- Station ID shows AOK.
- Groundspeed shows 100 knots.
- The test period is approximately 10 seconds.
- The TEST legend is reduced in size and shows in white when the self-test inactive.

CAUTION

Do not initiate a NAV/DME self test if a VOR or LOC is the current or preselect navigation source, as changes to the flight director guidance and active modes may occur.

3 To set or change a preset frequency value and/or identifier:

1. Enter a valid frequency value and/or identifier in the scratchpad. A valid identifier is from 1 to 13 characters long, starts with an alpha character, and may contain either alpha or numeric characters.

2. Push the appropriate line select key to transfer the preset frequency and/or identifier from the scratchpad to the numbered preset frequency field. The frequency value shows in white and its identifier (if present) shows in cyan.
3. Use the keyboard NEXT or PREV function keys to move between control pages.

NOTE

If an invalid frequency value is entered in the scratchpad, when the line select key is pushed an INVALID FREQUENCY message shows in the scratchpad and then the original scratchpad contents show.

NAV/DME RADIOS – SELECT NAVAID OPERATION

RATIONALE:

The entry of a station identifier with more than one VOR, DME, or localizer associated with it will cause the SELECT NAVAID page(s) to show unless one of the NAVAIDs is within 300 Nautical Miles (NM) of the aircraft FMS position. In that case, the frequency associated with that NAVAID is returned and the SELECT NAVAID page does not show.

SUMMARY:

The SELECT NAVAID page provides distinguishing information to allow the pilot to select the desired NAVAID. If there are more than two possible NAVAIDs, the data shows on subsequent SELECT NAVAID pages, which show the current page number of the total number of pages in the upper right corner separated by a / (slash). The NEXT and PREV function keys are used to move between SELECT NAVAID pages.

RULES:

The possible NAVAIDs are ordered by increasing distance from the aircraft FMS position.

▼ CHECKLIST:

1	To select a NAVAID, push the associated line select key. This will transfer the NAVAID frequency and identifier data to the NAV radio active frequency field.

POST CONDITIONS:

Upon completion of the NAVAID selection, the page that is in view automatically changes back to the page from where the NAVAID page was entered.

NAV/DME RADIOS – FMS DME TUNING

SUMMARY:

Each FMS will automatically tune the on-side VOR and associated DME channel when the AUTO NAV tuning mode is active, otherwise this DME channel will be tuned to the frequency for the associated manually tuned VOR, or the DME HOLD / directly tuned DME frequency. The remaining two channels of each DME will be automatically tuned by the on-side FMS regardless of the NAV tuning mode selection.

▼ CHECKLIST:

1	

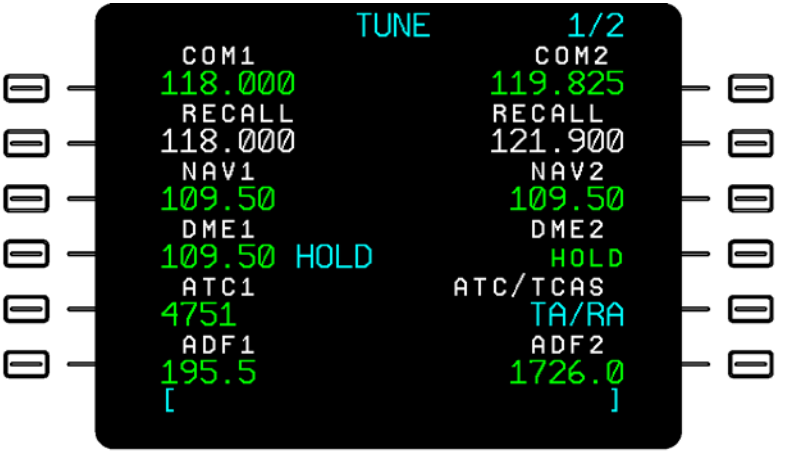
ADF RADIO TUNING AND CONTROL

RATIONALE:

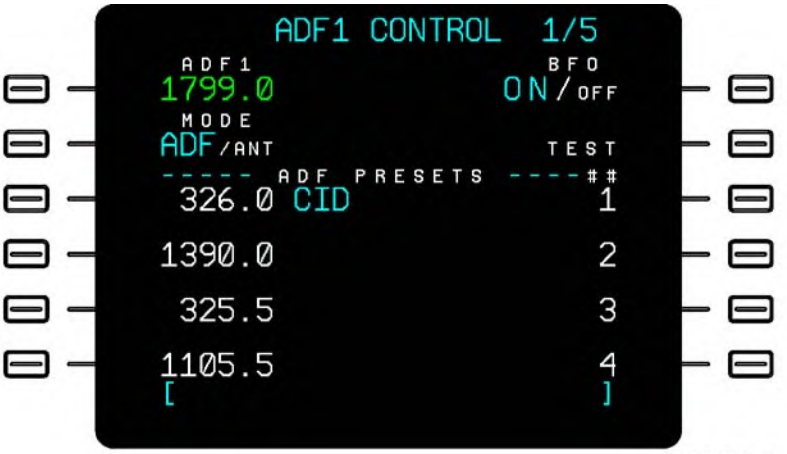
The ADF1 and ADF2 radio frequencies may be tuned from either the first TUNE page or from the ADF-specific control pages. ADF radio mode control and frequency presets are controlled from the ADF-specific control pages.

SUMMARY:

The active ADF frequencies are tuned from the ADF1 and ADF2 frequency lines on the first TUNE page or from the ADF-specific control pages. The ADF MODE, BFO, self test and present tuning selections are all controlled from the ADF-specific control pages.



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TPO4618_08

▼ CHECKLIST:

1	To show the first TUNE page, either push the TUN key on the CDU keyboard, or while on the ADF-specific control page, push the line select key next to the ADF1 or ADF2 active frequency line with the scratchpad empty.
2	To show the ADF-specific control page, while on the TUNE page push the line select key next to the ADF1 or ADF2 active frequency line with the scratchpad empty.
3	Push the PREV or NEXT keys on the CDU keyboard to advance forward or backward through pages.

ADF RADIO – FREQUENCY TUNING

PRECONDITIONS:

The active ADF frequencies are tuned from the ADF1 and ADF2 frequency lines on the first TUNE page or from the ADF-specific control pages.

If the value entered in the scratchpad is an invalid ADF frequency or preset number when the line select key is pushed, the scratchpad displays INVALID FREQUENCY for approximately 1 second and then displays the original scratchpad contents. The active ADF radio frequency remains unchanged.

RULES:

The active ADF frequency shows in green for approximately 3 seconds after entry and whenever the commanded CDU frequency matches the echo frequency from the radio. If the echo frequency from the radio is flagged as failed, missing, or does not match the commanded frequency, the active frequency value shows in yellow. The numbered preset data shows in green when it is selected and its active frequency is valid. The numbered preset data shows in yellow when it is selected

and its active frequency is invalid. The conditions that follow define a valid frequency or preset number entry:

- ADF tune increments are 500 Hz but show to a precision of 100 Hz.
- Three to five digits have been entered (for example, 1905 = 190.5).
- For the baseline ADF radio frequency range, the digits entered are between 190 and 17990 in increments of 0.5 (190.0 to 1799.0 kHz).
- If the optional ADF frequency range has been added, the digits entered are between 190 and 17990 in increments of 0.5 (190.0 to 1799.0 kHz), and between 21790 and 21850 (2179.0 to 2185.0 kHz).
- Valid preset numbers are from 1 to 20.

▼ CHECKLIST:

1 To change the active ADF radio frequency from a TUNE page, enter a valid ADF frequency or preset number in the scratchpad and then push the line select key next to the ADF radio legend. The frequency value or the frequency associated with the preset number is transferred to the ADF radio's active frequency field.

2 To change the active ADF radio frequency from the ADF-specific control page, perform one of the following steps:

1. Enter a valid ADF frequency or preset number in the scratchpad and then push line select key L1 next to the ADF radio legend.
2. With the scratchpad empty, push a line select key associated with a preset number/frequency to transfer the preset frequency to the ADF radio active frequency field.

ADF RADIO – OPERATION CONTROL

SUMMARY:

The ADF-specific radio control page is used to control ADF mode, BFO selection, self-test initiation, and preset tuning frequency definition.

▼ CHECKLIST:

1	To change the ADF operating mode between ADF and ANT, push the MODE line select key L2. The selected mode shows enlarged and in cyan. If the ANT mode is selected, a cyan ANT advisory shows on the TUNE page next to the ADF active frequency field.
2	To change the BFO mode between ON and OFF, push the BFO line select key R1. The selected mode shows enlarged and in cyan. If the BFO is selected ON, a cyan BFO advisory shows on the TUNE page next to the ADF active frequency field.
3	To initiate an ADF self test, push the TEST line select key R2. The TEST legend enlarges and shows in cyan while the self test is active. An ADF test cycle lasts approximately 10 seconds.
4	To set or change a preset frequency value and/or identifier: 1. Enter a valid frequency value and/or identifier in the scratchpad. A valid identifier is from 1 to 13 characters long, starts with an alpha character, and may contain either alpha or numeric characters.

2. Push the appropriate line select key to transfer the preset frequency and/or identifier from the scratchpad to the numbered preset frequency field. The frequency shows in white and its identifier (if present) shows in cyan. If an invalid frequency value is entered in the scratchpad, when the line select key is pushed an INVALID FREQUENCY message shows in the scratchpad and then the original scratchpad contents show.
3. Use the keyboard NEXT or PREV function keys to move between control pages.

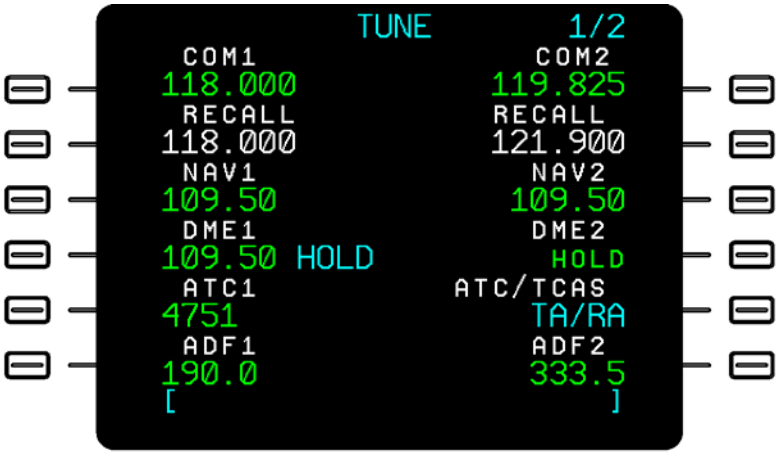
ATC TRANSPONDER CONTROL

RATIONALE:

The active ATC transponder ident beacon code selection function may be tuned from either the first TUNE page or from the ATC control page. ATC transponder ident beacon codes and control functions for the active transponder are controlled from the ATC control page.

SUMMARY:

The ident beacon code for the active ATC transponder can be set from the ATC# line (# indicates the active transponder) on the first TUNE page or the ATC control page. Active ATC transponder mode and status settings are controlled from the ATC control page.



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TPO4319_01

▼ CHECKLIST:

1	To show the TUNE page, either push the TUN key on the CDU keyboard or while on the ATC control page push line select key L1 (next to the active transponder ID) with the scratchpad empty.

2	To show the ATC control page while on the TUNE page, push line select key L5 (next to the active transponder ID) with the scratchpad empty.

ATC IDENT CODE SELECTION

PRECONDITIONS:

The active ATC ident beacon code selection is made with the ATC# line select key.

RULES:

A valid ATC transponder ident beacon code is defined as follows:

- One to four digits have been entered (for example, 1 = 0001).
- The digits entered are within a range of from 0 (zero) to 7777 (inclusive).

▼ CHECKLIST:

1	If a valid ATC beacon code value is entered in the scratchpad, operation of the ATC# line select key transfers the value from the scratchpad to the active beacon code display.
2	If the beacon code value entered in the scratchpad is invalid, operation of the ATC# line select key will cause the message INVALID CODE to show for approximately 1 second followed by the original scratchpad contents. The active beacon code value remains unchanged.

POST CONDITIONS:

The active ATC transponder beacon code is green for approximately 3 seconds after pilot entry and whenever the CDU-commanded beacon code matches the beacon code echoed from the active transponder. If the beacon code echoed back from the transponder is missing, flagged, or does not match the commanded beacon code, the beacon code

changes color to yellow. If the transponders are in STBY (Standby), the beacon code shows in white.

If a fault in the active transponder is detected, an XPDR FAIL message shows in yellow on the TUNE and ATC control pages to the right of the ATC legend and beacon code.

ATC MODE AND STATUS CONTROL

PRECONDITIONS:

To show the ATC control page, while on the TUNE page push line select key L5 (next to the active transponder ID) with the scratchpad empty.

▼ CHECKLIST:

1	An IDENT initiation is made by selecting the IDENT line select key L2. The IDENT legend will enlarge and change color to cyan when it is selected and active. The IDENT function remains active for approximately 18 seconds, and while it is active, an IDENT message shows to the right of the ATC legend on the TUNE page. Operation of the dedicated IDENT button on the CCP can also be used to select the ATC identification feature.
2	To select the ATC/TCAS mode, push line select key R1. The selected mode will cycle through STBY, TA/RA, TA, ALT ON, and ALT OFF, with the selected mode shown enlarged and in cyan.
3	To select the active ATC transponder (ATC1 or ATC2), push the SELECT ATC1/ATC2 line select key. The selected transponder (ATC1 or ATC2) shows enlarged and in cyan. The selected transponder will show to the right of line select key L5 on the TUNE page.

4 To initiate a self test of the TCAS and active ATC transponder, push the TEST line select key R2. The TEST legend shows in cyan while the self test is active. The test cycle lasts approximately 10 seconds. Successful completion of the test is indicated by a "TCAS SYSTEM TEST OK" aural.

5 The FLIGHT ID field on the ATC control page is used to enter the Flight Identification code. Up to eight alphanumeric characters may be entered into the FLIGHT ID field. The FLIGHT ID code may also be entered or changed on the TUNE 2/2 page.

NOTE

Once it has been entered, the Flight ID value shows in green if a transponder is active, otherwise it shows in white.

6 The altitude source (ADC1 or ADC2) shows in cyan when active.

7 The reported altitude shows in the center of the ATC control page display below the altitude source (ADC1 or ADC2) when the transponder is in Mode C. When valid, the altitude shows in white. If the altitude data is missing or flagged, it is replaced with four yellow dashes.

HF RADIO TUNING AND CONTROL

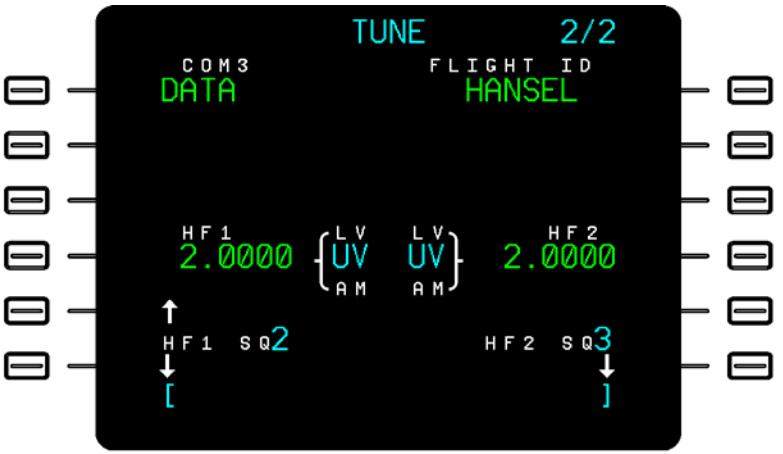
RATIONALE:

The HF1 and HF2 radio frequencies and emission mode (LV/UV/AM) may be selected from either the second TUNE page or from the HF-specific control pages. Other HF radio control, mode selections and frequency presets are made from the HF-specific control pages.

SUMMARY:

The active HF frequencies can be tuned from the HF1 and HF2 active frequency lines on the TUNE 2/2 page, the HF-specific control pages,

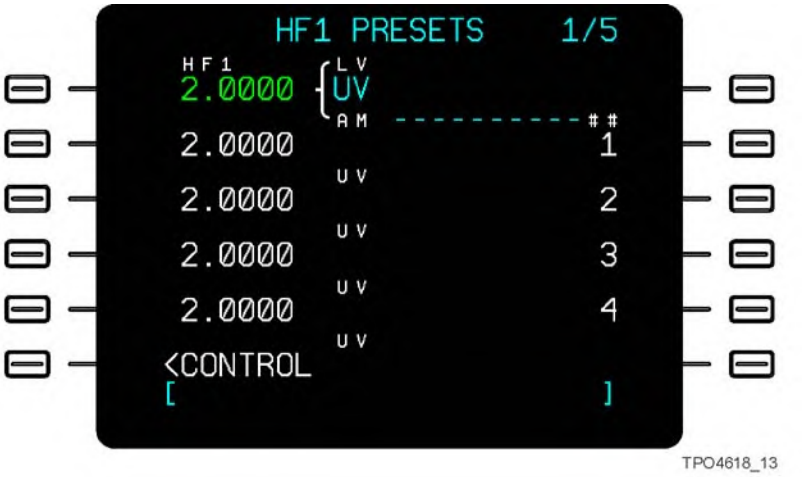
or the HF-specific preset pages. HF radio channel, mode control and power settings and access to frequency presets are controlled from the HF-specific control pages. Predefined frequency, emission mode and channel identifiers for up to 20 entries are controlled from the HF-specific presets pages.



TPO4322_02



TPO4618_12



PRECONDITIONS:

The active HF radio frequency can be changed from the TUNE 2/2 page, from the HF control page, or from any of the HF presets 1/X pages.

If the value entered in the scratchpad is an invalid HF frequency or preset number, an INVALID FREQUENCY (simplex/duplex mode) or INVALID CHANNEL (maritime or emergency mode) message shows in the scratchpad for approximately 1 second when the line select key is pushed. The active HF radio frequency will not be changed and the scratchpad will return to its original entry.

RULES:

The active HF frequency shows in green for approximately 3 seconds after entry and whenever the commanded CDU frequency matches the echo frequency from the radio. If the echo frequency from the radio is flagged as failed, missing, or does not match the commanded frequency, the active frequency value shows in yellow. The numbered preset data shows in green when it is selected and its active frequency is valid. The numbered preset data shows in yellow when it is selected and its active frequency is invalid. The conditions that follow define a valid frequency, maritime channel, emergency channel, or preset number entry:

- If a maritime channel has been selected, a small cyan MAR legend shows to the right of the active channel. If an emergency channel has been selected, a small EMER legend shows to the right of the active channel.

- One to six digits have been entered for a simplex frequency followed by the letter F (optional only for simplex mode). The decimal point is optional if no decimal digits are required (for example, 5 = 5.0000).
- The digits entered for frequency are between 2.0000 and 29.9999 in increments of 0.0100 (2.0 to 29.9999 MHz in 100-Hz increments).
- A duplex receive frequency has an R entered as the first character.
- A duplex transmit frequency has a T entered as the first character.
- Three to four digits, no decimal point, and the letter M have been entered for a maritime channel. (The letter M is optional when in the Maritime mode.)
- The range for a maritime channel is 0401 to 0429, 0601 to 0608, 0801 to 0837, 1201 to 1241, 1601 to 1656, 1801 to 1815, 2201 to 2253, and 2501 to 2510.
- One digit (from 1 to 6), no decimal point, and the letter E have been entered for the emergency channel.
- Two letters (LV, UV, or AM) have been entered for the emission mode. Emission mode is necessary only if a frequency (simplex/duplex) has been entered.
- Valid presets are from 1 to 20.

▼ CHECKLIST:

1	To show the TUNE 2/2 page, do one of the steps that follow: 1. Push the TUN key and then the NEXT key on the CDU keyboard. 2. On an HF-specific control page, push the line select key next to the HF1 or HF2 active frequency line. 3. On any of the HF-specific presets 1/X pages, push the line select key next to the HF1 or HF2 active frequency line.

2	To show the HF-specific control page, while on the TUNE 2/2 page push the line select key next to the HF1 or HF2 active frequency line with the scratchpad empty, or while on a HF-specific presets page, push the control line select key L6.

HF RADIO – DIRECT TUNING

PRECONDITIONS:

An HF radio active frequency can be changed from the TUNE 2/2 page, from the HF-specific control page, or from any of the HF-specific presets 1/X pages.

▼ CHECKLIST:

1	To change the active HF radio frequency from the TUNE 2/2 page, enter a valid HF frequency, channel, mode, or preset number in the scratchpad and then push the line select key next to the HF radio legend. The frequency value or the frequency associated with the preset number is transferred to the HF radio active frequency field.
2	To change the active HF radio frequency from the control page, enter a valid HF frequency, channel, mode, or preset number in the scratchpad and then push line select key L1 next to the HF radio legend.
3	To change the active HF radio frequency from any of the HF presets pages, enter a valid HF frequency, channel, mode, or preset number in the scratchpad and then push line select key L1 next to the HF radio legend.

HF RADIO – PRESET TUNING

SUMMARY:

An HF radio can be tuned to a preset frequency by using a line select key associated with a predefined entry on any of the HF-specific presets pages that contains frequency, channel, and mode information as appropriate. Preset entries can also be edited.

PRECONDITIONS:

The HF-specific presets 1/X page is accessed from the HF-specific control page with the presets line select key L6. Push the NEXT or PREV keys on the CDU keyboard to move forward or backward through the remaining presets pages.

▼ CHECKLIST:

1	With the scratchpad empty, push the line select key next to the desired numbered preset entry to transfer the numbered preset frequency, channel, and mode information to the active frequency field. **NOTE** The numbered preset entry information will change color to green if the frequency is valid. If the information is not valid, it will change color to yellow.
2	To set or change a preset entry: 1. Enter valid frequency, emission mode, and identifier information (in that order) in the scratchpad. A valid identifier is from 1 to 12 characters long, starts with an alpha character, and may contain either alpha or numeric characters.

2. Push the appropriate line select key to transfer the preset data from the scratchpad to the numbered preset field. The preset entry information shows in white and its identifier (if present) shows in cyan. If any of the information entered in the scratchpad is invalid, an INVALID FREQUENCY (simplex/duplex mode) or INVALID CHANNEL (maritime or emergency mode) message shows in the scratchpad for approximately 1 second when the line select key is pushed. The scratchpad will then return to its original entry.

HF RADIO – OPERATING MODE CONTROL

SUMMARY:

An HF radio operating mode is controlled from the HF-specific control page.

CHECKLIST:

1 To change the HF operating mode between FREQ (Frequency), EMER (Emergency), and MAR (Maritime), push the MODE line select key L5 as required to select the desired mode. The selected mode shows enlarged and in cyan.

NOTE

If the Maritime channel tuning option is not enabled, the MAR mode selection is not present.

HF RADIO – SQUELCH CONTROL

PRECONDITIONS:

An HF radio squelch level is controlled from either the TUNE 2/2 page or the HF-specific control page.

▼ CHECKLIST:

1 To change the HF radio's Squelch (SQ) level, use the line select keys next to the up (↑) and down (↓) arrows (when present) to set the squelch level at SQ0, SQ1, SQ2, or SQ3. The selected squelch level shows in cyan next to the HF radio legend.

NOTE

Up (↑) and/or down (↓) arrows show as appropriate for the direction(s) that the current squelch level may be adjusted. For example, if a squelch level of 3 (SQ3) shows, only a down arrow is present. If SQ0 shows, only an up arrow is present.

HF RADIO – POWER LEVEL CONTROL

PRECONDITIONS:

An HF radio transmit power level is controlled from the HF-specific control page.

▼ CHECKLIST:

1 To change the HF radio transmit power level, use the POWER line select key R1 to select LO (low), ME (medium), or HI (high).

1. Each push of line select key R1 will select the next power level.
2. The selected power level shows enlarged and in cyan.

HF RADIO – SELF TEST CONTROL

PRECONDITIONS:

An HF radio self-test mode is controlled from the HF-specific control page.

▼ CHECKLIST:

1 To initiate a self test of the HF radio, push the TEST line select key R2. During the self test, the TEST legend shows in cyan.

NOTE

When the self test is complete (after approximately 4 seconds), the TEST legend shrinks and shows in white.

HF RADIO – MESSAGES

▼ CHECKLIST:

1 A cyan TX advisory shows to the right of the HF radio legend on the TUNE 2/2, HF-specific control page, and each of the HF-specific presets pages when the HF radio is transmitting.

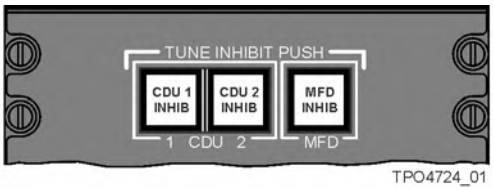
TUNING REVERSION

RATIONALE:

Tuning reversion allows the pilot to inhibit the pilot or copilot-side CDU radio tuning and/or the MFD radio tuning. When the pilot and copilot-side and MFD radio tuning is inhibited, an emergency tuning mode is automatically selected.

SUMMARY:

The CDU is the master tuner in normal operating mode (no inhibit buttons pressed). The on-side CDU is the master of the cross-side radios, when the cross side CDU is inhibited. When a CDU is inhibited, all tuning commands from that CDU are ignored. The MFD is the master tuner only when CDU1 and CDU2 inhibit is active. When the MFD is the master tuner, each MFD will tune the on-side radios. When both CDU inhibits and the MFD inhibit are selected, COM1 will be tuned to the 121.5 MHz emergency frequency.



▼ CHECKLIST:

1	Push the CDU1 INHIB button to disable the pilot side CDU tuning and configure the copilot-side CDU as the master controller for the pilot and copilot-side radios.
2	Push the CDU2 INHIB button to disable the copilot-side CDU tuning and configure the pilot side CDU as the master controller for the pilot and copilot-side radios.
3	Push the MFD INHIB button to remove the MFD radio tune window and menus.
4	Push the CDU1 and CDU2 INHIB buttons to disable tuning from both CDUs and configure the MFDs as the master controllers for the pilot and copilot-side radios.

5 Push the CDU1, CDU2 and MFD INHIB buttons to disable all of the tuning inputs and tune COM1 to 121.5 MHz.

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RADIO OPERATION

INTRODUCTION

The Cursor Control Panel (CCP) and Multifunction Display (MFD) provide single-point control of both on-side and cross-side radios from the pilot or copilot-side. A top level radio menu on the MFD provides an at a glance view of each of the on-side radios. Each radio has its own control page. The control pages provide control of all radio functions, including the primary functions available on the radio menu. A control page is available for each of the radios that follow:

- VHF COM radio tuning and control
- NAV Radio tuning and control
- ADF Radio tuning and control
- DME Radio tuning and control
- ATC Transponder selection, ident beacon code, and mode
- TCAS II control
- HF Radio tuning and control.

NOTE

When the FMS AUTO TUNE mode is active, the FMC may automatically tune the NAV and DME radios.

The RADIO button, RADIO ADV knob, DATA knob, PUSH SELECT button, FREQ ↕ Button, 1/2 Button, DME-H Button, IDENT button, and ATC button on the CCP are used for MFD radio tuning and control.

- The RADIO button is used to show the radio main menu on the MFD menu when another menu is in view. When the radio menu is already in view, operation of the RADIO button selects the radio control sub-menu (for example, COM control menu) for the radio highlighted by the selection box. When a radio control sub-menu is in view, operation of the RADIO button backs out of the control sub-menu and shows the radio main menu on the MFD.
- The RADIO ADV (advance) knob is used to move the selection box to the desired menu item. When tune mode is active, the RADIO ADV knob is used to set the most significant digits of the radio frequency.

- The DATA knob is used to set the frequency or channel highlighted by the selection box. When tune mode is active the DATA knob is used to set the least significant digits of the radio frequency.
- When the selection box is around a frequency or channel, the PUSH SELECT button is used to select and deselect tune mode. The selection box is replaced by a frequency tune cursor (which has a different shape and size) when tune mode is active. When the selection box is around a state (for example, on/off), the PUSH SELECT button is used to change the state.
- The 1/2 button is used to transfer the pilot-side and copilot-side radio menus.
- The DME-H (hold) button is used to select the DME hold feature for the selected DME when the active NAV source is VOR or LOC (push-on/push-off).
- The IDENT button is used to command the ATC IDENT feature.
- The ATC button is used to set the ATC Transponder mode to STBY if the current mode is not STBY and to TA/RA if the current mode is STBY.

Tuning reversion is available in the event of a controller failure. Refer to the Remote Switches section of this guide for details on tuning reversion.

The Audio Control Panel (ACP) installed at the pilot and copilot positions provides control of the microphone selection, volume controls, and voice/identifier selection at the installed position. Refer to the Audio Control Panel (ACP) section of this guide for details on operation of the ACP.

GENERAL MFD TUNING

RATIONALE:

Either pilot can tune all radios, on-side and cross-side, with the MFD as the master tuning control.

SUMMARY:

The MFD and CCP provide single point control of both the on-side and cross-side radios from the pilot or copilot position. The Radio Menu provides an at a glance summary of the radios as well as control for each. The RADIO button on the CCP shows the Radio Menu on the

MFD. The MFD sends signals to the aircraft radios to tune frequencies, as well as set operating modes, squelch levels, and voice/data modes. The MFD also responds to tune/control commands from the cross-side MFD or the CDU as either acts as the master tuning control.



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▼ CHECKLIST:

1	Push the RADIO button on the CCP to show the MFD Radio menu if another menu/list is active.
2	Push the 1/2 button to select the side 1 radios menu and the side 2 radios menu.
3	To tune a frequency/channel: 1. With the Radio Menu in view, operate the RADIO ADV knob on the CCP to move the selection box to the recall (bottom most) frequency or channel to be tuned. 2. Push the SELECT button on the CCP to change the selection box to a tune box. 3. Operate the RADIO ADV and radio DATA knobs to adjust the frequency/channel. The RADIO ADV knob changes the most significant digits and the radio DATA knob changes the least significant digits. 4. Push the FREQ ↕ (transfer) button to swap the active and recall frequency.
4	To directly tune the ATC Beacon Code, Flight ID number, DME hold frequency or HF Duplex frequency:

1. With the Radio Menu in view, operate the RADIO ADV knob on the CCP to move the selection box to the active code or frequency.
2. Push the SELECT button on the CCP to change the selection box to a tune box.
3. Operate the RADIO ADV and radio DATA knobs to adjust the code/frequency. The RADIO ADV knob changes the most significant digits and the radio DATA knob changes the least significant digits.

5 Single Digit Tuning is used to select HF COM frequency and to set the FLIGHT ID number. To set the HF COM frequency or Flight ID number:

1. With the Radio Menu in view, operate the RADIO ADV knob on the CCP to move the selection box to the necessary digit of the HF COM frequency or FLIGHT ID number.
2. Operate the radio DATA knob to set the necessary digit.

6 To change modes, adjust squelch, and other non-tuning functions of a radio:

1. Operate the RADIO ADV knob to move the selection box to the desired radio.
2. Push the RADIO button on the CCP to show the radio sub-menu for the selected radio.
3. Operate the RADIO ADV knob to move the selection box to the desired mode or option on the radio sub-menu.
4. Operate the radio DATA knob to change the mode or select an option.
5. Push the RADIO button to exit the sub-menu and return to the Radio Menu.

COM RADIO OPERATION

RATIONALE:

The COM control sub-menu provides control for COM radio modes other than those available through the Radio Menu.

SUMMARY:

The Radio Menu provides basic COM radio tuning control. The COM control sub-menu provides control display for other COM radio operation. Each COM radio installed on the aircraft has its own COM control sub-menu. The active frequency, recall frequency, and squelch status displays are identical to the display on the Radio Menu.





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▼ CHECKLIST:

1	Push the RADIO button on the CCP to show the Radio Menu on the MFD if another menu/list is active.
2	Operate the RADIO ADV knob on the CCP to move the selection box to the desired radio (COM1, COM2, or COM3) on the Radio Menu.
3	Push the RADIO button on the CCP to show the COM control sub-menu on the MFD.
4	The operations that follow are performed from the COM control sub-menu.

4.1 Change the SQUELCH mode.

1. Operate the RADIO ADV knob to move the selection box to the SQUELCH control.
2. Operate the radio DATA knob or the SELECT button to set the squelch to ON or OFF.

4.2 Change the frequency spacing for radio tuning.

1. Operate the RADIO ADV knob to move the selection box to the FREQ SEL control.
2. Operate the radio DATA knob or the SELECT button to set 8.33 or 25 kHz spacing.

NOTE

When the spacing is changed to 8.33 kHz, the recall COM radio frequency resolution is three decimal places for that radio. If the prior recall frequency was 8.33 kHz, no change is made to the recall until a new station is made. If the prior recall frequency had an 8.33 kHz frequency, the MFD changes the frequency to the next lower valid 25 kHz frequency value. If the prior recall frequency was a three-digit 25 kHz frequency, the third decimal digit is removed.

4.3 Select the COM TEST mode.

1. Operate the RADIO ADV knob to move the selection box to TEST.

2. Operate the radio DATA knob or SELECT button to select TEST.
   - The TEST legend enlarges when active and changes color to cyan.
   - A single aural beep at the conclusion of the test indicates pass. A double beep indicates fail.
   - Faults determined during self-test are stored in archival fault storage memory.
   - The self-test period is approximately 10 seconds.
   - The TEST legend is reduced in size and shows in white when the self-test inactive.

4.4 Change the COM 3 mode. This function is available when the Voice/Data option is enabled.

1. Operate the RADIO ADV knob to move the selection box to the MODE control.
2. Operate the radio DATA knob or the SELECT button to select VOICE or DATA.

NOTE

When DATA mode is selected, the squelch message, squelch status, transfer arrow, frequency selections 8.33 and 25 kHz, and TEST selection do not show on the COM3 control sub-menu.

NAV RADIO OPERATION

RATIONALE:

The NAV control sub-menu provides control for NAV radio modes other than those available through the Radio Menu.

SUMMARY:

The Radio Menu provides basic NAV radio tuning control. The NAV control sub-menu provides control display for other NAV radio operation. Each NAV radio installed on the aircraft has its own NAV control sub-menu. The active, recall, and DME frequency displays are identical to the display on the Radio Menu.



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▼ CHECKLIST:

1	Push the RADIO button on the CCP to show the Radio Menu on the MFD, if another menu/list is active.
2	Operate the RADIO ADV knob on the CCP to move the selection box to the NAV radio (NAV1 or NAV2) on the Radio Menu.
3	Push the RADIO button on the CCP to show the selected NAV control sub-menu on the MFD.
4	The operations that follow are performed from the NAV control sub-menu.
4.1	Tune the DME frequency. 1. With DME Hold set to on, operate the RADIO ADV knob to move the selection box to the DME frequency. 2. Push the SELECT button on the CCP to change the selection box to a tune box. 3. Operate the RADIO ADV and radio DATA knobs to adjust the frequency.

4.2 Select the marker sense mode.

1. Operate the RADIO ADV knob to move the selection box to MKR SENSE.
2. Operate the radio DATA knob or the SELECT button to select LO or HI.

CAUTION

Do not initiate a NAV/DME self test if a VOR or LOC is the current or preselect navigation source, as changes to the flight director guidance and active modes may occur.

4.3 Select the NAV TEST mode.

1. Operate the RADIO ADV knob to move the selection box to TEST.
2. Operate the radio DATA knob or the SELECT button to select TEST. When the test is active:
   - The TEST legend enlarges when active and changes color to cyan.
   - The VOR bearing shows 360 degrees and all Marker Beacons go active when a VOR frequency is tuned.
   - A fly up/fly left is indicated and all Marker Beacons go active when a LOC frequency is tuned.
   - DME distance shows 100 NM.
   - TTG shows 60 minutes.
   - Station ID shows AOK.
   - Groundspeed shows 100 knots.
   - The test period is approximately 10 seconds.
   - The TEST legend is reduced in size and shows in white when the self-test inactive.

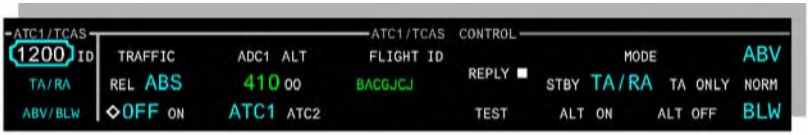
ATC/TCAS OPERATION

RATIONALE:

The ATC/TCAS control sub-menu provides control for ATC transponder and TCAS modes other than those available through the Radio Menu.

SUMMARY:

The Radio Menu provides basic ATC transponder Mode A beacon code (squawk code) control, TCAS mode control STBY (Standby), TA/RA (Traffic and Resolution Advisories), TA ONLY (Traffic Advisory only), ALT ON (Transponder only, mode C), ALT OFF (Transponder only, mode A) and altitude limits selection (ABV, NORM, BLW, ABV/BLW). The IDENT button on the CCP is used to activate the ATC identification feature. The ATC/TCAS control sub-menu provides control display for other ATC transponder and TCAS operations. The active beacon code, TCAS mode, and TCAS altitude limits selection that show on the left side of the ATC/TCAS control sub-menu are identical to the ATC/TCAS display on the Radio Menu.



TPO4620_70

▼ CHECKLIST:

1 Push the RADIO button on the CCP to show the Radio Menu on the MFD if another menu/list is active.

NOTE

Operation of the ATC button on the CCP changes the ATC Transponder mode to STBY if the current mode is not STBY and to TA/RA if the current mode is STBY.

NOTE

When a radio menu is in view and the TCAS/ATC mode is not STBY and ID is not active, operation of the ATC button on CCP causes the display to back out of a radio menu and place the selection box around the TCAS/ATC transponder code.

2 Operate the RADIO ADV knob on the CCP to move the selection box to ATC/TCAS on the Radio Menu.

3 Push the RADIO button on the CCP to show the ATC/TCAS control sub-menu on the MFD.

4 The operations that follow are performed on the ATC/TCAS control sub-menu.

4.1 Select TCAS traffic altitude mode.

1. Operate the RADIO ADV knob to move the selection box to TRAFFIC.
2. Operate the radio DATA knob or the SELECT button to select Relative (REL) or Absolute (ABS).

4.2 Select Other Traffic on or off.

1. Operate the RADIO ADV knob to move the selection box to Other Traffic (diamond symbol).
2. Operate the radio DATA knob or the SELECT button to set Other Traffic to ON (all traffic) or OFF (threat traffic only).

4.3 Select the active transponder.

1. Operate the RADIO ADV knob to move the selection box to ATC1 ATC2.

2. Operate the radio DATA knob or the SELECT button to select ATC1 or ATC2.

4.4 Set the Flight ID.

1. Operate the RADIO ADV knob to move the selection box to FLIGHT ID.
2. Push the SELECT button on the CCP to change the selection box to a tune box.
3. Operate the RADIO ADV and radio DATA knobs to set the Flight ID.

4.5 Select the ATC/TCAS Test.

1. Operate the RADIO ADV knob to move the selection box to TEST.
2. Operate the radio DATA knob or the SELECT button to select TEST. The message TCAS TEST shows in cyan when the TCAS self-test is active.

4.6 Select the TCAS mode.

1. Operate the RADIO ADV knob to move the selection box to MODE.
2. Operate the radio DATA knob or the SELECT button to select STBY (Standby), TA/RA (Traffic and Resolution Advisories), TA ONLY (Traffic Advisory only), ALT ON (Transponder only, mode C), ALT OFF (Transponder only, mode A).

4.7 Select the Other Traffic altitude window.

1. Operate the RADIO ADV knob to move the selection box to ABV NORM BLW.

2. Operate the radio DATA knob or the SELECT button to select ABV (the above altitude limit), BLW (the below altitude limit), ABV and BLW (unrestricted altitude mode), or NORM (ABV and BLW deselected).

ADF RADIO OPERATION

RATIONALE:

The ADF control sub-menu provides control for ADF radio modes other than those available through the Radio Menu.

SUMMARY:

The Radio Menu provides basic ADF radio tuning control. The ADF control sub-menu provides control display for other ADF radio operations. Each ADF radio installed on the aircraft has its own ADF control sub-menu. If only one ADF radio is installed, it will show on both the on-side and cross-side menus, and is labeled ADF control with no numerical designator. The active frequency, recall frequency, and antenna mode that show on the left side of the ADF control sub-menu are identical to the ADF display on the Radio Menu.



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▼ CHECKLIST:

1	Push the RADIO button on the CCP to show the Radio Menu on the MFD, if another menu/list is active.
2	Operate the RADIO ADV on the CCP to move the selection box to the ADF radio (ADF1 or ADF2) display on the Radio Menu.

3 Push the RADIO button on the CCP to show the ADF control sub-menu on the MFD.

4 The operations that follow are performed from the ADF control sub-menu.

4.1 Select the mode.

1. Operate the RADIO ADV knob to move the selection box to MODE.
2. Operate the radio DATA knob or the SELECT button to select ADF or ANT.

4.2 Select BFO mode.

1. Operate the RADIO ADV knob to move the selection box to BFO.
2. Operate the radio DATA knob or the SELECT button to select ON or OFF.

4.3 Select the ADF TEST.

1. Operate the RADIO ADV knob to move the selection box to TEST.
2. Operate the radio DATA knob or the SELECT button to select the TEST.
   - The TEST legend enlarges when active and changes color to cyan.
   - The Bearing pointer swings clockwise 90 degrees when the test is active.
   - The ADF test period is approximately 10 seconds.
   - The TEST legend is reduced in size and shows in white when the self-test is inactive.

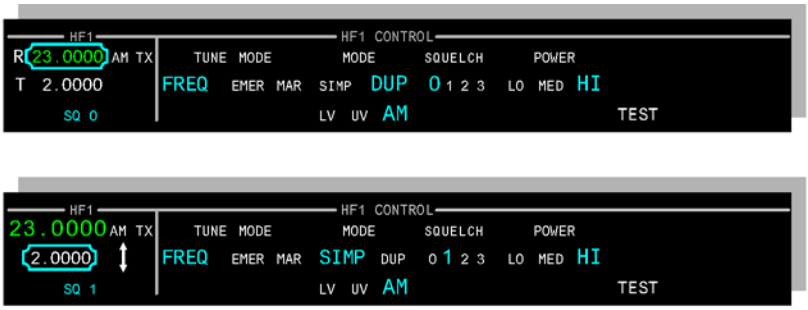
HF RADIO OPERATION

RATIONALE:

The HF control sub-menu provides control for HF radio modes other than those available through the Radio Menu.

SUMMARY:

The Radio Menu provides basic HF radio tuning control. The HF control sub-menu provides control display for other HF radio operations. Each HF radio installed on the aircraft has its own HF control sub-menu. The active frequency, recall frequency, and squelch status that show on the left side of the HF control sub-menu are identical to the HF display on the Radio Menu.



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▼ CHECKLIST:

1	Push the RADIO button on the CCP to show the Radio Menu on the MFD, if another menu/list is active.
2	Operate the RADIO ADV knob on the CCP to move the selection box to the HF radio (HF1 or HF2) on the Radio Menu.
3	Push the RADIO button on the CCP to show the HF control sub-menu on the MFD.
4	The operations that follow are performed on the HF control sub-menu.

4.1 Select the HF tune mode.

1. Operate the RADIO ADV knob to move the selection box to TUNE MODE.
2. Operate the radio DATA knob or the SELECT button to select FREQ (Frequency), EMER (Emergency), or MAR (Maritime).

4.2 Select the HF transmission mode.

1. Operate the RADIO ADV knob to move the selection box to MODE.
2. Operate the radio DATA knob or the SELECT button to select SIMP (Simplex) or DUP (Duplex).

4.3 Select the HF emission mode.

1. Operate the RADIO ADV knob to move the selection box to the emission mode.
2. Operate the radio DATA knob or the SELECT button to select LV (Lower Voice Single Sideband), UV (Upper Voice Single Sideband), or AM (Amplitude Modulation).

4.4 Select the HF squelch level.

1. Operate the RADIO ADV knob to move the selection box to SQUELCH.
2. Operate the radio DATA knob or the SELECT button to select 0 (zero), 1, 2, or 3.

4.5 Select the HF power level.

1. Operate the RADIO ADV knob to move the selection box to POWER.
2. Operate the radio DATA knob or the SELECT button to select LO, MED, or HI.

4.6 Select the HF TEST.

1. Operate the RADIO ADV knob to move the selection box to TEST.
2. Operate the radio DATA knob or the SELECT button to select TEST.
   - The TEST legend enlarges when active and changes color to cyan.
   - The HF test period is approximately 40 seconds.
   - At the conclusion of the test, a green PASS or yellow FAIL message replaces the TEST message. The PASS/FAIL message remains until the HF is tuned or there is a change to the squelch, mode, power, or test control.
   - The TEST legend is reduced in size and shows in white when the self-test inactive.

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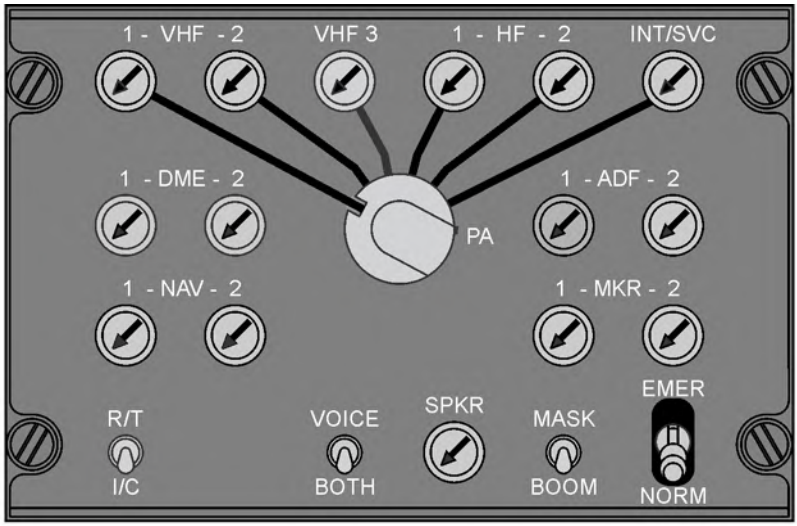
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MENUS AND DISPLAYS

AUDIO CONTROL PANEL (ACP)

Figure 6-1 Audio Control Panel (ACP)



The Audio Control Panel (ACP) installed at the pilot and copilot-side positions provides control of the microphone selection and audio volume at the installed position.

▼ SELECTIONS:

Microphone Selector Knob	The microphone selector knob routes the microphone audio to the selected transmitter or interphone. When VHF1, VHF2, VHF3, HF1 or HF2 is selected the microphone audio is connected to the transmitter of the associated radio. When the PA (Passenger Address) is selected, the microphone audio is connected to the PA system.

Audio Selector/ Volume Controls	The audio selector/volume controls VHF1, VHF2, VHF3, HF1, HF2, INT/SVC, DME1, DME2, ADF1, ADF2, MKR1, MKR2, NAV1, NAV2, and SPKR have a dual function of volume control and ON/OFF. When a selector/volume control is set to the in position the associated audio source is active and the volume control is functional. When a selector/volume control is set to the out position the associated audio source is off. The INT/SVC selector/volume control is used to adjust the volume of the interphone audio. The SPKR selector/volume control is used to adjust the volume of the associated speaker. Audio volume of aural warnings is not affected by SPKR volume control.
R/T / I/C PTT Switch	The R/T / I/C PTT switch is used to control the interphone system. The R/T position is momentary and spring loaded and when selected generates a Push-To-Talk (PTT) command. The center off position deselects the pilot and copilot interphone. The I/C position is detented and when selected provides a hot microphone condition for the interphone system.
VOICE/BOTH Switch	The VOICE/BOTH toggle switch controls the voice/identifier code audio input from the NAV and ADF receivers. In the VOICE position, only voice is heard from the NAV and ADF receivers.

MASK/BOOM Switch

The MASK/BOOM toggle switch controls the oxygen mask and headphone microphones. When the switch is in the BOOM position, the boom microphone is the active microphone. When the switch is in the MASK position, the oxygen mask microphone is the active microphone.

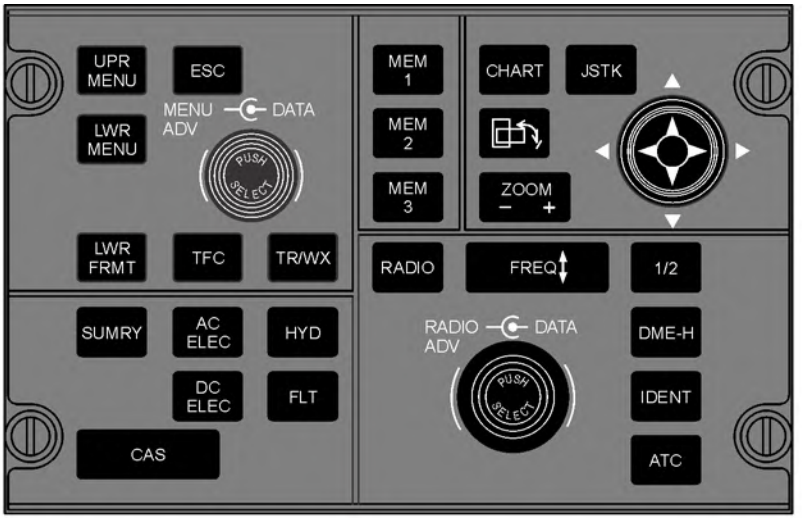
EMER/NORM Switch

The EMER/NORM toggle switch is a locking switch that is pulled out to select the emergency mode. When the NORM position is selected, audio is sent to the on-side headphones and speakers. When the EMER position is selected:

- Audio from the VHF COM, NAV, and Aural Warning Systems is routed to the headphones.
- The pilot-side radio audio is routed to the pilot-side headphones and the copilot-side radio audio is routed to the copilot-side headphones.
- The pilot-side boom and mask microphones XMIT PTT lines are routed directly to the VHF-1 transceiver.
- The copilot-side boom and mask microphones and XMIT PTT lines are routed directly to the VHF-2 transceiver.
- Aural warnings are routed to the speakers.

CURSOR CONTROL PANEL (CCP)

Figure 6-2 Cursor Control Panel (CCP)



The CCP provides MFD display control. The CCP is used to select and navigate the menus used to control the MFD display format. Quick access buttons on the CCP are used to store and display MFD display formats. The CCP provides dedicated controls for the optional IFIS Electronic Charts, Enhanced Maps, and Graphical Weather features. The CCP provides control of the Crew Alerting System displays and synoptic display formats. The CCP provides MFD radio tuning control. Two CCPs are installed, one for the pilot-side and one for the copilot-side.

NOTE

The MENU ADV knob, DATA knob, PUSH SELECT button, and ESC button on the CCP are used to navigate/operate the menus and sub-menus on the MFD. Refer to MFD Menu Navigation section in this chapter for details on MFD menu and sub-menu navigation and operation.

▼ SELECTIONS:

MFD Format and Menu Controls	The LWR FRMT button, UPPR MENU button, LWR MENU button, MENU ADV Knob, DATA knob, PUSH SELECT button, and ESC button are used to control the MFD display format/overlay state and to select and navigate the MFD menus.
LWR FRMT Button	When a split window format is active on the MFD, the LWR FRMT button is used to show the lower format direct access menu and to change the lower format to the next format on the list. Repeated operation of the FRMT button highlights the next format on the menu and shows the newly selected format on the MFD. When one of the synoptic pages is active in the lower MFD window, the LWR FRMT button is used to remove the synoptic page and show the previously selected format. When a full format is active on the MFD, the LWR FRMT button is used to show the previously active split window format and to show the direct access menu. The possible lower formats are PPOS, PLAN, 3D MAP, GWX, VIDEO, and TCAS.



TPO4620_79

UPR MENU Button

The UPR MENU button is used to select and deselect the UPR (main) MENU on the MFD. The UPR MENU is divided into a FORMAT and CONTROLS sub-menus. The FORMAT sub-menu is used to select the MFD display format. The possible selections are OFF, CHECKLIST, GWX, EICAS, SUMMARY, AC ELECTRIC, DC ELECTRIC, HYDRAULIC, and FLT CONTROL. The CONTROLS sub-menu contains controls and links to other sub-menus relating to the active MFD display format. As a result, the items contained on the CONTROLS sub-menu vary depending upon which MFD display format is active. Refer to the associated display format description (for example, MFD — GRAPHICAL WEATHER) in this chapter for details on the associated CONTROLS sub-menu.



TPO4620_94

LWR MENU Button

The LWR MENU button is used to select and deselect the LWR (main) MENU on the MFD. The LWR MENU is divided into a FORMAT and CONTROLS sub-menus. The FORMAT sub-menu contains controls for selecting the MFD display format. The possible selections are PPOS Map, Plan (Map), 3D Map, GWX, VIDEO, and TCAS. The CONTROLS sub-menu contains controls related to the active MFD display format. As a result the items contained on the CONTROLS sub-menu vary depending upon which MFD display format is active. The basic lower menu shows below. The basic lower menu has controls to select the overlays, map symbols, and PFD menus. Refer to the associated display format description (for example, MFD — GRAPHICAL WEATHER) in this chapter for details on the CONTROLS sub-menu that vary depending upon the active MFD format.



TPO4407_01

OVERLAYS Menu	The OVERLAYS menu is selectable by way of the MFD lower menu. The OVERLAYS menu is used to select the TCAS, Weather Radar, Terrain and Lightning overlays for display on the MFD. The TFC button on the CCP can also be used to access the TCAS overlay on the MFD. The TR/WX button on the CCP can also be used to select the Terrain/Radar/Lightning overlays on the MFD.
MAP SYMBOLS Menu	The MAP SYMBOLS menu is selectable by way of the MFD lower menu. The MAP SYMBOLS menu is used select map background symbology for display on the FMS PPOS, PLAN and 3D maps (for example, nearest airports). The map background symbology can also be selected by way of the FMS CDU. Refer to the MFD Map Displays section of the Bombardier Challenger 605/650 Pro Line 21™ Advanced Flight Management System (523-0822377) for details on selecting map background symbology by way of the CDU. Refer to the PFD — PPOS Map section in this chapter for details on map background symbology.
PFD Menu	The PFD menu is selectable by way of the MFD lower menu. The MFD PFD Menu is used to control the PFD using controls on the MFD. This is primarily used in the event of a DCP failure to retain full PFD functionality.



TPO4408_01

MENU ADV Knob

The MENU ADV knob is used to position the cyan selection box on a menu or sub-menu. A menu is a window that can be brought into view to show one or more control selections. The control selections may be separated into groups within a menu by way of sub-menus resident on the parent menu. Alternatively, sub-menu links may be provided to select sub-menus that are not resident on the parent menu.

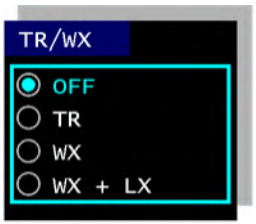
DATA Knob

The DATA knob is used to change the value highlighted by the cyan selection box between predefined limits, change the state of a highlighted item (for example, on/off), or select an item from a list.

PUSH SELECT Button	The PUSH SELECT button is used to change the state of a highlighted item, to select a item from a list, and to select a sub-menu for display.
ESC Button	The ESC button is used to back out of a sub-menu, clear a parent menu, and to abort a data entry action.
Direct Access Controls	The TFC button and TR/WX buttons are direct access buttons used to show the associated display format or overlay on the MFD without accessing a format menu.
TFC Button	The TFC button is used to select and deselect the TCAS Traffic overlay and the TCAS Only format on the MFD. When a compatible display format is active, momentary operation of the TFC button selects the TCAS Traffic overlay. When the current MFD format is not compatible with a Traffic overlay, or when a TCAS Traffic Advisory is active on the MFD, operation of the TFC button selects the TCAS Only format. The TCAS Only format is also selected when the TFC button is held in for more than 1 second. The display range is set automatically to 10 NM when the TCAS Only format is initially selected. Refer to PFD/MFD — TCAS Presentation in this chapter for details on the TCAS traffic overlay and TCAS Only format.

TR/WX Button

The TR/WX button is used to select/deselect the Terrain, Weather Radar and optional Lightning overlays on the MFD. When no Terrain warning or caution is active, the first push of the TR/WX button selects the next available overlay. When a Terrain warning or caution is active, the first push of the TR/WX button selects the Terrain overlay with a 10-NM range. When the TR/WX button is pushed for more than 1 second the Terrain overlay is selected with a 10-NM range. An appropriate display range is automatically selected for the Weather Radar overlay when necessary. Refer to the PFD/MFD — Terrain Overlay, PFD/MFD — Weather Radar Overlay, and PFD/MFD — Lightning Overlay sections in this chapter for details on the associated hazard avoidance overlays.



TPO4620_89

Quick Access Buttons

Three quick-access buttons (MEM1, MEM2, and MEM3) are used to store, then recall the MFD format/overlay configurations. Only combinations of split-display formats, for example, Checklist and PPOS Map, are stored. Full-display formats (for example, E-Charts and FMS Text Only) are accessed with dedicated controls and cannot be stored.

- To store the current MFD format/overlay configuration, push the desired quick access button for more than 3 seconds. The upper format, lower format, TERR/RDR/LX overlay state, and TFC overlay state are stored. The message STORE shows to indicate that the quick access button has been pushed for more than 3 seconds (for example, release the button). The message STORE COMPLETE shows when the button is released and the current MFD format/overlay configuration has been stored.
- To recall a stored MFD format/overlay configuration, momentarily push the desired quick access button. When a format/overlay configuration is recalled using the quick access buttons, any full formats that are currently in view are replaced with the appropriate split view (upper and lower formats). The message RECALL COMPLETE shows on the MFD when the recall action is complete.

Dedicated IFIS Controls

The CHART button, Orientation button, ZOOM button, joystick, and Joystick (JSTK) button are used to control the optional IFIS Electronic Charts (E-Chart), Enhanced Maps, and Graphical Weather features.

CHART Button

The CHART button is used to show the last Electronic Chart (E-Chart) in view. Refer to the MFD — E-Charts section of this guide for details on E-Charts.

Orientation Button	The orientation button located below the CHART button on the CCP is used to rotate the optional Electronic chart (E-Chart) 90 degrees (push-on/push-off). Refer to the MFD — E-Charts section of this guide for details on E-Charts.
Joystick	The joystick is used for zooming and panning the optional Electronic chart (E-Chart), to define a waypoint from the FMS PPOS or Plan Map, and for adjusting the viewing angle when the optional 3D Map format is active. When the MFD format is a split window and the joystick is a valid control for both windows, the joystick is positioned to the desired window with the JSTK button on the CCP. • E-Chart Zooming — With an E-Chart in view and not zoomed in, the joystick is used to select and position the pan/zoom indicator (box). Once the pan box has been moved to the desired position, the ZOOM button can be used to increase the size (x1.5) of the area inside the pan/zoom window (push-on/push-off). • E-Chart Panning — When zoomed-in on an E-Chart, the joystick is used to move the zoomed in area to the desired position on the chart.

	• Define Waypoints — The joystick cursor shows on the MFD when the joystick is deflected and the FMS Plan Map or PPOS Map is the active format. The latitude and longitude of the cursor position are sent to the FMS scratchpad upon operation of the CCP PUSH SELECT Button. The cursor is removed if the JSTK button is pressed to transfer the joystick control to the upper format in a split window MFD format, or if the joystick is not used for 60 seconds. • 3D Map Viewing Angle — The joystick is used to set the lateral and/or vertical viewing angles to any combination of 0 (zero) to 360 degrees laterally and 0 (zero) to 90 degrees vertically.
JSTK Button	When the MFD format is a split window and the joystick is a valid control for both windows, the joystick is positioned to the desired window with the JSTK button. A joystick control indicator (arrow) preceded by the legend JSTK shows temporarily in the lower right corner (pointing up) for the upper window or the upper right corner (pointing down) for the lower window.
Crew Alerting System and Synoptic Display Controls	The CCP provides control of the Crew Alerting System (CAS) and synoptic displays on the MFD. The CAS and synoptic controls include the CAS, SUMRY, AC ELEC, HYD, DC ELEC, and FLT buttons.

CAS Button	The CAS button is used to show the next page of CAS messages on the MFD. Each push of the CAS button shows the next page of CAS messages. When the CAS field is filled with warning messages, the list cannot be paged to show other non-warning messages. When less than a full page of CAS messages or the last page of CAS messages is in view, operation of the CAS button causes all non-warning messages to blank. Refer to the MFD — Crew Alerting section of this guide for details.
SUMRY Button	The SUMRY button is used to select the Summary Synoptic format on the bottom of the MFD. Refer to the MFD — Summary Display section of this guide for details.
AC ELEC Button	The AC ELEC button is used to select the AC Electrical system synoptic display on the MFD. Refer to the MFD — AC Electrical System Synoptic Display section of this guide for details.
HYD Button	The HYD button is used to select the Hydraulic system synoptic display on the MFD. Refer to the MFD — Hydraulic System Synoptic Display section of this guide for details.
DC ELEC Button	The DC ELEC button is used to select the DC Electrical system synoptic display on the MFD. Refer to the MFD — Flight Controls Synoptic Display section of this guide for details. Refer to the MFD — DC Electrical System Synoptic Display section of this guide for details.

FLT Button	The FLT button is used to select the Flight Controls system synoptic display on the MFD. Refer to MFD — Flight Controls Synoptic Display section of this guide for details.
Radio Tuning Controls	The RADIO button, RADIO ADV knob, DATA knob, PUSH SELECT button, FREQ ↕ Button, 1/2 Button, DME-H Button, IDENT button, and ATC button are used for MFD radio tuning and control. Refer to the Radio Operation chapter of this guide for details on MFD radio tuning and control.
RADIO Button	The RADIO button is used to show the radio main menu on the MFD menu when another menu is in view. When the radio menu is already in view, operation of the RADIO button selects the radio control sub-menu (for example, COM CONTROL menu) for the radio highlighted by the selection box. When a radio control sub-menu is in view, operation of the RADIO button backs out of the control sub-menu and shows the radio main menu on the MFD. Refer to the Radio Operation chapter of this guide for details on MFD radio tuning and control.
RADIO ADV Knob	The RADIO ADV (advance) knob is used to move the selection box to the desired menu item. When tune mode is active, the RADIO ADV knob is used to set the most significant digits of the radio frequency.

DATA Knob	The DATA knob is used to set the frequency or channel highlighted by the selection box. When tune mode is active, the DATA knob is used to set the least significant digits of the radio frequency.
PUSH SELECT Button	When the selection box is around a frequency or channel, the PUSH SELECT button is used to select and deselect tune mode. The selection box is replaced by a frequency tune cursor (which has a different shape and size) when tune mode is active. When the selection box is around a state (for example, on/off), the PUSH SELECT button is used to change the state.
FREQ ↕ Button	The FREQ ↕ button is used to swap the active and recall frequency. Refer to the Radio Operation chapter of this guide for details on MFD radio tuning and control.
1/2 Button	The 1/2 button is used to transfer the pilot-side and copilot-side radio menus. Refer to the Radio Operation chapter of this guide for details on MFD radio tuning and control.
DME-H Button	The DME-H (hold) button is used to select the DME hold feature for the selected DME when the active NAV source is VOR or LOC (push-on/push-off). Refer to the Radio Operation chapter of this guide for details on MFD radio tuning and control.
IDENT Button	The IDENT button is used to command the ATC IDENT feature. Refer to the Radio Operation chapter of this guide for details on MFD radio tuning and control.

ATC Button

The ATC button may be used to set the ATC Transponder mode to STBY if the current mode is not STBY and to TA/RA if the current mode is STBY. Refer to the Radio Operation chapter of this guide for details on MFD radio tuning and control.

CONTROL DISPLAY UNIT (CDU)

Figure 6-3 Control Display Unit (CDU)



TPO4637_01

The CDU provides centralized control and display functions for the Flight Management System (FMS) and the optional Datalink, FANS, HUD and SATCOM systems. The CDU is also the primary control/display interface for the Radio Sensor System (RSS). The CDUs datalink interface is also used to request graphical weather image uplinks that may then be selected for display on the MFDs via the IFIS or on the CDU, depending on the optional equipment fitted. The CDU also provides a means for control of FMS Map background symbology on

the MFD and PFD, though these functions are duplicated within the MFD and PFD menus.

NOTE

Flight Management System (FMS) operation is not described in this operator's guide. For information on operation of the FMS, refer to the Bombardier Challenger 605/650 Pro Line 21™ Advanced Flight Management System (523-0822377).

NOTE

Refer to the CDU Operation Chapter of this guide for information on operation of the Radio Sensor System by way of the CDU.

▼ SELECTIONS:

Line Select Keys	There are six line select keys on each side of the display (a total of 12 keys). Not every line select key is active for every display or menu page. Line select keys on the CDU are used in conjunction with the display and menu pages to: • Transfer data from the scratchpad to the appropriate data line. • Copy data from a data line to the scratchpad. • Select an option/mode from a menu of available options.
MSG Button	The MSG button is used to select/deselect the MESSAGES page on the CDU. The MESSAGES page shows new and old FMS messages.
DIR Button	The DIR button is used to select the DIRECT function of the FMS.
FPLN Button	The FPLN (flight plan) button is used to show the FMS FPLN pages.

LEGS Button	The LEGS button is used to show the FMS LEGS pages.
DEP ARR Button	The DEP ARR (Departure Arrival) button is used to show one of three pages, depending on the phase of flight. • DEPART page • ARRIVAL page • DEP/ARR INDEX page.
PERF Button	The PERF button is used to select the Performance function of the FMS.
DSPL MENU Button	The DSPL MENU button is used to show a menu of display options for the MFD. The menu page shows either Map Display options or Text Display options. The MFD DATA button controls which menu page shows on the CDU. • The Map Display options page may be used to select the navigation symbols that are shown on the MFD and PFD maps. The FMS text window on the MFD is also selected from this page. • The Text Display options is used to select various FMS text pages that show on the MFD in place of the map format. • For both the Map Display and Text Display options, the CDU line select keys are used to select and deselect the display options.

NOTE

For the FMS Text pages, only one page of text can show on the MFD at one time. Only one menu item can be selected at any one time on the Text Display page. Selecting one text display item automatically deselects the previously selected display item.

MFD DATA Button

The MFD DATA button is used to control which MFD MENU page shows on the MFD. The available MENU pages are the Map Display and the Text Display. Each push of the MFD DATA button alternately selects the Map Display and the Text Display (when the MFD MENU page shows on the CDU).

MFD ADV Button

The MFD ADV (Advance) button is used to move through MFD text pages or move the FMS Plan Map center waypoint on the MFD Plan Map. The MFD DATA button controls which page shows on the CDU.

- When the MFD MENU Map Display page is in view on the CDU, operation of the MFD ADV button causes the FMS Plan Map Center page to show. The line select keys are used to select a waypoint from the menu options as the FMS Plan Map center.
- When the MFD MENU Text Display page is in view on the CDU, operation of the MFD ADV button causes the Text Display page to show. The line select keys are used to select the next page or previous page of the FMS Text display on the MFD.

PREV Button	The PREV (Previous) button is used to show the previous page of a multi-page CDU display, for example, the TUNE page.
NEXT Button	The NEXT button is used to show the next page of a multi-page CDU display, for example the TUNE page.
EXEC Button	The EXEC (Execute) button acts as an enter or save function for some FMS flight plan edits and performance functions.
IDX Button	The IDX (Index) button is used to show the FMS INDEX pages. Pressing and holding the IDX button for 5 seconds will display the MCDU INDEX page, permitting access to other systems not provided on the FMS INDEX pages.
TUN Button	The TUN (Tune) button on the CDU is used to show the TUNE 1/2 page. The TUNE 1/2 and 2/2 pages provide tuning and mode control for both the on-side and cross-side radios. The PREV and NEXT buttons may be used to page between the TUNE 1/2 and 2/2.
CLR DEL Button	The CLR DEL (Clear/Delete) button is used to clear the scratchpad of data or to create a DELETE command that can be transferred to some data fields on CDU display pages. • With data in the scratchpad, the button acts as a backspace button, clearing characters from the scratchpad from right to left.

- With no data in the scratchpad, operation of the CLR DEL button enters a DELETE command into the scratchpad. To clear the DELETE command, push the CLR DEL button once.

BRT DIM Button

The BRT DIM (Bright/Dim) button is used to control the intensity of the CDU display.

CONTROL DISPLAY UNIT (CDU) – GRAPHICAL WEATHER

Figure 6-4 Control Display Unit (CDU) – Graphical Weather



TPO4638_01

Graphical weather images can be uplinked and selected for display on the CDU in aircraft with the required options. The images are superimposed on geopolitical boundaries. Six types of images are available for selection: NEXRAD Radar, US Storm Cell Identification and Tracking (SCIT) TOPS, Weather Depiction, Winds and Temperature, Turbulence, and Icing. Zoom keys on the CDU can enlarge an area of the image. Most image types are a single frame, however, NEXRAD

can consist of more than one frame. Frame keys enable sequential display of each frame, effectively animating the weather. The CDU Graphical Weather images are accessed through the Datalink pages from the CDU by selecting the WEATHER page and then selecting REQ on the GRAPHICAL WX line.

NOTE

The IFIS illustrations in this operator's guide are for illustrative purposes only.

▼ SELECTIONS:

Graphical Weather Regions

The table that follows lists the regions that must be selected from the CDU.

CDU Graphical Weather Regions

For NEXRAD:	Other than NEXRAD:	
USA	AFRICA	UK
NW US	AUS-TRALIA	S AMERICA
SW US	EUROPE	S PACIFIC
N CENTRAL US	INDIA	W PACIFIC
NE US	N ATLANTIC	N PACIFIC
SE US	N AMERICA	USA
S CENTRAL US		

Graphical Weather Types

The table that follows lists the types of Graphical Weather that can be selected from the CDU.

CDU Graphical Weather Types

Observations:	**Forecasts:**
NX	WINDS
TOPS	TURB WXDEP
WXDEP	ICING

NOTE

Flight Level and Time must be specified when requesting Forecasts.

VIEW GRAPHICAL WX Page

VIEW can be selected on the GRAPHICAL WX line of the WEATHER page or GRAPHIC can be selected on the MCDU page.

NOTE

The VIEW GRAPHICAL WX page lists the images received with the most recent first. Push the line select key adjacent to an image to display that image.

NEXT/PREV

When an image is in view the NEXT and PREV keys are used to show the next and previous image number. When the VIEW GRAPHICAL WX page is in view the NEXT and PREV keys displays additional images, when there is more than one page.

Frame Keys (← / →)

Each push of the → key progressively displays the next newer frame when multiple frames are stored. The ← key displays the frame in reverse order.

Zoom Key (– / +)

Each push of the Zoom key alternately enters or exits the Zoom mode. When Zoom mode is entered, five zoom boxes are overlaid on the graphic image. The zoom area CDU keys: 1, 3, 5, 7, 9 can be pressed to zoom a selected area of the full image.

RETURN

The RETURN select exits the weather image display.

NOTE

When the aircraft is on the ground and GNSS time is received, observation images older than 75 minutes and forecast images older than 3 hours are discarded.

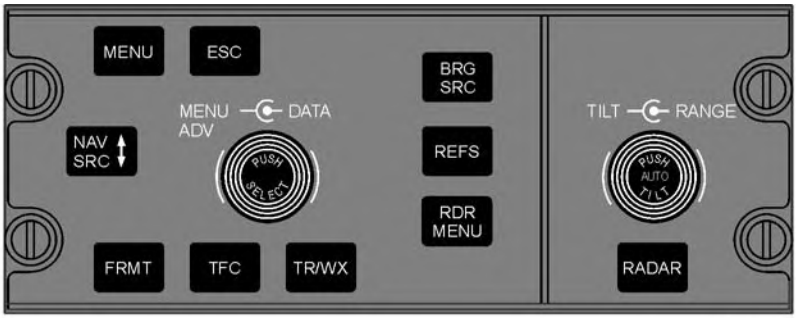
CDU Graphical Weather Messages

The messages that follow may be in view when the CDU Graphical Weather is active on the CDU:

- Image Type (NX, TOPS, WXDEP, WINDS, TURB, ICING)
- Image Issue Date/Time (forecasts only)
- Image Validity Date/Time
- Image Flight Level
- Image Frame Number
- Image Number
- Image Age (observations only).

DISPLAY CONTROL PANEL (DCP)

Figure 6-5 Display Control Panel (DCP)



TPO4645_01

The DCP provides PFD configuration and selection controls for Baro Set, V_{SPEED}/minimums references, PFD format, NAV/Bearing source selection, TCAS overlay, Terrain and Weather overlays for map, and PFD/MFD Terrain/Weather range. Two DCPs are installed, one for the pilot-side system and one at the copilot-side system.

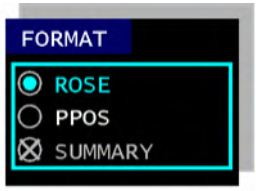
The MENU ADV knob, DATA knob, PUSH SELECT button, and ESC button on the DCP are used to navigate/operate the menus and sub-menus on the PFD. Refer to PFD Menu Navigation section in this chapter for details on PFD menu and sub-menu navigation and operation.

▼ SELECTIONS:

PFD Format and Menu Controls	The FRMT button, MENU button, MENU ADV Knob, DATA knob, PUSH SELECT button, and ESC button are used to select and control the PFD display format/overlay state and to select and navigate the PFD menus.

FRMT Button

The FRMT (Format) button is used to select the PFD display format. The first push of FRMT shows the direct access FORMAT menu. Each push of the FRMT button selects the next display format on the menu. The active format shows in larger, cyan text. The PFD display format can be Rose, PPOS Map, or Summary synoptic. Summary is only available when an on-side display is reverted.



TPO4620_88

MENU Button

The MENU button is used to select and deselect the PFD (main) MENU. The PFD MENU is divided into a FORMAT and CONTROLS sub-menus. The FORMAT sub-menu contains controls for selecting the PFD display format. The possible selections are ROSE and PPOS (Map). Refer to the associated section in this chapter for details on the associated format (for example, PFD/MFD — PPOS Map Format). The CONTROLS sub-menu contains controls for selecting the Navigation source, display range and links to other PFD-related sub-menus.

- Synthetic Vision System — Synthetic Vision System (SVS) menu item is used to select and deselect the display of the SVS.

- FLT PATH VCTR — The FLT PATH VCTR (Flight Path Vector) menu item is used to select and deselect the display of the flight path vector. The flight path vector selection affects the on-side system only. Refer to the PFD — Attitude section in this chapter for details on the flight path vector presentation.



TPO4409_01

NAV-SRC Control

The NAV-SRC (Navigation Source) menu item is used to change the active Navigation (NAV) source (for example, FMS1). Operation of the DATA knob when this item is highlighted changes the NAV source by way of a blind list. Refer to the PFD — Rose Format section in this chapter for details on the active NAV source and NAV-SRC menu.

RANGE Control	The RANGE menu item is used to set the display range for the on-side MFD and PFD. Operation of the DATA knob when this item is highlighted changes the display range by way of a blind list. The dedicated RANGE knob on the DCP can also be used to set the display range. The possible settings are: 5, 10, 25, 50, 100, 200, 300, and 600 NM. The 600-NM range is not available when the Weather Radar overlay or Terrain overlay is active on either on-side MFD or PFD. Refer to the associated sections in this chapter for details on specific range presentations and limits (for example, PFD/MFD — Weather Radar Overlay).
BRG SRC Menu	The BRG SRC (Bearing Source) menu is used to select the bearing source for the on-side MFD and PFD bearing #1 and bearing #2 pointers. The dedicated BRG SRC button on the DCP can also be used to show the BRG SRC menu on the PFD. Refer to the PFD — Rose Format section in this chapter for details on the bearing pointers and BRG SRC menu.



TPO4620_83

CONFIG Menu

The CONFIG (Configuration) menu is used to set the references listed below. All selections made on the CONFIG menu affect both the pilot and copilot-side system.

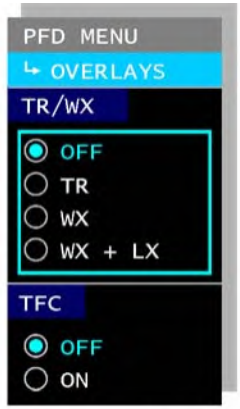


TPO4410_01

- PRESSURE — The PRESSURE menu item is used to set either inHg (IN) or Hecto Pascals (HPA) as the unit of measurement for the barometric pressure setting. Refer to the PFD — Altitude section in this chapter for details on the barometric pressure setting.
- HDG REF — The HDG REF menu item is used to set the heading reference to either Magnetic (MAG) or True (TRUE). Refer to the PFD — Rose Format in this chapter for details on the heading display.
- FLT DIRECTOR — The FLT DIRECTOR (Flight Director) menu item is used to select either a single cue (V-BAR) or split-cue (X-PTR) style flight director. Refer to the PFD — AFCS Message Field section in this chapter for details on the flight director display.
- METRIC ALT — The METRIC ALT (Metric Altitude) menu item is used to select a readout of metric barometric altitude and to change the readout of preselect altitude from feet to meters.
- FL ALERT — The FL ALERT (Flight Level Alert) menu item is used to select and deselect the flight level alert function. The altitude at which the alert will sound shows next to the FL ALERT legend on the CONFIG menu. The altitude value is received from the FMS. Refer to the PFD — Altitude Displays section in this chapter for details on the flight level alert function.

OVERLAYS Menu

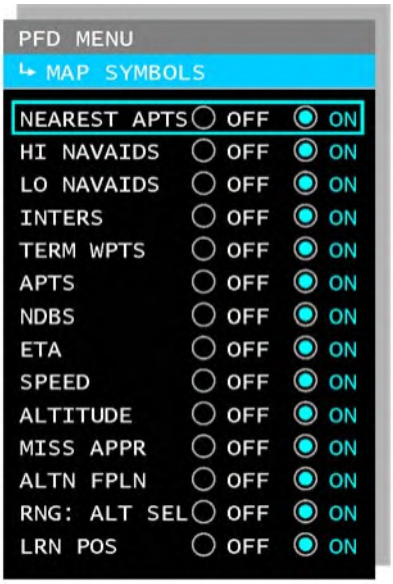
The OVERLAYS menu is used to select the TCAS, Weather Radar, Terrain and Lightning overlays for display on the PFD. The TFC button on the DCP can also be used to access the TCAS overlay on the PFD. The TR/WX button on the DCP can also be used to select the Terrain/Radar/Lightning overlays on the PFD. The OVERLAYS menu selection is not included on the PFD MENU when the active format is SUMMARY.



TPO4620_81

MAP SYMBOLS Menu

The MAP SYMBOLS menu is used select map background symbology for display on the FMS PPOS, PLAN and 3D maps (for example, nearest airports). The map background symbology can also be selected by way of the FMS CDU. Refer to the MFD Map Displays section of the Bombardier Challenger 605/650 Pro Line 21™ Advanced Flight Management System (523-0822377) for details on selecting map background symbology by way of the CDU. Refer to the PFD — PPOS Map section in this chapter for details on map background symbology.



TPO4620_87

RADAR Menu

The RADAR menu is used to select Weather Radar modes, stabilization, gain and other Weather Radar controls. The dedicated RDR MENU button on the DCP can also be used to select the RADAR menu (push-on/push-off). Refer to the PFD/MFD — Weather Radar Overlay section in this chapter for details on the Weather Radar controls and RADAR menu contents.



RTA-854 INSTALLATION
TPO4411_01



RTA-4114 INSTALLATION
TPO4411_02

REFS Menu

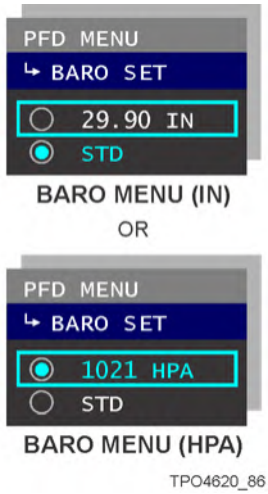
The REFS (References) menu is used to set and select for display the V_{SPEED} references and altitude minimums associated with take off and landing procedures. The dedicated REFS button on the DCP can also be used to select the REFS MENU on the PFD (push-on/push-off). Refer to the PFD — Airspeed section in this chapter for details on V_{SPEED} references and the REFS menu contents. Refer to the PFD — Altitude section of this guide for details on the altitude minimus and REFS menu contents.



TPO4412_01

BARO SET Menu

The BARO SET menu is used to set the barometric pressure setting. The barometric pressure setting can also be set with the remote-mounted installer-supplied BARO set knob. Refer to the PFD — Altitude section of this guide for details on the barometric pressure setting and BARO SET menu contents.



MENU ADV, DATA, PUSH SELECT and ESC Controls

The MENU ADV knob, DATA knob, PUSH SELECT button, and ESC button are used to navigate/operate the menus and sub-menus on the MFD. Refer to MFD Menu Navigation section in this chapter for details on MFD menu and sub-menu operation.

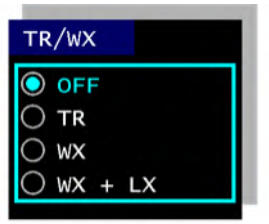
MENU ADV Knob

The MENU ADV knob is used to position the cyan selection box on a menu or sub-menu. A menu is a window that can be brought into view to show one or more control selections. The control selections may be separated into groups within a menu by way of sub-menus resident on the parent menu. Alternatively, sub-menu links may be provided to select sub-menus that are not resident on the parent menu.

DATA Knob	The DATA knob is used to knob to change the value highlighted by the cyan selection box between predefined limits, change the state of a highlighted item (for example, on/off), or select an item from a list.
PUSH SELECT Button	The PUSH SELECT button is used to change the state of a highlighted item, to select a item from a list, and to select a sub-menu for display.
ESC Button	The ESC button is used to back out of a sub-menu, clear a parent menu, and to abort a data entry action.
Direct Access Controls	The TFC and TR/WX buttons are direct access buttons used to show the associated display format or overlay on the PFD without accessing a format menu.
TFC Button	The TFC (Traffic) button is used to select and deselect the TCAS Traffic overlay and the TCAS Only format on the PFD. When a compatible display format is active, momentary operation of the TFC button selects the TCAS Traffic overlay. When the current PFD format is not compatible with a Traffic overlay, or when a TCAS Traffic Advisory is active on the PFD, operation of the TFC button selects the TCAS Only format. The TCAS Only format is also selected when the TFC button is held in for more than 1 second. The display range is set automatically to 10 NM when the TCAS Only format is initially selected. Refer to PFD/MFD — TCAS Map Format this chapter for details on the TCAS traffic overlay and TCAS Only format.

TR/WX Button

The TR/WX (Terrain/Weather Radar) button is used to select/deselect the Terrain, Weather Radar and optional Lightning overlays on the PFD. When no Terrain warning or caution is active, the first push of the TR/WX button selects the next available overlay. When a Terrain warning or caution is active, the first push of the TR/WX button selects the Terrain overlay with a 10-NM range. When the TR/WX button is pushed for more than 1 second the Terrain overlay is selected with a 10-NM range. An appropriate display range is automatically selected for the Weather Radar overlay when necessary. Refer to the PFD/MFD — Terrain Overlay, PFD/MFD — Weather Radar overlay, and PFD/MFD — Lightning Overlay sections in this chapter for details on the associated hazard avoidance overlays.



TPO4620_89

NAV SRC ↕ Button

The NAV SRC ↕ (swap) button is used to exchange the active NAV source for the preset NAV source. The DCP DATA knob is used to select the preset NAV source when the preselect NAV function is not active. Refer to the PFD — Rose Format section in this chapter for details on the preset NAV source.

BRG SRC Button

The BRG SRC (Bearing Source) button is used to show the BRG SOURCE menu on the PFD (push-on/push-off). The BRG SRC menu is used to select the bearing source for the on-side MFD and PFD bearing #1 and bearing #2 pointers. Refer to the PFD — Rose Format section in this chapter for details on the bearing pointers and BRG SRC menu.



TPO4620_83

REFS Button

The REFS (References) button is used to select the REFS MENU on the PFD (push-on/push-off). The REFS menu is used to set and select for display the V_{SPEED} references and altitude minimums associated with take off and landing procedures. Refer to the PFD — Airspeed section in this chapter for details on V_{SPEED} references and the REFS menu contents. Refer to the PFD — Altitude section of this guide for details on the altitude minimus and REFS menu contents.



TPO4412_01

RDR MENU Button

The RDR MENU (Weather Radar Menu) button is used to select the RADAR menu on the PFD (push-on/push-off). The RADAR menu is used to select Weather Radar modes, stabilization, gain and other Weather Radar controls. Refer to the PFD/MFD — Weather Radar Overlay section in this chapter for details on the Weather Radar controls and RADAR menu contents.



RTA-854 INSTALLATION
TPO4411_01



RTA-4114 INSTALLATION
TPO4411_02

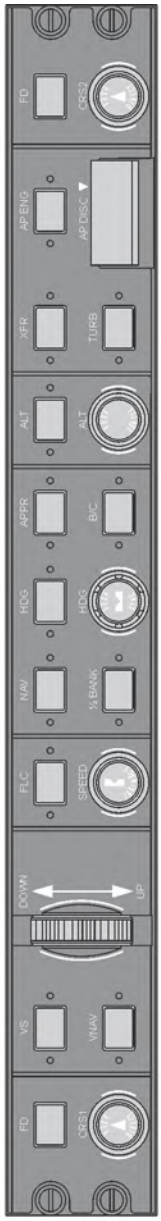
RADAR Button

The RADAR button is used to set the Weather Radar mode to on or standby. Refer to the PFD/MFD — Weather Radar Overlay section in this chapter for details on the RADAR button operation and the Weather Radar mode.

RANGE Knob	The RANGE knob is used to set the display range for the on-side MFD and PFD. The possible settings are: 5, 10, 25, 50, 100, 200, 300, and 600 NM. The 600-NM range is not available when the Weather Radar overlay or Terrain overlay is active. Refer to the associated sections in this chapter for details on specific range presentations and limits (for example, PFD/MFD — Weather Radar Overlay).
TILT Knob	The TILT knob is used to adjust the Weather Radar antenna vertical tilt angle. When the RTA-854 is installed, the AUTO TILT push selection is available. Refer to Appendix E– Weather Radar Techniques chapter of this guide for details on antenna tilt.

FLIGHT CONTROL PANEL (FCP)

Figure 6-6 Flight Control Panel (FCP)



TPO4646_01

The FCP contains most of the controls for operation of the AFCS. The FCP provides controls to engage/disengage the autopilot, transfer flight guidance, select/deselect turbulence mode, and to remove power from the aileron and elevator servos and the rudder linear actuators. The altitude, heading, airspeed, vertical speed, and pitch references are also controlled by way of the FCP. The half-bank control and both pilot and copilot-side FD buttons are also located on the FCP. The FCP is a single unit that controls both the pilot and copilot flight guidance. A non-Rockwell Collins Yaw Damper Panel provides controls for the yaw dampers. Other remote-mounted installer-supplied controls are used for AFCS control. Control inputs are supplied to both flight guidance channels of the selected FCC.

▼ SELECTIONS:

AP ENG Button	The AP ENG (Autopilot Engage) button is used to alternately select and deselect the autopilot. When selected, the autopilot engages if the AP DISC (Disconnect) bar is up, at least one of the yaw dampers is engaged, no excessive rates or attitudes are present, and FCC monitoring does not detect any autopilot faults. When engaged, the autopilot drives the servos in response to steering commands from the coupled side FCC module (pilot or copilot-side). The coupled side is selected with the XFR button on the FCP. Green lights on both sides of the AP ENG button show when the autopilot is engaged. The message AP ← (coupled to left side) or AP → (coupled to right side) shows in green on the PFD when the autopilot is engaged. Other conditions that affect the autopilot engage/disengage condition follow: • Operation of the AP DISC (Autopilot Disconnect) button, GA (Go Around) button, manual pitch trim control, or AP DISC bar disengage the autopilot.

- The autopilot automatically disengages if both yaw dampers are disengaged, if excessive rates or altitudes occur, or if the autopilot monitors in the FCC detect a failure. Operation of the remote AP DISC or GA button cancels the flashing and aural autopilot disengage warnings.

NOTE

Operation of the manual pitch trim control disengages the autopilot but not the flight director.

NOTE

If both yaw dampers disengage, the autopilot also disengages. Controls to engage the yaw dampers are located on the non-Rockwell Collins Yaw Damper Panel.

Automatic Pitch Trim

Automatic pitch trim is enabled whenever the autopilot is engaged. Automatic pitch trim off-loads sustained elevator-control forces. The autopilot cannot be engaged when a mistrim condition exists.

AFCS SEL Switch

The remote-mounted installer-supplied AFCS SEL (AFCS Select) switch is used to select the active and standby flight guidance computer. The AFCS select switch is located on the non-Rockwell Collins Reversion Switch Panel.

AP DISC Bar	The AP DISC (Autopilot Disconnect) bar is used to disengage the autopilot. The AP DISC bar disengages the Autopilot by removing power to the servo clutches. This control is typically only used when there is a failure in the Autopilot. When the bar is down, a red band becomes visible to indicate the disengage position. Manually raise the switch to allow the autopilot to be re-engaged.
TURB Button	The TURB (Turbulence) button is used to alternately select and deselect turbulence mode. The TURB button is functional only when the autopilot is engaged. When turbulence mode is selected, the FCC adapts autopilot gains for turbulent flight conditions. Green lights on both sides of the TURB button indicate TURB mode is selected/active. This mode automatically clears when the autopilot is disengaged or at LOC capture.
XFR Button	The XFR (Transfer) button is used to transfer flight guidance back and forth between the pilot and copilot-side. Normally, the pilot-side flight guidance drives the Autopilot and both the pilot and copilot-side flight directors. When flight guidance is transferred to the copilot-side, the copilot-side flight guidance drives the Autopilot and both flight directors. Green lights on both sides of the XFR button indicate XFR mode is selected/active.

FD Button	The on-side FD (Flight Director) button is used to alternately select and deselect the on-side flight director. When selected, the flight director command bars and AFCS mode messages show on the PFD. The basic modes pitch and roll are automatically selected when a flight director is activated by way of the FD button. Other conditions that affect the Flight director on/off state follow: • The coupled-side flight director is selected automatically when the Autopilot is engaged, or when a vertical or lateral mode is selected. • Operation of the flight guidance transfer (XFR button) automatically selects the coupled-side flight director when the Autopilot is engaged.
SYNC Switch	With the autopilot engaged, the control wheel SYNC (Synchronization) switch is use to disconnect the autopilot servos temporarily while the control wheel is repositioned. Upon release of the SYNC switch, the lateral and vertical references are set to the current conditions and the autopilot smoothly reengages. Regardless of whether the autopilot is engaged or disengaged, the lateral and vertical references for certain AFCS modes will synchronize to the current conditions when the SYNC switch is selected.
CRS Knob and PUSH DIRECT Button	The CRS (Course) knob is used to adjust the on-side VOR or LOC course pointer. The PUSH DIRECT button is used to select a course direct to the on-side VOR station, when VOR is the active NAV source.

Roll Mode	Roll mode is the basic lateral operating mode, and is automatically selected when a flight director is selected for display with the FD. Roll mode is also selected automatically when no other lateral mode is active. De-selecting the active lateral mode selects roll mode. No roll mode button is available on the FCP. When the roll attitude is more than 5 degrees from level when roll mode is selected, commands are generated to maintain the roll angle. When the roll attitude is less than 5 degrees (level), commands are generated to maintain the heading. SYNC switch operation synchronizes the roll reference to the current roll angle (or heading).
HDG Knob and PUSH SYNC Button	The HDG (hHeading) knob is used to adjust the selected heading reference. the PUSH SYNC (Synchronize) button is used to set the selected heading reference to the current heading. The selected heading reference is the target value used for heading mode. The selected heading reference is indicated by the position of the heading bug on the compass rose or arc. When the heading bug is not in view on the compass arc, a heading vector (dashed line) shows to indicate the position of the heading bug.

HDG Button	The HDG (Heading) button is used to alternately select and deselect heading mode. Heading mode is automatically selected before capture while operating in the navigation, approach, or back course mode. Manual selection of the heading mode overrides automatic selection. Heading mode generates commands to capture and track the selected heading reference. The selected heading reference is indicated by the position of the heading bug on the compass rose or arc. Operate the HDG knob to adjust the selected heading reference. The PUSH SYNC button is used to set the selected heading reference to the current heading. The heading reference is indicated by the position of the heading bug on the compass rose or arc. When the heading bug is not in view on the compass arc, a heading vector (dashed line) shows to indicate the position of the heading bug. Green lights on both sides of the HDG button indicate heading mode is selected/active. Deselecting heading mode with the HDG button selects roll mode.

NAV Button	The NAV (Navigation) button is used to alternately arm and clear navigation mode. Navigation mode generates commands to capture and track lateral guidance for en route navigation and non-precision approaches. En route guidance can be from a VOR or FMS. Approach guidance can be from a VOR, LOC (front or back course), or FMS. When TO is the active lateral mode and the FMS is the active navigation source, NAV will remain armed until the altitude has increased sufficient to permit capture, otherwise, the system operates in heading mode until capture of the active NAV source occurs. After capture, the system generates commands to track the lateral guidance. Dead reckoning operation is provided during VOR station passage. Green lights on both sides of the NAV button indicate NAV mode is armed/active. Deselecting NAV mode with the NAV button selects roll mode.

APPR Button

The APPR (Approach) button is used to alternately arm and clear Approach mode. Approach mode enables both lateral and vertical guidance. Lateral Approach mode generates commands to capture and track lateral guidance for LOC approaches and FMS non-precision approaches. If a NAV mode is not the active lateral mode when APPR is selected and the approach capture criteria are not met, HDG will become the active lateral mode. Vertical Approach mode generates commands to capture and track vertical guidance from Glide Slope (GS) or a FMS non-precision Glide Path (GP). Approach mode lateral capture operates the same as for NAV mode. Refer to the Navigation (NAV) Mode section this in this chapter for information on lateral capture. Upon lateral capture, the system arms for vertical capture. When a valid LOC is tuned and in range, GS is armed. When on an FMS non-precision approach with the FMS set up for VNAV operation, GP mode is armed automatically. With the FMS configured for ILS NAV-to-NAV operation, GS mode will be armed once LOC becomes the active lateral mode. Prior to vertical capture, the system operates in the currently active vertical mode. After capture, the system generates commands to track the active vertical guidance. Vertical capture is cleared by back course mode, loss of approach mode, or loss of the localizer. Green lights on both sides of the APPR button indicate APPR mode is armed/active. Deselecting APPR mode with the APPR button selects roll mode.

NOTE

When approach mode is selected, the flight guidance commands operate in a dual-independent configuration. The pilot side FCC channel supplies flight guidance to the pilot flight director, and the copilot-side flight guidance channel supplies flight guidance to the copilot-side flight director. The coupled side flight guidance channel continues to supply flight guidance to the autopilot.

B/C Button

The B/C (Back Course) button is used to alternately arm and clear back course mode. Back course mode generates commands to capture and track lateral guidance for Localizer (LOC) back course non-precision approaches. When selected, if a NAV mode is not the current active lateral mode and the B/C approach capture criteria are not met, HDG will become the active lateral mode. After capture, the system generates commands to track the lateral guidance. Green lights on both sides of the B/C button indicate back course mode is armed/active. Deselecting back course mode with the B/C button selects roll mode.

1/2 BANK Button	The 1/2 (Half Bank) button is used to alternately select and deselect half bank mode. Half bank mode is automatically selected when climbing through a transition pressure altitude of 31 600 feet pressure altitude, or if above this altitude when the flight director is selected. This automatic selection may be overridden by selecting the 1/2 bank button. Half bank is automatically deselected when descending through 31 600 feet. Half Bank mode is unlike the other lateral modes in that it does not calculate lateral guidance, but instead limits the magnitude of the lateral guidance to half the maximum bank angle. Half bank can be active concurrent with many of the other lateral modes. Some lateral modes preclude half bank and automatically deselect it. Half bank has no effect on roll mode operation.
Pitch Mode	Pitch mode is the basic vertical operating mode, and is automatically selected when no other vertical mode is active and the flight director is on. Pitch mode is automatically armed when VNAV is enabled and a vertical discontinuity exists. Pitch mode generates commands to hold the current pitch reference. Operation of the UP ↕ DOWN thumb wheel adjusts the pitch reference angle when pitch hold mode is active. Operation of the remote SYNC switch synchronizes the pitch reference to the current pitch attitude.

UP ↕ DOWN Thumb Wheel	The UP ↕ DOWN thumb wheel is used to adjust the vertical speed reference when vertical speed mode is active, and the pitch reference angle when pitch hold mode is active. Operation of the UP ↕ DOWN thumb wheel selects pitch mode and clears any active vertical mode (except glideslope capture) when a flight director is selected or the autopilot is engaged and the modes that follow are not active: vertical speed mode, vertical approach (capture or track), or overspeed mode.
VS Button	The VS (Vertical Speed) button is used to alternately select or deselect vertical speed mode. Vertical Speed mode generates commands to maintain the vertical speed reference. Upon selection of vertical speed mode, the vertical speed reference is synchronized to the current vertical speed. Operation of the UP ↕ DOWN thumb wheel adjusts the vertical speed reference. Operation of the remote SYNC switch synchronizes the vertical speed reference to the current vertical speed. The vertical speed reference can be set by the FMS when VNAV mode is active. Green lights on both sides of the VS button indicate vertical speed mode is selected/active. The vertical speed reference value shows on the vertical speed scale as a triangular bug, and as a digital readout with a direction arrow. Deselecting vertical speed mode with the VS button selects pitch hold mode.

VNAV Button	The VNAV (Vertical Navigation) button is used to alternately select and clear vertical navigation mode. VNAV enables the FMS to command various vertical modes, and to provide the references associated with those modes within the constraints of the preselect altitude setting. Step climbs may be initiated by setting the preselect altitude, then selecting the desired climb mode. The autopilot levels the aircraft once the preselect altitude is captured. During descents, VNAV computes a geographical path to each waypoint in the flight plan and gives guidance relative to that path, ensuring any descent altitude constraints are honored. The vertical modes that can be commanded by the FMS are pitch hold, preselect altitude, altitude hold, vertical speed, flight level change, and approach. The FMS can also command PATH mode which is a VNAV-only mode. PATH mode is armed automatically when ALTS mode is active, the FMS is set up for VNAV operation with an altitude contstraint or preselected altitude located below the current aircraft altitude. When armed, PATH mode is captured automatically when the FCS begins to capture the preselected altitude. The preselector overrides any VNAV commands generated by the FMS.

FLC Button	The FLC (Flight Level Change) button is used to alternately select and clear flight level change mode. Flight level change mode generates commands to acquire and track the IAS or Mach reference value, or climb/descend at 100 feet/minute towards the preselected altitude/Flight Plan Target Altitude, whichever yields the greater closure rate to the target altitude. Upon selection of flight level change mode, the airspeed or Mach reference value is synchronized to the current airspeed or Mach. Operation of the SPEED knob on the FCP adjusts the IAS or Mach reference value. Operation of the remote SYNC switch synchronizes the IAS or MACH reference to the current value. When VNAV is active, the FMS can control the IAS or Mach reference value. Green lights on both sides of the FLC button indicate flight level change mode is selected/active. Deselecting flight level change mode with the FLC button selects pitch hold mode.
SPEED Knob	The SPEED knob is used to adjust the selected airspeed reference (speed bug). The speed bug is a general purpose speed reference when it is not controlled by the AFCS or FMS. The speed bug indicates the speed reference when flight level change mode is active. The reference speed shows above the speed scale on the PFD.

IAS/MACH Button	The center IAS/MACH button is used to manually select the airspeed or Mach scale on the PFD. The Mach scale is only available when current Mach is greater than 0.40 Mach. The Mach scale is automatically selected when climbing through approximately 33 600 feet. The IAS scale is automatically selected when descending through approximately 33 600 feet. Alternatively, when VNAV is active and the FMS is commanding the speed target (the IAS/Mach reference on the PFD is shown in magenta), the IAS/Mach transition occurs per the active FMS VNAV speed schedule. Automatic transitions can be overridden with the IAS/MACH button.

ALT Knob	The ALT knob is used to adjust the Preselect Altitude (PSA) and arm altitude preselect mode. Altitude preselect mode is not available when altitude hold, vertical approach capture/track or go around are active. When armed, altitude preselect mode operates in the currently selected vertical mode (for example, VS). At the target altitude capture point, altitude preselect mode generates commands to level off and track the target altitude. There are two potential target altitudes, the preselect altitude, selected with the ALT knob on the FCP and the FMS Flight Plan Target Altitude (FPTA), entered by way of the CDU. The system uses the target altitude that is closest to the aircraft altitude in the vertical direction the aircraft is headed. Altitude hold mode is automatically selected following altitude capture. The target altitude is the ALTS or ALTV altitude that is closest to the aircraft altitude in the vertical direction the aircraft is headed. ALTS is armed when an altitude is preselected and the aircraft is not currently at the preselected altitude. ALTV is armed when VNAV is active and the FMS constraint altitude is more limiting than the pre-selector altitude.

ALT Button	The ALT (Altitude) button is used to alternately select and clear altitude hold mode. Altitude hold mode is automatically selected following capture of the preselect altitude. Operation of the remote SYNC switch synchronizes the altitude reference to the current value. Green lights on both sides of the ALT button indicate altitude hold mode is selected/active. Deselecting altitude hold mode with the ALT button selects pitch hold mode.
TOGA Switch	The thrust-lever TOGA (Takeoff/Go Around) switch is used to select takeoff mode or go around mode. • Operation of the TOGA switch on the ground selects takeoff mode. Takeoff mode is both a vertical and lateral mode. Vertical takeoff mode generates a fixed, 12-degree pitch-up command. Loss of an engine changes the pitch up command to 10 degrees. Lateral takeoff mode generates heading hold commands. Selecting takeoff mode selects both flight directors, disengages the autopilot (but not the yaw damper), selects altitude select mode, and clears all other lateral modes, with the exception of NAV when FMS is the active navigation source, which will remain armed. When takeoff mode causes an autopilot disengage, the visual and aural disengage warnings are cancelled by way of the TOGA button or AP DISC switch. Vertical takeoff mode is cleared by manual selection or automatic capture of another active mode, autopilot engagement, or SYNC switch operation.

- Operation of the TOGA switch in the air selects go around mode. Go around mode is both a vertical and lateral mode. Vertical go around mode generates a fixed, 10-degree pitch-up command for all engine and single engine inoperative operations. Loss of an engine changes the pitch up command to 12 degrees. Selecting the go around mode selects both flight directors, disengages the autopilot (but not the yaw damper), clears all other vertical modes except overspeed, and switches the flight guidance commands to a dual-independent configuration. If the FMS is the active lateral navigation source when go around mode is selected, that mode will be retained. Any other active lateral mode will be cleared. When vertical go around mode causes an autopilot disengage, the visual and aural disengage warnings are cancelled by way of the TOGA button or AP DISC switch. Go around mode is cleared by manual selection or automatic capture of another active mode, autopilot engagement, or SYNC switch operation. When go around mode is active, dual-independent flight guidance computations are supplied to the flight directors. The pilot side FCC supplies flight guidance commands to the pilot side flight director and the copilot-side FCC supplies flight guidance commands to the copilot-side flight director. Normally, the coupled-side FCC supplies flight guidance commands to the pilot and copilot flight directors.

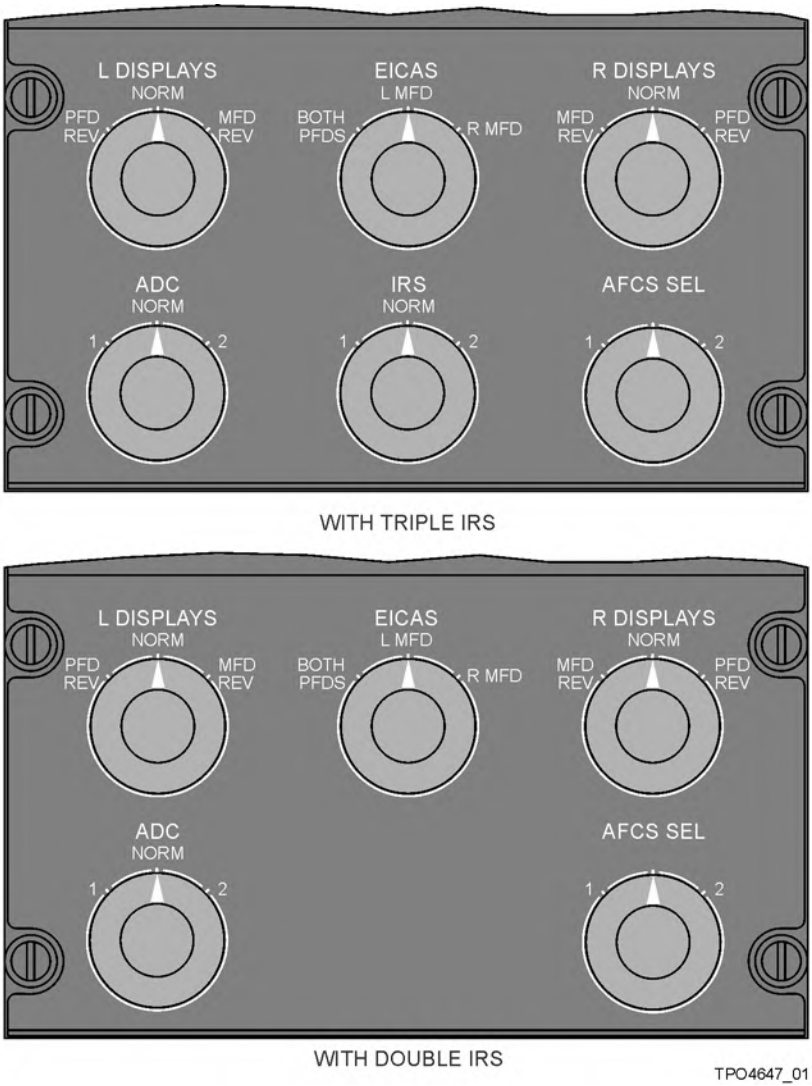
HEAD-UP GUIDANCE SYSTEM (HGS)/ ENHANCED VISION SYSTEM (EVS) CONTROLS

The HGS and EVS controls are found on the outside of the pilot glareshield. While the HGS also contains the combiner in front of the pilot, the overhead unit (also in front of the pilot), and the computer, none of the other items contain any controls.

For a full description of the HGS controls and operations, refer to the Rockwell Collins HGS Model 6605 for Bombardier CL-605 Aircraft Operator's Guide (CPN 523-0816693).

REVERSION SWITCH PANEL (RSP)

Figure 6-7 Reversion Switch Panel (RSP)



The non-Rockwell Collins Reversion Switch Panel (RSP) is used to control display reversion, the location of the primary and compressed EICAS presentations, attitude and heading data (IRS) reversion, and

air data (ADC) reversion. The Automatic Flight Control System select switch is also located on the RSP.

▼ SELECTIONS:

Display Reversion	The RSP is used to manage the basic configuration (either PFD or MFD) of the AFD and to control the location of EICAS. A PFD can be selected to show on either the inboard or outboard AFD. If a MFD fails, the PFD operation is not hindered. The L DISPLAYS and R DISPLAYS controls on the RSP are used to control the PFD and MFD presentations on the pilot and copilot-side AFD respectively. Each side DISPLAYS control has a NORM, PFD REV, and MFD REV control selection. • NORM — The inboard AFD acts as a MFD and the outboard AFD acts as a PFD. • PFD REV — The outboard AFD acts as a PFD and the inboard AFD is powered-down. • MFD REV — The inboard AFD acts as a PFD and the outboard AFD is powered-down.
EICAS Reversion	The EICAS control is used to manually control the location of the primary EICAS presentation. If the MFD with the primary EICAS presentation fails or is powered-down, the primary EICAS presentation is automatically transferred to the cross-side MFD. The three available positions of the EICAS control are L MFD, R MFD, and BOTH.

- L MFD — The primary EICAS presentation shows on the left inboard AFD.
  - When the left inboard AFD is a MFD (the AFD is not reverted) the MFD is locked into a split window configuration with the primary EICAS presentation in the upper window. The locked split window configuration does not allow the display of upper window menus. Refer to the MFD – Primary EICAS section in this chapter for details on the primary EICAS presentation.
  - When the left inboard AFD is a PFD (the AFD is reverted), a compressed EICAS presentation shows on the PFD. Refer to the MFD – Compressed EICAS section in this chapter for details on the compressed EICAS presentation.
  - When the left MFD is powered-down and the EICAS switch is set to L MFD, the EICAS presentation automatically shows on the right MFD.

- R MFD — The primary EICAS presentation shows on the right inboard AFD.
  - When the right inboard AFD is acting as a MFD (the AFD is not reverted) the MFD is locked into a split window configuration with the primary EICAS presentation in the upper window. The locked split window configuration does not allow the display of upper window menus. Refer to the MFD – Primary EICAS section in this chapter for details on the primary EICAS presentation.
  - When the right inboard AFD is a PFD (the AFD is reverted), a compressed EICAS presentation shows on the PFD. Refer to the MFD – Compressed EICAS section in this chapter for details on the compressed EICAS presentation.
- BOTH — A compressed EICAS presentation shows on any AFD that is configured as a PFD. Refer to the PFD — Compressed EICAS section in this chapter for details on the compressed EICAS presentation.

Air Data Reversion

The ADC control is used to select air data reversion. The three positions for the switch are 1, NORM, and 2.

- NORM — The pilot and copilot-side system each receives data from the on-side ADS.
- 1 — Both the pilot and copilot-side systems receive data from (the left) ADC 1. A common source annunciation, ADC1 (yellow) will be shown on both PFDs.

- 2 — Both the pilot and copilot-side systems receive data from (the right) ADC 2. A common source annunciation, ADC2 (yellow) will be shown on both PFDs.

AFCS SEL

The Automatic Flight Control System (AFCS) utilizes two dual-channel Flight Control Computers (FCC), one active and one standby. The remote-mounted installer-supplied AFCS SEL (AFCS Select) switch is used to select the active and standby FCC. Refer to Flight Control Panel (FCP) section in this chapter for details on operation of the AFCS.

REMOTE SWITCHES

The non-Rockwell Collins remote-mounted installer-supplied controls are located on the flight deck.

▼ SELECTIONS:

ALTM Knob	The ALTM (Altimeter) knob is used to set the barometric pressure setting. The PUSH STD button in the ALTM knob is used to set the standard barometric pressure setting of 29.92 inHg / 1013 hPa.
AP DISC Switch	The control wheel AP DISC (Autopilot Disconnect) switch is the primary means of disengagement of the autopilot. Operation of an AP DISC switch disengages the AP when engaged. The AP DISC switch is also used to cancel the flashing and aural autopilot disengage warning.
AURAL WARN TEST Switch	The AURAL WARN TEST switch is used to test the flight deck DCU aural warnings.
Checklist Select Switch	The optional checklist select switch is used to select the checklist format in the MFD upper window and the last selected lower format in the MFD lower window. Refer to the MFD — Checklist section in this chapter for details on checklist operation.
Checklist Line Advance Switch	The optional checklist line advance switch is used to command a line advance when a checklist is in view and the selection box is on the checklist. Refer to the MFD — Checklist section in this chapter for details on checklist operation.

DCU Disable Switch

The DCU 1 Disable and DCU 2 Disable switches are used to disable the audio warning of the associated DCU in event of a DCU failure.

FMS TUNE INHIB Switch

The FMS TUNE INHIB (flight management system tune inhibit) switch is used to disable the FMS remote tune function. When set to NORM, the radio sensor system accepts radio tuning data from the FMS. When set to INHIB, the radio sensor system does not accept remote tuning data from the FMS.

IRS Reversion

The left and right Inertial Reference System (IRS) reversion switches are used to select attitude and heading data reversion.



PANEL WITH 3RD IRS, TRIPLE FMS



PANEL WITH 3RD IRS, DUAL FMS



PANEL WITH DUAL IRS, TRIPLE FMS

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- Operation with 2 IRS sources (baseline) is as follows:
  - NORM — The pilot and copilot-side system each receives data from the on-side IRS.
  - 1 — Both the pilot and copilot-side systems receive data from (the left) IRS 1. A common source annunciation, IRS1 (yellow) will be shown on both PFDs.
  - 2 — Both the pilot and copilot-side systems receive data from (the right) IRS 2. A common source annunciation, IRS2 (yellow) will be shown on both PFDs.
- Operation with 3 IRS sources is as follows:
  - NORM — The pilot and copilot-side system each receives data from the on-side IRS.
  - ALT — The selected side system receives data from the center IRS and an alternate source annunciation is shown on the affect PFD.

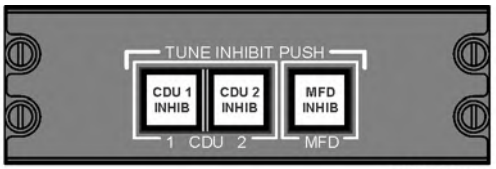
NOTE

In the event IRS reversion has been simultaneously selected for both L and R IRS in a triple IRS installation, the functional effect will be the same as selecting the alternate source for the L IRS only. This is implemented in aircraft wiring to prevent the operation of certain flight control system functions on a single, common reference.

FMS 3 Switch	The FMS 3 switch is used to select the optional third Flight Management System (FMS). • NORM — The number 1 and 2 flight management computers provide FMS data to the displays. • 1 — The input from the FMS 1 is disconnected and the input from FMS 3 is connected. FMS 3 operates as a left side FMS. • 2 — The input from the FMS 2 is disconnected and the input from FMS 3 is connected. FMS 3 operates as a right side FMS.
Manual Pitch Trim Control	The manual pitch trim control is used to apply manual pitch trim commands. Operation of the manual pitch trim control disengages the autopilot (but not the yaw damper).
Overspeed Test Switch	The optional remote-mounted installer-supplied overspeed test switch is used to test the flight deck OVERSPEED WARNING. This function is available with weight-on-wheels only.

Radio Tuning Reversion

The radio tuning reversion controls are used to select radio tuning reversion. Tuning reversion is used to inhibit the pilot or copilot-side CDU radio tuning and/or the MFD radio tuning. The CDU is the master tuner in normal operating mode (no inhibit buttons pressed). The on-side CDU is the master of the cross-side radios when the cross side CDU is inhibited. When a CDU is inhibited, all tuning commands from that CDU are ignored. The MFD is the master tuner only when CDU1 INHIB and CDU2 INHIB are active. When the MFD is the master tuner, each MFD will tune the on-side radios. When CDU1 INHIB, CDU2 INHIB, and MFD INHIB are active, an emergency tuning mode is automatically selected.



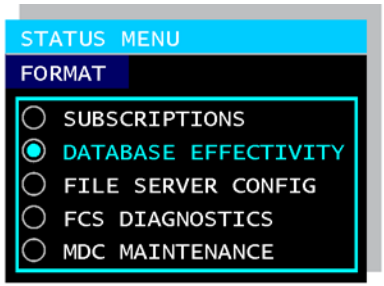
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- CDU 1 INHIB — This selection is used to disable the pilot side CDU tuning and configure the copilot-side CDU as the master controller for the pilot and copilot-side radios.
- CDU 2 INHIB — This selection is used to disable the copilot-side CDU tuning and configure the pilot side CDU as the master controller for the pilot and copilot-side radios.
- CDU 1 INHIB and CDU 2 INHIB — This configuration is used to make the MFD the master radio tuner.

- MFD INHIB — This selection is used to remove the MFD Radio Tune window and menus.
- CDU 1 INHIB, CDU 2 INHIB, and MFD INHIB — This configuration is used to tune the left COM radio to 121.5 kHz. The message 121.5 EMER shows in the COM1 position on each CDU and the MFD.

MAINT Switch

The momentary MAINT switch is used along with the UPR or LWR MENU button on the CCP to show the status format menu on the pilot or copilot-side MFD. When the MAINT switch is selected to MFD1 or MFD2, the last STATUS format in view shows on the selected MFD. When the UPR or LWR MENU button on the same-side CCP is selected, a menu with the available STATUS formats shows on the MFD. The STATUS menu provides a means to navigate to all the infrequently used secondary display formats (for example, DATABASE EFFECTIVITY).

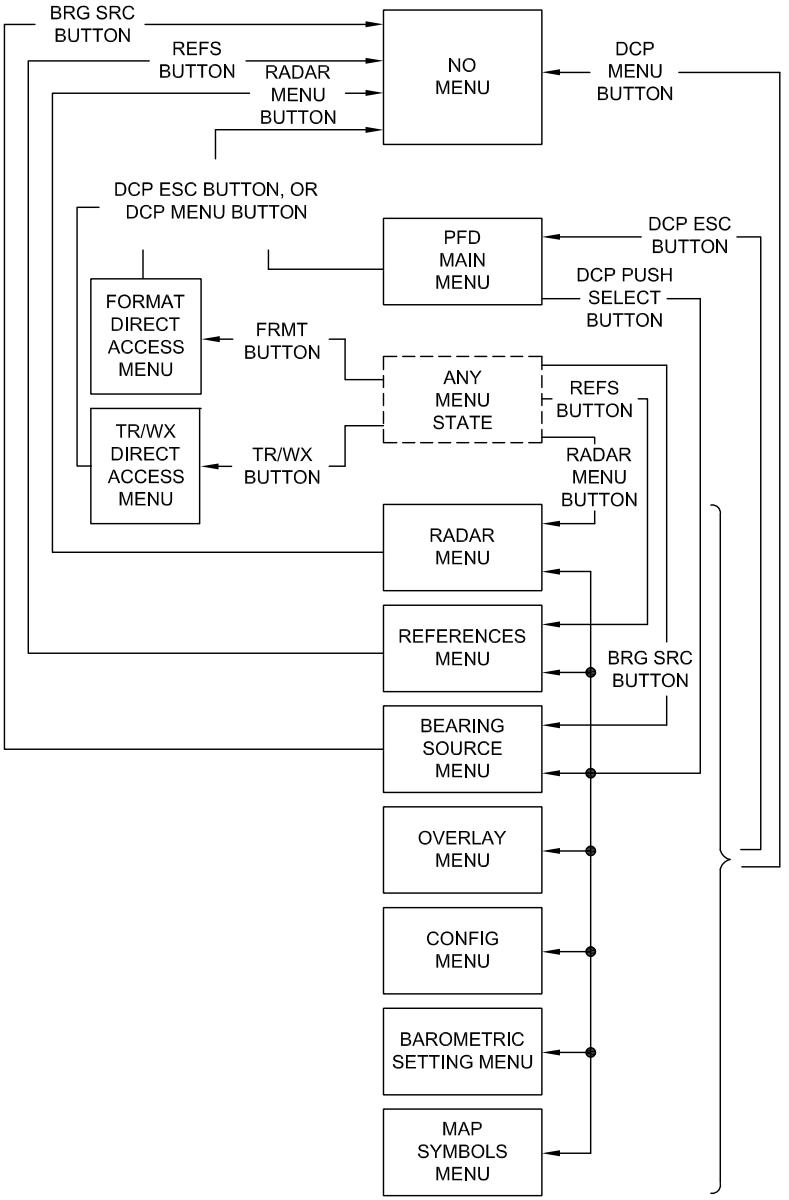


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SYNC Switch	With the autopilot engaged, the control wheel SYNC (Synchronization) switch is used to disconnect the autopilot servos temporarily while the control wheel is repositioned. Upon release of the SYNC switch, the lateral and vertical references are set to the current conditions and the autopilot smoothly reengages. The lateral and vertical references also synchronize to the current conditions when the autopilot is not engaged and the SYNC switch is selected.
TOGA Switch	The thrust lever TOGA (Takeoff Go-Around) switch is used to select flight director takeoff mode when on the ground or Go-Around mode when in the air. Refer to Flight Control Panel (FCP) section in this chapter for details on TOGA switch operation.
Trim Disconnect Control	The trim disconnect control is used to disengage the non-Rockwell Collins horizontal stabilizer trim control unit.

PFD/MFD – MENU NAVIGATION

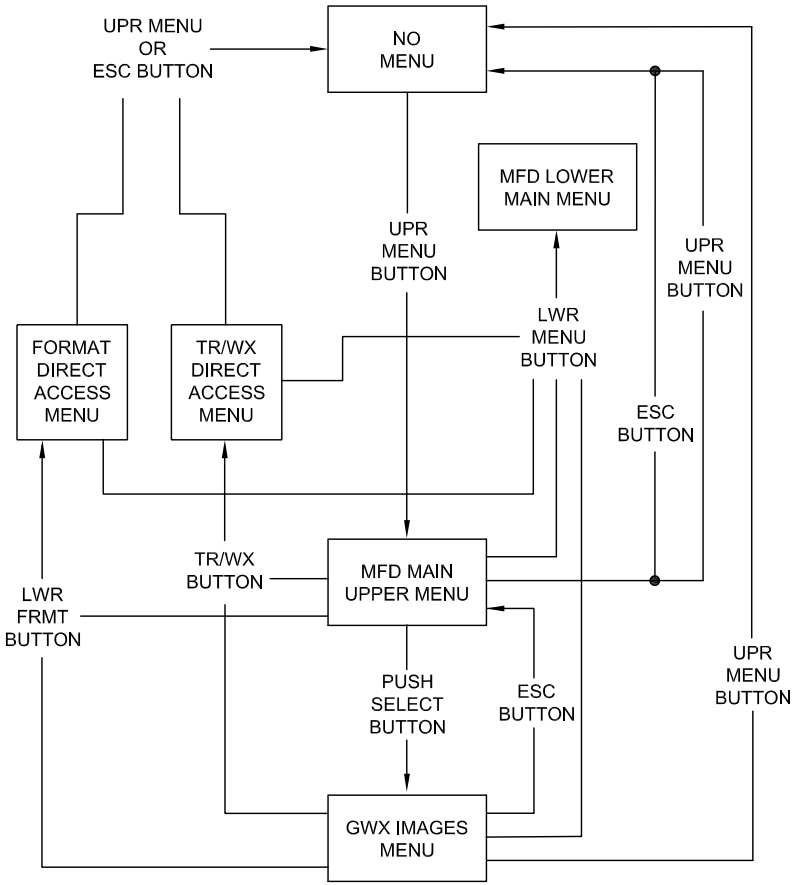
Figure 6-8 PFD/MFD Menu Navigation



PFD MENU NAVIGATION

TPO4320_01





MFD UPPER WINDOW NAVIGATION

A menu is a window that can be brought into view to show one or more control selections. The control selections may be separated into groups within a menu by way of sub-menus resident on the parent menu. Alternatively, sub-menu links may be provided to select sub-menus that are not resident on the parent menu. Various control methods are used to operate the menus and sub-menus (for example, single state toggle, action/command, numeric data entry, etc.). The MENU ADV knob, DATA knob, PUSH SELECT button, and ESC button are used to navigate/operate the menus and sub-menus on the MFD. The MENU ADV knob is used to position the cyan selection box when a menu or sub-menu is in view. The DATA knob is used to change the value highlighted by the cyan selection box between predefined limits, change the state of a highlighted item (for example, on/off), or select an item from a list. The PUSH SELECT button is used to change the state of a highlighted item, to select a item from a list, and to select a sub-menu for display. The ESC button is used to back out of a sub-menu, clear a parent menu, and to abort a data entry action.

▼ SELECTIONS:

Sub-menus	Sub-menus are used to group sets of controls within a menu. Sub-menu links are used to access sub-menus that do not show on the parent menu. Resident sub-menus are indicated by a sub-menu header on the parent menu. Non-resident sub-menus are indicated by a cyan triangle on the parent menu. Sub-menus that are not selectable (inactive) do not show a cyan triangle. When a non-resident sub-menu is selected, the sub-menu title shows below the parent menu title. The parent menu title shows in grey and the sub-menu (active menu) title shows in blue. When available, operation of a direct access control can also be used to show a sub-menu.

Single State Toggle	This function is used toggle a function back and forth between on and off. An empty white box shows when the control state is off. A grey X shows when the control is disabled (not selectable). Operate the associated Advance (ADV) knob and PUSH SELECT button to select on or off. When available, operation of a direct access control has the same effect as operation of the ADV knob PUSH SELECT button. A cyan check mark shows in the box when the control state is on.
Action/Command	This function is used to initiate some action or event that is self evident (for example, test). A small filled diamond with a diamond shaped outline indicates an item has not been selected. Operate the associated ADV knob and PUSH SELECT button to initiate the desired action. A filled cyan diamond indicates an active selection. In some cases, after a selection is activated the current menu and all parent menus are automatically exited.

Multi State Toggle	This function is used for items with more than one control state when only one state can be active at a time (for example, off, on, and auto). A horizontal or vertical list may be used to show the control states. An empty white circle shows for control states that are not selected. A grey X shows when a control state is disabled (not selectable). Operate the associated advance ADV knob and PUSH SELECT button to select the desired state. When available, operation of a direct access control has the same effect as operation of the ADV knob and PUSH SELECT button. A filled cyan circle shows for the active control state.
Select List	This function is used to select one item from a list of similar items. This differs from the multi-state toggle in that once the item is selected the menu closes and a predefined action takes place (for example, select a chart for display). When more than one page of items exist, a NEXT PAGES and/or PREV PAGE control shows in the title bar. Operate the ADV knob and PUSH SELECT button to show the desired list page. Operate the ADV knob and PUSH SELECT button to select the desired item and close the list.

Numeric Data Entry	This function is used when a value must be entered (for example, antenna tilt value). The data entry field shows in cyan when the control is active (not disabled). The control name and data entry field show in grey when the control item is disabled (not active). Operate the associated advance ADV knob and DATA knob to change the desired value between predefined limits. When available, operation of a direct access control (for example, antenna tilt knob) has the same effect as operation of the ADV knob and DATA knob. The direct access control allows the associated value to be changed without needing to bring the menu in to view.
Single Toggle Numeric Data Entry	This function is used when a value must be entered and the selection of that value has an on/off state associated with it (for example, V_{SPEEDS}). An empty white box shows when the control state is off. A grey X shows in the box when the control is disabled (not selectable). Operate the associated ADV knob and DATA knob to change the desired value between predefined limits. Operate the PUSH SELECT button to toggle the control back and forth between on and off. A cyan check mark shows in the box when the control state is on.

Multi Toggle Numeric Data Entry	This function is used when more than one display value can be set, but only one of the values can be active (in view) at any time (for example, RA MIN and BARO MIN). An empty white circle shows when a value/control state is not selected. A grey X shows when the control state is disabled (not selectable). Operate the associated advance ADV knob and DATA knob to change the desired value between predefined limits. Operate the PUSH SELECT button to toggle the control back and forth between on and off. When available, operation of a direct access control has the same effect as operation of the ADV knob and DATA knob. A filled cyan circle shows for the value/control state that is active.
Enumerated (one after another) Data Entry	This function is used when there is a list of items or states that can be selected, but only the currently selection shows on the menu (for example, active NAV source). The current selection shows in cyan when the control is not disabled. Two grey dashes show when the control is disabled. Operate the associated ADV knob and DATA knob to change selected item. When available, operation of a direct access control has the same effect as operation of the ADV knob and DATA knob.

Edit Link - Select List Control	This function is used to generate a link to a piece of information that needs to be recalled later. The information is selected from a sub-menu list (for example, E-Charts). The edit link function is indicated by the presence of cyan brackets. Operate the associated ADV knob to position the cyan selection box to the desired link. To show the linked item, push the PUSH SELECT button for less than 1 second. To edit the link, push and hold the PUSH SELECT button for more than 1 second. This action shows a sub-menu list of other link items. Operate the ADV knob and PUSH SELECT button to update the link to the desired list item. Push the Escape (ESC) button to abort the edit link action. The bookmark links can be automatically populated by the FMS (for example, a Terminal Area E-Chart, determined by the FMS flight plan is used to populate the appropriate link item). The text is magenta when the link has been populated automatically by the FMS.

Edit Link - Alphanumeric Control

This function is used to generate a link to a particular piece of information that needs to be recalled later. The information is not selected from a list but is recalled from a database by way of the alphanumeric identifier (for example, KCID). The edit link function is indicated by the presence of cyan brackets. Operate the ADV knob to position the cyan selection box to the desired link. To show the linked item, push the PUSH SELECT button for less than 1 second. To edit the link, push and hold the PUSH SELECT button for more than 1 second. This action activates data entry mode which brings the data entry cursor into view. The data entry cursor is a cyan box that fits around a single character in the data entry field. Operate the associated ADV knob to move the cursor to the character to be changed. Operate the DATA knob to cycle through the list of available characters. Repeat this process until the desired character string is entered. Push the ESC button to abort the data entry mode.

PFD – ANGLE OF ATTACK

Figure 6-9 PFD – Angle of Attack



An Angle of Attack (AOA) indicator shows full-time to the left of the airspeed scale on the PFD. The AOA indication is normalized to correct for aircraft configuration effects, and therefore shows as a value between 0.0 and 1.0. The value of 1.0 represents the stall angle of attack.

▼ SELECTIONS:

AOA Pointer and Scale	Current normalized AOA is indicated by a white, left facing pointer. The open pointer symbol points to the right edge of the AOA scale. The AOA pointer is positioned dynamically as normalized AOA changes. The AOA scale is made up of a vertical double width line with grey tick marks at every 0.10 AOA increment. Labels show on the scale at the even AOA indices (for example, .0, .2, .4).

NOTE

K1 and K2 are normalized AOA breakpoint values stored in the Data Concentrator Unit (DCU). The DCU passes these values to the AFD in order to draw appropriate color bands on the AOA scale.

PFD – AIRSPEED

Figure 6-10 PFD – Airspeed



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Indicated airspeed shows on the PFD as a moving airspeed or Mach scale with a fixed-position pointer. Both scales provide airspeed trend, overspeed, impending stall speed, a STALL message, an AFCS speed bug, and V_{SPEEDS} references.

▼ SELECTIONS:

Airspeed Display	The airspeed display is made up of a vertical scale and fixed-position pointer. A digital readout of Mach shows part-time below the scale. The range of the scale is 40 to 400 knots IAS. Short tick marks on the scale show at intervals of 5 knots, and longer tick marks show every 10 knots. Scale numbers show every 20 knots. A readout of the current airspeed shows inside the pointer. The airspeed readout changes to yellow for an overspeed pre-alert and to red for an actual overspeed. The airspeed readout changes to yellow for a Impending Stall Speed Low Speed Cue (ISS LSC) Pre-Warn. The airspeed readout changes to red for an ISS LSC Warn. A 3-digit readout of Mach shows below the airspeed scale when Mach is .450 or greater. The Mach readout is removed when Mach is less than .400. The color of the Mach readout follows the color of the airspeed readout for overspeed and Impending Stall Speed Low Speed conditions. The IAS scale is automatically selected when descending through approximately 33 600 feet. Alternatively, when VNAV is active and the FMS is commanding the speed target (the IAS/Mach reference on the PFD is shown in magenta), the Mach/IAS transition will occur per the active FMS VNAV speed schedule. Automatic transitions can be overridden with the IAS/MACH button on the FCP.

Mach Display	The Mach display is made up of a vertical Mach scale and a fixed-position pointer. A readout of the current Mach shows inside the pointer. The range of the scale is 0.2 to 1.0 Mach. The Mach scale has grey tick marks every 0.01 Mach with the even Mach values longer than the odd Mach values and white numerals every 0.04 Mach. The Mach readout changes to yellow for an overspeed pre-alert and to red for an actual overspeed. The Mach readout changes to yellow for a Impending Stall Speed Low Speed Cue (ISS LSC) Pre-Warn. The Mach readout changes to red for an ISS LSC Warn. A digital readout of Airspeed shows part-time below the Mach scale. The Mach scale is only available when current Mach is greater than 0.40 Mach. The Mach scale is automatically selected when climbing through approximately 33 600 feet. Alternatively, when VNAV is active and the FMS is commanding the speed target (the IAS/Mach reference on the PFD is shown in magenta), the IAS/Mach transition will occur per the active FMS VNAV speed schedule. Automatic transitions can be overridden with the IAS/MACH button on the FCP.
Airspeed Trend Vector	The airspeed trend vector is a vertical, variable length magenta line that shows to the right of the airspeed/Mach scale. The tip of the airspeed trend vector points to the airspeed/Mach that will be (approximately) in 10 seconds at the present rate of acceleration or deceleration.

Maximum Limit Airspeed Marker	A maximum limit speed marker (V_{MAX}) shows on the airspeed scale to indicate the maximum speed for the current configuration. V_{FE}, V_{MO}, or M_{MO} may show. When not in an overspeed condition, the V_{MAX} marker shows as a triple-wide vertical red line on the Airspeed scale. The marker is constantly positioned from the top of the scale down to the appropriate limiting speed on the scale. V_{MAX} is computed as the lessor of the speeds that follow: V_{MO}, M_{MO}, and V_{FE} (current flap extended limit speed). The marker becomes emphasized and the airspeed digital readout changes color for an overspeed pre-alert and condition.
Overspeed Pre-Alert	An overspeed pre-alert occurs when the airspeed trend vector exceeds the V_{MAX} by 3 knots for 5 seconds or more. The overspeed marker is emphasized by changing to a wider red and red outlined dim blue checkerboard pattern. The airspeed digital readout changes color to yellow. If the Mach readout is in view, the readout changes to yellow. The pre-alert ends when the airspeed and airspeed trend vector is less than or equal to V_{MAX}.

An airspeed alert occurs when the current airspeed becomes greater than V_{MAX} by 0.5 knots. The flight deck aural OVERSPEED WARNING message sounds and the overspeed marker is emphasized by changing to a wider red and red outlined, dim blue checkerboard pattern. The overspeed pre-alert will typically already have the checkerboard in view. The airspeed digital readout changes color to red. If the Mach readout is in view, the Mach readout changes color to red. The overspeed symbology returns to previous symbols and/or colors, and the aural alert becomes inactive, when the airspeed is less than or equal to V_{MAX}.

NOTE

The V_{MO} default value of 348 knots is used when a failure occurs such that V_{MO} cannot be calculated.

NOTE

The optional remote-mounted installer-supplied overspeed test switch is used to test the flight deck OVERSPEED WARNING. This function is available with weight-on-wheels only.

Low Speed Cues

Low speed cues (LSC) show on the airspeed/Mach scale and are based on calculations from the angle of attack vane. There are three distinct low speed cues, the V_{MIN} marker, the V_{MIN} pre-warning, and the V_{MIN} warning.

V Minimum Marker	A red V_{MIN} speed marker indicates the minimum airspeed/Mach that the aircraft can reach without stalling. The calculation changes with aircraft configuration changes (flaps). The V_{MIN} calculation is also sensitive to aircraft load factor (g), and therefore increases as load factor is applied in maneuvering flight. The V_{MIN} marker is available 3 seconds after a ground to air transition. The V_{MIN} marker is a triple-thick, red vertical line that extends from the bottom of the airspeed scale up to the V_{MIN} value. The V_{MIN} marker becomes emphasized and the airspeed digital readout changes color when airspeed trend or current airspeed gets below the V_{MIN} value.
V Minimum Pre-Warning	A V_{MIN} pre-warning occurs when the airspeed trend vector goes below the top of the V_{MIN} speed marker by 3 knots for 5 seconds or more. The V_{MIN} marker becomes emphasized by changing to a wider red and red outlined, dim blue checkerboard pattern. The airspeed digital readout changes color to yellow. If the Mach readout is in view, the readout changes to yellow. The pre-warning ends when the airspeed trend vector is greater than or equal to V_{MIN}.

V Minimum Warning	The V_{MIN} warning occurs when the airspeed becomes less than the top of the ISS LSC marker by 3 knots. The ISS LSC Marker is emphasized by changing to a wider red and red outlined dim blue checkerboard pattern. The V_{MIN} pre-warning will typically already have the checkerboard in view. The Airspeed digital readout changes color to red. If the Mach readout shows, the readout changes to red. The V_{MIN} warning is cleared and the airspeed symbology returns to previous symbols and/or colors when the airspeed is greater than or equal to the V_{MIN} value.
STALL Message	The flight deck aural warning STALL horn sounds and the message STALL shows in red to the left of the attitude display when commanded by the Stall Protection Computer.
Speed Bug	The speed bug is a general purpose speed reference when it is not controlled by the AFCS or FMS. The speed bug is used to display the reference speed for Flight Level Change (FLC) mode. The speed bug can be commanded by the FMS in VNAV sub modes. The speed bug shows on the airspeed scale, and as an icon and digital readout above the airspeed scale and on the AFCS field above the attitude display. The SPEED Knob on FCP is used to set the value of the speed bug. The pilot setting of the speed bug with the SPEED knob and IAS/Mach button has priority over control from the FMS and AFCS. The speed bug, readout, and icon are cyan when set with the SPEED knob and magenta when set by the FMS.

V_{SPEEDS}	V_{SPEED} reference bugs show on the airspeed scale as a visual cue of the takeoff and approach speeds. The V_{SPEEDS} are selected for display using the References (REFS) menu on the PFD. V_{SPEEDS} are grouped on the REFS menu pages into takeoff speeds and approach speeds. Power-up (default) V_{SPEED} values, FMS-calculated V_{SPEED} values (FMS option), or manually adjusted V_{SPEED} values may be selected to show on the airspeed scale. When selected for display, the V_{SPEED} values show at the associated airspeed on the airspeed scale.

On the ground, when the REFS menu first shows, the default V_{SPEED} values show on the menu in small white digits. The small white digits indicate that the V_{SPEED} references are set to OFF. When a reference is set to OFF, the V_{SPEED} value does not show on the airspeed scale. V_{SPEED} values set to ON by way of the REFS menu show in cyan characters on the MENU. V_{SPEED} values set to ON by way of the FMS show in magenta characters on the menu. FMS-generated V_{SPEED} data is transmitted to the PFD when the SEND control is selected on the FMS CDU Performance Page. When two FMSs are fitted, selecting SEND from one CDU causes both FMSs to send the V_{SPEEDS} to the on-side PFD. Upon initial transmission of V_{SPEED} data from the FMS, the reference values show in magenta. When V_{SPEED} data is transmitted from the FMS, the values are set to ON. When the DCP DATA knob is operated while a value is magenta, the V_{SPEED} changes to OFF and the color changes accordingly. This color change is used to signal that the value in the field is no longer the value set by the FMS.

Each V_{SPEED} reference bug shows as a horizontal line with a label on the airspeed scale at the associated airspeed. V_1, V_R, V_2, and V_T also show in a list in the lower half of the airspeed scale when the airspeed is less than 40 knots.

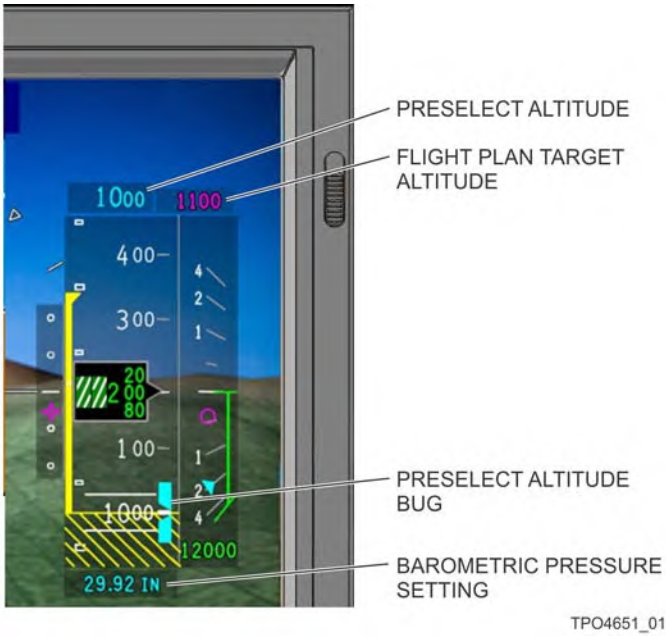
Each V_{SPEED} is settable between 100 knots and V_{MO}. The lower range of each V_{SPEED} value is bumped to the selected value of the previous V_{SPEED} in the order that follows: V_1, V_R, V_2, and V_T. V_{SPEED} values that have been bumped are set to OFF (for example, if V_R is set to 120 knots and then subsequently V_1 is set to 130 knots, V_R is automatically deselected and increased to 130 knots.) The V_{SPEEDS}, V_{RF} and V_{AC} are settable between 100 knots and V_{MO} and are not bumped.

The V_{SPEED} power-up values default to the minimum V_{SPEED} value of 100 knots. The power-up V_{SPEED} state is all V_{SPEEDS} deselected (OFF).

The takeoff V_{SPEEDS} V_1 V_R and V_2 V_{SPEEDS} are removed from the display at 225 knots. The approach V_{SPEEDS} (V_{RF} and V_{AC}) are removed from the display when on the ground and the airspeed drops below 50 knots.

PFD – ALTITUDE

Figure 6-11 PFD – Altitude



The altitude display is a moving altitude scale with a fixed-position pointer. A digital readout of altitude shows inside the pointer. The equivalent metric value of barometric altitude can show in addition to the display of altitude in feet.

▼ SELECTIONS:

Altitude Display	The altitude scale is made up of a coarse and fine moving scale. The fine altitude scale shows 540 feet of altitude in view at even 100-foot increments. The 500 foot marks have white horizontal lines above and below the numerical labels. The 000 foot mark has a longer white horizontal line above and below the numerical labels. The coarse altitude scale is made up of large and small white rectangular symbols along the left side of the altitude scale. The large rectangles represent integral one thousand-foot altitude marks and the small rectangles represent intermediate five-hundred foot altitude marks. Current barometric altitude shows inside a pointer indicating the current altitude on the altitude scale. Green stripped shutters cover the ten thousands and thousands digits when a leading zero digit would be in view, except for negative altitudes.

Metric Altitude

A display of metric barometric altitude and metric preselect altitude can be selected to show in addition to the display of barometric altitude and preselect altitude in feet. The readout of metric barometric altitude shows inside a grey box located above the barometric altitude pointer on the altitude scale. The readout of metric barometric altitude is followed by the letter M. When selected, the readout of metric preselect altitude shows inside a grey box located above the display of preselect altitude in feet. The readout of metric preselect altitude is followed by the letter M. The display of metric barometric altitude and metric preselect altitude is selected by way of the PFD CONFIG menu.

Barometric Pressure Setting

Barometric pressure setting shows in cyan below the altitude scale on the PFD. The barometric pressure setting is set by way of the BARO SET menu on the PFD.

- A preset value of barometric pressure shows part-time below the active barometric pressure value. The preset is in view when the active barometric pressure is set to the standard value of 29.92 inHg / 1013 hPa and when the preset value is changing. This preset value can be swapped with the standard value to become the active value. When the active value is set to the standard value, the legend STD is appended to the displayed 29.92 / 1013 digital readout. In this condition, the remote-mounted installer-supplied BARO is used to change the preset value. The PUSH STD button in the BARO knob is used to swap the preset value with the STD value.

- The pilot and copilot-side pressure settings are not synchronized. When the pressure settings for the pilot and copilot are different by more than 0.02 inHg, a yellow line is drawn below the pressure setting digits/legend.
- The unit of measurement for the barometric pressure value can be set to inches of mercury (IN) or hectoPascals (HPA). The unit of measurement is synchronized between the pilot and copilot-side displays. The unit of measurement for barometric pressure is set by way of the PFD CONFIG menu.

Preselect Altitude

The preselect altitude value shows as a digital readout and as a reference bug on the altitude scale. The preselect altitude value is both a visual cue and a reference value used by the AFCS preselect altitude mode. Automatic changing of the preselect altitude readout to flashing, or yellow, or yellow underline provides visual cues of altitude alert or failure conditions (respectively).

- A digital readout of preselect altitude in feet shows in cyan above the barometric altitude scale. A display of metric preselect altitude can be selected to show in addition to the display of preselect altitude in feet. The readout of metric preselect altitude shows inside a grey box located above the readout of preselect altitude in feet. The readout of metric barometric altitude is followed by the letter M. When a split is detected between the pilot and copilot-side preselect altitude a flashing yellow underline shows below the preselect altitude readout for 5 seconds. The preselect altitude bug on the altitude scale is removed when the preselect altitude readout underline is flashing yellow.
- The preselect altitude bug shows in cyan on the course altitude scale and/or the fine altitude scale. The preselect altitude bug comes into view on the course altitude scale when the preselect altitude is within 1350 feet of the current barometric altitude. The preselect altitude bug comes into view on the fine altitude scale when the preselect altitude is within 270 feet of the current barometric altitude. The bug moves out of view behind the altitude pointer in conjunction with the bug moving onto the fine scale. The preselect altitude bug brackets the altitude pointer when the preselect altitude is equal to the barometric altitude. The ALT knob on the Flight Control Panel (FCP) is used to adjust the preselect altitude on both the pilot and copilot-side and arm AFCS altitude preselect mode.

Preselect Altitude Alert

The preselect altitude alert indicates when the barometric altitude is approaching the preselect altitude value. The preselect altitude readout flashes and the flight deck aural warning chime sounds when the barometric altitude is within the acquisition limit. The readout stops flashing when the barometric altitude is within the deviation limit.

- The preselect altitude readout flashes cyan and the flight deck aural warning sounds when the barometric altitude closes within ±1000 feet of the preselect altitude.
- The preselect altitude readout stops flashing when the barometric altitude closes to within ±200 feet of the preselect altitude.
- The ALT knob on the FCP can be used to manually cancel the preselect altitude alert.
- The preselect altitude readout flashes yellow and the flight deck aural warning sounds when the barometric altitude exceeds ±200 feet of the preselect altitude after it has been within ±200 of the preselect altitude. The readout shows in cyan when the barometric altitude returns to within ±200 the preselect altitude, or a new preselect altitude is selected, or the ALT knob on the FCP is pushed.

- When the aircraft barometric altitude closes to within ±1000 feet of the preselect altitude, but does not cross within ±200 feet of the preselect altitude, then exits more than ±1000 the preselect altitude, the flashing readout changes to flashing yellow. The readout returns to flashing cyan when the barometric altitude again closes to within ±1000 the preselect altitude. The flashing continues until either the barometric altitude returns to within ±200 the preselect altitude, or a new preselect altitude is selected, or the ALT knob on the FCP is pushed. The aural warning sounds momentarily each time the ±1000 of the preselect altitude mark is crossed, whether entering or exiting.
- The preselect altitude visual and aural alerts are automatically inhibited during vertical approach modes which are in the capture state.

Radio Altitude

The radio altitude value shows as a digital readout and as an analog reference on the altitude scale. The digital readout of radio altitude shows in the lower part of the attitude scale. Analog radio altitude shows in the altitude scale area to improve ground awareness.

- The digital readout of radio altitude shows on the altitude scale when the radio altitude is 2500 or less AGL. The radio altitude readout normally shows in green. The radio altitude readout changes to yellow when the Radio Altitude Minimum (RA MIN) value is in view on the altitude scale.

	• Analog radio altitude shows as a yellow hatch line with a yellow separation line on top, which comes into view at the bottom of the barometric altitude scale when the radio altitude is approximately 270 feet. The bottom of the altitude scale is 270 feet below the current barometric altitude. The scale moves up to meet the 0 (zero) foot tick mark when the radio altitude is 0 (zero) feet. • A white 0 (zero)-foot tick mark is added to the left of the barometric altitude pointer whenever the digital radio altitude readout is in view on the altitude scale.
RA/BARO MIN	Either the radio altitude based Radio Altitude Minimum (RA MIN) or barometric altitude based Barometric Altitude Minimum (BARO MIN) can be selected to show on the altitude scale and to trigger the Minimum Alert (MIN ALERT). The RA MIN and BARO MIN values are set by way of the PFD REFS menu. The RA or BARO MIN altitude that triggers the MIN alert is also set by way of the PFD REFS menu.



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- The RA analog reference is a cyan pole with a flag at the top, which extends up from the analog radio altitude reference along the left side of the altitude scale. The RA analog reference extends up toward the radio altitude 0 (zero)-foot tick mark as the radio altitude decreases. The RA MIN digital readout shows below the altitude scale when RA MIN is selected and the radio altitude is 2500 feet or less AGL. The RA MIN digital reference is preceeded by the MIN legend and followed by the RA legend. At the selected RA MIN alert altitude, the RA MIN analog reference changes color from cyan to yellow, flashes for 5 seconds and then shows steady.

- The BARO MIN analog reference is a cyan backwards sigma shaped symbol that shows on the fine altitude scale at the selected BARO MIN altitude. The bug moves up the fine altitude scale as the barometric altitude decreases. The BARO digital readout shows below the altitude scale when BARO MIN is selected and the barometric altitude is within 2500 of the BARO MIN altitude. The BARO MIN digital reference is preceeded by the MIN legend and followed by the BARO legend. At the selected BARO MIN alert altitude, the BARO MIN analog reference changes color from cyan to yellow, flashes for 5 seconds and then shows steady.

NOTE

REFS menu items are synchronized. When either side RA MIN or BARO MIN value is changed, the same values show on both PFD.

Minimums Alert	The minimums alert MIN shows boxed and in yellow to the right of the pitch scale during a descent when the radio altitude or barometric altitude has reached the selected RA or BARO MIN altitude respectively. In addition, the MIN readout and MIN analog reference changes from cyan to yellow. The MIN alert message, MIN digital readout and analog reference flash for 5 seconds, and then show steady. The MIN alert also triggers the minimums aural warning. The MIN alert is discontinued when the RA/BARO MIN is changed to a value more than 50 feet below the current barometric altitude and when the barometric altitude is more than 50 feet above the current MIN setting. The MIN alert function is inhibited on the ground and disabled until the aircraft climbs higher than 50 feet above the RA/BARO MIN readout.
FMS Flight Plan Target Altitude	A readout of the FMS Flight Plan Target Altitude (FPTA) shows part-time above the altitude scale. The FPTA readout shows in magenta. The © symbol follows the FPTA when temperature compensation is selected by way of the FMS CDU. The FMS Temperature Compensation feature is used to correct for non-standard day temperature conditions. The FPTA shows on the PFD under the conditions that follow: • FMS is the active NAV source • The active flight plan includes VNAV altitudes • Advisory VNAV is enabled on the CDU.

The FPTA shows during a NAV-to-NAV capture until the point of glideslope capture. When an LPV approach has been selected and the active vertical leg terminates at the missed approach point, the displayed FPTA is "RWY", as is shown on the FMS LEGS page. This is done to emphasize that LPV is not a barometric altitude approach. When the PFD indicates "RWY" and the active vertical AFCS mode annunciation is either VGP or VPATH, the FMS provides vertical guidance to the LPV approach and any remaining barometric altitude constraints as set on the LEGS page or the altitude preselector will not be applied.

FLXXX Alert On/Off

The active barometric pressure setting flashes until acknowledged when Flight Level (FL) alert is set to on and the barometric altitude passes through the FL alert altitude. For power-up on the ground the FL alert is set to 180 and the color is cyan. For non-US flights the ability exists to change the FL alert altitude. The alert altitude is changed by way of the FMS CDU. When a new FL alert altitude is received from the FMS the color is magenta. The FL alert is set to on or off by way of the PFD CONFIG menu. Swap the active and preset barometric pressure settings or change the FL alert altitude by way of the FMS CDU to acknowledge the alert.

PFD – ATTITUDE

Figure 6-12 PFD – Attitude



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Attitude is represented by a wall to wall sky/ground attitude display with the blue sky and brown earth separated by a white horizon line. Automatic Flight Control System (AFCS) Flight Director (FD) command bars show on the attitude display. The FD command bars present pitch and roll guidance information from the AFCS. An aircraft symbol is superimposed over the center of the attitude display. Chevrons show for excessive pitch angles. The display is decluttered when the aircraft is flown in an unusual attitude.

▼ SELECTIONS:

Roll Scale	The roll scale indicates the current roll attitude. Symmetrically positioned white tick marks indicate 10, 20, and 30 degrees of roll. Symmetrically positioned unfilled white triangular markers indicate 0 (zero) and 45 degrees of roll. Symmetrically positioned white ticks located to the outside of the 45-degree markers indicate 60 degrees of roll. A white triangular pointer at the top of the scale is used to indicate the current roll angle.
Pitch Scale	The pitch scale is centered on the attitude ball. The horizon line represents 0 (zero) degrees of pitch. The pitch scale moves vertically in pitch with respect to the aircraft symbol. Pitch marks are in increments of 2.5 degrees up to ±30 degrees. When the attitude exceeds ±30 degrees of pitch, the scale expands to ±90 degrees. Red excessive pitch chevrons show for excessive pitch conditions.
Slip/Skid Indicator	The slip/skid indicator at the base of the roll pointer is used to show a slip or skid condition. The slip/skid indicator uses the base of the roll pointer as its reference. A displacement equal to the width of the rectangle at the base of the roll pointer is approximately equivalent to one ball displacement of a conventional inclinometer.

Aircraft Symbol

A V-shaped aircraft symbol or a split T-shaped aircraft symbol shows in the center of the attitude display. The V-shaped aircraft symbol shows when the single cue (V-Bar) FD style is selected. The split-T aircraft symbol shows when the cross-pointer (X-PTR) style Flight Director (FD) is selected. Wing tip symbols show in line with the apex of the V-shaped aircraft symbol. A small white square shows at the attitude center of the split T aircraft symbol. Both styles of aircraft symbol are black with white borders. The flight director style is selected by way of the CONFIG menu on the PFD.



V-BAR



X-PTR

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Unusual Attitude Indication

The PFD is decluttered when the pitch angle is greater than 30-degrees nose up or 20-degrees nose down, or roll angle exceeds 65 degrees. Nonessential information is removed from the PFD (the display is decluttered) and warning chevrons show on the pitch scale for an unusual attitude condition. All information except CAS warning, CAS caution, engine information, attitude, airspeed, altitude, vertical speed, compass, slip/skid, ADC Reversion, IRS Reversion, YD disengage, AP engage/disengage, and AOA is removed from the PFD. The display returns to normal when the excessive attitude condition is corrected. Excessive pitch attitude chevrons will also come into view at varying attitudes, depending on the selected PFD format, as follows:

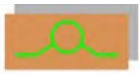
- In full view without Engine Data displayed, at 27° +/-1°, slides out of full view at 23° +/- 1°.
- In full view without Engine Data displayed, at -18° +/- 1°, slides out of full view at -13° +/-1°.
- In full view with Engine Data displayed at 28° +/-1°, slides out of full view at 22° +/- 1°.
- In full view with Engine Data displayed at -19° +/-1°, slides out of full view at -13° +/- 1°.

Excessive Pitch Chevrons

Red excessive pitch attitude chevrons come into view at 30 degrees of nose up pitch and 20 degrees of nose down pitch. The excessive pitch attitude chevrons go out of view at when the nose up pitch is 25 degrees or less and the nose down pitch is less than 15 degrees.

Flight Path Vector

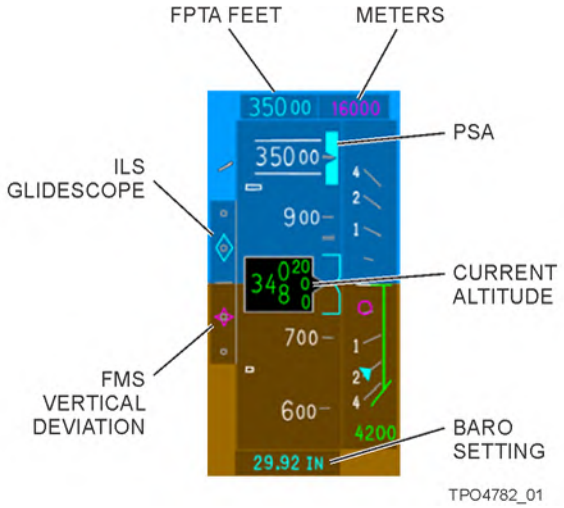
The Flight Path Vector symbol indicates the instantaneous direction of the aircraft earth frame velocity relative to the aircraft's longitudinal axis. The Flight Path symbol displays the actual flight path vector in space where the lateral position represents the azimuth of the ground track reference to the heading, and the vertical position represents the climb or descent angle relative to the horizon. In the symbol, the swept wings of the symbol are angled downward 30 degrees from horizontal so that in a standard turn, the appropriate wing will overlay the horizon line. The Flight Path Vector symbol is set to on or off by way of the CONFIG sub-menu of the PFD menu.



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PFD – VERTICAL DEVIATION DISPLAY

Figure 6-13 PFD – Vertical Deviation Display



The vertical deviation scale shows directly to the left of the altitude scale. A vertical deviation pointer shows against a scale of two white dots on either side of the center tick mark. Glideslope (GS) or VNAV deviation can show. The deviation scale is displayed when LOC or FMS are the active navigation source and conditions are met for display. The scale goes out of view when Localizer back course logic is active.

▼ **SELECTIONS:**

ILS Glideslope (G/S) Deviation	Glideslope (G/S) deviation shows when LOC is the active NAV source and the Glideslope signal is active. The vertical deviation pointer is a green (yellow if cross side LOC has been selected) diamond-shaped symbol. Each dot of the scale represents approximately a ¼-degree displacement. The pointer indicates the position of the glideslope relative to the aicraft. The G/S is below the aircraft when the pointer is below the center mark, and above the aircraft when the pointer is above the center mark. The pointer goes out of view when LOC is the active NAV source and back course logic is true. • The deviation scale consists of two gray dots above and below a center tick mark. • The deviation scale shows when LOC is the active NAV source. • The scale goes out of view when Back Course localizer logic is active. • The deviation pointer is a diamond shape and is the same color as the LOC source. • Each of the inner dots represents approximately a ¼ degree displacement. • The pointer indicates the position of the GS relative to the aircraft. The GS is below the aircraft when the pointer is below the center mark, and above the aircraft when the pointer is above the center mark.

ILS Glideslope (G/S) Preselect Deviation	The G/S preselect deviation pointer is a cyan diamond that is displayed during NAV-to-NAV approach operations when the aircraft enters the terminal area, the FMS has tuned the localizer frequency, there is a glideslope associated with the approach and the glideslope becomes active.
FMS Vertical Deviation	FMS VNAV deviation shows when FMS is the active NAV source and the FMS commands VNAV, or when the NAV-to-NAV capture function is active. The VNAV deviation pointer is a magenta (yellow if the cross side FMS has been selected) waypoint symbol for enroute and RNP approach operations. For LPV approach, the VNAV deviation pointer is a magenta diamond-shaped symbol. The scaling is variable and controlled by the FMS, ranging from 500 feet for full scale (2 dot) deflection for en-route operations to 150 feet for RNP approaches. During LPV approaches, the deviation is angular based, providing equivalent behavior to an ILS approach. • The FMS vertical scaling is 250 feet per dot, except when the message is APPR, RNP APPR, or Required Navigation Performance Authorization Required (RNP AR) APPR. When APPR, RNP APPR, RNP AR APPR. messages are displayed, vertical scaling is 75 feet per dot after the Final Approach Fix (FAF).

- The FMS can also command angular vertical scaling. This is controlled by the FMS and shows LPV APPR in the FMS message field below the Airspeed scale. The standard default scaling is 500 feet = two dots.

Vertical Excessive Deviation Alert (RNP AR)

The FMS vertical excessive deviation alert is activated when the aircraft has sequenced the FAF during an RNP AR APPR operation and the deviation pointer becomes greater than 1 dot off center. The deviation alert is a yellow outline around the side of the scale that contains the deviation pointer. It flashes for 5 seconds and then is steady.

PFD – LATERAL DEVIATION DISPLAY

Figure 6-14 PFD – Lateral Deviation Display



A scale of two white dots on either side of a white center tick mark shows above the compass readout. The scale shows only when LOC/ILS or FMS is the active NAV source.

The lateral deviation distance scaling is as follows:

- VOR — One dot deflection = a 5-degree deviation from centerline.
- LOC — One dot deflection = a 1-degree deviation from centerline.
- FMS Enroute and RNP Approach — The scaling is variable, with the full scale (2 dot) deflection equal to the RNP value (nautical miles) displayed to the right of the Lateral Deviation Scale.
- FMS LPV Approach — The deviation is angular based, providing equivalent behavior to a LOC approach. ANG is annunciated in the RNP field to the right of the Lateral Deviation Scale.

An expanded lateral-deviation scale shows above the compass replacing the lateral deviation scale when LOC is the active NAV source and both PFDs detect CAT2 conditions. CAT2 conditions are AGL ≤ 600 feet and selected RA < 200 feet, and RA MIN alert selected on. The scale is made up of two white squares on either side of the center tick mark.

▼ SELECTIONS:

LOC Deviation Pointer	The LOC deviation pointer is a diamond that is the same color as the selected NAV source (cyan for on-side, yellow for cross-side). The pointer movement is reversed when LOC is the active NAV source and in Back Course mode.

FMS Deviation Pointer	The FMS deviation pointer is a waypoint symbol that is the same color as the selected FMS (magenta for on-side, yellow for cross-side). The FMS pointer changes to a magenta diamond when a SBAS LPV approach is active. The FMS deviation pointer movement is reversed when FMS is the active NAV source and the track is greater than 110 degrees from the present heading.
FROM Annunciation	The magenta FROM annunciation shows to the left of the lateral deviation scale when FMS is the NAV source and the FMS is flying away from a waypoint.
RNP Readout	The Required Navigation Performance (RNP) readout shows on the right of the lateral deviation scale. The color matches the FMS source (magenta for on-side, yellow for cross-side) when FMS is the active NAV source. RNP shows as X.XX, where any value greater than 9.99 shows as dashes. RNP shows as ANG when LPV approaches are active. When FMS is the active source and a RNP change occurs, the RNP display flashes for 5 seconds.
Lateral Excessive Deviation Alert	The lateral excessive deviation alert is activated when the aircraft is flying a leg with an RNP value of less than 1.0 nm as indicated in the navigation database, or if RNP AR APPR or RNP AR MA is displayed on the PFD FMS message line and the deviation pointer is displaced by 2 or more dots. The excessive deviation alert is a yellow outline around the side of the lateral deviation scale that contains the deviation pointer. It flashes for 5 seconds and then is steady.

Preselect LOC deviation pointer	The LOC preselect deviation pointer is a cyan diamond that is displayed during NAV-to-NAV approach operations when the aircraft enters the terminal area, the FMS has tuned the localizer frequency, and the localizer signal associated with the selected ILS, Localizer or Back Course Localizer approach becomes active.

PFD – VERTICAL SPEED

Figure 6-15 PFD – Vertical Speed



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The Vertical Speed (VS) display is made up of a vertical analog scale and pointer. A part-time digital readout of VS shows above or below the VS scale. A selected VS bug and a FMS vertical speed required advisory pointer also show on the VS scale when commanded by the AFCS or MFD. TCAS II Resolution Advisory (RA) vertical speed cues show on the PFD VS scale when the TCAS computer detects an RA.

▼ SELECTIONS:

VS Pointer	The VS pointer moves points to the current VS on the VS scale. As the pointer moves, a drag line appears to emphasize the VS. The normally green VS pointer and drag line show in red when the current VS is within a red portion of a TCAS Resolution Advisory (RA).

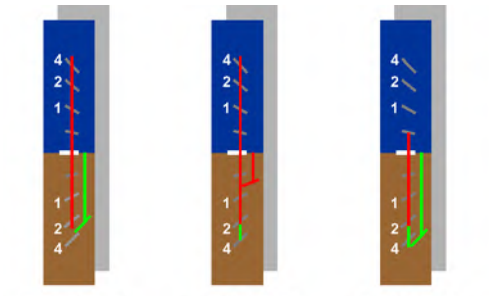
VS Digital Display	A part-time digital readout of VS shows at the top or bottom of the VS scale. When the current VS is greater than +300 feet/minute, the digital readout shows at the top of the VS scale. When the current VS is more than -300 feet/minute, the digital readout shows at the bottom of the VS scale. The digital readout is removed when the ascent/descent is less than ±100 feet/minute respectively.
Selected VS Display	Selected VS is a reference value used for Automatic Flight Control System (AFCS) VS mode. The UP ↕ DOWN thumb wheel on the FCP is used to set the selected VS. When VS mode is active, a selected VS reference bug shows on the VS scale and digital readout of the selected VS value shows in the AFCS vertical mode field. The VS bug shows a right-pointing triangle on the VS scale. The selected VS readout is preceded by a VS bug icon, and followed by an arrow. Arrow direction is up for a selected climb and down for a selected descent. The VS bug, readout and icon show in cyan when commanded by the UP ↕ DOWN thumb wheel on the FCP. The VS bug, readout and icon show in magenta when the FMS is commanding the VS reference. Refer to the AFCS VS Mode section of this guide for more information on the selected VS.

FMS Vertical Speed Required Advisory Pointer

The FMS Vertical Speed Required (VSR) advisory pointer indicates the minimum average VS required to climb (or descend) from the present position to the next climb (or descent) VNAV altitude. The FMS VSR advisory pointer is a magenta open circle that shows on the VS scale. The specific conditions under which the FMS VSR advisory pointer shows are controlled by the FMS. The FMS must be the active NAV source for the advisory pointer to show.

TCAS II RA VS Cues

TCAS II Resolution Advisory (RA) vertical speed cues show on the PFD vertical speed scale when the TCAS computer detects an RA. The RA VS cues advise the pilot of the VS necessary to satisfy a vertical RA. RAs are issued by the TCAS computer when corrective maneuvering is required in order to avoid a collision. The message NO VSI RA shows in yellow in the TCAS data field when the TCAS VS cues cannot be computed or if VS is not in view on the PFD. Two types of RAs can be issued by the TCAS computer: preventive and corrective.



- Preventive RAs are issued for threat traffic which the TCAS computer has determined current pitch will resolve the threat situation. The VS range to avoid shows as a red vertical band on the VS scale. To comply with an RA, avoid flying vertical speeds within the red-banded areas. Separate cues are provided for climb and descend restrictions. RA guidance that results in both climb and descend restrictions cause both climb and descend cues to show simultaneously. The normally green VS pointer and drag line change to red when the pointer is pointing to a red portion of the vertical band.
- Corrective RAs are issued for threat traffic for which the TCAS computer has determined that corrective action needs to be taken to avoid the traffic. The vertical speed range to seek and avoid shows on the PFD vertical speed scale. The VS range to avoid shows as a red vertical band on the VS scale. The vertical speed range to seek shows as a green band. To comply with a corrective RA, avoid flying vertical speeds within the red-banded areas and do fly the vertical speeds within the green-banded areas. The normally green VS pointer and drag line change to red when the pointer is pointing to a red portion of the vertical band.

PFD – AFCS MESSAGE FIELD

Figure 6-16 PFD – AFCS Message Field



TPO4454_01

The Automatic Flight Control System (AFCS) is a combination of Autopilot, Yaw Damper, and automatic pitch trim system. Lateral and vertical modes selected by the pilot are used by the Flight Guidance Computers to generate steering commands. The steering commands are supplied to the flight directors and the autopilot. The Yaw Damper provides control signals to drive the rudder actuator for dutch roll damping and turn coordination. The automatic pitch trim function interfaces with the aircraft pitch trim system to automatically off-load the forces on the elevator and the control column. The AFCS is made up of two dual-channel Flight Guidance Computers (one active and one standby), a Flight Control Panel, a Yaw Damper Panel, two primary servos and two rudder actuators. The Primary Flight Displays and Display Control Panels are considered part of the EFIS but have AFCS functions. Remote-mounted switches on the control wheels and thrust levers have AFCS functions. Other inputs to the AFCS include those from the Air Data System, Inertial Reference System, navigation receivers, Data Concentrator Unit, and the aircraft pitch trim system.

The Automatic Flight Control System (AFCS) provides information to the pilot through the PFD, as well as the Flight Control Panel (FCP) and other warning and messaging systems. The supplied information includes the selected flight guidance modes, flight director steering commands, autopilot and Yaw Damper status. The flight control status, flight director status, the lateral and vertical arm and capture modes and warnings, Autopilot, and Yaw Damper operational status messages show above the attitude display on the PFD. AFCS messages flash for a period of time upon presentation, unless otherwise noted. AFCS messages flash for a period of time and then go out of view upon loss of drive data from the FCC.

▼ **SELECTIONS:**

Dead Reckoning Message	The message DR shows in white during dead reckoning operation. Dead reckoning operation is provided during VOR station passage.
Synchronization Message	The message SYNC shows in yellow when synchronization is active. Selecting the synchronization function synchronizes the compatible lateral and vertical references to those currently being flown. When the autopilot is engaged, selecting the synchronization function releases the aileron and elevator servo clutches so the aircraft can be flown manually.
Half Bank Message	The message 1/2BANK shows in white when half bank is active. A curved half bank arc shows in green along the top of the roll scale when half bank is active. Half bank limits the magnitude of the lateral guidance to half the maximum bank angle. Half bank can be active concurrent with many of the other lateral modes.
Lateral Approach Mode Message	The message APPR shows in white in the lateral Approach mode arm field when approach mode is armed. The message APPR shows in green in the lateral Approach mode capture field when approach mode is captured. Lateral Approach mode generates commands to capture and track lateral guidance for ILS/LOC approaches and FMS non-precision approaches.

Lateral Arm Mode Field

The lateral arm mode field shows the mode message for the lateral mode that is armed for capture. Armed lateral modes show in white in the lateral arm mode field. The lateral mode messages are as follows:

Lateral Arm Mode Messages

Mes-sage	Meaning
VOR1	VOR1 is the coupled NAV source and NAV or APPR is armed for capture.
VOR2	VOR2 is the coupled NAV source and NAV or APPR is armed for capture.
LOC1	LOC1 is the coupled NAV source and NAV or APPR is armed for capture.
LOC2	LOC1 is the coupled NAV source and NAV or APPR is armed for capture.
B/C1	LOC1 is the coupled NAV source and B/C is armed for capture.
B/C2	LOC2 is the coupled NAV source and B/C is armed for capture.
LNV1	FMS1 is the coupled NAV source and NAV or APPR is armed for capture.

LNV2	FMS2 is the coupled NAV source and NAV or APPR is armed for capture.
LNV3	FMS3 is the coupled NAV source and NAV or APPR is armed for capture.

Lateral Capture Mode Field

The lateral capture mode field shows the captured lateral mode message. Captured mode messages show in green in the lateral capture field. The lateral mode messages are as follows:

Lateral Capture Mode Messages

Mes-sage	**Meaning**
VOR1	VOR1 is the coupled NAV source and NAV or APPR is captured.
VOR2	VOR2 is the coupled NAV source and NAV or APPR is captured.
LOC1	LOC1 is the coupled NAV source and NAV or APPR is captured.
LOC2	LOC1 is the coupled NAV source and NAV or APPR is captured.
B/C1	LOC1 is the coupled NAV source and B/C is captured.

B/C2	LOC2 is the coupled NAV source and B/C is captured.
LNV1	FMS1 is the coupled NAV source and NAV or APPR is captured.
LNV2	FMS2 is the coupled NAV source and NAV or APPR is captured.
LNV3	FMS3 is the coupled NAV source and NAV or APPR is captured.
ROLL	Roll mode is the active lateral mode.
HDG	Heading hold is the active lateral mode.
GA	Go Around mode is the active lateral mode.
TO	Take Off mode is the active lateral mode.
NOTE: The lateral capture mode message shows in yellow when the input is invalid and a FMS mismatch exists.	

Lateral Capture Degradation Line

The lateral capture degradation line is a horizontal line that overwrites the lateral capture mode field. The lateral capture degradation line indicates a degradation of the flight control system's lateral operation.

Autopilot Message Field

The Autopilot mode message and couple arrow show in the Autopilot message field. The Autopilot mode message indicate that the autopilot is engaged, engaged during synchronization, or disengaged. The couple arrow points to the left or right to indicate that the autopilot is, or would be, engaged to the pilot or copilot-side flight guidance channel. The color coding for the Autopilot message is as follows:

Autopilot Message

Message	**Meaning**
AP (green)	Autopilot is engaged
AP (yellow)	Autopilot is engaged and synchronization is active
AP (red)	Autopilot has disengaged. Continuous flashing indicates a disengage due to a failure. Momentary flashing that changes to steady indicates a manual disengage.

NOTE

The Autopilot disconnect alert can be cancelled with the Autopilot disconnect button or the takeoff/go around switch. Reengaging the autopilot also cancels the disengage alert.

Flight Director Single Source Message

The flight director single source message FD1 or FD2 shows in green on the non-coupled side (as controlled by the XFR button on the FCP) and only appears when the FCCs are not in the independent mode. The message indicates the Flight Director source is from the cross-side FCC as defined in the table that follows. The message is cleared when both FCCs are operating in the independent mode.

Flight Director Single Source Message

Message	**Meaning**
FD1 (on PFD2)	Copilot-side command bars are driven by the pilot-side FCC.
FD2 (on PFD1)	Pilot-side command bars are driven by the copilot-side FCC.

Vertical Arm Mode Field

The vertical arm mode field shows the mode message for the vertical mode that is armed for capture. Armed vertical modes show in white in the vertical arm mode field. The vertical mode messages are further subdivided into the sections that follow:

- Altitude arm mode field
- VNAV arm mode field
- Vertical approach arm mode field
- Vertical capture mode field.

Altitude Arm Mode Field

The target altitude shows in the altitude arm mode field. The target altitude may be either the Preselect Altitude (PSA) or the Flight Plan Target Altitude (FPTA).

Altitude Arm Mode Field

Message	Meaning
*ALTS (white)	The preselect altitude (PSA) is the target altitude.
ALTV (white)	The Flight Plan Target Altitude (FPTA) is the target altitude.
*The ALTS message flashes and shows in yellow when ALTS is armed and then de-selected or aborted.	

VNAV Arm Mode Field

The VNAV arm mode field indicates the VNAV modes that are armed for capture. The possible VNAV mode messages are PATH, PTCH, and VNAV.

VNAV Arm Mode Messages

Message	Meaning
PATH (white)	The VNAV only mode PATH has been armed automatically.
PTCH (white)	The default vertical mode Pitch is armed as a result of a vertical discontinuity.
FLC (white)	VNAV has automatically armed FLC mode. Note that while FLC may be the capture mode either with or without VNAV active, it cannot be manually armed.
VNAV (yellow)	A VNAV error exists.

Vertical Approach Arm Mode Field

The vertical approach arm mode field represents the approach mode that is armed for capture. The vertical approach arm mode field shows when a vertical approach mode is armed. The possible vertical approach arm mode messages are GS and GP.

Vertical Approach Arm Mode Messages

Message	**Meaning**
GS (white)	A valid LOC is tuned and in range and approach mode is active.
GP (white)	The FMS set up for VNAV operation and approach mode is active.

Vertical Capture Mode Field

The vertical capture mode field represents the current vertical mode. Captured mode messages show in green in the vertical capture field. The possible vertical captured mode messages are as follows:

Vertical Capture Mode Messages

Mes-sage	**Meaning**
ALTS CAP (green)	The AFCS begins to capture the preselected altitude.
ALTV CAP (green)	The AFCS begins to capture the FMS FPTA.
ALTS (green)	The AFCS is tracking the preselected altitude.
ALTV (green)	The AFCS is tracking the FMS FPTA.
FLC (green)	FLC has been selected by way of the FLC button or has transitioned from armed to active and the AFCS is tracking toward the preselected altitude.
*FLC (yellow)	FLC is captured and an overspeed condition exists.
ALT (green)	ALT has been selected by way of the ALT button and the AFCS is tracking the current altitude, or the aircraft was in ALTS or ALTV and the preselect or FMS FPTA has been changed away from the tracked altitude.
**VS (green)	VS has been selected by way of the VS button.

Mes-sage	Meaning
PATH (green)	VNAV is active with an altitude constraint or preselected altitude located below the current aircraft altitude and the AFCS has either initiated the capture or is tracking the FMS guidance to the constraint altitude.
PTCH (green)	Pitch mode is the default vertical mode. All other vertical modes are de-selected.
GA (green)	Go around mode has been selected (TO/GA button pressed while airborne) and the AFCS is generating fixed pitch-up and heading hold commands.
TO (green)	Takeoff mode has been selected (TO/GA button pressed while on the ground) and the AFCS is generating fixed pitch-up and wings level commands.
GS (green)	The AFCS has automatically began to capture the glideslope.
GP (green)	The AFCS has either initiated or is tracking the FMS provided vertical approach guidance.

Mes-sage	Meaning
	* A selected airspeed bug icon and a readout of the selected airspeed show to the right of the FLC mode message. The FLC mode message color is cyan when selected manually. The FLC mode message color is magenta when controlled by the FMS.
	** A right facing triangle and a readout of the selected vertical speed show to the right of the VS mode message. An up or down arrow shows to the right of the selected vertical speed readout. The arrow represents the direction of the selected vertical speed. A vertical speed bug corresponding to the selected vertical speed shows on the vertical speed scale. The VS triangle, readout, arrow, and bug show in cyan.
	NOTE: When VNAV is enabled the vertical capture mode message is prefixed with the letter V. As PATH, ALTV, ALTV CAP and GP are only available when VNAV is active, these modes will always be displayed as VPATH, VALTV, VALTV CAP and VGP.

Vertical Capture Degradation Line

The vertical capture degradation line is a horizontal red line that overwrites the vertical capture mode field. The vertical capture degradation line indicates a degradation of the flight control system's vertical operation.

Yaw Damper Disengage Message

The message YD shows in yellow on the PFD when both Yaw Damper's have disengaged.

Half Bank Mode Arc	A green arc shows above the roll scale to indicate Half bank mode is active. The arc extends out 15 degrees either side of the 0 (zero)-degree triangle on the roll scale.
Flight Director V-Bar or Cross Pointer	Flight Director (FD) steering commands show in the attitude ball as either single cue (V-Bar) or split cue cross pointer (X-PTR). The aircraft symbol superimposed over the center of the attitude ball can be a V-shaped single cue or an optional split T shaped split cue. The aircraft symbol selection follows the selection of the flight director steering commands. The single cue or split cue is selected from the CONFIG menu on the PFD.

PFD – ROSE FORMAT

Figure 6-17 PFD/MFD — Rose Format



The Rose format is a conventional electronic-HSI presentation that shows a full compass rose, heading readout, lubber line, selected heading bug, course pointer, lateral deviation bar, lateral deviation scale, and to/from arrow. Two bearing pointers are available for display on the Rose format. The Rose format supports the display of hazard avoidance overlays. When an overlay is selected for display, a half-range ring is added to the rose format. Controls located on the DCP are used to select the Rose format for display on the PFD.

▼ SELECTIONS:

Compass Card	The Rose format is a full 360-degree compass card with the familiar white triangular pointers at ±45 and ±135 degrees from the lubber line. White tick marks show at 90, 180, and 270 degrees from the lubber line. The lubber line pointer contains a digital readout of the current heading. The legend TRUE shows in cyan when true heading has been selected for display. No legend shows when the heading type is magnetic. The heading bug is a rectangular shaped symbol that is positioned on the periphery of the compass rose at the selected heading. The bug shows in cyan when the value is set manually. A heading vector line shows when the heading bug is off scale and when the selected heading is changing. A three-digit readout of the selected heading shows to the left of the lubber line pointer. The source of heading information is normally the on-side Inertial Reference System (IRS1 or IRS2). Controls located on the FCP are used to change the selected heading value and to set the heading bug under the lubber line.

Navigation Block

The Navigation Block provides information for the navigation source selected for the on-side PFD and MFD. Controls located on the DCP along with a menu on the PFD are used to select the active NAV source. The Navigation Block is displayed full time to the left of both the PFD ROSE and PPOS formats. The block consists of up to six lines of navigation data. The sixth line, EPU is only populated when an FMS is the active navigation source, and is not shown in the compressed PFD format.

- A NAV source legend shows at the top of field to indicate the active NAV source. The active NAV source can be VOR1, VOR2, LOC1, LOC2, FMS1, FMS2, or FMS3. The onside VOR/LOC and FMS NAV source legends are shown in green and magenta, respectively. Cross side sources are shown in yellow. When the active navigation source is a VOR or LOC a readout of the VOR/LOC frequency is shown to the right of the NAV source legend.

- A legend and readout of the selected course/track shows below the NAV source legend. The legend is CRS (course) when the active NAV source is a VOR or LOC. The identifier is B/C (back course) when a localizer frequency is tuned, the active NAV source is LOC, and the difference between current heading and the selected course is more than 110 degrees. The identifier is DTK (desired track) when the active NAV source is FMS. The identifier is HDG when the active NAV source is FMS and a heading leg is active.
- The station or waypoint Identifier (ID) shows below the course/desired track readout. When DME Hold is selected and the active NAV source is VOR or LOC, the station ID does not show. When the active NAV source is FMS and a lateral navigation alert occurs, the waypoint ID and distance readout flash.
- Time-To-Go (TTG) in hours and minutes shows below the station/waypoint ID. The TTG readout is from the same source as the distance readout. Dashes show in place of the TTG readout when the source of TTG data is not received. TTG does not show when the an active NAV source frequency is not tuned.

- Distance in nautical miles shows below the time-to-go readout. The distance readout is normally associated with the active NAV source. This provides distance to the next waypoint for FMS, distance to the VOR station for VOR, and distance to the runway for LOC. When DME Hold is active and the active NAV source is VOR or LOC, the distance readout is followed with a cyan H and the station identifier is removed. The FMS provides the distance and waypoint ID when FMS is the active NAV source. If the active NAV source is FMS and a Lateral Navigation (LNAV) alert occurs, the distance readout and waypoint IF flash.

Estimated Position Uncertainty (EPU)

EPU is shown below the distance display when an FMS is the active navigation source. The EPU value is provided by the FMS and is calculated based on the sensors currently being used in the navigation solution. EPU is a measure based on a defined scale in nautical miles (NM), which conveys the current horizontal position estimation performance. It is not an estimate of the actual error, but a defined statistical indication providing the 95% bound on horizontal position. The FMS compares this value to the RNP value displayed next to the lateral deviation scale in the ADI portion of the PFD. FMS messages alert the crew when the EPU value exceeds the RNP value.

Preset NAV Source

Preset NAV source information shows part-time below the active NAV source information on the PFD. The preset NAV source, course, and station frequency show surrounded by a grey box. The preset NAV source can be swapped with the Active NAV source. The preset NAV course can also be swapped when active NAV source is VOR or LOC. The preset NAV source shows when the FMS has not armed a Preselect NAV source. The NAV swap button on the DCP is used swap the values.

NOTE

Some illustrations show EPU when FMS is not the active navigation source, this is inacurate, but does not affect operation and will not be updated at this time.



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- Preset NAV Source — The preset NAV source shows to the right of the SRC legend in cyan. When preset NAV is active, the DATA knob on the DCP is used to scroll through the list of installed NAV sources. When the active NAV source is changed by way of the PFD or MFD menu the previous active NAV source becomes the new preset NAV source. The preset NAV source is changed to FMS following a NAV-to-NAV transfer.

- Preset Course — The preset Course (CRS) value shows to the right of the CRS legend in cyan. The preset CRS function is active when the preset NAV source is VOR or LOC, and the cyan selection box surrounds the preset CRS value. With the selection box around the preset CRS value, the DATA knob on the DCP is used to set the preset course value. The preset course is blank when the source is FMS.
- Preset Frequency — The preset frequency shows to the right of the FREQ legend in green when the preset source is VOR or LOC. The frequency that shows is from the VHF NAV radio and does not have an associated control. The frequency shows in green. The preset frequency is blank for source of FMS. The preset frequency is blank when the echo frequency is invalid or missing.

Preselect NAV Source

The Preselect NAV source is the destination LOC entered at the end of the FMS flight Plan. The Preselect NAV source and frequency show in cyan below the active NAV source information on the PFD. A dashed, double-bar preselect course pointer shows in cyan on the PFD. The preselect lateral deviation bar uses the active NAV source deviation scale. When the on-side FMS is the active NAV source and the NAV-to-NAV capture function is enabled, the FMS can command the Preselect NAV source (destination LOC) to become the active NAV source. Upon LOC capture, the FMS commands a change in the active NAV source from FMS to LOC. The conditions that follow are required to enable the NAV-to-NAV capture function:



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Controls and menus located on the CDU are used to select the Preselect NAV source.

Course/Desired Track Pointer	A single-bar course/desired track pointer, to/from arrow, lateral deviation/cross-track distance bar, and scale shows for all NAV sources (FMS, VOR, and LOC). The pointer is a course pointer when VOR or LOC is the active NAV source, and a track pointer when the active NAV source is FMS. The pointer shows in the same color as the active NAV source. Controls located on the FCP are used to set the desired course when the active NAV source is VOR or LOC. The FMS selects the course when the active NAV source is FMS.
Lateral Deviation/Cross-track Distance Bar	The lateral deviation/cross-track distance bar is the center portion of the course/desired track pointer. The bar moves left or right of the course/track pointer to indicate the position of the selected course/track relative to the aircraft position. Lateral deviation shows when the NAV source is VOR or LOC. Cross-track distance shows when the NAV source is FMS. The magnitude of the course deviation or cross-track distance is measured against the lateral deviation/cross-track distance scale.
Lateral Deviation/Cross-track Distance Scale	The lateral deviation/cross-track distance scale shows perpendicular and intersects the lateral deviation/cross-track distance bar. The lateral deviation/cross-track distance scaling is as follows: • VOR — One dot deflection = a 5-degree deviation from centerline. • LOC — One dot deflection = a 1-degree deviation from centerline.

- FMS Enroute and RNP Approach — The scaling is variable, with the full scale (2 dot) deflection equal to the RNP value (nautical miles) displayed to the right of the Lateral Deviation Scale.
- FMS LPV Approach -- The deviation is angular based, providing equivalent behavior to a LOC approach. ANG is annunciated in the RNP field to the right of the Lateral Deviation Scale.

Expanded Lateral Deviation Scale

An expanded lateral-deviation scale shows above the compass replacing the lateral deviation scale when LOC is the active NAV source and both PFD detect CAT2 conditions. CAT2 conditions are AGL ≤ 600 feet and selected RA < 200 feet, and RA MIN alert selected on. The scale comes into view automatically when the preselect NAV source becomes the active NAV source. The scale is made up of two grey squares on either side of the grey center tick mark. The pointer is green (yellow if cross side LOC has been selected) and diamond-shaped. Lateral deviation scaling is as described for the lateral deviation/cross-track distance scale.

To/From Arrow

The to/from arrow is a solid triangle on the course/desired track pointer. The arrow points toward the head of the pointer to indicate a to condition and toward the tail to indicate a from condition. The arrow shows in the same color as course/desired track pointer.

Track Pointer

The track pointer is positioned on the compass rose at the current aircraft track over the earth. The track pointer is a green open-circle shape. The difference between the track pointer position and the lubber line is the drift angle.

Bearing Pointers

Two bearing pointers can be selected to show on the MFD and PFD. One from number 1 bearing sources and one from number 2 bearing sources. The bearing pointer from the number 1 sources is a cyan single-lined pointer. The bearing pointer from the number 2 source is a white double-lined pointer. When selected, a bearing source menu shows vertically along the left of the PFD. The bearing source menu shows a list of the available bearing sources. Controls located on the CCP are used to show the bearing source menu.



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When a bearing pointer is active, a bearing source data fields show in the lower left corner of the PFD. The bearing source, bearing pointer icon, bearing distance, and bearing identifier show in the bearing pointer data field.

- When available, bearing distance shows in the same color as the bearing pointer. For VOR bearing, the bearing distance is blank when DME Hold is active. Bearing distance does not show when the bearing source is also the active NAV source.
- When available, the bearing identifier shows in the same color as the bearing pointer. For FMS, the identifier is the next (to) waypoint. For VOR, the identifier is either the station ident or frequency. Frequency shows for a minimum of 10 seconds when a VOR bearing is first selected or a new frequency is selected. When the ident is not available, the VOR Frequency remains. When DME Hold is selected, the VOR Frequency shows full-time. The ADF identifier is always the ADF frequency. The identifier does not show when the bearing is also the active NAV source.

Controls located on the DCP are used to select and deselect the bearing pointers, as well as the source for each bearing pointer.

GS/TAS/ISA/ SAT/TAT /Wind	Groundspeed (GS), True Airspeed (TAS), International Standard Atmosphere Delta (ISA), Static Air Temperature (SAT), Total Air Temperature (TAT), and wind magnitude and direction show in a field on the left side of the display. GS and TAS shows in knots. ISA, SAT, and TAT show in degrees Celsius. The source of TAS, ISA, SAT, and TAT is the currently selected Air Data Computer. The source of GS and wind is the on-side FMS unless the cross-side FMS is selected as the active NAV source, then the cross-side FMS is the source.

PFD – COMPRESSED EICAS

Figure 6-18 PFD – Compressed EICAS



TPN7591_01

The compressed Engine Indicating and Crew Alerting System (EICAS) format shows above the primary flight instrument presentations on both PFD when selected by way of the Reversion Switch Panel (RSP).

The compressed EICAS presentation is the same as the primary EICAS presentation, with the exceptions that follow. Refer to the MFD — Primary EICAS section in this chapter for details on the EICAS presentation.

- The ITT and N2 gauges do not show (the digital readouts remain).
- The high vibration message VIB is moved to the N1 gauge.
- The fuel summary display is replaced by a single readout of total fuel quantity.
- The cabin environment display does not show on the compressed EICAS.
- The CAS message field operates the same but has room for fewer messages.

PFD – SYNTHETIC VISION SYSTEM/FLIGHT PATH VECTOR

Figure 6-19 PFD – Synthetic Vision System



The optional Synthetic Vision System (SVS) display replaces the standard blue sky/brown ground of the ADI. The foreground items (for example, altitude tape, airspeed tape) remain unchanged. The terrain and position are calculated by the Synthetic Vision Computer (SVC), using inputs from the aircraft navigation sources, ADS, and attitude sources. The SVS display is heading-centered, and the zero pitch line accurately shows the terrain level with the aircraft. When turned on, the SVS horizon and ADI zero pitch line are distinct. The higher the altitude is, the higher the separation is.

CAUTION

Do not use the SVS as the sole means of navigation. It is also not a replacement for TAWS. If a terrain caution or warning annunciates, follow the TAWS display.

Terrain coloring presents a synthetic equivalent of the pilot looking out of the window. Terrain elevation is shown by color-coded altitude bands.

Texture patterns, shading, shadowing and grid lines are overlaid on the terrain to enhance the interpretation of data. Blue is used to indicate oceans and other large water masses, but is distinct from the sky. If terrain data is not available for a location, the missing data is shown in black at sea level. The terrain texture, shadow, and shading help to interpret the contours of the terrain in view. Grid lines are displayed on terrain and bodies of water to increase the sense of motion and depth perception. Refer to the illustration above for individual colors.

NOTE

SVS image may be removed when flying toward a missing terrain area.

The known missing terrain areas are:
S72°-S73°, W103° 48'-W106° 12'
S72°-S73°, W79° 48'-W81° 0'
S66°-S67°, E162° 48'-E164° 0'
S67°-S68°, E164° 0'-E165° 12'

The SVS display is selected on the REFS 2/4 menu. Selection of SVS ON automatically turns on the Flight Path Vector (FPV). On the SVS is removed when any of the items that follow occur:

- SVS is selected OFF.
- SVS fail detected.
- Compressed PFD is selected.
- When an unusual attitude is detected.
- When an attitude fail flag shows.
- Pitch, Roll, or HDG sensors fail.
- Altitude or airspeed fail.
- Video image error detected.
- GPS integrity is insufficient.
- AFD, altitude, airspeed, attitude, pitch, roll, or miscompare detected.

TIP

The SVS image is based on barometric altitude. On the ground, an incorrect baro setting may not allow the SVS image to show.

On take off, with SVS selected, the runway is drawn up to 20 seconds after liftoff. When performing an approach with SVS selected, the airport

shows at 15 NM, and the runway centerline shows at 10 NM. The SVC also includes a 7.5 NM extended centerline on the arrival runway. The distance is relative to the Airport Reference Point (ARP). The extended centerline is a dashed magenta line. The departure runway is outlined in cyan, and the arrival runway is outlined in magenta. When a runway has a displaced threshold, the SVC uses the actual location of the threshold as the runway starting point.

CAUTION

If an airport is not in the WGS-84 reference system, it may not be drawn correctly in the SVS image. Verify location and orientation using other means.

The date of the databases can be checked on the same MFD effectivity page as the IFIS dates. The databases are loaded through the DBU. The terrain database is not released on a schedule, while the airport database changes every 28 days. The terrain database has a 15 arc second resolution worldwide, and a six arc second resolution within 6 NM of an airport area with a runway greater than 2000 feet.

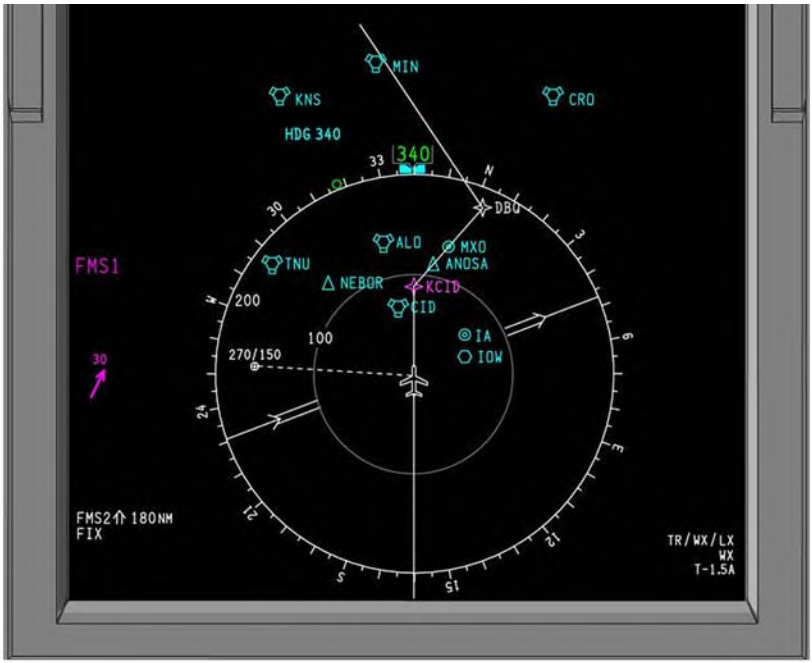
TIP

The IFIS effectivity page also shows the dates for the two SVS databases (terrain and airport/runway).

The FPV shows automatically when the SVS is selected ON. The FPV shows as a green circle with two wings to show bank. It shows where the nose of the aircraft is in relation to the SVS background. The wings of the FPV are angled so that when the aircraft is in a standard rate turn, the angled portion overlaps the horizon line. The FPV is selected ON or OFF on the REFS 2/4 menu. FPV data comes from the IRS. The circle shows over all annunciations except for fault annunciations and deviation scales. It shows laterally between the airspeed tape and the altitude tape, and vertically between the FD/AP annunciator field and the HSI field.

PFD/MFD – PPOS MAP FORMAT

Figure 6-20 PFD/MFD – PPOS Map



TPO4650_07

The Present Position (PPOS) Map format is a 360-degree moving map used to display the active FMS flight plan. The active flight plan is represented by flight plan waypoints and flight plan legs. Background NAVAID symbology, for example airports, is available for display on the PPOS Map. Enhanced Map (E-Map) features (for example, geopolitical boundaries, airspace, and airways) may be available on the MFD when the optional Integrated Flight Information System (IFIS) is installed. Waypoints can be defined on the FMS PPOS Map in the same manner as on the Plan Map. The map radius is selected with the RANGE knob on the DCP. An extended PPOS Map format, which is available on the MFD, provides an increased forward viewing area. Three submodes are available on the MFD PPOS Map: no data window, data window on, and data window on with VNAV data. Weather Radar, Lightning, Terrain, and TCAS overlays are available for display on the PPOS Map format. The PPOS Map is selected for display on the MFD with controls

located on the CCP. The PPOS Map is selected for display on the PFD with controls located on the DCP.

▼ SELECTIONS:

MFD Map Source	The map source is the same as the NAV source when the NAV source is a FMS, otherwise, the map source is the on-side FMS.
PFD Map Source	The PFD map source is set from the NAV SOURCE menu on the PFD. The NAV SOURCE menu is selected with the NAV SRC button on the DCP.

NOTE

With a PPOS Map showing, if any NAV source other than FMS1/FMS2 is selected from the NAV SOURCE menu, or if an automatic transition of PRESELECT CRS to active course occurs during a NAV-to-NAV capture, the FMS Flight Plan symbology is removed and replaced by the Course/Deviation display.

Flight Plan Symbology	Flight plan legs show as solid lines connecting the flight plan waypoints. Waypoint symbology includes waypoints, user-defined waypoints, runway thresholds, holding patterns, and procedure turns. Flight plan waypoints are identified with a string of up to seven characters. The next (to) waypoint shows in magenta.

Map Symbols

Background symbols are made up of user-selected symbology that shows geometrically correct on the map. The selection of map symbology is available in two places: the CDU and the AFD (by way of the MAP SYMBOLS menu). The MAP SYMBOLS menu is used select map background symbology for display on the FMS PPOS, PLAN and 3D maps (for example, nearest airports). The map background symbology can also be selected by way of the FMS CDU. Refer to the MFD Map Displays section of the Bombardier Challenger 605/650 Pro Line 21™ Advanced Flight Management System (523-0822377) for details on selecting map background symbology by way of the CDU. Refer to the PFD — PPOS Map section in this chapter for details on map background symbology.



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- A white icon shows on the map, appropriately positioned along the current FMS track line, which depicts the range to altitude intercept for the current VNAV leg. This icon is called the Altitude Intercept Arc.
- When the FMS declares waypoint passage, the next (to) waypoint symbology on the map flashes until the LNAV alert condition is no longer asserted.
- The active waypoint symbol and identifier are magenta. The previous waypoint and identifier are cyan. The adjoining track line, and other flight plan waypoints, identifiers, and associated track lines, are white.
- Modifications to the active flight plan, alternative flight plan information, and a secondary flight plan may show. Modified and alternative flight plan information shows as white dashes. The secondary flight plan information shows in cyan.
- If a parallel offset exists, the PPOS and PLAN Maps show the offset as a magenta dashed line parallel to the flight plan path. If no parallel offset exists for an active flight plan, an offset that is part of a modified flight plan shows with a white dashed line. An offset that is part of the active flight plan shows with a magenta dashed line. Only one parallel offset can show graphically, with the active flight plan offset taking precedence.

- FMS Map symbols may be accompanied by identifier data. The number of characters, size, color, rotation angle of the character line, and the number of character lines that show are controlled by the FMS.

Aircraft Symbol

The Aircraft symbol is a representation of the aircraft in the center of the compass with nose toward the lubber line.

Extended Map (MFD only)

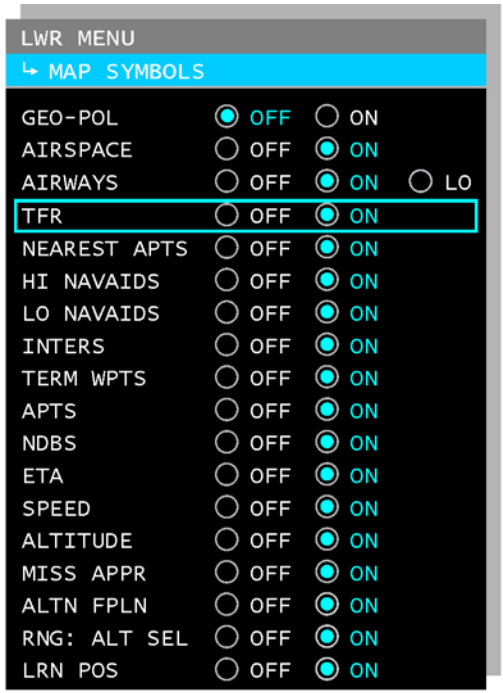
The area above the MFD navigation display can be used to show an extended Map format that provides approximately 50 percent more forward viewing area. Three submodes are available when the extended Map format is active, no data window, data window on, and data window on with VNAV data. Selection of the extended Map submodes is controlled by way of the FMS CDU.

- The text window on submode shows multiple lines of FMS text above the FMS Plan Map. The text window typically shows data associated with the current or upcoming flight legs.
- The data window with VNAV submode can show up to seven rows of text associated with the vertical waypoints.

Define Waypoints	The joystick cursor (+ symbol) shows on the MFD when the joystick is deflected and the FMS Plan Map or PPOS Map is the active format. The latitude and longitude of the cursor position is sent to the FMS upon operation of the PUSH SELECT button on the CCP. The cursor is removed if the JSTK button is pressed to transfer the joystick control to the upper format in a split window MFD format or if the joystick is not used for 60 seconds.
Range Symbology	A full-range ring and a half-range ring show full-time. The full-range in nautical miles shows inside the range ring on the left side. One half of the full-range value shows in the left portion of the half-range ring. The RANGE knob on the DCP is used to select the display range.
Weather Radar, Lightning, Terrain, and TCAS overlays	Weather Radar, Lightning, Terrain, and TCAS overlays are available for display on the PPOS Map. Controls located on the CCP are used to select overlays on the MFD. Controls located on the DCP are used to select overlays on the PFD.
Enhanced Maps (optional)	Enhanced Maps (E-Maps) features are available when the EMAPS IFIS option is activated. The map menu on the MFD allows selection of the E-Maps.

NOTE

A blank TFR may be displayed when selecting a TFR Text Report from the TFR Text Report list. If this occurs, re-request the TFR by re-selecting to TFR Text Report List and re-selecting the TFR.

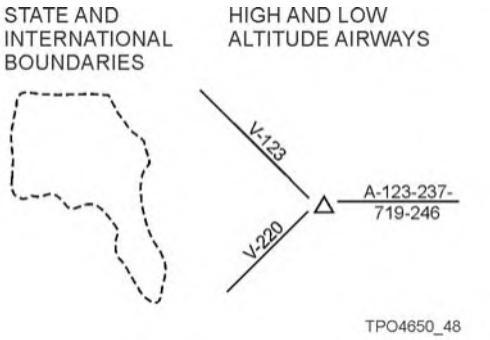


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- First Order Administrative boundaries (defined as State/Province boundaries) are depicted using dotted grey lines for the U.S.A., Canada, Russia, China, and Australia.
- International boundaries are depicted using dotted grey lines.

  NOTE: Some First Order Administrative and International boundaries will appear to have a blue background at ranges of 200 nautical miles or greater.

- Airways are depicted using a series of straight grey lines from the start to the finish of each airway segment. A triangle symbol is placed at the end of each leg of an airway. The identifier for the airway shows in the middle of each segment of the airway line. The identifier does not show when the actual text length of the identifier is too great to be drawn in the middle of the segment.



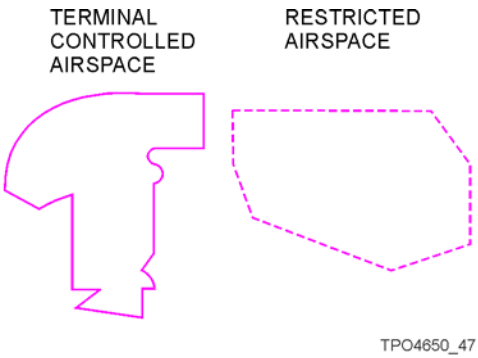
- Rivers are depicted using dark-blue solid lines.
- Lakes are depicted using a dark-blue filled area.
- Coastlines/Oceans are depicted using a dark-blue filled area.



Controlled airspace that is depicted on the map includes Class B and C, CTA, and TMA/TCA airspace.

- Controlled airspace is depicted using solid magenta lines.
- Restricted airspace is depicted using dotted magenta lines.

  NOTE: Restricted airspace boundaries may appear as solid magenta lines rather than dashed magenta lines depending on the selected map range when the boundaries of multiple restricted areas overlap.



Graphical Weather (optional)

Activation of the XM or Datalink (Universal) graphical IFIS option adds the NEXRAD/Universal GWX overlay to the FMS Plan map. The NEXRAD feature allows an uplinked NEXRAD digital image to be overlaid on the Plan map. The status of the Universal GWX image shows on the MFD when the GWX overlay is active on the Plan map. The PLAN menu on the MFD allows selection of the NEXRAD/Universal graphical weather on the MFD Plan map.

NOTE

The NEW GWX message flashes initially when the FSU receives a new image while using Universal GWX. The message remains in view until it is cleared by going to the image list. Press the MENU button and to show the list of images. Additional new images received by the FSU do not cause the message to flash until the message is cleared.

NOTE

When displaying TFRs and GWX simultaneously on the Plan Map, GWX is always layered on top of the TFR information. If the TFR information is parially blocked by the GWX image, deselecting GWX via the MFD menu allows for full display of TFR information.

NOTE

If the FSU INOP message is displayed in the lower left hand corner of an MFD, refrain from making selections on the GRAPHICAL WEATHER menu as the IFIS system may be attempting to recover and may react inconsistently to commands received during this period.

GWX Status (optional for Universal GWX)

The status of the GWX image shows on the MFD when the GWX overlay is active on the Plan map. The status shows on the right side of the MFD as shown below. Refer to the MESSAGES AND ANNUNCIATORS chapter for detailed information. The status message shows as one of the items that follow:

- NEXRAD AGE HH:MM (in white). The HH:MM readout is the age of the NEXRAD image in hours and minutes.
- NEXRAD AGE HH:MM (in yellow).
- NO USA NEXRAD AVAILABLE.



TPO4723_01

Auto Map Declutter

Airspace, Airways, and geopolitical E-Map symbology is removed automatically based on the selected range as described below:

- Airways symbology is automatically removed when a range above 50 NM is selected.
- Airspace symbology is automatically removed when a range above 100 NM is selected.
- Geopolitical symbology is automatically removed when ranges above 300 NM are selected.

Temporary Flight Restriction

TFR geographic boundaries are depicted in yellow with yellow alphanumeric identifiers.

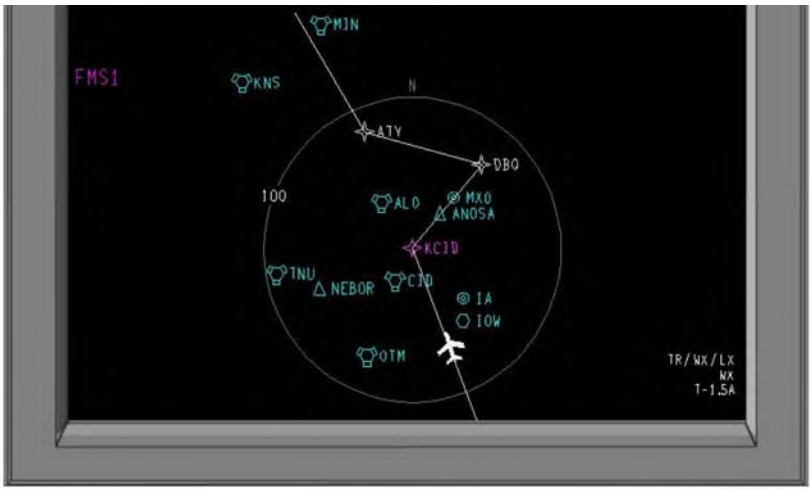
Fault Messages

Faults that prevent the proper display of the E-Map overlays are identified with specific fault messages. The fault messages show in the lower center portion of the MFD. The E-Map symbology is removed when an E-Map fault message is in view. The E-Map fault messages are as follows:

- CHECK DATABASE STATUS (yellow)
- AIRSPACE FAULT (white)
- AIRWAYS FAULT (white)
- GEO-POL FAULT (white)
- TFR FAULT (white)
- NO AIRSPACE AVAILABLE (white)
- NO AIRWAYS AVAILABLE (white)
- NO GEO-POL AVAILABLE (white)
- PROCESSING REQUEST (white)
- NO TFRS AVAILABLE (white)

MFD – PLAN MAP FORMAT

Figure 6-21 MFD – FMS Plan Map

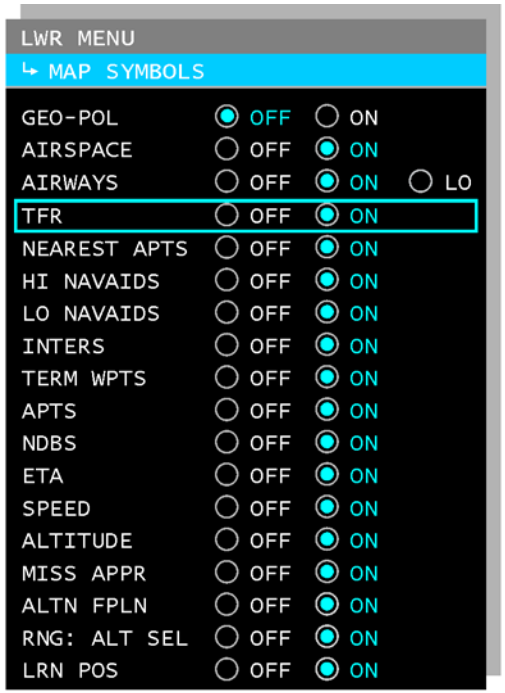


TPO4473_01

The FMS Plan Map is a fixed, true north-up stationary map used to display the active FMS flight plan. The active flight plan is represented by flight plan waypoints and flight plan legs. Background NAVAID symbology, for example airports, is available for display on the Plan Map. Enhanced Map (E-Map) features (for example, geopolitical boundaries, airspace, and airways) may be available when the optional Integrated Flight Information System (IFIS) is installed. Graphical Weather (GWX) features may be available when the optional IFIS is installed. Waypoints can be defined on the FMS Plan Map in the same manner as on the PPOS Map. A moving aircraft symbol is positioned geographically on the map and oriented to the aircraft heading. The map center is a user-selected latitude and longitude. The map radius is selected with the RANGE knob on the DCP. An extended Plan Map format is available for display that provides increased forward viewing area. Three submodes are available when the extended Plan Map format is active: no data window, data window on, and data window on with VNAV data. The FMS Plan Map is selected for display on the MFD with controls located on the CCP.

▼ **SELECTIONS:**

Map Source	The map source is the same as the NAV source when the NAV source is a FMS, otherwise, the map source is the on-side FMS.
Flight Plan Symbology	Flight plan legs show as solid lines connecting the flight plan waypoints. Waypoint symbology includes waypoints, user-defined waypoints, runway thresholds, holding patterns, and procedure turns. Flight plan waypoints are identified with a string of up to seven characters. The next (to) waypoint show in magenta.
Map Symbols	Background symbols are made up of user-selected symbology that shows geometrically correct on the map. The selection of map symbology is available in two places: the CDU and the AFD (by way of the MAP SYMBOLS menu). The MAP SYMBOLS menu is used to select map background symbology for display on the FMS PPOS, PLAN and 3D maps (for example, nearest airports). The map background symbology can also be selected by way of the FMS CDU. Refer to the MFD Map Displays section of the Bombardier Challenger 605/650 Pro Line 21™ Advanced Flight Management System (523-0822377) for details on selecting map background symbology by way of the CDU. Refer to the PFD — PPOS Map section in this chapter for details on map background symbology.



TPO4472_01

Aircraft Symbol

The moving aircraft symbol is positioned geographically on the map and oriented to the current aircraft heading.

Map Center

When the FMS Plan Map is initially selected, the next (to) waypoint is the map center. Thereafter, the flight plan center can be centered about a position or waypoint selected by way of the FMS.

Extended Map

The area above the MFD navigation display can be used to show an extended Map format that provides approximately 50 percent increased forward viewing area. Three submodes are available when the extended Map format is active: no data window, data window on, and data window on with VNAV data. Selection of the extended Map submodes is controlled by way of the FMS CDU.

- The text window on submode shows multiple lines of FMS text above the FMS Plan Map. The text window typically shows data associated with the current or upcoming flight legs.
- The data window with VNAV submode can show up to seven rows of text associated with the vertical waypoints.

Define Waypoints

The joystick cursor (+ symbol) shows on the MFD when the joystick is deflected and the FMS Plan Map or PPOS Map is the active format. The latitude and longitude of the cursor position is sent to the FMS upon operation of the button on the CCP. The cursor is removed if the JSTK button is pressed to transfer the joystick control to the upper format in a split window MFD format or if the joystick is not used for 60 seconds.

Range Symbology

When the optional 3D Map is not installed, the range symbology includes a full-range ring and a full-range readout. When 3D is installed, the range ring shows half the selected range except for the top range selection, which will put the map in full-range. The RANGE knob on the DCP is used to select the display range.

Enhanced Maps (option)

Enhanced Maps (E-Maps) features are available when the EMAPS IFIS option is activated. E-Map symbology includes GEO-POL boundaries, airspace, airways, graphical weather radar, and Temporary Flight Restriction (TFR). The map menu on the MFD allows selection of the E-Maps symbology.

NOTE

A blank TFR may be displayed when selecting a TFR Text Report from the TFR Text Report list. If this occurs, re-request the TFR by re—selecting to TFR Text Report List and re-selecting the TFR.

Graphical Weather (option)

The optional Integrated Flight Information System (IFIS) adds the NEXRAD feature to the FMS Plan Map. The NEXRAD feature allows an uplinked NEXRAD digital image to be overlaid on the Plan Map. The status of the GWX image shows on the MFD when the GWX overlay is active on the Plan Map. The NEXRAD graphical weather image is selected for display by way of the PLAN Map menu on the MFD. Refer to the Graphical Weather section in this chapter for a description of the NEXRAD graphical weather image on the FMS Plan Map.

NOTE

The NEW GWX message flashes initially when the FSU receives a new image while using Universal GWX. The message remains in view until it is cleared by going to the image list. Press the MENU button and to show the list of images. Additional new images received by the FSU do not cause the message to flash until the message is cleared.

NOTE

When displaying TFRs and GWX simultaneously on the Plan Map, GWX is always layered on top of the TFR information. If the TFR information is parially blocked by the GWX image, deselecting GWX via the MFD menu allows for full display of TFR information.

NOTE

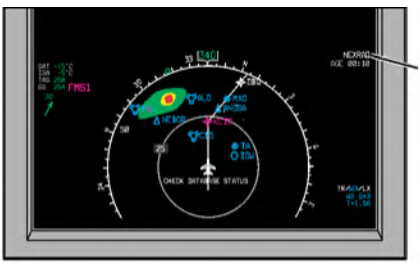
If the FSU INOP message is displayed in the lower left hand corner of an MFD, refrain from making selections on the GRAPHICAL WEATHER menu as the IFIS system may be attempting to recover and may react inconsistently to commands received during this period.



TPO4620_75

Graphical Weather Status (optional)

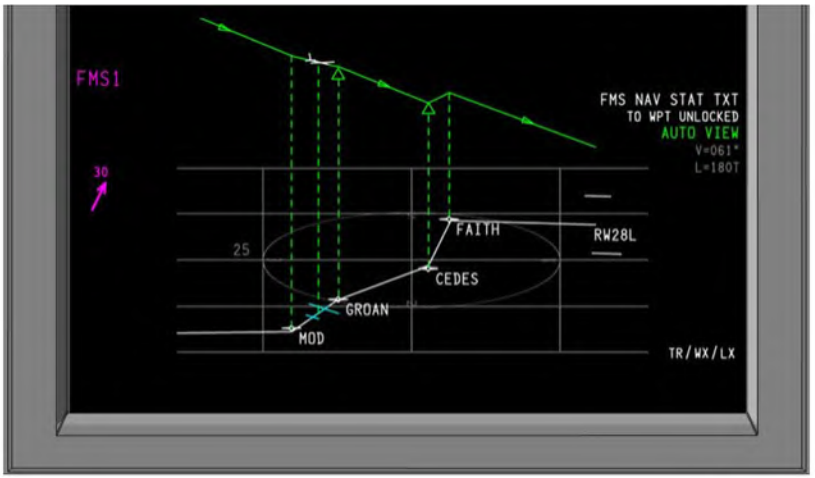
The status of the GWX image shows on the MFD when the GWX overlay is active on the Plan Map. The status shows on the right side of the MFD. Refer to the MESSAGES AND ANNUNCIATORS chapter for detailed information.



TPO4620_50

MFD – 3D MAP FORMAT

Figure 6-22 MFD – 3D Map



TPO4650_08

3D Map is an optional, advanced FMS feature that provides lateral, vertical, and performance-predicted flight plan information in a single, Three-Dimensional (3D) format. The map data is a combination of what would typically be presented as two separate map formats: a vertical profile and a Plan Map. The 3D Map has an adjustable viewing orientation. The 3D Map allows predicted flight path views that are referenced from the ground (for example, a vertical profile view), referenced directly over a map center position (for example a Plan Map view), or referenced from an intermediate point in between. All lateral flight plan waypoints, map symbols, associated legs, and background data show for the selected map range. Vertical information is limited to Top-of-Climb (TOC) and Top-of-Descent (TOD) indications, and waypoint speed/altitude constraints along the flight plan route. The aircraft symbol represents the aircraft present position. In addition to the traditional map controls for range and map center, angle of view controls provide for rotating the map presentation about two axes. Controls located on the CCP are used to select the 3D Map format for display on the MFD. Controls located on the MFD 3D Map Controls menu are used to control 3D Map features.

▼ SELECTIONS:

Lateral Viewing Angle	The lateral viewing angle of the 3D Map can be adjusted to any value between 0 and 360 degrees. The lateral viewing angle is independent of the vertical viewing angle. The joystick on the CCP is used to control the lateral and vertical viewing angle. The two axes of rotation are: • A line through the map center parallel to the display's vertical plane, allowing various vertical viewing perspectives. • A line through the map center perpendicular to the map center lateral plane, allowing various lateral viewing perspectives.
Vertical Viewing Mode	There are three vertical viewing modes: top, side, and shoulder. Each viewing mode provides a specific combination of FMS lateral and vertical flight plan information.



TPO4620_90

- All lateral flight plan waypoints, map symbols, associated legs, and background data show for the selected map range. Vertical information is limited to Top-of-Climb (TOC) and Top-of-Descent (TOD) indications, and waypoint speed/altitude constraints along the flight plan route. The aircraft symbol represents the aircraft present position.

- For vertical viewing angles between 0 and 90 degrees, all lateral and vertical flight plan information shows. The range ring and cardinal directions rotate with the vertical viewing angle and are positioned on a planar grid representing a flat surface projection of waypoints, airports, etc., on the earth's surface. The lateral position for any vertical symbol, including the aircraft symbol, is located on the grid through a vertical line from the vertical symbol to the grid. Vertical flight plan information, for example, waypoint altitude constraints, shows as triangles at the appropriate altitudes above the grid. Performance waypoints, for example, TOC and TOD indications, also show at the appropriate altitudes above the grid. A vertical reference line shows for each altitude constraint and performance waypoint.
- A vertical auto scale sets the vertical dimension of the 3D picture based on the highest vertical point above the grid. This may be a point along the flight path, an altitude constraint, or the present altitude of the aircraft, provided the aircraft is within viewing range on the map.

- A series of lines representing the aircraft performance-predicted flight path shows above the grid. The lateral position for each predicted altitude shows along the flight path, creating a connect-the-dots display of the predicted altitudes for the flight plan. A small arrow shows at the halfway point of each path segment to represent the forward direction of the flight plan. A 3D aircraft symbol (created by adding a vertical tail to the aircraft symbol) shows in white at altitude above the grid, and is oriented to present aircraft heading, pitch, roll, and viewing angle. A second aircraft symbol representing the shadow of the aircraft is depicted in cyan directly below (on the grid) and is oriented to the current viewing angle. A vertical reference line connects the cyan aircraft shadow to the white 3D aircraft.

NOTE

If a flight plan altitude constraint is less than 1500 feet AGL and is more than 10 NM from the departure runway, predicted flight trajectory for the 3D Map ignores the altitude constraint. The FMS divides a flight plan into four phases: takeoff, climb, cruise, and descent. The FMS takeoff phase ends and the climb phase begins at 1500 feet AGL. Because of this, the FMS assumes all altitude constraints beyond the takeoff phase are above 1500 feet AGL. An altitude constraint below 1500 feet AGL more than 10 NM from the runway would be outside of the takeoff phase for the 3D Map performance calculations. However, only the 3D Map graphical depiction of the flight path shows incorrectly. The FMS still correctly honors and flies the flight plan altitude constraint. As a result, in those cases where a waypoint climb altitude constraint of less than 1500 feet AGL is specified on the CDU during preflight, the 3D Map display may show contradictory information for predicted path while the aircraft is on the ground.

NOTE

For the 3D Map, no performance calculations are computed for destination airport SIDS in the flight plan. As a result, the 3D Map display does not plot vertical predicted path information for SIDS at the destination airport.

- With the vertical viewing angle set to 90 degrees, (the profile or side view), all vertical flight plan and minimal lateral information shows. The grid shows as a horizontal line. The range ring, digital range readout, and cardinal directions are replaced with a distance/direction line directly below the horizontal line. A digital distance readout is centered below the distance/direction line. Arrows pointing outward are on the ends of the distance/direction line. Adjacent to the arrows are digital readouts in units of degrees representing two angles that are perpendicular to the current lateral viewing angle. A vertical scale shows on the left side of the display with the bottom of the scale aligned with the horizontal grid line. The digital readouts at the top and bottom represent the maximum and minimum values of the scale.
- Display of lateral flight plan information is limited to the waypoint identifiers immediately below the horizontal line. All vertical flight plan information, TOC and TOD waypoints and the predicted flight path show at appropriate altitudes above the horizontal line. Only one aircraft symbol shows and no aircraft shadow shows.

3D Map Controls

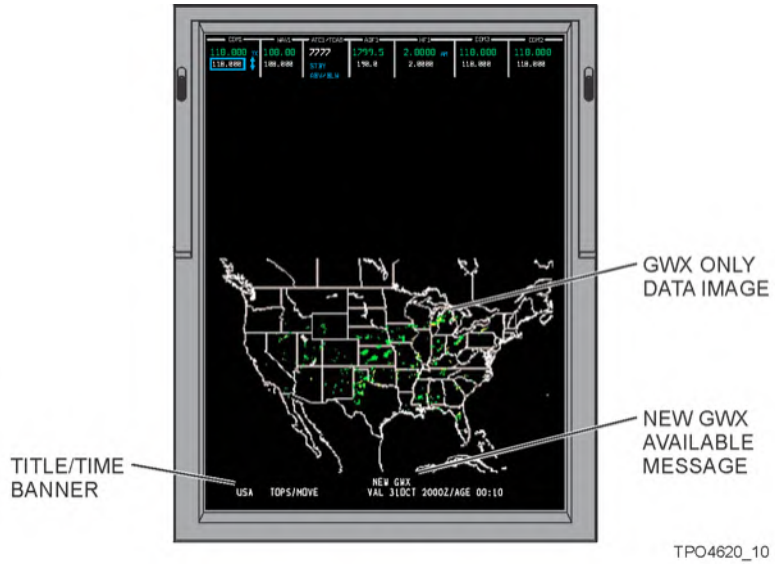
The 3D Map controls provide for manual control of the viewing modes and preset viewing angles. Manual control allows the pilot to set the view to any desired angle within the available limits. By positioning the CCP joystick, the pilot may set the lateral and/or vertical viewing angles to any combination of 0 to 360 degrees laterally and 0 to 90 degrees vertically. View presets are one of two forms of automated viewing control. Presets allow the pilot to select fixed viewing modes with minimal workload. There are three view presets available through the CCP controls: Top View, Shoulder View, and Side View.

- Joystick — The joystick is used to set the view to any desired lateral and/or vertical viewing angle combinations between 0 to 60 degrees laterally and 0 to 90 degrees vertically.
- Top View— The top view is the familiar top-down viewing angle.
- Shoulder View — The shoulder view shows both lateral and vertical flight plan information. The shoulder view is a 3D Map presentation that is set to 22 degrees lateral and 75 degrees vertical viewing angle relative to the inbound course of the map center waypoint.
- Side View — The side view shows a full vertical view with a lateral viewing angle perpendicular to the inbound course of the map center waypoint.

- Auto View — When auto view is active and the next (to) FMS flight plan waypoint is the map center, the inbound course is referenced to the point abeam of the aircraft present position instead of the map center waypoint location. This allows the view to automatically rotate to track multiple course changes which occur in FMS radius, arc, holding pattern, and procedure turn leg types. The mode remains active until it is deselected. The message AUTO VIEW shows in the lower left corner of the display when auto view is active. The joystick control and the top, shoulder, and side view selections remain active when auto view is active.

MFD – GRAPHICAL WEATHER (GWX-5000)

Figure 6-23 GWX-5000 Graphical Weather



The Graphical Weather (GWX) format provides the ability to show one GWX image at a time on the MFD. New GWX images are requested by the pilot by way of controls on the CDU. Refer to the Rockwell Collins Corporate Datalink System CMU-4000/RIU-40X0 Operator's Guide, CPN 523-0790499, for detailed information on using the CDU to request and view GWX images.

NOTE

The IFIS illustrations in this operator's guide are for illustrative purposes only.

The GWX images are uplinked by way of VHF datalink system from the Information Service Provider (Universal Weather). A list of saved and available GWX images shows on the MFD when requested by the pilot. Using controls on the CCP, the pilot selects the desired GWX image to show on the MFD.

The GWX image can be a forecast or an observation image. Forecast images include WINDS ALOFT, ICING, and TURBULENCE. Observation images include: NEXRAD, TOPS/MOVEMENT, and

Weather (WX) DEPICTION. Each datalinked GWX image is paired with a corresponding geopolitical background image. A title/time banner shows for each GWX image.

NOTE

Datalinked GWX is for informational purposes and should not be used for tactical decision making. By its very nature, datalinked GWX is delayed from real-time weather conditions.

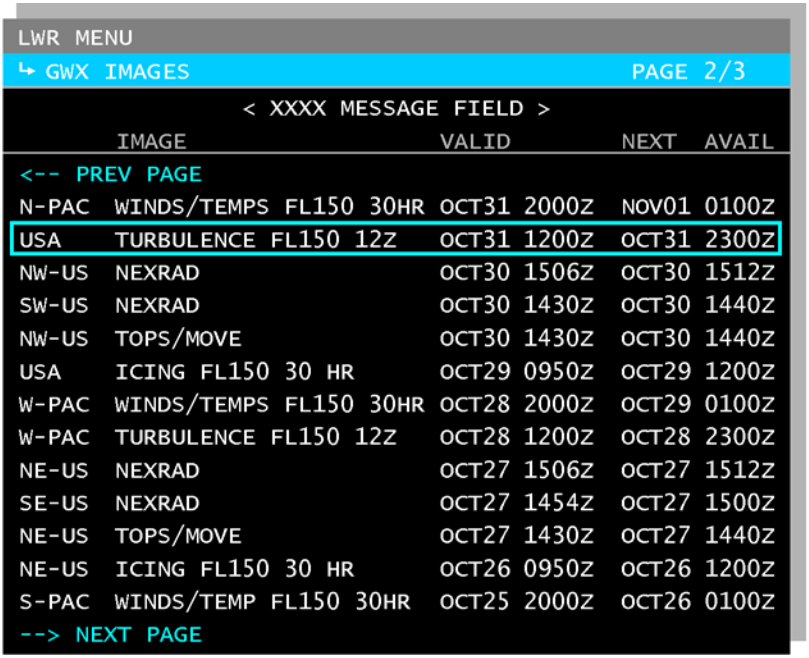
NOTE

The USA NEXRAD precipitation image is the only image available as an overlay on the Plan Map format. The CAN NEXRAD item shown on is applicable only to the GWX-3002 XM Graphical Weather Installation.

NOTE

On occasion, the National Weather Service incorrectly identifies test data that will exhibit unrealistic weather patterns. Examples are: 1) donut-shaped pattern, 2) pie slice-shaped pattern, 3) circle-shaped pattern, and 4) bull's eye-shaped pattern. If an image of this nature is encountered and the pilot suspects the data is in error, wait until the next image update cycle, request a new image, and make sure the test pattern is removed.

GWX IMAGE LIST



TPO4630_93

The GWX Image menu is selected by way of the MFD LWR MENU. Controls located on the CCP are used to select and operate the MFD LWR MENU. GWX images are selected for display by way of the GWX IMAGE list. The GWX Image menu is selected by way of the MFD LWR MENU. Controls located on the CCP are used to select the GWX IMAGES list.

For each GWX image on the list, a region abbreviation, name of the image, date, month and time of the image show. The date, month, and time of the next available image also show for each GWX item on the list. New graphical weather images can be added to the list by way of the CDU REQ GWX page. The newest image to be received shows at the top of the list. The GWX region name and region abbreviations are:

GWX Region Abbreviations

GWX Abbreviation	Region Description
AFR	Africa
AUS	Australia
EUR	Europe
INDIA	India
N-AM	North America
N-ATL	North Atlantic
N-PAC	North Pacific
S-AM	South America
S-PAC	South Pacific
W-PAC	West Pacific
UK	United Kingdom
NC-US	North Central United States
NE-US	Northeast United States
NW-US	Northwest United States
SC-US	South Central United States
SE-US	Southeast United States
SW-US	Southwest United States
USA	United States

NOTE

The list will hold a maximum of 50 items. The oldest item will be removed from the list as new items are added.

▼ SELECTIONS:

Graphical Images	The corresponding graphical weather image is in view on the MFD when the image is selected from the GWX Image list. An example of each graphical weather image type and an accompanying description are as follows: • NEXRAD — Refer to the figure on page 6-195 • TOPS/MOVEMENT — Refer to the figure on page 6-196 • WX DEPICTION — Refer to the figure on page 6-198 • WINDS ALOFT — Refer to the figure on page 6-200 • TURBULENCE — Refer to the figure on page 6-202 • ICING — Refer to the figure on page 6-203.
New Images	New graphical weather images can be added to the list by way of the CDU. When the system receives a new (uploaded) image, the text message NEW GWX shows on the MFD. The message is cleared by entry into the image list. The message GWX RECEIVED shows on the CDU. The newest image to be received shows at the top of the list. Refer to the applicable CMU Operator's Guide for details on uploading images and adding to the image list. **NOTE** As long as images in the entry list remain unviewed, the message GWX RECEIVED continues to show on the CDU.

New images are available according to the schedule that follows:

New Image Schedules

Image Type	Schedule
NEXRAD	Every 6 minutes
US SCIT TOPS/ Movement	Every 30 minutes (Typically available at 10 and 40 minutes past the hour)
WX Depiction	Every 60 minutes (Typically available at 20 minutes past the hour)
Winds Aloft	0Z, 6Z, 12Z, 18Z, 30 hr, 36 hr, 42 hr, 48 hr, 72 hr
Turbulence	0Z, 6Z, 12Z, 18Z, 30 hr, 36 hr, 42 hr, 48 hr, 72 hr
Icing	0Z, 6Z, 12Z, 18Z, 30 hr, 36 hr, 42 hr, 48 hr, 72 hr



TECH DETAIL

The Zulu (Z) forecast times refer to that time in the future for weather forecasts. Therefore, if the current time is 15Z, an 18Z forecast is for 18Z today and a 12Z forecast is for 12Z tomorrow. The hour HR forecast times refer to the number of hours since the last model run for weather forecasts. The model runs are completed at 0Z and 12Z each day. Therefore, if the current time is 15Z, a 30HR forecast is for 18Z tomorrow (for example, 30 hours from the last model run, which is 12Z today).

Retention of Images

NEXRAD, forecast, and observation images are retained as follows:

- For each NEXRAD region, up to the five most current unique images are retained.
- When not overwritten by a newer image, images are purged at the time intervals that follow when on the ground.

Image Purge Schedules

IMAGE TYPE	PURGE TIME
NEXRAD	Valid time + 90 minutes
WX DEPICTION	Valid time + 90 minutes
TOPS/MOVEMENT	Valid time + 90 minutes
WINDS ALOFT	Valid time + 9 hours
ICING	Valid time + 9 hours
TURBULENCE	Valid time + 9 hours

Old Images

The menu item shows in yellow within a yellow box when the image is old. Images are considered old when the current time exceeds the product's next available time by more than 30 minutes.

Geopolitical Background Image

Each datalinked GWX image is paired with a corresponding geopolitical background image. The geopolitical background image shows geographic and political borders. The borders show in white with a black background.

Title/Time Banner

A title/time banner shows along the bottom of the display when a GWX image is in view. Two different formats are available on the title/time banner, one for a forecast image and one for an observation image.

- For a forecast image, the region name, image name, valid date/month/time of the image, and the issue day/month/time of the image show on the title/time banner. Forecasts include Winds Aloft, Icing, and Turbulence images.
- For an observation image, the region name (for example, USA), image name (for example, NEXRAD), and the valid day/month/time, and age of the image in hours and minutes show on the title/time banner. Observations include NEXRAD, Tops and Movement, and Weather Depiction images.

GWX Messages

The messages that follow may be in view on the MFD when the GWX format is active.

- GWX FAULT
- SUBSCRIPTION EXPIRED
- NOT RECEIVING DATA
- SIGMET OVERLAY INCOMPLETE - CHECK TEXT.

GRAPHICAL WEATHER IMAGES

Figure 6-24 NEXRAD Image



A NEXRAD image is in view on the lower display area of the MFD when selected from the GWX image list and the GWX format is active.

NOTE

On occasion, the National Weather Service incorrectly identifies test data that will exhibit unrealistic weather patterns. Examples are: 1) donut-shaped pattern, 2) pie slice-shaped pattern, 3) circle-shaped pattern, and 4) bull's eye-shaped pattern. If an image of this nature is encountered and the pilot suspects the data is in error, wait until the next image update cycle, request a new image, and make sure the test pattern is removed.

▼ SELECTIONS:

NEXRAD Image Colors	The table that follows lists the colors used on NEXRAD images to represent areas of precipitation and the relative intensity.

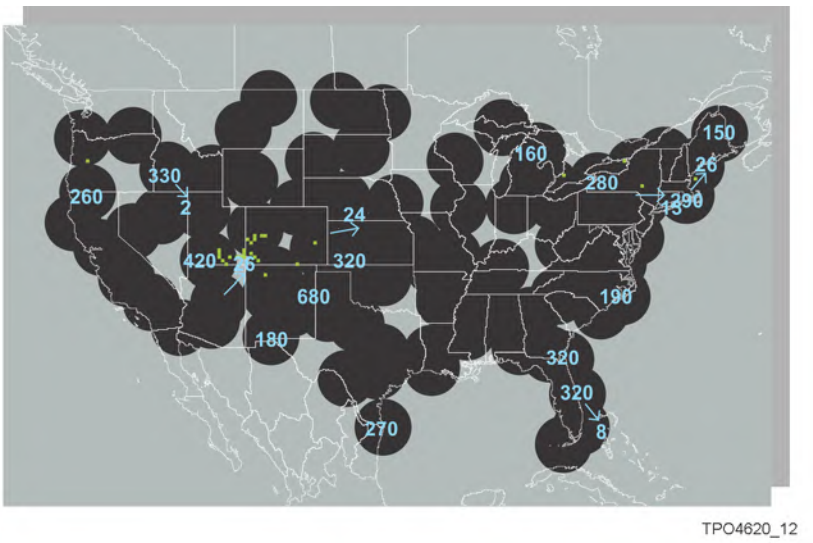
NEXRAD Color Coding (RAIN)

Precipitation Intensity	Color
20 ≤ dBZ < 30	Green ( )
30 ≤ dBZ < 40	Yellow ( )
dBZ ≥ 40	Red ( )

NOTE

The same color coding is used to show the intensity of snow and rain.

Figure 6-25 TOPS/MOVEMENT Image



A TOPS/MOVEMENT image is in view on the lower display area of the MFD when selected from the GWX image list and the GWX format is active.

NOTE

On occasion, the National Weather Service incorrectly identifies test data that will exhibit unrealistic weather patterns. Examples are: 1) donut-shaped pattern, 2) pie slice-shaped pattern, 3) circle-shaped pattern, and 4) bull's eye-shaped pattern. If an image of this nature is encountered and the pilot suspects the data is in error, wait until the next image update cycle, request a new image, and make sure the test pattern is removed.

▼ SELECTIONS:

TOPS/MOVEMENT Image Symbology

Precipitation symbology, US SCIT TOPS, and cell movement information shows on TOPS/MOVEMENT images. US SCIT TOPS and cell movement shows for major storm cells. The altitude of the Weather Radar tops in hundreds of feet shows in blue centered over the storm cell. If more than one storm type is indicated for a single storm cell, only the most severe type shows. For storm cells with measurable groundspeed, a track arrow is drawn from the center point of the storm cell out in the direction of the true track of the cell. A groundspeed readout in knots shows at the end of the track arrow.

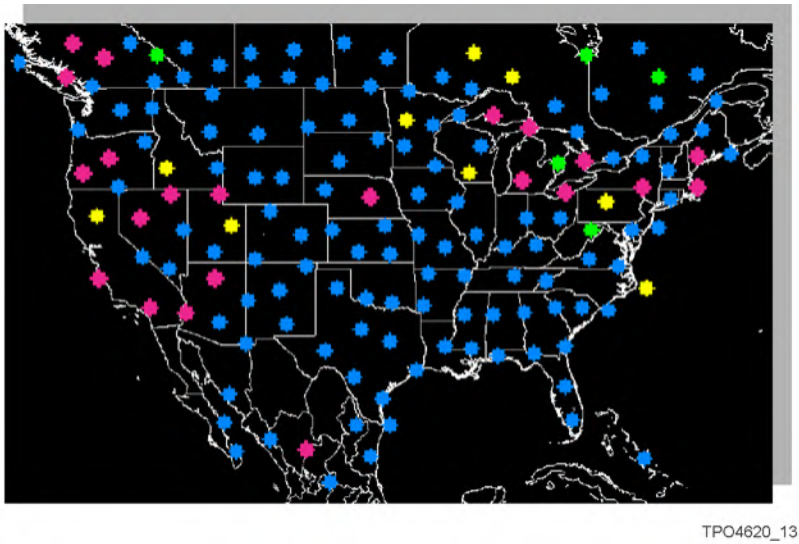


The table that follows lists the colors used on TOPS/MOVEMENT images to represent areas of precipitation and the relative intensity.

TOPS/MOVEMENT Color Coding

Precipitation Intensity	Color
20 ≤ dBZ < 30	Green ( )
30 ≤ dBZ < 40	Yellow ( )
dBZ ≥ 40	Red ( )

Figure 6-26 WX Depiction Image



A WX DEPICTION image is in view on the lower display area of the MFD when selected from the GWX image list and the GWX format is active. The WX DEPICTION chart shows forecast visibility and ceiling conditions for the selected region.

▼ SELECTIONS:

WX DEPICTION Image Colors	The table that follows lists the categorical outlook definitions used by Universal Weather. The bottom table lists the FAA categorical outlook definitions.

Flight Rule Condition Definitions (Universal Weather)

Flight Rule Condition	Definition
VFR	Ceiling > 3000 ft and Visibility > 5 statute miles (8.05 km or 6400 m)
Marginal VFR	Ceiling ≤ 3000 ft or Visibility ≤ 5 sm (8.05 km or 6400 m)
IFR	Ceiling < 1000 ft or Visibility < 3 sm (4.83 km or 5400 m)
Low IFR	Ceiling < 500 ft or Visibility < 1 sm (1.61 km or 1609 m)

The table that follows list the colors used on WX DEPICTION images.

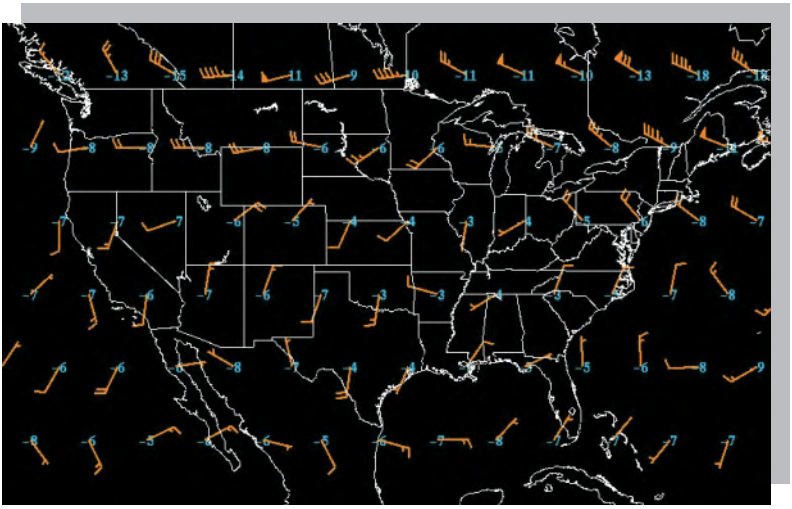
WX DEPICTION Color Coding

Flight Rule Condition	Color
VFR	Sky Blue ( )
Marginal VFR	Green ( )
IFR	Yellow ( )
Low IFR	Magenta ( )

NOTE

WX DEPICTION is based on a finite number of reporting stations. For finer weather detail, refer to other sources, for example, NEXRAD or TAF reports.

Figure 6-27 Winds/Temps Image



TPH5384_14

A WINDS ALOFT image is in view on the lower display area of the MFD when selected from the GWX image list and the GWX format is active. The specified altitudes available for winds aloft images (x 1000 feet) are: 5, 10, 18, 24 30, 34, 39, 45, and 53.

▼ SELECTIONS:

WINDS ALOFT Image Symbology	The Winds Aloft charts depict forecast wind direction, wind speed and temperature at a specified altitude. Wind speed is indicated with the use of pennants and barbs on an arrow. The pennants represent 50 knots, barbs represent 10 knots, and half-barbs represent 5 knots. The total wind speed is obtained by the summation of all the pennants and barbs on the arrow. The head points to the wind direction. The temperatures are depicted next to the grid points in degrees Celsius. A missing wind is represented with the letter M.

The illustration below shows the symbology used in WINDS ALOFT images.

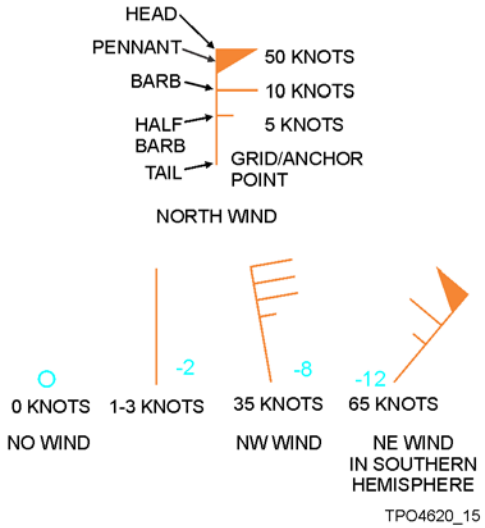
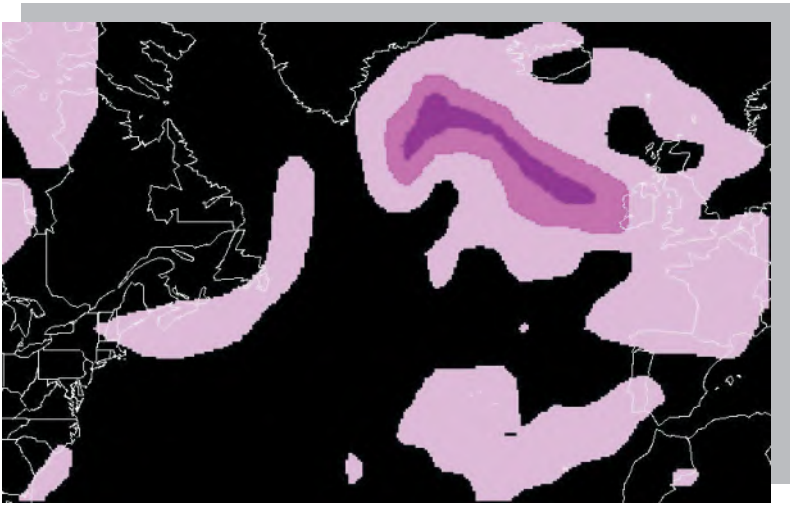


Figure 6-28 Turbulence Image



TPH5384_16

A TURBULENCE image is in view on the lower display area of the MFD when selected from the GWX image list and the GWX format is active. The specified altitudes available for turbulence images (x 1000 feet) are: 5, 10, 18, 24 30, 34, 39, 45, and 53.

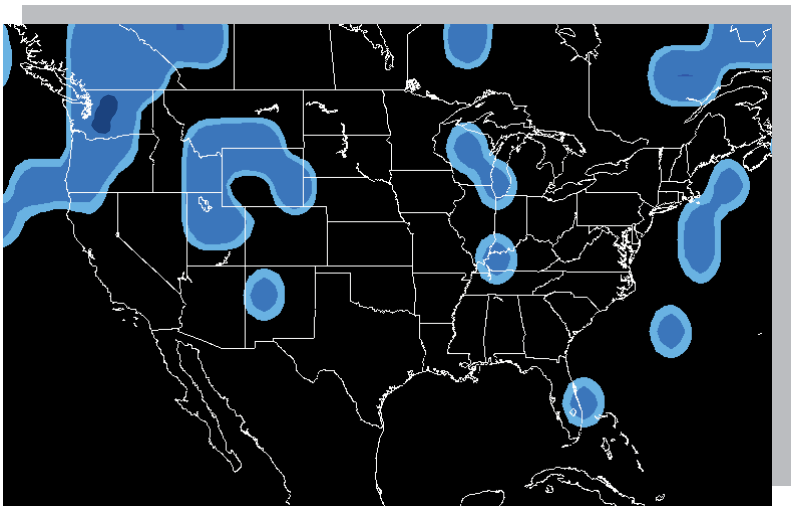
▼ SELECTIONS:

TURBULENCE Image Colors	The table that follows lists the colors used on TURBULENCE images to represent areas of likely turbulence.

TURBULENCE Color Coding

Turbulence Probability	**Color**
Low probability	Light Magenta ( )
Moderate probability	Medium Magenta ( )
High probability	Dark Magenta ( )

Figure 6-29 Icing Image



TPH5384_17

An ICING image is in view on the lower display area of the MFD when selected from the GWX image list and the GWX format is active. The specified altitudes available for icing images (x 1000 feet) are: 0, 5, 10, 18, 24 and 30.

▼ SELECTIONS:

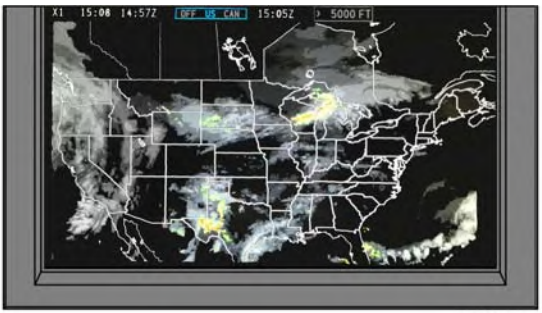
ICING Image Colors

The table that follows lists the colors used on ICING images to represent areas of potential icing.

ICING Color Coding

Icing Potential	Color
Trace rime	Light Blue ( )
Trace clear – Light rime or clear	Medium Blue ( )
Light or greater mixed ice	Dark Blue ( )

Figure 6-30 Satellite Image

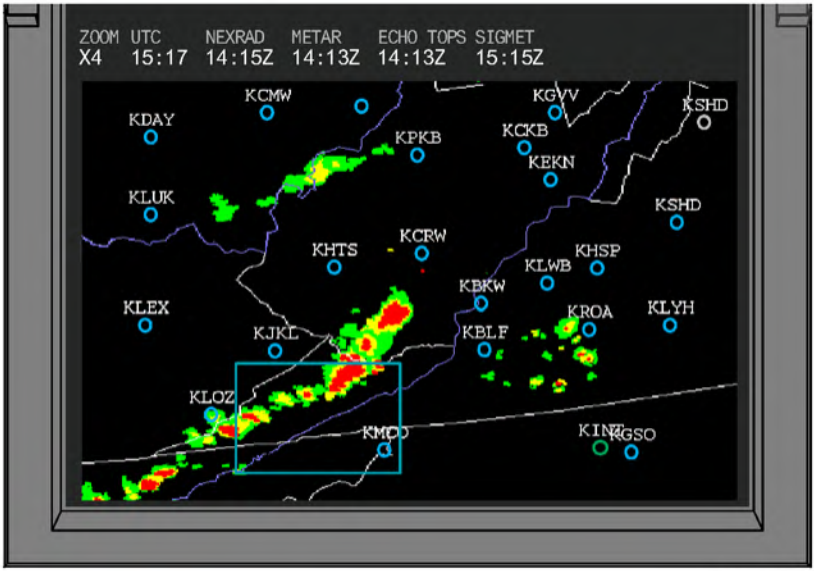


TPO4503_01

An satellite image is in view on the lower display area of the MFD when selected from the GWX image list and the GWX format is active.

MFD – GRAPHICAL WEATHER (GWX-3002)

Figure 6-31 GWX-3002 Graphical Weather



TPO4620_18

The Graphical Weather (GWX-3002) format provides the ability to show GWX images or reports on the MFD. The GWX images are provided by Baron Services by way of a satellite Information Service Provider (XM Satellite Weather Service).

The GWX image can be a textual weather report/forecast or a graphical image. Weather reports include Significant Meteorological (SIGMET) and Airman's Meteorological (AIRMET) advisories and Aviation Routine Weather Reports (METAR). Weather forecasts are Terminal Area Forecast (TAF). Observation images include US NEXRAD, Canadian NEXRAD, and US SCIT TOPS. Selection of the desired image is by way of the GRAPHICAL WEATHER menu. Controls located on the CCP are used to select the GRAPHICAL WEATHER menu. Refer to the figure on page 6-210.

NOTE

The graphical images are for the whole of the contiguous USA and southern Canada only. No other regions of the world are covered by this weather service provider.

NOTE

GWX is for informational purposes and should not be used for strategic decision making. By its very nature, GWX is delayed from real time weather conditions.

Weather products are transmitted by the satellite on the intervals listed in the table that follows:

XM Weather Products Transmission Schedule

Product	Transmit Interval	Typical Information Update Interval
NEXRAD	5 minutes	5 minutes
METAR	12 minutes	On the hour ± 10 minutes unless special (SA)
US SCIT TOPS	90 seconds	30 minutes
SIGMETs	12 minutes	As weather changes
AIRMETs	12 minutes	As weather changes
TAF	12 minutes	Every 6 hours (0Z, 6Z, 12Z, 18Z) unless amended (ADMT)

▼ SELECTIONS:

ZOOM	The ZOOM display shows the current selected zoom level under the ZOOM legend. The table below shows the zoom level with the associated relative zoom and the MFD text display.

Zoom Levels

Zoom Level	Relative Zoom Factor	Zoom Text	Relative Map Scale
0	1	X1	Entire CONUS
1	4	X4	1/4 CONUS (≈ 640 NM)
2	4	X16	1/16 CONUS (≈ 160 NM)

Operation of the ZOOM button on the CCP cycles the zoom level between the values listed in the table above. The zoom level selected when GWX format is reselected is the last commanded. The area of the image to be zoomed is indicated by the pan indicator.

UTC

The current system time shows under the UTC legend.

GWX Product Legend/Time Stamp

The GWX Product Legend and Time Stamp shows when the associated graphical overlay is selected ON from the OVERLAYS menu. The time in view is the time of the observation or report.

Pan Indicator

The pan indicator shows on the graphical display when the joystick on the CCP is moved and the GWX format is active and the zoom level is zero or one. The pan indicator is a green box. Movement of the pan indicator is controlled by the joystick.

Stale Images

A graphical weather image is considered stale when the current time exceeds the product time stamp by the amounts that follow:

Stale Image Times

Image Type	Stale Time
NEXRAD	15 minutes
US SCIT TOPS	4.5 minutes
METAR	36 minutes
SIGMET	36 minutes

The Plan Map age readout shows in yellow within a yellow box if the NEXRAD product is stale. The GWX product time stamp shows in yellow within a yellow box if the associated product is stale. Images are deleted if the current time exceeds the product's time by more than 2 hours.

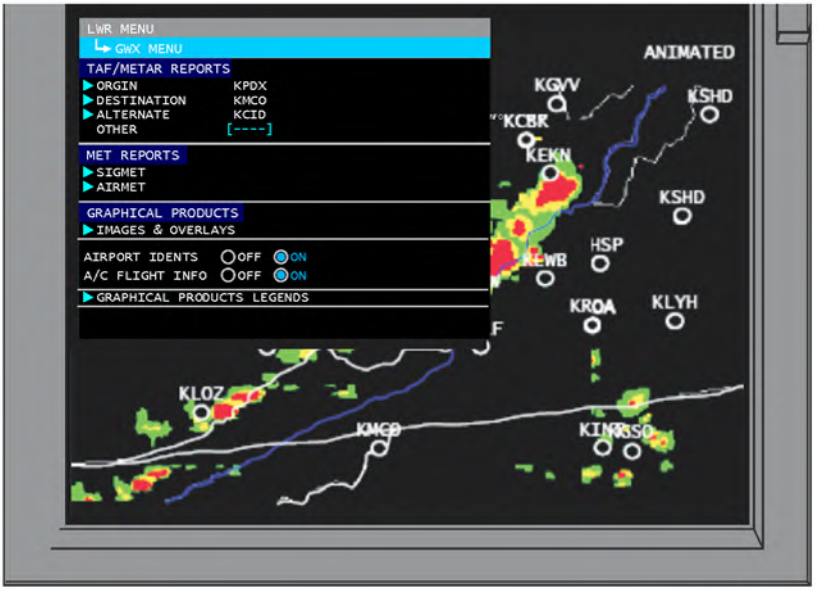
GWX Message Field

The messages that follow may be in view on the MFD in the GWX message field when the GWX format is selected.

- GWX FAULT
- SUBSCRIPTION EXPIRED
- NOT RECEIVING DATA
- SIGMET OVERLAY INCOMPLETE - CHECK TEXT.

NOTE

If the XM Receiver does not receive a valid satellite signal at power-up (for example, while the aircraft is in a hangar), the XM Receiver will be unable to authorize an XM Weather subscription, resulting in the display of the SUBSCRIPTION EXPIRED message. The SUBSCRIPTION EXPIRED message will be in view in the GWX message field on the MFD until the XM Receiver receives a valid satellite signal and completes subscription authorization processing.



TPO4475_01

The GRAPHICAL WEATHER menu provides controls for selection of textual weather reports and forecasts, as well selection of graphical image overlays for the lower display area of the MFD. An OVERLAYS legend provides a key to interpreting the symbology used on the graphical weather overlays.

▼ SELECTIONS:

GRAPHICAL WEATHER menu	The GRAPHICAL WEATHER menu is selected by way of the MFD LWR MENU. Controls located on the CCP are used to select and operate the MFD LWR MENU. Refer to the IFIS Menu Navigation appendix of this guide for details on navigating to and operating the GWX menus.



TPO4661_01

TAF/METAR REPORTS

The TAF/METAR REPORTS menu provides selections for the ORIGIN, DESTINATION, ALTERNATE, and OTHER airports. When an airport is selected, a list of the three most recent METAR reports available for that airport shows on the first page, in order of most recent to oldest. Refer to the figure on page 6-213. A second page for the airport TAF/METAR reports shows the current TAF report available for that airport. Refer to the figure on page 6-214.

NATIONAL MET REPORTS

The NATIONAL MET REPORTS menu provides selections to display SIGMET and AIRMET reports for the whole of the contiguous USA. When a report type is selected (SIGMET or AIRMET), the appropriate reports page shows, listing all reports in order of the listing that shows below. Each SIGMET and AIRMET is in view on a separate page. Refer to the figure on page 6-215.

- SIGMETs — Listed in alphabetical order by identifier

- AIRMETs — Listed in the order received.

IMAGES & OVERLAYS

The OVERLAYS menu shows a list of overlays that are available to be in view on the MFD when the GWX format is selected. Refer to the OVERLAYS entry.

NOTE

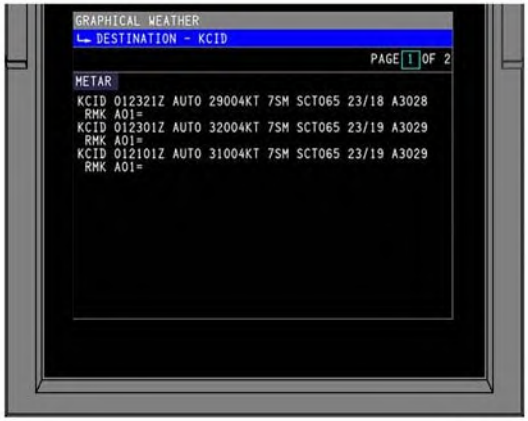
Selecting GWX MENU clears any previously selected images from the GWX format as soon as the GWX MENU is displayed.

GRAPHICAL PRODUCTS LEGENDS

The OVERLAY LEGENDS page shows all the symbology used on the overlays to assist the operator with interpreting the information that shows on the overlay. Refer to the figure on page 6-229.

METAR/TAF REPORTS

Figure 6-32 METAR Reports Page



TPO4620_20

The METAR Reports page displays up to the last three METAR reports for the associated airport.

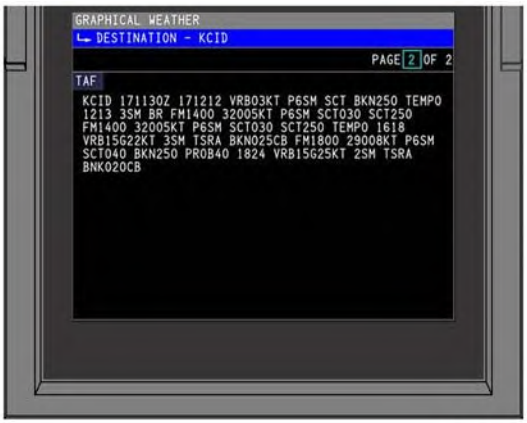
▼ SELECTIONS:

Title	The title for the METARS Report page will be one of the items that follow: • ORIGIN <airport ICAO identifier> • DESTINATION <airport ICAO identifier> • ALTERNATE <airport ICAO identifier> • OTHER <airport ICAO identifier>.
METAR Reports	The METAR Reports section shows the three most recent METAR reports for the associated airport. If no METAR text exists for the associated airport, the message NO DATA AVAILABLE shows instead of a METAR report.

NOTE

The only METARS that are available are those received since aircraft power-up or that follow an airport change.

Figure 6-33 TAF Report Page



TPO4620_21

The second page of the TAF/METAR Reports page for the selected airport shows the current TAF report for the airport. The TAF report page shows the TAF report in its entirety.

SIGMET/AIRMET REPORTS

Figure 6-34 SIGMET/AIRMET Reports Page



TPO4620_22

The SIGMET/AIRMET Report page title will be either SIGMET or AIRMET, depending on the selection made on the page. When there are multiple reports available, additional pages are available to display the SIGMETs or AIRMETs.

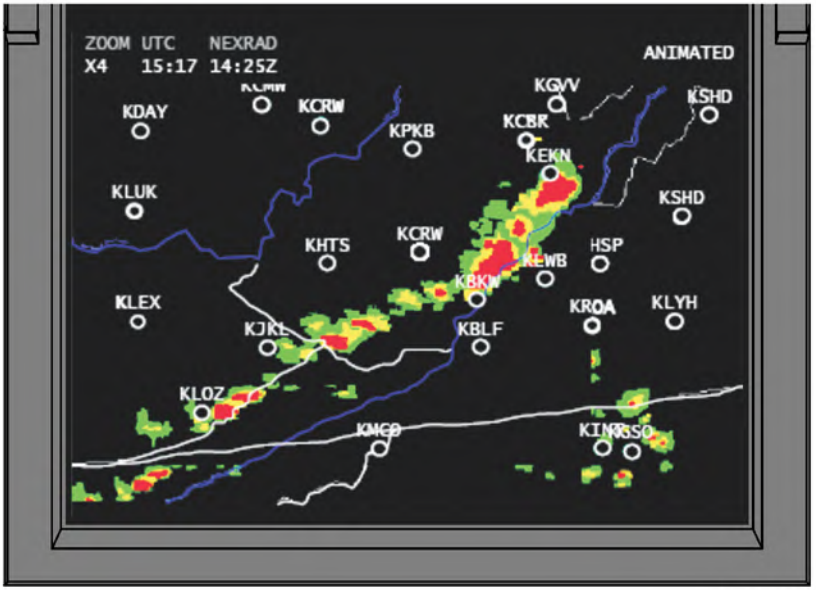
If no graphical SIGMET data is available, the message NO GRAPHICAL SIGMET appears in the upper right corner of the SIGMET Report page.

TECH DETAIL

A maximum of 50 active SIGMET reports and 75 active AIRMET reports can be presented.

ANIMATED NEXRAD

Figure 6-35 Animated NEXRAD



The Animated NEXRAD state allows the crew to view a time-elapsed sequence of weather images. The sequence of images runs in a loop using a minimum of three, to a maximum of five valid NEXRAD images received from the datalink. The loop sequence runs until cancelled. During normal operation the system stores a snapshot of the NEXRAD and associated time stamp, every 15 minutes or greater, as soon as available. A maximum of five snapshots are stored. When Animated NEXRAD mode is selected, the stored snapshots are in view as a series of frames in time sequence (oldest first) with a 0.7-second interval between each frame.

NOTE

Animated NEXRAD cannot be selectable until three or more valid frames of data have been stored by the system.

The messages that follow can show when Animated NEXRAD is active on the display.

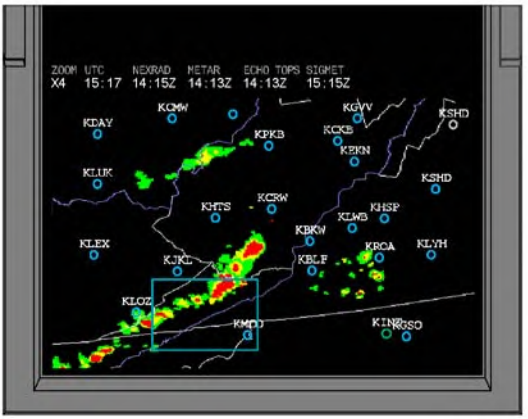
- ACQUIRING IMAGES
- AVAILABLE.

OVERLAYS

Condition(s):

When US or CANADIAN NEXRAD is selected from the OVERLAYS menu, the NEXRAD image is in view on the MFD when the GWX format is selected.

Figure 6-36 NEXRAD Overlay



TPO4620_23

▼ SELECTIONS:

Rain Depiction

The table that follows lists the colors used on NEXRAD images to represent regions of rain precipitation and the relative intensity.

NEXRAD Rain Precipitation Color Coding

Rain Intensity	Color
20 ≤ dBZ < 30	Green ( )
30 ≤ dBZ < 40	Yellow ( )
dBZ ≥ 40	Red ( )

NOTE

Mixed Precipitation Depiction not applicable to Canadian NEXRAD image

Mixed Precipitation Depiction

The table that follows lists the colors used on NEXRAD images to represent regions of mixed (rain and snow) precipitation and the relative intensity.

Mixed Precipitation Intensity	Color
10 ≤ dBZ < 30	Light Magenta ( )
30 ≤ dBZ < 40	Medium Magenta ( )
dBZ ≥ 40	Dark Magenta ( )

NOTE

Mixed Precipitation Depiction not applicable to Canadian NEXRAD image

Snow Depiction

The table that follows lists the colors used on NEXRAD images to represent regions of snow precipitation and the relative intensity.

B

Snow Intensity	Color
10 ≤ dBZ < 30	Light Blue ( )
30 ≤ dBZ < 40	Medium Blue ( )
dBZ ≥ 40	Dark Blue ( )

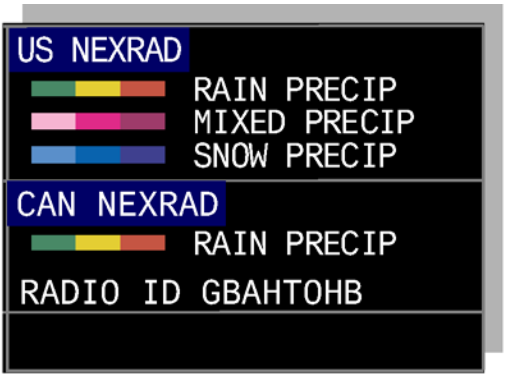
Other Conditions

The table that follows lists the colors used on NEXRAD images to represent other conditions.

C

Other Condition	Color
No significant precipitation	Black ( )
Missing data or no RADAR coverage	Dark Grey ( )

The illustration that follows shows the NEXRAD precipitation depictions and the associated colors.



TPO4620_24

Condition(s):

When US SCIT TOPS is selected from the OVERLAYS menu, the US SCIT TOPS image is in view on the MFD when the GWX format is selected.

▼ SELECTIONS:

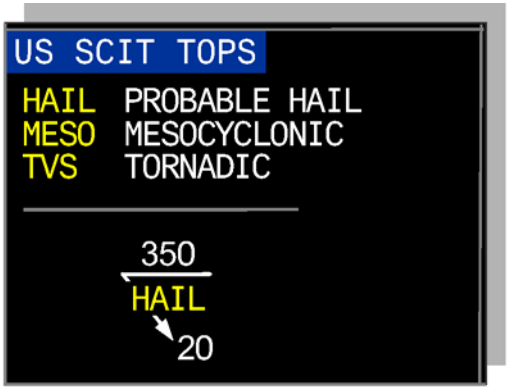
US SCIT TOPS Symbology	Precipitation symbology, US SCIT TOPS, and cell movement information shows on US SCIT TOPS images. If more than one storm type is indicated for a single storm cell, only the most severe type shows. The US SCIT TOPS legend shows a white horizontal line drawn with the left-most end of the line located at the latitude/longitude of the storm cell. The storm top altitude in hundreds of feet shows above the line in white. A storm type legend is listed under the line. The storm type and color will be one of the items that follow:

US SCIT TOPS Storm Type

Storm Type Legend	Color
No storm type	<blank>
HAIL	Yellow
MESO	Yellow
TVS	Yellow

A white track arrow is drawn from the latitude/longitude of the storm cell in the direction of the true track of the cell when track is known. A groundspeed readout in knots shows at the end of the track arrow when the groundspeed of the cell is known.

The illustration that follows shows the US SCIT TOPS symbology.



TPO4620_25

Condition(s):

When US or CAN METAR is selected from the OVERLAYS menu, the selected graphical METAR image is in view on the MFD when the GWX format is selected.

Figure 6-37 METAR Overlay



TPO4620_26

▼ SELECTIONS:

METAR	The METAR overlay is made up of flight rule icons that are colored according to the IMC Flight Rules data received from the datalink for the associated airport. If no data is received, or the received data is corrupted or cannot be interpreted, a grey circle shows for the associated airport. To reduce display clutter, the number icons that show is filtered based on the selected zoom level. The table that follows lists the categorical outlooks as defined by the FAA. METAR selections are: • OFF • US

- CAN

Flight Rule Condition Definitions (FAA)

Flight Rule Condition	Definition
VFR	Ceiling > 3000 ft and Visibility > 5 statute miles
Marginal VFR	1000 ≤ Ceiling ≤ 3000 ft and/or 3 ≤ Visibility ≤ 5 miles
IFR	500 ≤ Ceiling < 1000 ft and/or 1 ≤ visibility < 3 miles
Low IFR	Ceiling < 500 ft and/or Visibility < 1 mile

The illustration that follows shows the Flight Rule condition and the associated icon color.

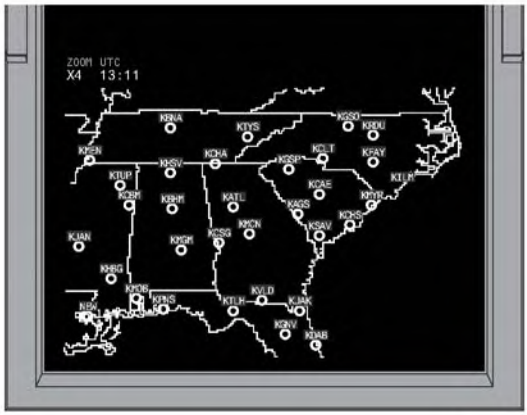


TPO4620_27

Condition(s):

When AIRPORT ICONS is selected ON from the OVERLAYS menu and the GWX format is selected, the current graphical image on the MFD is overlaid with airport icons.

Figure 6-38 Airport Idents Overlay



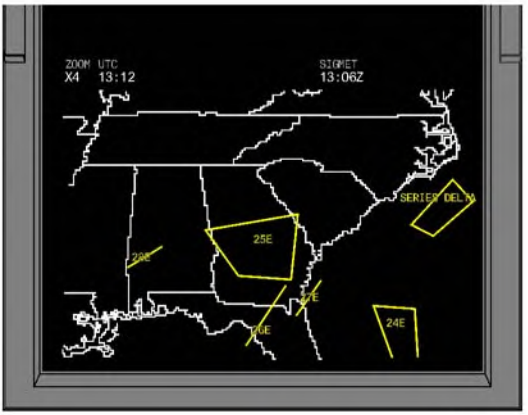
TPO4481_01

The AIRPORT ICONS depicts airports and reporting stations that are within the current image. The AIRPORT ICONS consist of a small circle with the ICAO identifier for the airport above the circle.

Condition(s):

When SIGMET is selected from the OVERLAYS menu, the graphical SIGMET images are in view on the MFD when the GWX format is selected.

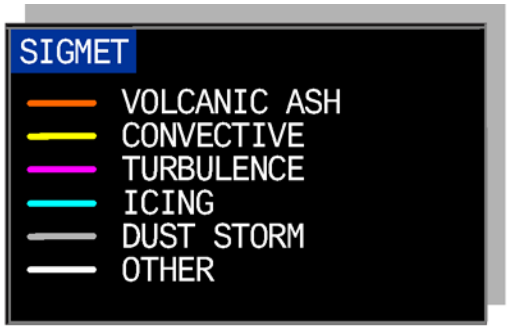
Figure 6-39 SIGMET Overlay



The SIGMET is made up of a point, line, or unfilled polygon that defines the geographic extent of the report and its identifier.

▼ SELECTIONS:

SIGMET	The illustration that follows shows the SIGMET condition and the associated display color.



TPO4620_30

Condition(s):

When A/C FLIGHT INFO is selected from the OVERLAYS menu, an aircraft symbol, showing aircraft location, is in view on the MFD when the GWX format is selected.



TPO4482_01

▼ SELECTIONS:

A/C FLIGHT INFO

The A/C FLIGHT INFO overlay is made up of an aircraft symbol that shows the location of the aircraft on the map. Also included in this overlay are the origin and destination airports, which are highlighted above.

NOTE

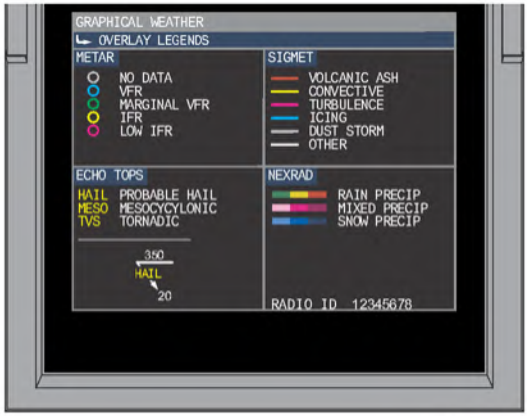
The symbol is yellow when a geographically-referenced chart is in view and GNSS present position data is not available.

GEO-REFERENCED
NO GNSS PPOS DATA

TPO4546_01

OVERLAY LEGENDS

Figure 6-40 Overlay Legends Page



TPO4505_01

The OVERLAY LEGENDS page displays all the symbology used in the graphical weather overlays.

MFD – GWX OVERLAYS MENU

Figure 6-41 manMFD – GWX Images & OverlaysMenu



TPO4929_01

The GWX OVERLAYS menu is part of the GRAPHICAL WEATHER menu. The GWX OVERLAYS menu is used to control the various layers of information including NEXRAD, US SCIT TOPS, METARs, US SIGMETs and US LIGHTNING. Once an overlay is selected for display on the format, it remains on until it is selected off. The GRAPHICAL WEATHER menu is selected and operated with controls on the CCP.

MFD – ELECTRONIC CHARTS

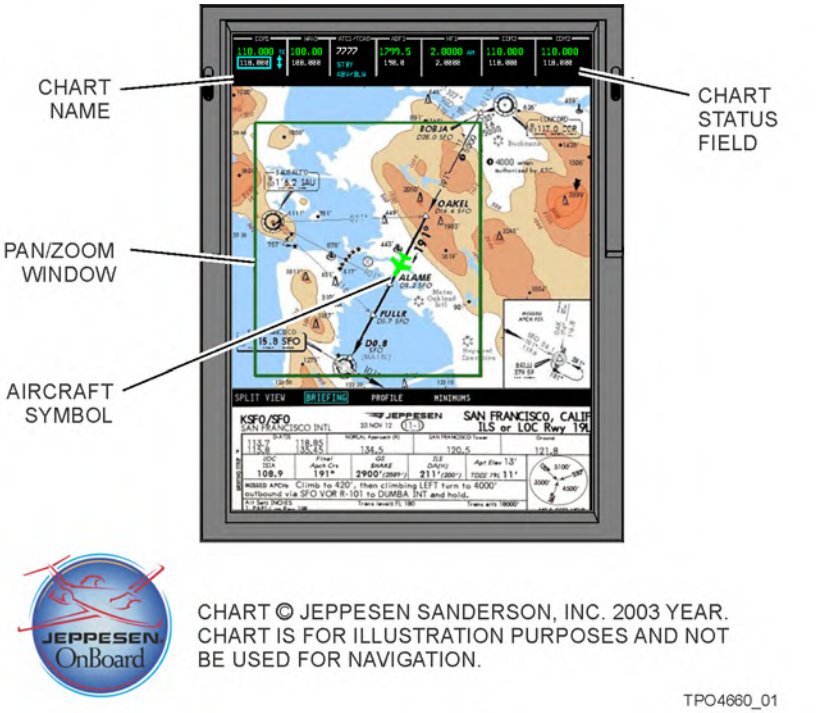
The Electronic Chart (E-Chart) format provides the ability to show an electronic version of a conventional paper instrument chart on the MFD. The E-Charts are linked automatically by the FMS when a flight plan is entered and can also be selected manually by the pilot. The available charts are listed on the Chart Main Index. Controls for chart selection are on the CCP. When aircraft position data is available, a moving aircraft symbol shows on E-Charts that are geographically-referenced. A non-geographically-referenced chart will have a green aircraft symbol with a circle and slash on the top right hand corner of the chart. Refer to the IFIS Menu Navigation appendix of this guide for details on navigating to and operating the E-Chart index.

NOTE

The use of E-Charts with the geographically-referenced aircraft symbol does not relieve the pilot of responsibility for navigation.

DISPLAY CHART

Figure 6-42 Display Chart Format



The selected Jeppesen E-Chart shows when selected by the pilot. Controls on the CCP are used to select a chart for display, pan around the chart, zoom in or out on the chart, and change the orientation of the chart.

A moving aircraft symbol shows on the chart when the chart is geographically-referenced, as determined by the Jeppesen database and the aircraft position is within the geographically-referenced part of the chart.

▼ SELECTIONS:

Chart Name	The name of the chart in view shows immediately above the chart. The chart name includes the airport ID, chart index number and procedure ID.
Aircraft Symbol	A moving aircraft symbol shows on the chart when the chart is geographically-referenced, as determined by the Jeppesen database, and the aircraft position is within the geographically-referenced part of the chart. The moving aircraft symbol shows the current aircraft position latitude and longitude as determined by the same-side Global Navigation Satellite System (GNSS). The nose of the aircraft symbol is oriented toward the aircraft true heading. When the same-side GNSS data is absent or invalid, the opposite-side GNSS is used.

NOTE

GNSS is constantly calculating position and is accurate within a window. The aircraft symbol is positioned accordingly on the chart. The result may be that the aircraft symbol may move slightly, which will be most noticeable on the airport charts when the aircraft is not moving.

NOTE

The aircraft symbol shows for any ICAO airport identifier that is World Geodetic System 1984 (WGS84) compliant.

A crossed-out aircraft symbol shows above the chart to indicate when the aircraft position is not available to show on the chart. The crossed-out aircraft symbol shows above and to the right of the chart name. The symbol is yellow when a geographically-referenced chart is in view and GNSS present position data is not available. The symbol is a green aircraft with a white circle/slash when the chart in view is not geographically-referenced.

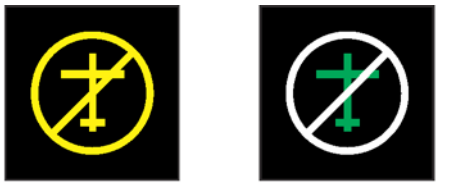


Chart Cycle Messages

Early/late cycle chart messages show in the chart status field to the right of the chart name. The message that shows will be one of the items that follow (in order of priority).

- OUT OF DATE
- CHECK DATE.

NOTE

The CHECK DATE message may appear for a short period of time while the GNSS tries to acquire satellites. If satellites are not in view (for example, while the aircraft is in a hangar), the CHECK DATE message may persist and the system will be unable to automatically check for database effectivity.

Chart Orientation

When a new chart is selected, the orientation is set to its default (portrait or landscape) as determined from the Jeppesen database. The first push of the orientation button (located below CHART button on the CCP) rotates the chart 90 degrees. The second push rotates the chart back to its original orientation.

Chart Scaling

When a new chart is selected, the chart is initially scaled such that the width of the chart fills the display. Operate the joystick to move the pan/zoom window to the area of the chart the pilot wishes to view. Push the ZOOM + button to expand the region of the chart enclosed within the pan indicator to fill the chart format. Push the ZOOM – button to return the chart to normal scaling.

Chart Panning

When zoomed in, operate the joystick to move the desired area of the chart into view on the MFD. Moving the joystick in the horizontal moves the pan/zoom window horizontally on the chart. Moving the joystick in the vertical axis moves the pan/zoom window vertically.



TECH DETAIL

The last viewed orientation, zoom, and pan positions are recalled the next time each chart is in view.

SPLIT VIEW

When the SPLIT approach chart view is selected from the chart main index, three SPLIT VIEW modes are selectable using the MENU ADV knob on the CCP. All SPLIT VIEW modes include the graphic portion of the chart which may be panned and/or zoomed.

- BRIEFING — The breifing portion of the chart is shown. When the split chart view is active and go-around is selected using the TO/GA buttons, it will be displayed if not already shown.
- PROFILE — The vertical profile portion of the chart is shown.
- MINIMUMS — The minimums data associated with the approach are shown.

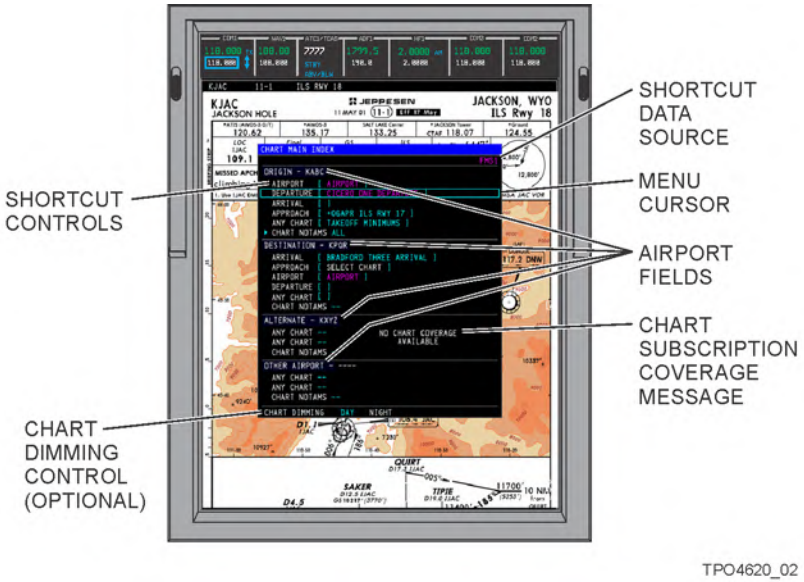
Fault Messages

The fault messages that follow may be in view on the MFD when an E-Chart is in view:

- OUT OF DATE
- CHECK DATE
- CHART FAULT
- NO CHART AVAILABLE.

CHART MAIN INDEX

Figure 6-43 Chart Main Index



The Chart Main Index shows the list of shortcuts to the charts needed for the procedures associated with the origin, destination, and alternate airport. Selecting one of these shortcuts causes the associated chart to show. The origin, destination, and alternate airport entries are supplied by the FMS and a flight plan is entered. The Other Airport entry is always entered by the pilot on this menu.

Controls on the CCP are used for data entry. Data entry mode is used to enter airport IDs. A dimming control is provided at the bottom of the Chart Main Index.

▼ SELECTIONS:

Airport Fields	The ICAO identifier (ID) of the origin, destination, alternate, and one other airport shows next to the associated headers (for example, ORIGIN – KCID). The airport IDs are supplied by the FMS and based on the active flight plan. The ICAO identifier of the other airport always shows as a data entry field next to the header. If the on side FMS has failed or the origin or destination airports have not been specified in the FMS flight plan, the data entry fields allow the flight crew to manually enter the desired airport.
Shortcut Controls	Shortcut controls show below the origin, destination, alternate, and other airport fields/airport IDs on the Chart Main Index. The shortcut controls identify the selected chart and provide a quick means of selecting the indicated chart. • ORIGIN — Controls for the AIRPORT, DEPARTURE, ANY CHART and CHART NOTAMS ALL shortcuts show below the ORIGIN header and airport ID. • DESTINATION — Controls for the ARRIVAL, APPROACH, AIRPORT, ANY CHART and CHART NOTAMS ALL shortcuts show below the DESTINATION header and airport ID. • ALTERNATE — Controls for two ANY CHART shortcuts and one CHART NOTAMS ALL shortcut show below the ALTERNATE header and airport ID.

- OTHER AIRPORT — Controls for two ANY CHART shortcuts and one CHART NOTAMS ALL shortcut show below the OTHER AIRPORT header and airport ID.

Shortcut Names

The shortcut name shows next to the associated shortcut control when a chart has been selected automatically by the FMS or manually by the pilot. The shortcut naming convention is as follows:

NOTE

Refer to the Chart Name entry in the menu table on page 6-233 for the chart naming convention.

- When the chart has been selected automatically by the FMS, the shortcut name is the name of the chart. The shortcut name shows in magenta.
- When the FMS has not found a matching procedure or more than one matching procedure has been found, the shortcut name is SELECT CHART. The shortcut name shows in white.
- When the chart has been selected by the pilot, the shortcut name is the name of the chart. The shortcut name shows in cyan.

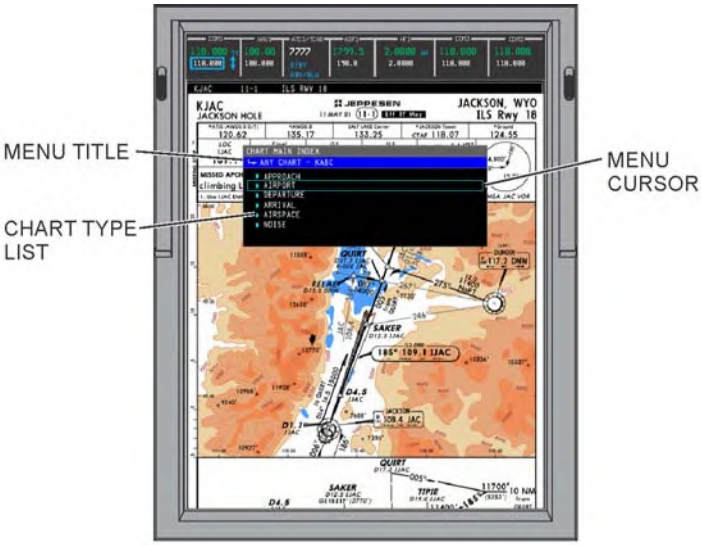
Shortcut Data Source

The shortcut data source shows in the menu status field on the Chart Main Index. The menu status field is located in the top right corner of the Chart Main Index. The data source message is one of the items that follow:

	• FMSn (for example, FMS1 or FMS2) shows in magenta when the selected map source is valid and an active flight plan exists. • FMSn NO DATA AVAILABLE (for example, FMS1 or FMS2) shows in white when the selected map source is invalid.
NO CHART COVERAGE AVAILABLE Message	When the origin, destination, or alternate airports are not covered by the regions contained within the current chart subscription, the message NO CHART COVERAGE AVAILABLE shows for that airport. The message shows in white to the right of the associated airport header (for example, origin, destination, alternate, or other airport).
Data Entry Field	Data entry mode is used to enter airport IDs for the airport search function.
Chart Dimming Control	A dimming control is provided at the bottom of the Chart Main Index, which allows the pilot to select the day or night color.

CHART TYPE MENU

Figure 6-44 Chart Type Menu

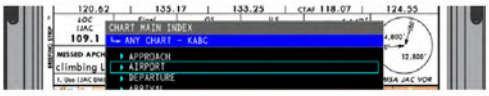


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The Chart Type menu is accessed by way of the ANY CHART selection on the Chart Main Index. The Chart Type menu shows a list of the chart types available for the selected airport. Selecting an item from the list causes the Chart List to come into view. The Chart List shows a list of the charts that are available of the selected type (for example, Approach) for the selected airport.

▼ SELECTIONS:

Menu Title	The menu title shows below the CHART MAIN INDEX header at the top of the menu. The menu title allows the user to know where they are in the menu tree. The menu title shows as ANY CHART followed by the airport ID (for example, ANY CHART KABC). The airport ID identifies the selected airport.



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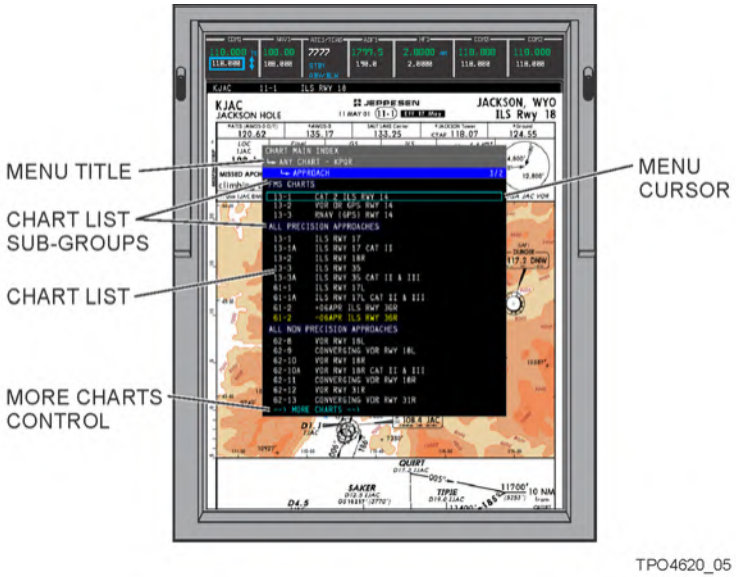
Chart Type List

A list of the chart types that are available for the selected airport shows below the airport ID. The possible chart types are as follows:

- Approach
- Airport
- Departure
- Arrival
- Airspace
- Noise.

CHART LIST

Figure 6-45 Chart List



The Chart List is accessed by way of the Display Chart, the Chart Main Index, or the Chart Type menus. The Chart List shows a list of charts of the selected type (for example, APPROACH) for the selected airport. When a chart is selected by way of the Display Chart method, the newly selected chart comes into view on the MFD. When a chart is selected, the associated shortcut on the Chart Main Index is updated to the new selection.

▼ SELECTIONS:

Menu Title	The menu title shows below the CHART MAIN INDEX header at the top of the menu. The menu title identifies the current location within the menu tree. The menu title shows as ANY CHART followed by the airport ID. The airport ID identifies the selected airport.



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Chart List Sub-Groups

The charts that show on the Chart List are listed in order of their sub-groups. The sub-groups, in order, are as follows:

- The sub-group FMS CHARTS shows when the FMS flight plan includes a procedure that matches the type of Chart List in view.
- The sub-group ALL PRECISION APPROACHES shows when the chart type is APPROACH and at least one precision approach is available for the selected airport.
- The sub-group ALL NON-PRECISION APPROACHES shows when the chart type is APPROACH and at least one non-precision approach is available for the selected airport.
- The sub-group ALL plus the chart type (for example, ALL ARRIVALS) shows when the chart type is ANY CHART.

Chart List

The charts are listed below their sub-group in order of their index numbers, with the lowest index number at the top of the menu.

Chart Names

The chart names show to the right of the chart index numbers on the chart list. The chart name for any current chart is the procedure ID (for example, ILS RWY 9).

Chart Updates

Jeppesen allows for charts to be updated midway through the effective dates of a database. Under these conditions, the database will contain additional charts that become effective on a given date. A chart in the list shows in white if the effectivity date indicates it is a current chart. An item in the list shows in yellow when its effectivity date indicates it is not current. A chart is considered not current when one of the items that follow occurs:

- There is a new chart and its effectivity date has not yet occurred.
- The chart is to be deleted and its effectivity date has passed.
- The chart is the early cycle of a mid-cycle revision and the current date/time is the late cycle of the database.
- The chart is the late cycle of a mid-cycle revision and the current date/time is the early cycle of the database.



TECH DETAIL

The switch-over time for Jeppesen E-Charts is 09:01 Coordinated Universal Time (UTC). If the current UTC is earlier than 09:01 of the chart's begin date, or later than 09:01 of the chart end date, the chart is considered noncurrent.

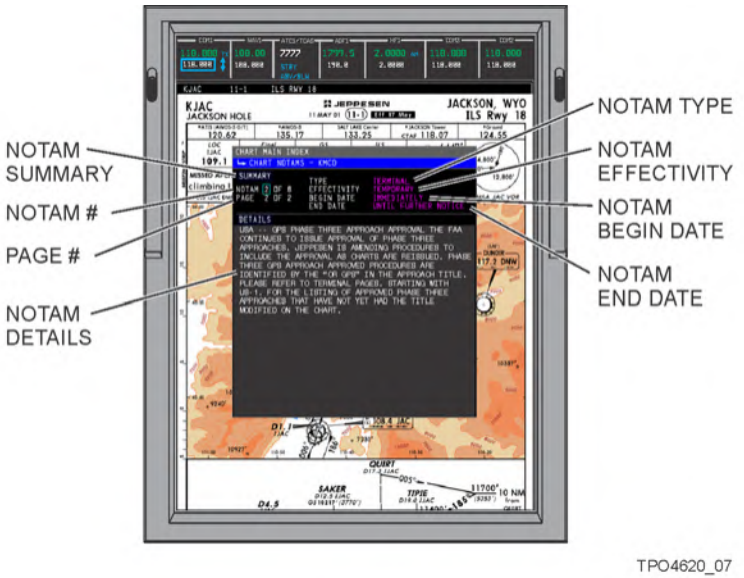
A chart that is being revised mid-cycle is identified as follows:

- A chart name with a minus sign and a date preceding the procedure ID (for example, -06APR ILS RWY 36R) is effective up to the associated date.

- A chart name with a plus symbol and a date preceding the procedure ID (for example, +06APR ILS RWY 36R) is effective on and after the associated date.

NOTAMS MENU

Figure 6-46 NOTAMS Menu



The Chart NOTAMS menu shows the chart NOTAMS available for the selected airport. The page is broken into two fields: the NOTAM summary and NOTAM details. When more than one NOTAM is available for the selected airport, the selected NOTAM and total number of NOTAMs shows in the summary field. The selected NOTAM readout is also a data entry field that allows the user to select another NOTAM for viewing. The NOTAM type, effectivity, begin date, and end date show in the summary field. The NOTAM text as defined in the Jeppesen charts database shows in the details field.

NOTE

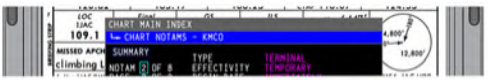
When there are temporary changes to en route or terminal charts, these changes are included in Jeppesen's Chart NOTAMs. The Chart NOTAMs are sometimes used for last-minute changes that missed the cutoff dates for changes. These NOTAMs are usually produced every two weeks. It is important to note that the Chart NOTAMs only highlight changes to the Jeppesen charts and do not substitute for the NOTAMs issued by the Federal Aviation Administration (FAA).

NOTE

New NOTAMs can be issued at any time by the FAA. These new NOTAMs may not be in the database installed in the FSU until the next database update. Prior to departure, always make sure that the latest NOTAMs are available either by requesting them using the onboard datalink or as hard copy.

▼ SELECTIONS:

Menu Title	The menu title shows below the CHART MAIN INDEX header at the top of the menu. The menu title allows the user to know where they are in the menu tree. The menu title shows as: CHART NOTAMS followed by the airport ID. The airport ID identifies the selected airport.
NOTAM Type	The NOTAM type, either TERMINAL or GENERAL, shows next to the TYPE header in the Summary field.



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NOTAM Effectivity	The NOTAM effectivity shows as TEMPORARY or PERMANENT as defined in the Jeppesen charts database. The NOTAM effectivity shows next to the EFFECTIVITY header in the Summary field.
NOTAM Begin Date	The begin date, as defined in the Jeppesen charts database, shows next to the BEGIN DATE header in the Summary field. The begin date shows as one of the items that follow: • IMMEDIATELY • NO BEGIN DATE • DD MMM YY (specified date).
NOTAM End Date	The end date, as defined in the Jeppesen charts database, shows next to the END DATE header in the Summary field. A text message or date shows for the end as one of the items that follow: • UNTIL FURTHER NOTICE • NO END DATE • DD MMM YY (specified date).
NOTAM #	When more than one NOTAM is available for the selected airport, the number of available NOTAMs shows next to the NOTAM legend. The first number is the selected NOTAM and the second number is the total number of NOTAMS for this airport. The NOTAMS are listed in the order that follows: 1. By type: Terminal NOTAMS first, then General. 2. By effectivity: Permanent NOTAMS first, then Temporary.

3. By begin date: IMMEDIATELY first, then by date, earliest to latest.

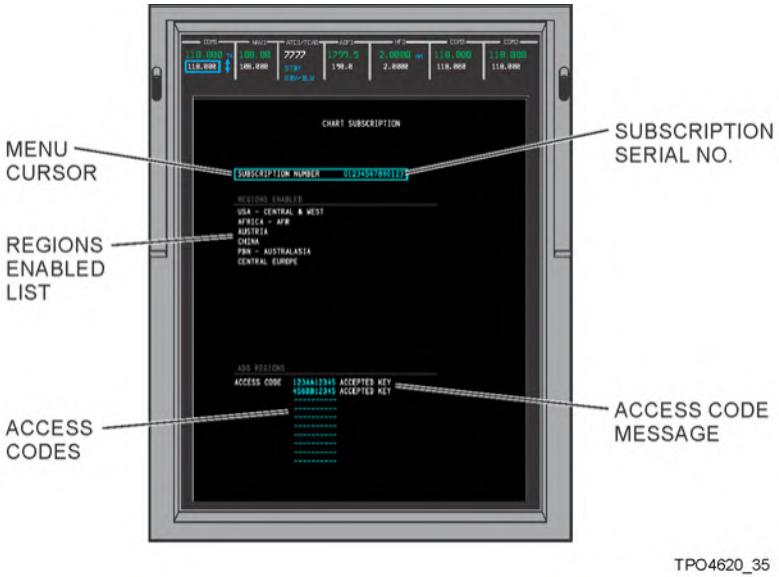
NOTAM Messages

The messages that follow may be in view on the MFD when the NOTAMs menu is active.

- PROCESSING REQUEST
- NOTAM DATA FAULT.

MFD – CHART SUBSCRIPTION PAGE

Figure 6-47 MFD – Chart Subscription Page



The Chart Subscription page allows the pilot or maintenance personnel to enter information to change the chart region coverage of the database. The Chart Subscription page contains a subscription number data entry field, a list of enabled regions, and an add regions data entry field. Regions are added to the list by entering temporary access codes. Up to seven temporary access codes can be entered. Both the

subscription number and access codes are obtained from Jeppesen. The Chart Subscription page can only be modified on the ground.

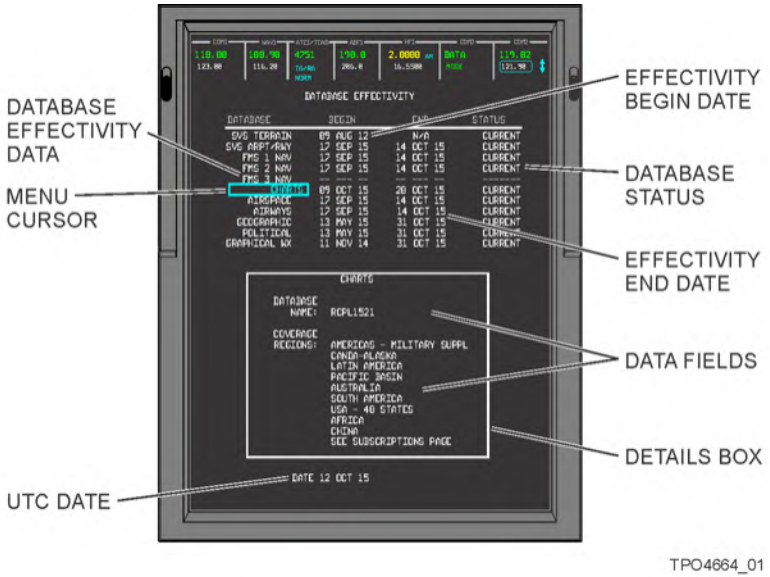
The Chart Subscription page is selected from the status format menu on the MFD. The status menu is enabled with a remote-mounted installer-supplied MAINT switch. Refer to the Remote Switches section in this chapter for details on MAINT switch and status format menu.

▼ SELECTIONS:

Menu Title	The Menu title shows below the CHART MAIN INDEX header at the top of the menu. The menu title allows the user to know where they are in the menu tree. The menu title shows as CHART NOTAMS followed by the airport ID. The airport ID identifies the selected airport.
Regions Enabled List	Once the subscription serial number has been entered, a list of the enabled regions associated with the subscription serial number shows below the subscription serial number.
Regions Enabled Messages	The text messages that follow are in view in the regions enabled list: • NO REGIONS AVAILABLE FOR CURRENT SUBSCRIPTION • PROCESSING REQUEST • SUBSCRIPTION FAULT.
Chart Subscription Messages	The text messages that follow are in view in the add regions field: • ACCEPTED KEY • INVALID ENTRY • PROCESSING • CODE FAULT.

MFD – DATABASE EFFECTIVITY PAGE

Figure 6-48 MFD – Database Effectivity Page



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The optional Database Effectivity page provides the aircraft operator with a means to check to see if the installed databases are current. A list of the installed databases shows on the Database Effectivity page. The list includes the name, begin date, end date, and status (current or not current) for enabled databases. The Database Effectivity page is also used to navigate between the installed databases and to provide specific information regarding the installed databases. The Database Effectivity page is selected from the status format menu on the MFD. The status format menu is enabled with a remote-mounted installer supplied MAINT switch. Refer to the Remote Switches section in this chapter for details on MAINT switch and status format menu.

The MFD shows database information from the Synthetic Vision Computer (SVC), Flight Management Computers (FMCs), and the File Server Unit (FSU).

NOTE

If UTC time is unavailable when the FSU initially powers up, all databases are marked as expired. When UTC becomes valid, the database effectivity are checked accordingly.

NOTE

The database information for the optional third FMC is shown dashed until FMS 3 REV is selected out of STBY. FMS 1 NAV or FMS 2 NAV will then be shown as dashed, depending on the FMS 3 REV selection made.

▼ SELECTIONS:

Database List

The database list shows all databases that support enabled applications. The list shows below the DATABASE legend. The possible list items and the update time of the associated database shows in the table that follows:

FREQUENCY OF DATABASE UPDATES

Database	Update Periods
SVS TERRAIN	As Required
SVS ARPT/RWY	28 days
FMSx (1–3)NAV database	28 days
CHARTS database	14 days
AIRSPACE* database	28 days
AIRWAYS* database	28 days
GEOGRAPHIC* database	As required
POLITICAL* database	As required
GRAPHICAL WX databases	As required

NOTE

* AIRSPACE, AIRWAYS, GEOGRAPHIC and POLITICAL databases are all included in an "E-Maps" database that is updated every 28 days.

Database Status

The database status messages that follow are in view on the Database Effectivity page.

- CURRENT (white)
- NOT CURRENT (yellow).

NOTE

The associated database begin date and end date also show in yellow when the database is NOT CURRENT.

Details Window

The database details window provides additional information regarding the selected database. Various databases provide different levels of detail information in the detail window. Most provide regional coverage information. Other databases provide additional information (for example,subscriber name).

Messages

The text messages that follow may be in view on the Database Effectivity page:

- PROCESSING REQUEST — Displayed when the display is waiting for IFIS database information to be provided.
- DATABASE EFFECTIVITY FAULT — Displayed when IFIS database data is not available after 30 seconds.

The text messages that follow may be in view in the Details Window for the CHARTS database:

- SEE SUBSCRIPTIONS PAGE — Displayed in the 10th position when the number of regions enabled in the subscription exceeds 10.
- NO REGIONS AVAILABLE FOR CURRENT SUBSCRIPTION — Displayed when the entered subscription does not provide coverage for any region.

UTC Date

The current date shows at the bottom of the Database Effectivity page when the master date is available from the avionics system. When the current date is failed or missing, dashes show in place of the UTC data.

PFD/MFD – TCAS PRESENTATION

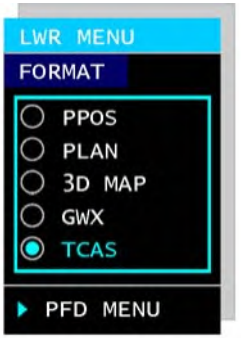
Figure 6-49 PFD/MFD – TCAS Map



TCAS Traffic information shows on the AFD to aid in the identification of nearby traffic and potential collision threats. A TCAS traffic overlay is available on the MFD and PFD. A TCAS Only format is available on the MFD. The TFC button on the DCP is used to select the TCAS Traffic display on the PFD. The TFC button on the CCP is used to select the TCAS Traffic display on the MFD. The TCAS mode is set by way of the ATC/TCAS menu on the MFD or CDU.

▼ SELECTIONS:

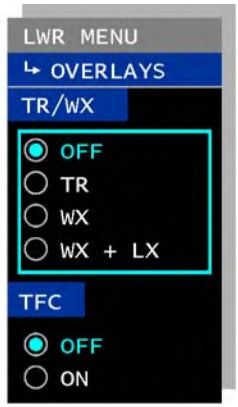
TCAS Only Format	The TCAS Only format is a heading-up, 360-degree moving map that graphically depicts the relative bearing, range, altitude, and vertical direction of nearby transponder equipped aircraft that reply to TCAS interrogations. The TCAS Only format is only available for display on the MFD. The TCAS Only format is not compatible with other hazard avoidance overlays. Radar and Terrain information if previously in view are removed from display when TCAS Only is selected. Selection of Radar or Terrain deselects TCAS Only mode. The TCAS Only format can be selected by way of the MFD lower menu.



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TCAS Overlay

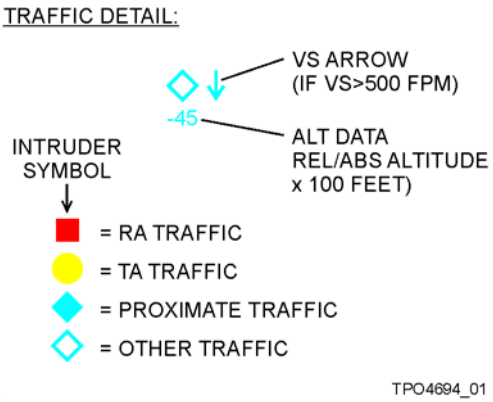
The TCAS overlay is a TCAS traffic map, similar to the TCAS Only format, that can be added to the Rose or PPOS Map formats on the MFD and PFD. The TCAS overlay can be selected for display on the MFD by way of the overlays sub-menu of the MFD lower menu. The TCAS overlay can be selected for display on the PFD by way of the overlays sub-menu of the PFD menu. The PFD overlays menu shows below. Refer to the PFD/MFD — Menu Navigation of this guide for details on menu navigation.



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TCAS Traffic Symbols

The TCAS computer classifies nearby aircraft into one of four types: Other Traffic (OT), Proximate Traffic (PT), Traffic Advisory (TA), and Resolution Advisory (RA). Four different traffic shapes and colors are provided to represent aircraft of different threat levels. Up to 32 intruders can show at one time on the display.



Other Traffic (OT)

The symbol for Other Traffic is a hollow cyan diamond. Other Traffic is intruder traffic that does not yet pose a threat to your aircraft because the intruder relative altitude is greater than 1200 feet or the intruder range is greater than 6 NM. The display of Other Traffic can be set to on or off by way of the ATC/TCAS sub-menu on the MFD or CDU.

Proximate Traffic (PT)

The symbol for Proximate Traffic is a solid cyan diamond. Proximate Traffic is intruder traffic that does not yet pose a threat. The intruder relative altitude is 1200 feet or less, and the intruder range is within 6 NM. Proximate traffic cannot be deselected and always shows on the traffic display as an aid to the pilots when visually acquiring TAs and RAs.

Traffic Advisory (TA)

The symbol for TA traffic is a solid yellow circle. TAs represent potential threat aircraft that should be monitored because, depending upon their continued flight path, they may become RA traffic. TAs show on the MFD starting 20 - 48 seconds before the calculated closest point of approach (based on own aircraft altitude). For a TA, the audio message TRAFFIC, TRAFFIC is broadcast over the aircraft audio system and the message TRAFFIC shows in yellow on the PFD. TAs are normally issued 15 seconds before an RA is issued.

TA ONLY mode is used to disable the RA traffic detection. When TA ONLY mode is set, RA traffic shows as TA traffic. The message TA ONLY shows in white on the PFD when the system is operating in TA ONLY mode and the traffic overlay is not selected for display, cyan if the traffic overlay is active. The message TA ONLY shows in yellow when the system is operating in TA ONLY mode and TA traffic is detected. TA ONLY mode is set by way of the ATC/TCAS sub-menu on the MFD or CDU.

Resolution Advisory (RA)

The symbol for Resolution Advisory traffic is a solid red square. In addition to the RA traffic symbology that shows on the MFD and/or PFD map, vertical speed commands show on the PFD vertical speed scale. The message TRAFFIC shows in red below the vertical speed scale on the PFD and aural commands are broadcast over the flight deck audio system. RA traffic is traffic that the TCAS computer has determined to be potential threats and that require immediate pilot action. There are two types of RAs, corrective and preventive.

- Corrective RAs are issued for threat traffic when the TCAS computer has determined that corrective action needs to be taken to avoid the traffic. The vertical speed range to seek and/or avoid shows on the PFD vertical speed scale. To comply with the RA, avoid flying vertical speeds within the red band and fly the vertical speeds within the green band. Several different types of aurals may be broadcast over the flight deck audio system during a corrective RA. Refer to the TCAS Audio appendix of this guide for details on TCAS Audio.

- Preventive RAs are issued for threat traffic for which the TCAS computer has determined that the current vertical speed will resolve the threat situation. The vertical speed range to avoid shows on the PFD vertical speed scale. To comply with the RA, avoid flying vertical speeds within the red band. Several different types of aurals may be broadcast over the flight deck audio system during a corrective RA. Refer to the TCAS Audio appendix of this guide for details on TCAS Audio.

Traffic Altitude

The Traffic altitude readout accompanies the intruder symbol and gives the relative or absolute altitude of the intruder in a format that matches the TCAS altitude readout. The readout is the same color as the TCAS intruder symbol it is accompanying. When an intruder aircraft is at a higher altitude, the readout shows above the associated TCAS symbol. When the intruder aircraft is at a lower altitude, the readout shows below the associated TCAS symbol. The altitude shows in relative or absolute intruder altitude in 100s of feet. The altitude format (relative or absolute) is set from the ATC/TCAS sub-menu on the MFD or CDU.

NOTE

Altitude data does not show for intruder aircraft that are not reporting mode C (altitude) data.

- When relative altitude is selected (therefore, no TCAS altitude readout shows on the MFD/PFD), or absolute altitude is inoperative, the altitude readouts that show with each intruder are relative to the current aircraft altitude, and the value has a range of -99 to +99. The two-digit field represents thousands and hundreds of feet. The two-digit field is preceded by a plus (+) or minus (-) sign. The plus or minus sign indicates that the intruder is above or below own aircraft altitude (for example, +22 = 2200 feet above and -02 = 200 feet below). Values below -99 show as -99. Values above +99 show as +99.
- For TCAS absolute altitude modes, the altitudes that show with the intruder are absolute altitudes with a range of 000 to 999. The 3-digit field represents tens of thousands, thousands and hundreds of feet (for example, 230 = 23 000, 234 = 23 400 feet). When the calculated value is less than zero, the three digits are replaced with the letters XXX. When the calculated value is greater than 999, the readout shows as 999.

Traffic Vertical Direction

An arrow representing the vertical direction of the intruder shows to the right of each traffic symbol when the actual (not relative) vertical speed of the intruder is equal to or greater than ±500 feet/minute. An upward pointing arrow shows for climbing traffic, and a downward pointing arrow shows for descending traffic.

No Bearing Traffic	The two most threatening TA or RA intruders for which TCAS cannot calculate a bearing show below the ABOVE/NORM/BELOW selection in the TCAS data field on the right side of the MFD and/or PFD. The intruders type (TA or RA), distance (NM), and altitude (if available) is listed in the table. An upward or downward-pointing arrow shows to the right of the altitude data to indicate the vertical-direction of the traffic. The color of the entire row is yellow for TA, red for RA.
Out of Range Traffic	TCAS targets show out to a maximum of 40 NM. TA or RA traffic outside the selected display range show at the correct bearing and at the maximum selected range with only half of the symbol in view.

Display Range	A full-range ring and a half-range ring show in the TCAS Only format. The full-range ring is a white circle with fixed inward tick marks at the 12 cardinal clock positions. The full-range value shows in white in the full-range ring near the 10 o'clock position. A half-range ring shows with hash marks like the Rose/TCAS but no half-range value shows. There is no half-range ring when the full-range is 5 NM. The 3-NM range ring is formed by grey tick marks at the 12 cardinal clock positions. A circle bisecting each line is the 3-NM range ring. The 3-NM range ring shows when the full-range is 5, 10, or 25 NM. The RANGE knob on the DCP is used to select the display range. The standard (general aviation) display ranges are: 5, 10, 25, 50, 100, 200 and 300 NM. When the TR and WX overlays are not selected, a 600-NM range is available.
TFC Legend	The Traffic (TFC) legend shows at the top of the TCAS data field on the MFD and PFD. The TFC legend indicates a TCAS traffic display has been selected. When traffic is not active or the current format uses the legend display area the legend is removed. The TFC legend is cyan when the format is compatible with the display of traffic (for example, PPOS Map, TCAS). The TFC legend is white when the format is not compatible with the display of traffic (Plan).

TFC Data Field

The TCAS data field shows below the TFC legend when the TCAS traffic map is active. Only one TCAS status or mode can show at a time. The priority order is: TCAS TEST, TCAS FAIL, TCAS OFF, TA ONLY, NO VSI RA, TA/RA (blank field).



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- TA/RA mode is the normal TCAS mode. No mode message shows when TA/RA mode is the active mode. TA/RA mode is used when full TCAS functionality is desired. When TA/RA mode is set, the TCAS computer generates intruder traffic symbology and Resolution Advisory (RA) vertical speed fly-to commands. The intruder traffic symbology shows on the PFD and/or MFD when selected.
- The message TCAS FAIL shows in yellow when a TCAS fault is detected. No TCAS traffic symbols or range information show on the PFD or MFD and no RAs show on the PFD vertical speed scale. Failures related exclusively to the traffic display cause the message when TFC is active.

- The message TCAS TEST shows in cyan when the TCAS self-test is active. The 10-second self-test routine tests the items that follow: TCAS R/T or TSS, TDR(s), TCAS antennas, RAD ALT data, heading data, and the AFD. A test pattern shows on the traffic displays and PFD vertical speed scale. The test patterns include the four types of traffic symbols. The messages TCAS TEST and TCAS FAIL shows on the MFD. Red and green TCAS RA VS advisory bands show on the VS scale. The message TCAS TEST shows on the PFD. After successful completion of the test, the system returns to the selected operating modes, and the audio message TCAS SYSTEM TEST OK is broadcast over the flight deck audio system. If the set operating mode is not STBY, TCAS traffic and resolution advisories are available for display in approximately 5 seconds. If a fault is detected in the TCAS self-test, the message TCAS FAIL shows on the PFD and MFD and the audio message TCAS SYSTEM TEST FAIL is broadcast over the flight deck audio system.
- The message TCAS OFF shows in cyan when the TCAS system is set to STBY mode. The TCAS is automatically set to STBY mode when altitude reporting is set to off. The message ALT OFF shows on the ATC/TCAS Radio menu when altitude reporting is set to off.

- The message NO VSI RA shows in yellow when the TCAS vertical speed cues cannot be computed or vertical speed is not in view on the PFD.

Other Traffic Off Message

When the display of Other (non-threat) Traffic is set to OFF, the message other traffic (the diamond shaped OT symbol) OFF shows in cyan below the TCAS status/mode message.

Own Airplane TCAS Altitude

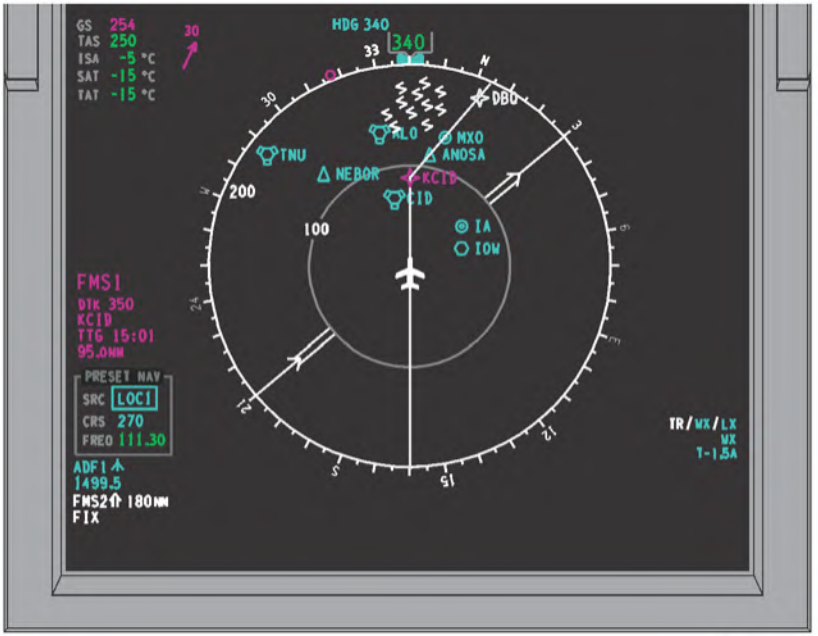
The TCAS altitude readout shows in the TCAS data field to help the pilot compare the own aircraft altitude with the intruders altitude when above 18 000 feet MSL and ABS altitude is selected. The TCAS altitude readout shows in cyan on two lines to the left of the ABOVE/BELOW message. The TCAS altitude display is pressure altitude, the same as used by the transponder. When ABS altitude is active below 18 000 feet, there is a 15-second maximum display for the TCAS Altitude. After 15 seconds, the message changes to ABS INOP.

ABOVE/BELOW/ BOTH Message

The message ABOVE, BELOW, or BOTH shows in cyan in the TCAS data field when ABOVE, BELOW, or ABOVE and BELOW are selected. This message indicates that the altitude window for monitoring OTHER Traffic is expanded to 9900 feet ABOVE, or 9900 feet BELOW, or both. Normally, OTHER Traffic within ±2700 feet shows.

PFD/MFD – LIGHTNING OVERLAY

Figure 6-50 PFD/MFD – Lightning Overlay



TPO4650_26

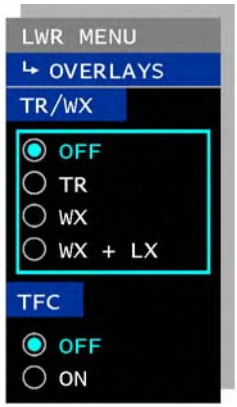
The Lightning Detection System (LDS) detects various levels of lightning activity and processes that information for display on the PFD and MFD. When the LDS overlay is selected for display and the LDS detects lightning activity, lightning bolt icons show on the display at the relative bearing and distance to the aircraft. Lightning detection is limited to approximately 200 NM. The TR/WX button on the DCP is used to select the LDS overlay on the PFD. The TR/WX button on the CCP is used to select the LDS overlay on the MFD.

NOTE

The LDS is a non-Rockwell Collins avionics unit. Since LDS information is not generated by Rockwell Collins avionics units, it is subject to change by the vendor without notification to Rockwell Collins avionics. The Rockwell Collins avionics units described in this pilot's guide show typical operation and capabilities. The non-Rockwell Collins avionics units may or may not cause typical operation. Refer to the appropriate vendor documentation for specific details on the non-Rockwell Collins avionics units.

▼ SELECTIONS:

Lightning Symbology	The lightning symbology is displayed in white. **NOTE** The lightning icons are removed from the display when an LDS fault is reported when the LDS overlay is in view.
Overlays Menu	The Lightning overlay can be selected for display on the MFD by way of the overlays sub-menu of the MFD lower menu. The Lightning overlay can be selected for display on the PFD by way of the overlays sub-menu of the PFD menu. The MFD overlays menu shows below. Refer to the PFD/MFD — Menu Navigation of this guide for details on menu navigation.



TPO4620_78

TR/WX/LX Legend

The TR/WX/LX legend on the PFD and MFD indicates the active overlay(s). The legend for the active overlay(s) shows in larger cyan text. The inactive overlay legends show in smaller white text. The TR/WX/LX legend is removed for some formats that do not allow the legend to show.

Display Range

Lightning detection is generally limited to approximately 200 NM from the aircraft by the Lightning Detection System. Range symbology includes a full-range ring, a half-range ring, and a half-range readout. The RANGE knob on the DCP is used to set the MFD and PFD display range.

PFD/MFD – TERRAIN OVERLAY

Figure 6-51 PFD/MFD – Terrain Overlay and TAWS Alerts



TPO4650_15

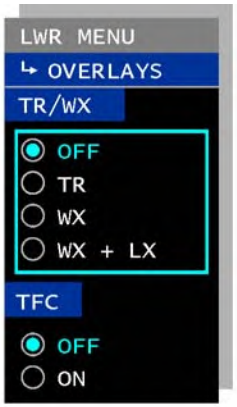
The optional TAWS uses aircraft geographic position, aircraft altitude, and an internal terrain database to predict potential conflicts between the aircraft flight path and terrain. When selected, graphical terrain data shows on the Rose and PPOS Map format. The optional terrain pop-up feature allows automatic display of terrain data when a terrain hazard is encountered. Visual and audio warnings of excessive deviation below an ILS glideslope or LPV glidepath, severe windshear conditions, altitude callouts during final approach, alerts for descent below Decision Height (DH), and alerts for excessive bank angle are available. TAWS alerts show on the PFD when commanded by the TAWS computer. The TR/WX button on the DCP is used to select the Terrain overlay on the PFD. The TR/WX button on the CCP is used to select the terrain overlay on the MFD.

NOTE

The TAWS is a non-Rockwell Collins avionics unit. Since TAWS information is not generated by Rockwell Collins avionics units, it is subject to change by the vendor without notification to Rockwell Collins. The Rockwell Collins avionics units described in this operator's guide show typical operation and capabilities. The non-Rockwell Collins avionics units may or may not cause typical operation. Refer to the appropriate vendor documentation for specific details on the non-Rockwell Collins avionics units.

▼ SELECTIONS:

Terrain Symbology	Graphical Terrain symbology shows in a heading-up ±90-degree arc. The scan starts out from the center of the display and moves out to both sides simultaneously. After moving out to both edges the sweep starts over at the lubber line. For standard operation, only terrain within 2000 feet of the aircraft shows graphically. For Peaks operation, terrain that is within a display range that has a highest and a lowest elevation shows graphically.
Overlays Menu	The Terrain overlay can be selected for display on the MFD by way of the overlays sub-menu of the MFD lower menu. The Terrain overlay can be selected for display on the PFD by way of the overlays sub-menu of the PFD menu. The MFD overlays menu shows below. Refer to the PFD/MFD — Menu Navigation of this guide for details on menu navigation.



TPO4620_78

Terrain Colors

The display colors/patterns represent various terrain elevations with respect to the altitude of the aircraft. Refer to the Terrain Colors Appendix of this guide for details on terrain colors.

TAWS Messages

The TAWS is effectively three independent systems in one: Ground Proximity, Terrain, and Windshear Systems. Each subsystem has distinct aural alerts system messages. The TAWS is also capable of providing altitude related callouts and/or an excessive bank angle callout. Refer to the TAWS manufacturer operations manual for functional details. The TAWS messages that follow are available for display on the PFD.

TAWS Messages

Message	Meaning
WINDSHEAR (red)	Windshear warning
PULL UP (red)	GPWS warning
PULL UP (red)	TAWS warning
GND PROX (yellow)	GPWS alert
GND PROX (yellow)	TAWS caution
WINDSHEAR (yellow).	Windshear caution

NOTE

Refer to the TAWS manufacturer Operations Manual for functional details.

Peaks Operation	For standard TAWS operation, only terrain within 2000 feet of the aircraft shows graphically. For Peaks operation, terrain that is within a display range that has a highest and a lowest elevation shows graphically. The two elevation values show below a PEAKS Legend above the upper right side of the heading arc. The top number is the highest elevation of the graphical display range. The lowest elevation shows below the highest elevation number. The highest and lowest elevation values show in the same color as the highest and lowest terrain color pattern on the display, or dashes if there is unknown terrain in the display area, or one elevation may be blanked if the highest and lowest elevation are the same. Different elevation ranges can show on each side of the flight deck because the range selected can be different on each side. The range is -99 to 511 (hundreds) of feet. Peaks operation allows white, cyan and blue (for water) colors.
Terrain Pop-Up	Terrain Pop-Up allows graphical terrain data to show automatically on the PFD when a terrain caution or warning is encountered. The Terrain overlay is added to the current PFD display format if the Terrain overlay is not already in view on the on-side MFD. If the current PFD format is incompatible with Terrain, the format is changed to PPOS with a 10-NM range. After Terrain Pop-Up initially occurs, the normal control of the format, range, and selected overlays is restored. The PFD will not automatically change the format and range again unless a new Terrain Pop-Up is received.

TR/WX/LX Legend	The TR/WX/LX legend on the PFD and MFD indicates the active overlay(s). The legend for the active overlay(s) shows in larger cyan text (for example, TR). The inactive overlay legends show in smaller white text. The TR/WX/LX legend is removed for some formats that do not allow the legend to show.
Terrain Mode/ Status	Terrain status messages show the current mode/condition when Terrain is active. The status messages show below the TR/WX/LX legend in the space used for Weather Radar status. When a Terrain compatible format is active on the MFD, the status message shows in cyan. When the MFD format is changed so it is not Terrain compatible, the status message shows in white. The available fault messages show in the center of the map when commanded by the TAWS computer. • TERRAIN mode message shows when the mode is set to Terrain. • TERRAIN OFF mode message shows Terrain is inhibited. The graphical Terrain display is removed when this message is in view. • TERRAIN TEST shows when the Terrain test is active. When the test is active, each of the TAWS failure messages is sequentially activated, in addition to a graphical Terrain test pattern that shows on the display. The graphical Terrain display is removed when this message is in view.

- TERRAIN RANGE XXXNM shows below the map center when the display range and TAWS range do not agree. XXXNM is the TAWS range. The graphical Terrain display is removed when this message is in view.
- TERRAIN NOT AVAILABLE shows in yellow above the aircraft symbol in the center of the map when Terrain is not failed. The graphical Terrain display is removed when this message is in view.
- TERRAIN FAIL shows in yellow above the aircraft symbol when the TAWS computer detects a fault. The message flashes for 5 seconds and then shows steady. The graphical Terrain display is removed when this message is in view.

Terrain Range

Terrain data is available in ranges up to 300 NM. The default display range for the Terrain overlay is 10 NM. Range symbology includes a full-range arc, a half-range ring, and a half-range readout. The RANGE knob on the DCP is used to select the display range.

Terrain Test

Terrain test is activated with a remote-mounted installer-supplied Test switch. The Terrain test is not available when airborne. The Terrain test shows a graphical Terrain test pattern, if possible, regardless if a Terrain failure is present. The MFD will respond to the Terrain Pop-Up commands when Terrain test is active. If Terrain test changes the PFD/MFD format or range, normal control returns immediately after the Terrain test is complete.

PFD/MFD – WEATHER RADAR OVERLAY

Figure 6-52 PFD/MFD – Weather Radar Overlay



TPO4650_32

The Weather Radar System (WXR) locates and depicts the bearing, range, and precipitation rate of all detectable precipitation within the scan area and display range. The WXR locates precipitation-related weather targets as well as precipitation-related turbulence targets. The WXR can be used for ground mapping. The WXR is made up of a Receiver/Transmitter Antenna (RTA), four AFD, two Display Control Panels (DCP), and two Cursor Control Panels (CCP). The RTA contains an antenna, RF assembly, pedestal assembly, and base assembly. The RF assembly contains the receiver and transmitter. The pedestal contains tilt/scan motors and optical feedback switches. The base contains the power supply and signal processing circuits. A sweep mark, representing the position of the Weather Radar sweep, shows in cyan. A Weather Radar data field shows on the display to indicate the selected operating mode and system status. The weather and map information can be overlaid on most of the navigation formats on either or both MFD and PFD. The TR/WX button on the DCP is used to select the Weather Radar overlay for display on the PFD. The TR/WX button on the CCP is used to select the Weather Radar overlay on the MFD. The RADAR menu on the PFD is used to select the system operating mode. The RDR MENU button on the DCP is used to show the RADAR menu.

The Weather Radar System installed on the aircraft is either an RTA-854 or the RTA-4114 Multiscan system. Operational differences are described where applicable below.

▼ SELECTIONS:

WXR Colors (RTA-854)

Detectable weather shows in four colors: green, yellow, red, and magenta. The lowest precipitation rates shows in green. The highest precipitation rates and turbulence (TWR only) show in magenta. Refer to the table that follows for details on the relationship between precipitation rates and display colors.

WXR Colors

Display Color	**Precipitation Rate**
No display	Less than 0.03 in/hr
Green	0.03 to 0.07 in/hr
Yellow	0.07 to 0.20 in/hr
Red	0.20 to 0.52 in/hr
Magenta	0.52 in/hr and greater
NOTE: Turbulence also shows in magenta.	

WXR COLORS (RTA-4114)

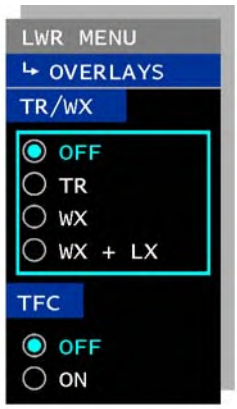
The MultiScan weather radar shows precipitation in three colors, green, yellow, and red. The lowest precipitation rates show in green. The highest precipitation rates show in red, and turbulence shows in magenta. Refer to the list that follows for details on the relationship between precipitation rates and display colors:

WXR Colors

Display Color	Precipitation Rate
No display	Less than 0.04 in/hr
Green	0.04 to 0.157 in/hr
Yellow	0.157 to 0.47 in/hr
Red	0.47 in/hr and greater
NOTE: Turbulence shows in magenta.	

Overlays Menu

The Weather Radar overlay can be selected for display on the MFD by way of the overlays sub-menu of the MFD lower menu. The Weather Radar overlay can be selected for display on the PFD by way of the overlays sub-menu of the PFD menu. The MFD overlays menu shows below. Refer to the PFD/MFD — Menu Navigation section of this guide for details on menu navigation.



TPO4620_78

Sweep Mark	A sweep mark, representing the position of the Weather Radar sweep, shows in cyan. The sweep mark scan is normally ±60 degrees. The scan sweep is reduced to ±30 degrees when the Sector Scan feature of the RTA-854 is active. Sector increases the scan rate but decreases the scan sweep.
PAC Alert Arc	A yellow PAC alert arc shows at the perimeter of the WXR display to identify where Path Attenuation Compensation (PAC) has been exceeded. The PAC feature compensates automatically for heavier rainfall rates, and the Weather Radar beam absorption they cause. This results in a more accurate display of the rainfall rates. The compensation provided by the PAC alert function can be exceeded. The PAC alert arc indicates that the Weather Radar beam has been completely absorbed by heavy rainfall rates and that areas of unknown rainfall rates may exist beyond the PAC alert arc.
TR/WX/LX Legend	The TR/WX/LX legend on the PFD and MFD indicates the active overlay(s). The legend for the active overlay(s) shows in larger cyan text (for example, WX). The inactive overlay legends show in smaller white text. The TR/WX/LX legend is removed for some formats that do not allow the legend to show.
Weather Radar (RTA-854)	The following data covers RTA-854 operation.

Weather Radar Data Field

The WXR data field shows along the right side of the display. WXR mode, receiver gain setting, antenna tilt angle readout, Auto Tilt indication, Ground Clutter Suppression indication, and Fault indications. The RTA-854 also provides the USTB, Weather Radar Target (TGT) alert, and Turbulence (TRB) alert messages.

- The first line shows the Weather Radar mode and, when other than NORM, the receiver gain setting. The mode can be STBY, WX, WX+TURB, TURB, MAP, and TEST. The message GCS replaces the active Weather Radar mode message when the GCS feature is active. Turbulence is only operational when the display range is set at or below 50 NM. When the display range is set to more than 50 NM, the +TURB message changes to white, flashes for 5 seconds and then shows steady. Receiver gain shows the letter G with a plus/minus sign and 1, 2, or 3.

- The second line shows the tilt angle readout and USTB message. The tilt angle readout is made up of the letter T followed by a plus or minus sign and up to three digits for the readout, except when the tilt is 0 (zero). When the commanded tilt value and the tilt value returned from the Weather Radar disagree for more than 5 seconds, the tilt value changes to yellow. The tilt angle readout is followed by the letter A when Auto Tilt is enabled. When the Auto Tilt state reported by the Weather Radar disagrees with the commanded Auto Tilt state for more than 5 seconds, the tilt angle readout is followed by a letter A in yellow. The message USTB (unstabilized) shows in yellow in place of the tilt angle readout when an attitude failure to the Weather Radar occurs. WX FAULT is displayed in yellow in place of the tilt angle readout when the weather radar reports a failure.
- The third line shows the Target (TGT) mode message when the Target feature is active.



TPO4655_07

Weather Radar Menu

The RADAR menu on the PFD is used to select the system operating mode. The RDR MENU button on the DCP is used to show the RADAR menu. The Menu Advance Knob (MENU ADV), DATA knob, and PUSH SELECT button on the DCP are used to operate the RADAR menu. Refer to the Menu Navigation section in this chapter for details on RADAR menu navigate and operation.



RTA-854 INSTALLATION
TPO4411_01

Weather Radar Mode

The possible Weather Radar modes are Standby (STBY), Weather (WX), Weather Plus Turbulence (WX+TURB), Turbulence Only (TURB), Ground Mapping (MAP), and Test (TEST).

- Standby (STBY) mode is the power-up mode. The Weather Radar System does not transmit when STBY mode is active. STBY mode is automatically selected 60 seconds after air/ground transition to ground. Selecting STBY on either side causes both sides to go to STBY. When both sides are in STBY and one side selects an active operational mode, then both sides will go to that operational mode (for example, if STBY is active and the copilot selects WX, WX would be the current Weather Radar mode on both the pilot and copilot-side. Independent Weather Radar modes may then be selected by each side.
- Weather (only) mode is recommended as the normal operating mode. When weather (WX) mode is active, the Weather Radar System is optimized to detect precipitation. Detectable precipitation shows as one of four colors: green, yellow, red, or magenta (least to greatest precipitation rate).

- Weather Radar Plus Turbulence (WX+TURB) mode is available when a Turbulence Weather Radar System is installed. WX+TURB mode is used to detect precipitation and precipitation-related turbulence targets. Detectable precipitation shows as one of four colors: green, yellow, red, or magenta (least to greatest precipitation rate). The highest precipitation rates and turbulence show in magenta. WX+TURB mode is only active out to 50 NM for the RTA-854. When a display range greater than 50 NM is selected, WX mode is automatically selected. This is indicated by the +TURB turning white and flashing for 5 seconds before being displayed steady.
- Turbulence only (TURB) mode shows precipitation-related turbulence targets only. This is useful for closely analyzing areas of precipitation-related turbulence that have been detected while in WX+TURB mode. TURB mode automatically defaults to WX+TURB mode after a 30 second time-out. TURB mode is only active out to 50 NM for the RTA-854.

- Map mode provides the most detailed ground returns and should be used when purposely mapping the terrain. Signal processing and target colors are changed to accentuate ground features in MAP mode. Ground targets show in cyan, green, yellow, and magenta (least reflective to most reflective). The Path Attenuation Compensation (PAC), Ground Clutter Suppression (GCS), Sensitivity Time Control (STC), and optional Turbulence detection features used for weather detection are disabled when MAP mode is selected. For this reason, MAP mode should not be used for weather detection.
- Test mode is used to make sure the displays ability to display the necessary colors. A test pattern made up of six colored arcs in a rainbow-like pattern show on the display when test mode is active. The rainbow-like pattern is made up of the colors green, yellow, red, and magenta. The fifth arc changes between red and magenta on alternate scans. Normal Weather Radar returns do not show when test mode is active.

Target Alert	The Target alert feature provides an alert to the pilot when a Weather Radar target is detected and all on-side Weather Radar displays have been deselected. When target mode is active, the Weather Radar searches the returns from a ±15 sector in front of the aircraft within the range of 7 to 200 NM. If a target is detected between 7 and 200 NM, a yellow TGT message shows on the PFD. If a TWR is installed and turbulence is detected between 7 and 50 NM, a yellow TRB message shows on the PFD. As soon as Weather Radar is selected for display, the target function on that side is suspended, and the Weather Radar returns to its previously selected operational mode. The WXR provides Target alert only and the TWR provides both Target alert and Turbulence alert.
Sector Scan	Sector scan reduces the antenna scan angle from the normal ±60-degree sweep (120 degree total) to a ±30-degree sweep (60 degree total). By reducing the antenna scan angle, the amount of time needed for the antenna to complete its sweep is also reduced. This effectively increases the Weather Radar update rate. Selecting sector scan on one display affects all displays.
Ground Clutter Suppression	Ground Clutter Suppression (GCS) reduces the intensity of ground returns in WX, WX+TURB, and TURB modes, which assists in the interpretation of rainfall rates. Refer to Weather Radar Techniques appendix for details on GCS.

Antenna Tilt	Proper use of the TILT control allows the operator to achieve the best picture of storm cell size, height, and relative direction of movement. Refer to Weather Radar Techniques appendix for details on Antenna Tilt.
Auto Tilt	Auto Tilt is designed to reduce the number of times the pilot needs to adjust the TILT control whenever the aircraft altitude or the Weather Radar range setting changes. The Auto Tilt feature automatically adjusts the antenna tilt angle to attempt to keep the Weather Radar scanning approximately the same region of space when the aircraft climbs and descends. Refer to Weather Radar Techniques appendix for details on Auto Tilt.

Receiver Gain

A variety of gain settings allow the crew to increase or decrease the receiver gain to analyze Weather Radar returns. Normally, receiver gain is set to the 0 (NORM) position. Six other positions, including three of increased gain for the utmost in sensitivity, are available for analyzing the details of the returns. The higher settings are useful in identifying the lightest levels of precipitation, while the lower settings are useful for more in-depth studies of the most intense weather targets. Each of the minus settings (–1, –2, and –3) reduces the sensitivity of the Weather Radar System by approximately one color level. This results in a total reduction of three color levels when operating with the GAIN set to the –3 position. Therefore, if a target was magenta (RTA-854) or red (RTA-4114) in NORM, the same target at –3 will be almost totally removed from the display. Because of this effect, the Weather Radar should not be operated in one of the reduced gain settings for an extended period of time.

NOTE

The pilot should always return the gain to the NORM position when the display analysis is complete. Failure to do so may result in missing significant targets when the Weather Radar is operating in one of the minus gain settings.

Range Knob

The RANGE knob on the DCP is used to set the MFD and PFD display range. Weather Radar targets can show in up to a 300-NM range. When Weather Radar is selected, display ranges greater than 300 NM are not available.

NOTE

Extending outward from the aircraft symbol is an area that does not display any Weather Radar targets. The size of this blanked range area equals one-eighth of the selected range. Depending on the selected range, the blanked area can be as large as 37 NM.

Weather Radar

The following data covers RTA-4114 operation.

Weather Radar Data Field

The first line shows the Weather Radar mode and, when other than NORM, the receiver gain setting. The mode can be STBY, WX, WX+TURB, TURB, MAP, and TEST. Receiver gain shows the letter G with a plus/minus sign and 1, 2, or 3.

- The second line shows the tilt angle readout. The tilt angle readout is made up of the letter T followed by a plus or minus sign and up to three digits for the readout, except when the tilt is 0 (zero). When the radar is operating in Manual control mode, and the commanded tilt value and the tilt value returned from the Weather Radar disagree for more than 5 seconds, the tilt value changes to yellow. The tilt angle readout is followed by the letter A when Automatic control mode is enabled.

- The third line shows the Ground Clutter Suppression (GCS) indication. This feature is normally active when the radar is operating in the Automatic control mode. It can be deselected, but will be re-activated after 30 seconds. GCS is not available in Manual control mode.

Faults reported by the weather radar are displayed in place of the GCS annunciation. The faults include:

- WX FAULT
- AUTO FAULT
- TURB FAULT
- CTRL FAULT.



TPO4655_07

Weather Radar Menu

The RADAR menu on the PFD is used to select the system operating mode. The RDR MENU button on the DCP is used to show the RADAR menu. The Menu Advance Knob (MENU ADV), DATA knob, and PUSH SELECT button on the DCP are used to operate the RADAR menu. Refer to the Menu Navigation section in this chapter for details on RADAR menu navigate and operation.



RTA-4114 INSTALLATION
TPO4411_02

Weather Radar Mode

The possible Weather Radar modes are Standby (STBY), Weather (WX), Weather Plus Turbulence (WX+TURB), Turbulence Only (TURB), Ground Mapping (MAP), and Test (TEST).

- Standby (STBY) mode is the power-up mode. The Weather Radar System does not transmit when STBY mode is active. STBY mode is automatically selected 60 seconds after air/ground transition to ground. Selecting STBY on either side causes both sides to go to STBY. When both sides are in STBY and one side selects an active operational mode, then both sides will go to that operational mode (for example, if STBY is active and the copilot selects WX, WX would be the current Weather Radar mode on both the pilot and copilot-side. Independent Weather Radar modes may then be selected by each side.
- Weather (only) mode is recommended as the normal operating mode. When weather (WX) mode is active, the Weather Radar System is optimized to detect precipitation. Detectable precipitation shows as one of four colors: green, yellow, red, or magenta (least to greatest precipitation rate).

- Weather Radar Plus Turbulence (WX+TURB) mode is available when a Turbulence Weather Radar System is installed. WX+TURB mode is used to detect precipitation and precipitation-related turbulence targets. Detectable precipitation shows as one of four colors: green, yellow, red, or magenta (least to greatest precipitation rate). The highest precipitation rates and turbulence show in magenta. WX+TURB mode is only active out to 50 NM for the RTA-854. When a display range greater than 50 NM is selected, WX mode is automatically selected. This is indicated by the +TURB turning white and flashing for 5 seconds before being displayed steady.
- Turbulence only (TURB) mode shows precipitation-related turbulence targets only. This is useful for closely analyzing areas of precipitation-related turbulence that have been detected while in WX+TURB mode. TURB mode automatically defaults to WX+TURB mode after a 30 second time-out. TURB mode is only active out to 50 NM for the RTA-854.

- Map mode provides the most detailed ground returns and should be used when purposely mapping the terrain. Signal processing and target colors are changed to accentuate ground features in MAP mode. Ground targets show in cyan, green, yellow, and magenta (least reflective to most reflective). The Path Attenuation Compensation (PAC), Ground Clutter Suppression (GCS), Sensitivity Time Control (STC), and optional Turbulence detection features used for weather detection are disabled when MAP mode is selected. For this reason, MAP mode should not be used for weather detection.
- Test mode is used to make sure the displays ability to display the necessary colors. A test pattern made up of six colored arcs in a rainbow-like pattern show on the display when test mode is active. The rainbow-like pattern is made up of the colors green, yellow, red, and magenta. The fifth arc changes between red and magenta on alternate scans. Normal Weather Radar returns do not show when test mode is active.

Target Alert (RTA-854 Only)	The Target alert feature provides an alert to the pilot when a Weather Radar target is detected and all on-side Weather Radar displays have been deselected. When target mode is active, the Weather Radar searches the returns from a ±15 sector in front of the aircraft within the range of 7 to 200 NM. If a target is detected between 7 and 200 NM, a yellow TGT message shows on the PFD. If a TWR is installed and turbulence is detected between 7 and 50 NM, a yellow TRB message shows on the PFD. As soon as Weather Radar is selected for display, the target function on that side is suspended, and the Weather Radar returns to its previously selected operational mode. The WXR provides Target alert only and the TWR provides both Target alert and Turbulence alert.
Sector Scan (RTA-854 Only)	Sector scan reduces the antenna scan angle from the normal ±60-degree sweep (120 degree total) to a ±30-degree sweep (60 degree total). By reducing the antenna scan angle, the amount of time needed for the antenna to complete its sweep is also reduced. This effectively increases the Weather Radar update rate. Selecting sector scan on one display affects all displays.
Ground Clutter Suppression (GCS) (RTA-854)	Ground Clutter Suppression (GCS) reduces the intensity of ground returns in WX, WX+TURB, and TURB modes, which assists in the interpretation of rainfall rates. Refer to Weather Radar Techniques appendix for details on GCS.

Ground Clutter Suppression (GCS) (RTA-4114)	Ground Clutter Suppression (GCS) reduces the intensity of ground returns in WX, WX+TURB, and TURB modes, which assists in the interpretation of rainfall rates. GCS is automatically activated for any of these modes when the AUTO control is selected. GCS can be disabled in AUTO mode. It will be reactivated after 30 seconds. GCS is not available when MAN is the selected control. Refer to Weather Radar Techniques appendix for details on GCS.
Antenna Tilt	Proper use of the TILT control allows the operator to achieve the best picture of storm cell size, height, and relative direction of movement. Refer to Weather Radar Techniques appendix for details on Antenna Tilt.
Auto Tilt (RTA-854)	Auto Tilt is designed to reduce the number of times the pilot needs to adjust the TILT control whenever the aircraft altitude or the Weather Radar range setting changes. The Auto Tilt feature automatically adjusts the antenna tilt angle to attempt to keep the Weather Radar scanning approximately the same region of space when the aircraft climbs and descends. Refer to Weather Radar Techniques appendix for details on Auto Tilt.
Antenna Tilt (RTA-4114)	Proper use of the TILT control allows the operator to achieve the best picture of storm cell size, height, and relative direction of movement when the MAN control is selected. Refer to Weather Radar Techniques appendix for details on Antenna Tilt.

Receiver Gain

A variety of gain settings allow the crew to increase or decrease the receiver gain to analyze Weather Radar returns. Normally, receiver gain is set to the 0 (NORM) position. Six other positions, including three of increased gain for the utmost in sensitivity, are available for analyzing the details of the returns. The higher settings are useful in identifying the lightest levels of precipitation, while the lower settings are useful for more in-depth studies of the most intense weather targets. Each of the minus settings (–1, –2, and –3) reduces the sensitivity of the Weather Radar System by approximately one color level. This results in a total reduction of three color levels when operating with the GAIN set to the –3 position. Therefore, if a target was magenta (RTA-854) or red (RTA-4114) in NORM, the same target at –3 will be almost totally removed from the display. Because of this effect, the Weather Radar should not be operated in one of the reduced gain settings for an extended period of time.

NOTE

The pilot should always return the gain to the NORM position when the display analysis is complete. Failure to do so may result in missing significant targets when the Weather Radar is operating in one of the minus gain settings.

Range Knob	The RANGE knob on the DCP is used to set the MFD and PFD display range. Weather Radar targets can show in up to a 300-NM range. When Weather Radar is selected, display ranges greater than 300 NM are not available.

NOTE

Extending outward from the aircraft symbol is an area that does not display any Weather Radar targets. The size of this blanked range area equals one-eighth of the selected range. Depending on the selected range, the blanked area can be as large as 37 NM.

▼ SELECTIONS:

Weather Radar Control (RTA-4114)	The possible Weather Radar Control selections are Standby (STBY), Automatic (AUTO), Manual (MAN), and Test (TEST). These control selections are synchronized across the cockpit. • Standby (STBY) is the power-up control selection. The Weather Radar System does not transmit when STBY mode is active. STBY mode is automatically selected 60 seconds after air/ground transition to ground.

- Automatic (AUTO) control is recommended as the normal control selection. When AUTO is selected, the Multiscan radar automatically manages the antenna tilt angle and provides full time Ground Clutter Supression in applicable modes. the AUTO FAULT annunciation in the weather radar status lines alerts the crew when this mode is not available and Manual (MAN) mode should be selected instead.
- Manual (MAN) control is provided as a backup means of operation when AUTO mode is not available. GCS is not available when MAN is selected. The crew is responsible for setting the tilt angles when MAN is selected.
- Map mode provides the most detailed ground returns and should be used when purposely mapping the terrain. Signal processing and target colors are changed to accentuate ground features in MAP mode. Ground targets show in cyan, green, yellow, and magenta (least reflective to most reflective). The Path Attenuation Compensation (PAC), Ground Clutter Suppression (GCS), Sensitivity Time Control (STC), and optional Turbulence detection features used for weather detection are disabled when MAP mode is selected. For this reason, MAP mode should not be used for weather detection.

MFD – ENHANCED VISION SYSTEM

Figure 6-53 MFD – Enhanced Vision System



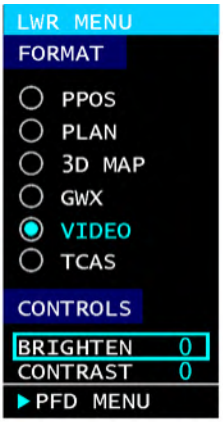
TPO4697_12

The optional Enhanced Vision System (EVS) uses a Forward Looking Infrared (FLIR) camera to generate a video display in the cockpit. The EVS system can increase pilot situational awareness of the terminal environment during low light, night, or low visibility conditions.

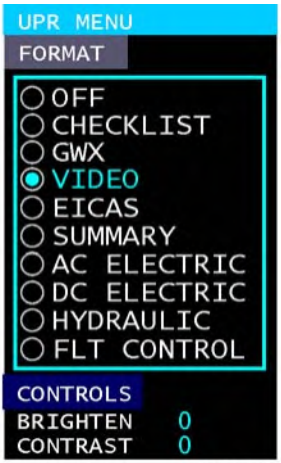
The EVS mode is selected through the MFD Lower or Upper Format Menus. The EVS display may only be shown on either the Upper or Lower MFD window. EVS will be shown on the first selected window. The controls for the HUD are on the glareshield, and can be found in the Rockwell Collins HGS Model 6605 for Bombardier CL-605 Aircraft Operator's Guide (CPN 523-0816693).

▼ SELECTIONS:

VIDEO	Selecting this control activates FLIR cooling mode and enables EVS.



TPO4697_13



TPO4697_14

BRIGHTEN

A highlighted value indicates the brightness level being commanded. The highlighted control is the one that will be acted upon by the data knob and/or select. Brightness may be set between MIN (-10) and MAX (+10).

CONTRAST

A highlighted value indicates the contrast level being commanded. The highlighted control is the one that will be acted upon by the data knob and/or select. Contrast may be set between MIN (-10) and MAX (+10).

MFD – PRIMARY EICAS

Figure 6-54 MFD – Primary Engine Indicating System



TPO4650_13

The primary Engine Indicating and Crew Alerting System (EICAS) format normally shows on the pilot-side MFD. The primary EICAS is a full-sized (approximately 1/2 a MFD) format. The primary EICAS is divided into three sections: the engine indicating displays, primary flight control/cabin environment/auxiliary power unit displays, and crew alerting system messages. The engine indicating displays include displays of N1, N2, ITT, oil pressure and temperature, fuel flow, fuel quantity, engine start, reverse thrust, ground and flight spoilers, and engine vibration messages as well as engine related flags. The primary flight control display includes displays of trim, gear, and flaps messages. The cabin environment display is made up of digital readouts of cabin altitude, cabin altitude rate, cabin differential pressure, cabin air temperature, and crew oxygen pressure. The auxiliary power unit display includes digital readouts of APU RPM and EGT. The primary EICAS can be selected to show on the pilot or copilot-side MFD or a compressed EICAS can be selected to show on both the pilot and copilot-side PFD. If the MFD with the primary EICAS fails or is powered-down, the primary EICAS is automatically moved to the

remaining MFD. Controls located on the RSP are used to manually select the location of the primary EICAS format and to select and deselect the display of the compressed EICAS format on the pilot and copilot-side PFD.

▼ SELECTIONS:

N1 Display	The N1 display for each engine is made up of a circular gauge with a digital readout for N1. N1 shows as a percentage of maximum allowable N1. An N1 reference bug, N1 reference digital readout, thrust code, and thrust reverser status message also show for on the N1 display. • The N1 scale is an arc with the 0 (zero) percent tick mark located at the 90-degree position and ending at the N1 redline tick mark. • The N1 pointer shows when the N1 digital readout is not dashed. The N1 pointer moves clockwise for increasing values of N1 and varies linearly with the N1 (percent) value. • A pointer-following arc graphically extends the N1 scale to the tip of the N1 pointer to indicate an exceedance value. The N1 pointer-following arc moves clockwise for increasing values of N1 and varies linearly with the N1 (percent) value.

- The N1 digital readout shows inside the associated N1 scale. The N1 digital readout shows in green when N1 is less than the N1 redline value. The N1 digital readout shows as four yellow dashes when the associated N1 engine RPM data is invalid. The N1 digital readout shows in red when the N1 input value is greater than the N1 redline value. When the digital readout turns red, the readout flashes and then shows steady.
- The N1 reference digital readout shows between the left and right N1 scales. The N1 reference digital readout shows the selected N1 reference value. The color of the N1 reference digital readout matches the color of the associated thrust code. The N1 reference digital readout does not show when the selected N1 reference is invalid.
- The thrust code message shows below the N1 reference digital readout. The possible thrust codes are: TO, CLB, MCT, CRZ, TGT, and APR. The color of all N1 thrust codes, except TGT, is magenta. The color of the TGT thrust code is cyan. The N1 thrust code does not show when the N1 thrust code input is invalid.

- The N1 reference pointer shows as a bug on the outside of the N1 scale. The N1 reference pointer bug is an open circle when N1 THRUST CODE is equal to CRZ. The N1 reference pointer bug is an open ended triangle when N1 THRUST CODE is not equal to CRZ. The N1 reference pointer bug shows on the outside of the N1 scale next to the equivalent N1 reference digital readout value. The N1 reference pointer adopts the color of the N1 reference digital readout.
- The APR message shows in green. The APR message has display priority over N1 pointer.
- Thrust reverser status message REV. The thrust reverser status message has display priority over N1 pointer and APR. The thrust reverser status message is removed when the discrete input is invalid. The color is a function of the thrust reverser condition as follows:
  - The color of the thrust reverser status message is green when discrete inputs indicate normal thrust reverser deployment.
  - The color of the thrust reverser status message is yellow when the thrust reverser unlocked discrete input is set and normal reverser deployment is not indicated.

ITT Display

The ITT display for each engine is made up of a circular gauge and a digital readout. Engine ITT shows in degrees Celsius. The ITT scale has three different redline positions. A hot start message shows on the ITT display.

- The ITT scale provides a linear analog scale for the respective ITT pointer. The ITT scale is an arc with the 0-degree Celsius tick mark located at the 90-degree arc position and ending at the ITT redline tick mark. The redline mark will correspond to one of three different values, depending on which scale is active. The ITT scale and tickmark will remain the same, but the scaling can be different depending on the aircraft's operating state. The ITT scale is green from the 0 percent tick mark to the ITT redline tick mark. The 0-degree Celsius tick mark is green. The ITT redline tick mark is positioned such that the leading edge is at the ITT redline value. The ITT redline tick mark is red.

- The ITT pointer shows when the ITT digital readout is not dashed. The ITT pointer, in conjunction with the ITT scale, graphically depict the ITT digital readout value. The ITT pointer moves clockwise for increasing values of the ITT digital readout and varies linearly with the ITT (degrees Celsius) value. The ITT pointer has a display range of 0-degrees Celsius (90 degrees of arc position) to the maximum input value (360 degrees of arc position). The ITT pointer is pegged at the maximum input value. The ITT pointer adopts the color of the ITT digital readout. The ITT pointer adopts the flashing status of the ITT digital readout.
- The ITT pointer-following arc graphically extends the ITT scale to the tip of the ITT pointer to indicate an exceedance value. The ITT pointer-following arc shows when ITT digital readout value > ITT redline. The ITT pointer-following arc moves clockwise for increasing values of the ITT digital readout and varies linearly with the ITT (degrees Celsius) value. The ITT pointer-following arc adopts the color of the ITT digital readout.

- The color of the ITT digital readout is green when the ITT input value is less than the ITT redline value. The ITT digital readout shows as three yellow dashes (---) when ENGINE ITT is invalid. The ITT digital readout shows the input value. The color of the ITT digital readout is red when the ITT input value is greater than the ITT redline value. When the digital readout turns red, the readout flashes and then shows steady.
- The ITT hot start message HOT shows in red. The ITT hot start message has display priority over ITT pointer.

N2 Display

The N2 display for each engine is made up of a circular gauge with a digital readout. N2 shows as a percentage of maximum allowable N2. A high vibration message shows on the N2 display.

- The N2 yellow band shows when WING ANTI-ICE is ON. The N2 yellow band scale provides a linear analog scale for the respective N2 pointer. The N2 yellow band scale is an arc with the 0 (zero) degree tick mark located at the 90-degree arc position and ending at the N2 redline tick mark. The N2 yellow band scale is yellow from the 0 (zero) percent tick mark to the green tick mark. The N2 yellow band scale is green from the green tick mark to the N2 redline tick mark. The 0 (zero) percent tick mark is yellow. The N2 redline tick mark is red.

- The N2 scale shows when N2 yellow band scale is not in view. Each N2 scale provides a linear analog scale for the respective N2 pointer. The N2 scale is an arc with the 0 (zero) percent tick mark located at the 90-degree arc position and ending at the N2 redline tick mark. The N2 scale is green from the 0 (zero) percent tick mark to the N2 redline tick mark. The 0 (zero) percent tick mark is green.
- The N2 pointer-following arc shows when N2 digital readout value > N2 redline. The N2 pointer-following arc graphically extends the N2 scale or N2 yellow band scale to the tip of the N2 pointer to indicate an exceedance value. The N2 pointer-following arc moves clockwise for increasing values of the N2 digital readout and varies linearly with the N2 (percent) value. The N2 pointer-following arc adopts the color of the N2 digital readout.
- The N2 digital readout shows the input value. The N2 digital readout shows as three yellow dashes with a decimal point (---) when ENGINE TURBINE RPM N2 is invalid. The N2 digital readout shows for values > = 100 percent. The color of the N2 digital readout is red when the N2 input value is greater than the N2 redline value. When the digital readout turns red, the readout flashes and then shows steady. The color of the N2 digital readout is yellow when the input value is in the caution range. The color of the N2 digital readout is green when the input is in the normal range.

High Vibration Message	The high vibration message VIB shows in yellow when high vibration is reported. The high vibration message has display priority over N2 pointer.
Fuel Flow	The Fuel Flow (FF) display shows the fuel flow to the engine normally in PPH. The FF display is made up of a digital readout without units of measurement for each engine. The FF readout is always green. Yellow dashes replace the readout if fuel flow data is invalid. The FF PPH may optionally be configured to show fuel flow in kilograms per hour (KPH). If fuel flow is set to readout in KPH, the display shows FF KPH.
Oil Pressure	The Oil Pressure (OIL PRESS) display shows the oil pressure of each engine in PSI. The OIL PRESS display is made up of a digital readout without units of measurement for each engine. The color of the oil pressure digital readout is red when the input value is in the warning range. The color of the oil pressure digital readout is green when the input value is in the normal range. The color of the oil pressure digital readout is yellow when the input value is in the caution range. Yellow dashes replace the readout when oil pressure data is invalid.

Oil Temperature	The Oil Temperature (OIL TEMP) display shows the oil temperature of each engine in degrees Celsius. The OIL TEMP display is made up of a digital readout without units of measurement for each engine. The color of the oil temperature digital readout is green when the input value is in the normal range. The color of the oil temperature digital readout is yellow when the input value is in the caution range. The color of the oil temperature digital readout is red when the input value is in the warning range. The readout flashes when the color changes to red. Yellow dashes replace the readout when oil temperature data is invalid.
Engine Vibration	The engine vibration (FAN VIB) display for each engine is made up of two digital readouts with a legend. Vibration shows in mils. Yellow dashes replace the readout when engine vibration data is invalid. The color of the engine vibration digital readout is green when the input value is in the normal range. The color of the engine vibration digital readout is yellow when the input value is in the caution range.
Fuel Summary	The fuel quantity display is made up of four individual tank quantities and total quantity. The fuel quantity units of measurement can be kilograms (KGS) or pounds (LBS). The fuel quantity digital readouts are color-coded according to the operating range. When a valid input is not present, yellow dashes show in place of the digital readout.

- Left Wing Fuel Tank — The digital readout for the left wing fuel tank shows on the left and in green when the fuel level is not low and the quantity is balanced between the two sides. The color is yellow when the fuel level is low or the quantity is not balanced between the two sides. The color is magenta when the color conditions for green or yellow are not met.
- Right Wing Fuel Tank — The digital readout for the right wing fuel tank shows on the right and in green when the fuel level is not low and the quantity is balanced between the two sides. The color is yellow when the fuel level is low or the quantity is not balanced between the two sides. The color is magenta when the color conditions for green or yellow are not met.
- Auxiliary Fuel Tank — The digital readout for the auxiliary fuel tank shows between the left and right wing fuel tank readouts. The color is white when the tank is empty. The color is yellow when the tank is considered heavy. The color is green when the tank is not empty and not heavy. The color is magenta when the color conditions for white, yellow, or green are not met.

- Tail Fuel Tank — The digital readout for the tail fuel tank shows below the auxiliary fuel tank readout. The color is white when the tank is empty. The color is yellow when the tank is considered heavy. The color is green when the tank is not empty and not heavy. The color is magenta when the color conditions for white, yellow, or green are not met.
- Total Fuel — The digital readout for total fuel shows next to the TOTAL legend. The color is green when the input value is normal. The color is yellow when the input value is in the caution range. The color is magenta when the conditions for green or yellow are not met.

Horizontal Stabilizer Trim

The horizontal stabilizer trim display is made up of a pointer that moves vertically on a scale and a digital readout. The legend STAB identifies the horizontal stabilizer trim display. The top of the scale represents full Nose Up (NU) and the bottom of the scale full nose down trim. A green takeoff band shows on the vertical scale when the aircraft is on the ground. The trim pointer shows in green when the pointer is inside the takeoff trim band and the takeoff trim band is green. The trim pointer shows in white whenever the pointer is outside the takeoff trim band and the takeoff trim band is green. A digital readout of the trim position shows within the pointer. The digital readout value matches the color of the pointer.

Aileron Trim	The aileron trim display is made up of a straight line pointer that moves like the wings rolling against an arc-shaped scale. The scale is labeled Left Wing Down (LWD) and Right Wing Down (RWD).
Rudder Trim	The rudder trim display is made up of a triangular pointer that moves on a horizontal scale. The rudder trim scale is made up of a horizontal line divided by five equally spaced tick marks. The trim pointer is removed whenever the input is outside valid range or is reported as invalid.
Landing Gear Position Indicators	The landing gear position indicator is made up of the landing gear legend and three indicator boxes that represent left, nose, and right landing gear. The landing gear position indicators and GEAR legend are removed after 30 seconds when the gear is not in transit, not down and locked, or unsafe, and the flaps are no longer in transit and are fully retracted. Each landing gear shows graphically with a unique position indicator box that reflects current state. The possible positions are Down (DN), UP, or IN TRANSITION. • The landing gear position indicator box shows as a white box containing the message UP in white when the landing gear is up. The landing gear indicator box shows as a white box containing a yellow flashing UP legend, when LANDING GEAR HORN has been set for 10 seconds or more and the landing gear is up.

- The landing gear indicator box shows as a yellow box containing yellow hash marks when the landing gear is in transit. The landing gear indicator box shows as a yellow box containing yellow flashing hash marks, when LANDING GEAR HORN has been set for 10 seconds or more and the landing gear is in transit.
- The landing gear indicator box shows as a green oval containing black DN legend when the landing gear is down.
- The landing gear indicator box shows as a red box containing red hash marks when the landing gear is in an unsafe position. The landing gear indicator box shows as a red box containing red flashing hash marks, when LANDING GEAR HORN has been set for 10 seconds or more and the landing gear is in an unsafe position.
- The landing gear indicator box is replaced with two yellow dashes when any of the previous color conditions are not met or input is invalid.

Flap Position	The flaps display is made up of the flaps legend, a digital readout of flap position, and a corresponding horizontal scale. The flaps position scale is made up of four horizontally placed detent markers. There are detent positions at 20, 30, and 45 degrees. The flaps position bar shows when the readout is above 0 (zero). The flaps position bar, anchored on the 0 (zero) degrees extends to the right for increasing values of FLAP POSITION. The flaps position digital readout shows in green. The flaps position digital readout shows as two yellow dashes when the input is invalid. The entire flaps position display is removed shortly after the flaps are fully retracted and gear is in the UP position. Yellow dashes replace the digital readout when flap position data is invalid.
Spoiler Position	The spoilers display depicts the position of flight spoilers and ground spoilers. The flight spoilers are the outside pair of spoiler pointers. The ground spoilers are represented by the inside pair of pointers. Left and right flight spoiler positions and ground spoiler positions show by means of a pair of pointers extending toward their limit lines. The flight spoiler limit lines have two limit lines: one at the fully extended value and one at the mid-travel extended value. Increasing flight spoiler position values result in upward flight spoiler pointer motion. The ground spoilers show in two positions only: fully extended or fully retracted. The fully extended position is indicated by a pointer extending to its limit line. Absence of the pointer indicates the fully retracted position.

Cabin Environment Display

The cabin environment display is made up of digital readouts of cabin altitude, cabin altitude rate, cabin differential pressure, cabin air temperature, and crew oxygen pressure. Yellow dashes replace the digital readout when the input data is invalid.

- Cabin altitude shows next to the ALT legend. The value is color-coded to indicate the normal, caution, and warning range for Cabin Altitude. The message MAN shows next to cabin altitude when the cabin pressurization system is in manual mode. Values that are within the warning range show in red. Values that are within the caution range show in yellow. Values that are within the normal range show in green.
- Cabin rate shows in green next to the RATE legend. A cabin rate arrow graphically depicts the increasing or decreasing cabin rate.
- Cabin differential pressure shows next to the △P legend. The value is color-coded to indicate the normal and warning range. The color is green when the input value is in the normal range. The color is red when the input value is in the warning range.
- Cabin Temperature shows in green next to the TEMP legend.
- Oxygen pressure shows next to the OXY legend. The value is color-coded to indicate the normal range. The color is green when the input value is in the normal range. The color is yellow when the input value is outside the normal range.

APU RPM	The APU RPM shows next to the RPM legend. The APU RPM value is measured as a percentage of maximum allowable RPM. The value is color-coded to indicate the normal, caution, and warning range. Values within the warning range show in red. Values within the caution range show in yellow. Values within the normal range show in green. The APU display is removed when the APU is not powered-up.
APU EGT	The APU EGT shows next to the EGT legend. The value is color-coded to indicate the normal, caution, and warning range. Values within the warning range show in red. Values within the caution range show in yellow. Values within the normal range show in green. The APU display is removed when the APU is not powered-up.

Crew Alerting System (CAS)

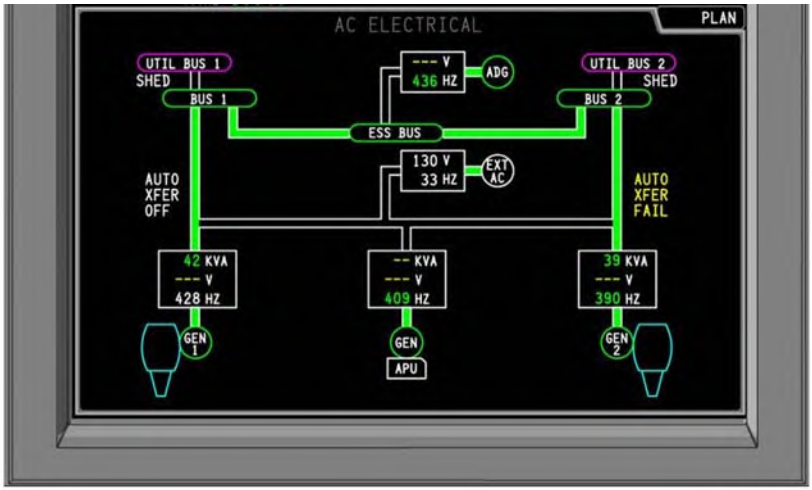
The CAS function shows four categories of system messages in the order that follows: warning, caution, advisory, and status. The aircraft master warning system is used to indicate that new warning/caution messages are in view on the AFD. The message MSGS shows on the first message line following any warning messages. When a new message appears, the new message shows before any existing messages of its category not hidden. The message list containing the new messages must be examined in order to hide the list again. The CAS message field normally shows on the MFD, but can also show on the compressed PFD. Controls located on the CCP are used to page through all of the caution, advisory, and status messages, and hide any existing old non-warning messages.

- Warning messages take precedence over all other types, and always show in red at the top of the CAS display. The most recent warning message shows at the top of the warning message list. Warning messages show on one page only. A new warning message causes both aircraft master warning lights to come on, accompanied by an aural tone or message.

- Caution messages show in yellow and follow any warning messages. The most recent caution message shows at the top of the caution message list. Caution messages may appear on more than one page. A new caution message causes both aircraft master warning lights to come on, accompanied by an aural tone.
- Advisory messages show in green and follow the caution messages. The most recent advisory message shows at the top of the advisory message list. Advisory messages may appear on more than one page.
- Status messages show in white and follow the advisory messages. The most recent status message shows at the top of the status message list. Status messages may appear on more than one page.

MFD – AC ELECTRICAL SYSTEM SYNOPTIC

Figure 6-55 MFD – AC Electrical System Synoptic



TPO4650_02

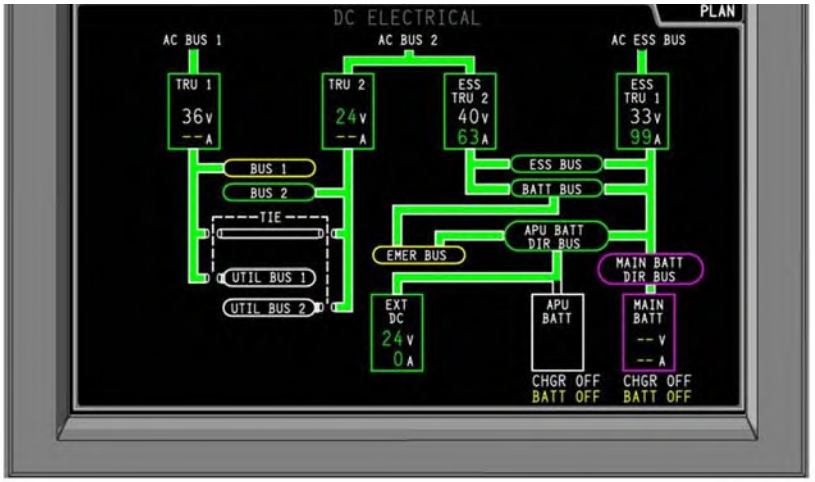
The AC electrical system synoptic display graphically depicts the operational status of the five sources of AC electrical power. The three main sources are generator 1, generator 2 and the auxiliary power unit generator, which show continuously on the AC electrical system display. The air-driven generator shows only when it is deployed. The external AC power input shows when it is electrically connected to the aircraft. The AC electrical system synoptic display can be selected by way of the UPR (main) MENU on the MFD or with the dedicated AC ELEC button on the CCP.

▼ SELECTIONS:

System Components	Graphic depictions of generators 1 and 2, the APU generator, the air-driven generator, the external AC utility busses 1 and 2, and AC busses 1 and 2 show on the AC Electrical System Synoptic display. The system components are connected by flow lines. The system component graphics are color-coded according to their operating state. The flow lines show in green when conditions are normal. When operating outside of normal conditions, the flow lines are not filled or colored.
Digital readouts	Digital readouts of voltage and frequency shows for the five sources of AC electrical power. Digital readouts of the electrical load shows for the three main sources of AC electrical power. The digital readouts are color-coded according to their operating range. Yellow dashes show in place of the digital readout when a valid input is not present.
Status Messages	The status messages that follow show on the AC electrical system synoptic display: • SHED (white) • AUTO XFER OFF (white) • AUTO XFER FAIL (yellow).

MFD – DC ELECTRICAL SYSTEM SYNOPTIC

Figure 6-56 MFD – DC Electrical System Synoptic Display



TPO4650_03

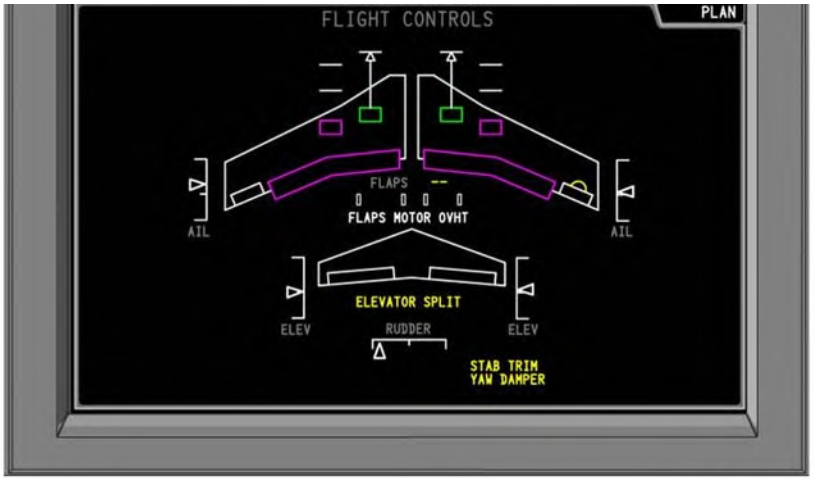
The DC electrical system synoptic display graphically depicts the operational status of the DC power sources and accompanying DC busses. Graphic depictions of system components and flow lines, digital readouts and system status messages show on the display. The DC electrical system synoptic display can be selected by way of the UPR (main) MENU on the MFD or with the dedicated DC ELEC button on the CCP.

▼ SELECTIONS:

System Components	Graphic depictions of the Transformer Rectifier Units (TRU) 1 and 2, essential TRUs 1 and 2, the main battery, the APU battery, and the external DC input show on the DC electrical system synoptic display. The system components are connected by flow lines. The system component graphics are color-coded according to their operating state. The flow lines show in green when conditions are normal. When operating outside of normal conditions, the flow lines are not filled or colored.
Digital Readouts	Digital readouts of DC voltage and load show for TRU 1 and 2, essential TRU 1 and 2, external DC, and the main battery. The digital readouts are color-coded according to their operating range. Yellow dashes show in place of the digital readout when a valid input is not present.
Status Messages	The availability and charger status messages that follow show for the APU battery and main battery: • CHGR OFF (white) • BATT OFF (yellow).

MFD – FLIGHT CONTROLS SYNOPTIC DISPLAY

Figure 6-57 MFD – Flight Controls Synoptic



TPO4650_04

The flight controls synoptic page graphically depicts the position of the ailerons, elevator, rudder, flaps, ground spoilers, and flight spoilers. Digital readouts of system parameters and system status messages also show on the display. The flight controls synoptic page can be selected by way of the UPR (main) MENU on the MFD or with the dedicated FLT button on the CCP.

▼ SELECTIONS:

System Components	A scale/pointer shows to indicate left and right Aileron (AIL) position, left and right Elevator (ELEV) position, flap position and rudder position. Pointers show to indicate the position of the ground spoilers and flight spoilers. The aileron, elevator, and rudder pointers show in white and the flap pointer shows in green. The outlines of the system components are color-coded to indicate their operating state.

Digital Readouts

A digital readout shows for flap position. The flaps digital readout is color-coded to indicate the flap operating state. Two yellow dashes show in place of the digital readout when a valid input is not present.

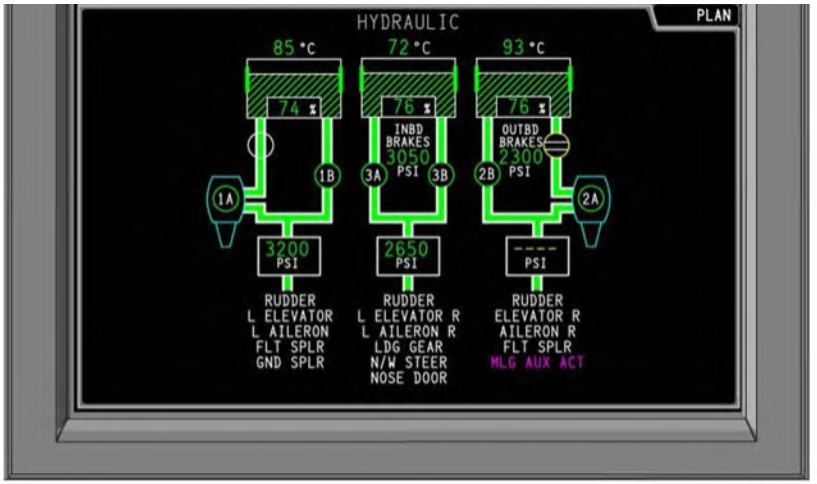
Status Messages

The status messages that follow show on the flight controls synoptic display:

- YD 1 INOP (white or yellow in accordance with the operating state)
- YD 2 INOP (white or yellow in accordance with the operating state)
- YAW DAMPER (white or yellow in accordance with the operating state)
- STAB CH 1 INOP (white or yellow in accordance with the operating state)
- STAB CH 2 INOP (white or yellow in accordance with the operating state)
- STAB TRIM (white or yellow in accordance with the operating state)
- FLAPS MOTOR OVHT (white)
- ELEVATOR SPLIT (yellow).

MFD – HYDRAULIC SYNOPTIC

Figure 6-58 MFD – Hydraulic Synoptic



TPO4650_01

The hydraulic system synoptic display graphically depicts the operational status of the threes sources of hydraulic pressure. The status of the reservoirs, the hydraulic fluid quantity, temperature, and pressure, pump status, valve positions, hydraulic flow, and hydraulic power users status for three systems, and the inboard and outboard brake pressures show on the hydraulic system synoptic display.

▼ SELECTIONS:

System Components	The hydraulic system synoptic display shows the engines, engine pumps, pump parameters, firewall shut-off-valves, and hydraulic systems 1, 2 and 3. Each of the system component graphics is color-coded according to their operating state. Flow lines connect the different hydraulic sources, engine pumps, engines, and parameter boxes. The flow lines are color-coded according to their operating state as follows:

- Green — Normal
- Yellow — Low hydraulic pressure
- Red — An engine fire is present and the Shut Off Valve (SOV) needs to be closed.

NOTE

When operating outside of normal conditions the flow lines are not colored.

Digital Readouts

The hydraulic system synoptic page shows digital readouts for the hydraulic system temperature, quantity, and pressure, and the inboard brake pressure and outboard brake pressure. The digital readouts are color-coded according to their operating range. When a valid input is not present, yellow dashes show in place of the digital readout.

Hydraulic System Quantity

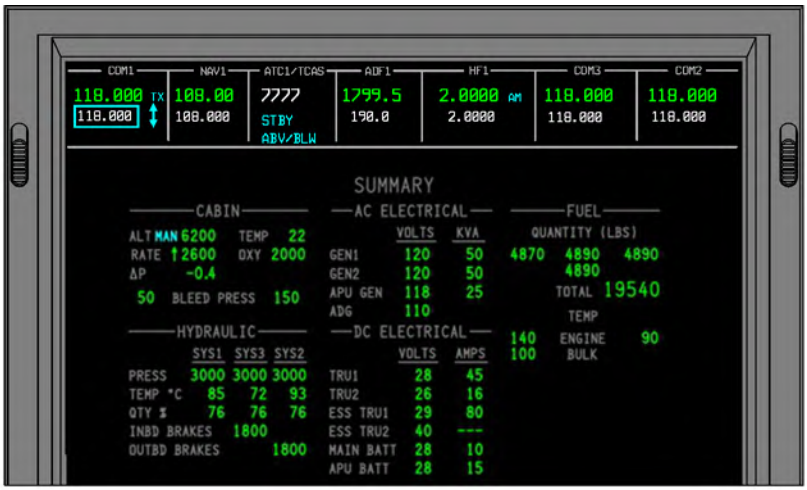
A digital readout, scale, and pointer shows for each of the three hydraulic systems on the hydraulic system synoptic page. The hydraulic quantity pointer and scale graphically depicts the hydraulic quantity for each system. The pointer moves upward for increasing values hydraulic quantity. The pointer and diagonal lines show in the same color as the digital readout.

Hydraulic Power Users Legends

A list of the individual systems that are powered by each hydraulic system show on the hydraulic system synoptic page below the graphic depiction of the associated hydraulic system. The color of the individual system messages are color-coded according to their operating state.

MFD – SUMMARY SYNOPTIC

Figure 6-59 MFD – Summary Synoptic



TPO4650_17

The summary synoptic page provides a single page summary of selected aircraft system synoptics. Digital readouts of key information for the environmental control system, cabin pressurization system, AC and DC electrical systems, fuel system, and hydraulic system show on the summary synoptic page. The compass rose or arc does not show when the summary synoptic page is active. Instead, a horizontally-oriented compass tape shows above the summary synoptic information. The familiar selected heading bug and track pointer show on the compass tape. The summary page is the default display on the upper half of the copilot-side MFD, when checklist and maintenance display mode are not selected to show on the copilot-side MFD. The summary synoptic page can be selected to show on the lower half of the MFD with the SUMRY button on the CCP. The summary page can be selected to show on the PFD with the FRMT Button on the DCP when the PFD is in the compressed format.

▼ SELECTIONS:

Cabin Summary	Cabin altitude, cabin pressure rate, cabin differential pressure, cabin temperature, oxygen pressure, and bleed pressure show in the cabin summary section of the summary display. • Cabin altitude — A digital readout of the cabin Altitude (ALT) shows color-coded according to the operating range. The cabin altitude digital readout shows in green when the value is in the normal range, yellow when the value is in the caution range, and red when the value is in the warning range. When a valid input is not present, yellow dashes show in place of the digital readout. The message MAN shows in cyan when the cabin pressurization system is operating in manual mode. • Cabin pressure rate — A digital readout of the cabin pressure rate (RATE) shows with a preceding arrow pointing up for positive values, and pointing down for negative values. The cabin rate digital readout and arrow show in green. When a valid input is not present, yellow dashes show in place of the digital readout. • Cabin differential pressure — A digital readout of the cabin differential pressure (CAB ΔP) shows color-coded according to the operating range. The cabin differential pressure shows in green when the value is in the normal range and in red when the value is in the warning range. When a valid input is not present, yellow dashes show in place of the digital readout.

- Cabin temperature — A digital readout of the cabin Temperature (TEMP) shows in green. When a valid input is not present, yellow dashes show in place of the digital readout.
- Oxygen pressure — A digital readout of the oxygen pressure (OXY) shows color-coded according to the operating range. The oxygen pressure shows in green when the value is in the normal range and in yellow when the value is outside the normal range.
- Bleed pressure — Pilot and copilot-side Bleed Pressures (BLEED PRESS) show in green. When a valid input is not present, yellow dashes show in place of the digital readout.

AC Electrical Summary

Voltage and load show for generator 1, generator 2, and the APU generator in the AC electrical section of the Summary display. Voltage shows for the Air-Driven Generator (ADG). The digital readouts are color-coded according to their operating range. When a valid input is not present, yellow dashes show in place of the digital readout.

Fuel Quantity Summary

The fuel quantity display is made up of four individual tank quantities and total quantity. The fuel quantity units of measurement can be kilograms (KGS) or pounds (LBS). The fuel quantity digital readouts are color-coded according to the operating range. When a valid input is not present, yellow dashes show in place of the digital readout.

- Left Wing Fuel Tank — The digital readout for the left wing fuel tank shows on the left and in green when the fuel level is not low and the quantity is balanced between the two sides. The color is yellow when the fuel level is low or the quantity is not balanced between the two sides. The color is magenta when the color conditions for green or yellow are not met.
- Right Wing Fuel Tank — The digital readout for the right wing fuel tank shows on the right and in green when the fuel level is not low and the quantity is balanced between the two sides. The color is yellow when the fuel level is low or the quantity is not balanced between the two sides. The color is magenta when the color conditions for green or yellow are not met.
- Auxiliary Fuel Tank — The digital readout for the auxiliary fuel tank shows between the left and right wing fuel tank readouts. The color is white when the tank is empty. The color is yellow when the tank is considered heavy. The color is green when the tank is not empty and not heavy. The color is magenta when the color conditions for white, yellow, or green are not met.

- Tail Fuel Tank — The digital readout for the tail fuel tank shows below the auxiliary fuel tank readout. The color is white when the tank is empty. The color is yellow when the tank is considered heavy. The color is green when the tank is not empty and not heavy. The color is magenta when the color conditions for white, yellow, or green are not met.
- Total Fuel — The digital readout for total fuel shows next to the TOTAL legend. The color is green when the input value is normal. The color is yellow when the input value is in the caution range. The color is magenta when the conditions for green or yellow are not met.
- Fuel Temperature — Left and right engine fuel temperature and bulk fuel temperature show below fuel quantity on the summary synoptic page.

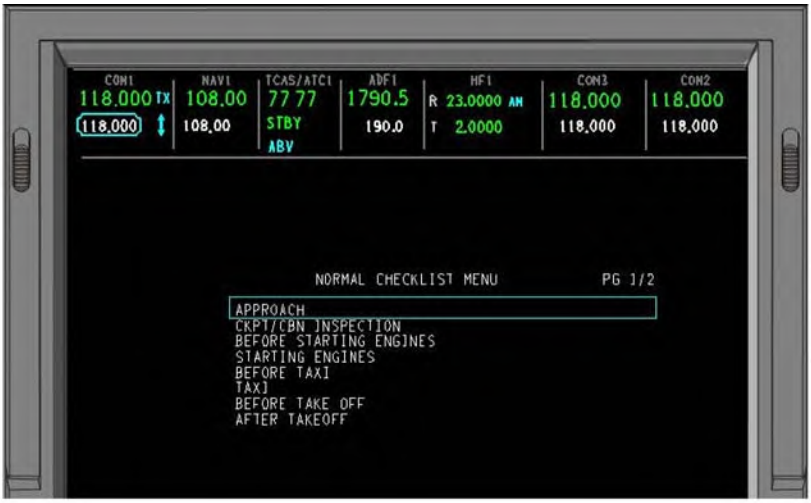
Hydraulic Summary

The key readouts of hydraulic pressure, hydraulic temperature in degrees Celsius, and hydraulic quantity show on the Summary synoptic page. The pressure status for the Inboard (INBD) and Outboard (OUTBD) brakes also shows on the summary synoptic page. The readouts are color-coded according to the operating range. When a valid input is not present, yellow dashes show in place of the digital readout.

DC Electrical Summary	Digital readouts for Transformer Rectifier Unit (TRU) 1 and 2, Essential (ESS) TRU 1 and 2, inboard brake pressure, and outboard break pressure show on the Summary synoptic page. The readouts are color-coded according to the operating range. When a valid input is not present, yellow dashes show in place of the digital readout.

MFD – CHECKLISTS

Figure 6-60 MFD – Checklists



TPO4650_33

The checklist function is used to store, retrieve, and view aircraft-specific checklists. Completed checklists are periodically loaded into the Maintenance Data Computer (MDC) from a portable memory device. There is a Preamble page, which requires acknowledgement, and up to four checklist types. On-aircraft capability to create, edit, or delete checklists does not exist. Checklist pages show on the right MFD only. Controls located on the CCP are used to select and operate the checklists.

▼ SELECTIONS:

Checklist Location	The checklists can show on either MFD (but not both MFD at the same time), as long as the selected MFD does not have EICAS locked on and the MFD is not reverted to a PFD. The checklist previously in view is recalled on either MFD. It is not possible to have the left MFD and right MFD recall different checklists. A checklist can be moved from the left MFD to the right MFD and the last active checklist is recalled, whether that checklist was in view on the left or right MFD.
Checklist Index	The available checklist types show on the checklist index. The possible checklist types are: normal, abnormal, emergency, and user. The checklist index shows on the MFD when the checklist format is selected and if, when the checklist was previously exited, the display was a checklist with no unchecked items. The checklist format is selected by way of the MFD upper menu. An optional remote checklist select control may also be available.
Checklist Menus	When a specific type of checklist (for example normal) is chosen from the checklist index, the menu for that checklist shows on the display. From the specific checklist menu, an individual checklist is selected.

Preamble Page	The preamble page is used to make sure that the correct version of the checklist is loaded. The preamble page shows the first time the normal, abnormal, or emergency menu is selected from the checklist index after power-up. After verifying the checklist version number, acknowledge the preamble page. The PUSH SELECT button on the CCP are used to acknowledge the preamble page.
Specific Checklist	When a specific checklist is in view when the checklist format is deselected, reselecting the checklist format recalls that specific checklist. The checklist format is deselected when another MFD format is selected. The specific checklist (previously in view) is recalled on either MFD. Only one checklist function is performed, even though checklists can be selected to show in multiple places. Consequently, there is only one recall specific checklist. A specific checklist can be moved from the left MFD to the right MFD and the last active checklist is recalled, whether that checklist was in view on the left or right MFD.
Checklist Control	Controls located on the CCP are used to operate the checklist function. An optional remote checklist select and line advance controls may also be available.

- When a checklist index is in view, operation of the MENU ADV (menu advance) knob counterclockwise (clockwise) moves the selection box up (down) to a specific checklist menu. Operation of the PUSH SELECT button selects the highlighted checklist menu or preamble page. When the selection box surrounds the RESET CHECKLIST COMPLETE HISTORY item, operation of the PUSH SELECT button resets the checklist history. Resetting the checklist complete history sets all checklist items to unchecked.
- When a checklist menu is in view, operation of the MENU ADV knob counterclockwise (clockwise) moves the selection box up (down) to select a specific checklist. Operation of the PUSH SELECT button shows the selected checklist. Operation of the ESC button shows the associated checklist index.

- When a checklist is in view, operation of the PUSH SELECT button checks off the item highlighted by the selection box and moves the selection box to the next unchecked item on the current page, the following page, and finally to the next unchecked item on a previous page. When the checked item was the last unchecked item within the checklist title, the selection box shows around the message CHECKLIST COMPLETE on the final page of the checklist. For normal checklist titles only, subsequent operation of the PUSH SELECT button shows the first page of the next normal checklist on the normal checklist menu. For other checklists, or when no additional normal checklist titles remain unchecked, subsequent operation of the PUSH SELECT button shows the associated checklist menu.
- When a specific checklist is in view, operation of the MENU ADV knob counterclockwise (clockwise) moves the selection box up (down) without checking off the items. When the current item is the first checkable item, counterclockwise (up) operation of the MENU ADV knob wraps the selection box to the last selectable item on the current section of the checklist. When the current item is the last item in the checklist, but there are other unchecked items within the checklist section, operation of the MENU ADV clockwise (down) moves the selection box to the unchecked item that is closest to the top of that checklist.

- When a checklist is in view, operation of the optional remote checklist line advance control while a checklist is in view has the same function as operation of the PUSH SELECT button.
- When a checklist is in view, operation of the ESC button shows the associated checklist menu.

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MESSAGES AND ANNUNCIATORS

AURAL ANNUNCIATIONS

▼ : Autopilot Disconnect Warning

Autopilot Disconnect Warning Cavalry Charge Aural	⇒ Push either the AP/SP DISC switch or the TOGA switch on either control wheel to cancel the autopilot disconnect aural. Re-engage the autopilot if able. The autopilot has been disconnected, either manually by the pilot or automatically by the AFCS. If the autopilot was disconnected manually, the autopilot disconnect aural sounds momentarily. If the autopilot was disconnected automatically, the autopilot disconnect aural sounds continuously until cancelled by the pilot. The autopilot automatically disconnects if: • The yaw damper is disengaged for any reason. • The flight guidance computer detects an internal failure. • A stall condition is detected. • The extreme attitude limits are reached.

▼ : Datalink Alert

Datalink Alert 2 Level Tone	⇒ View datalink message/clearance on the CDU FANS/ATN/TCAS Aural occurs when a datalink message is received.

▼ : Overspeed Alert

Overspeed Alert OVERSPEED Voice	⇒ Reduce aircraft speed. The Overspeed Alert occurs when the current airspeed exceeds V_{MO}/M_{MO} by 3 knots.

▼ : Preselect Altitude Alert

Preselect Altitude Alert C Chord Aural	⇒ Monitor aircraft altitude to make sure the aircraft levels off/maintains the PSA. The PSA alert notifies the pilot that the aircraft barometric altitude is approaching the PSA. PSA alert occurs when any one of the following items is true: • Whenever the aircraft altitude passes ±1000 feet of the PSA, entering or exiting. • When the aircraft altitude exceeds ±200 feet of the PSA after it has been within ±200 feet of the PSA.

▼ : SELCAL Alert

Selcal Alert SELCAL voice	⇒ No response is required. SELCAL received.

▼ : Stall Alert

Stall Alert STALL Voice	⇒ Correct the stall condition. The stall alert occurs when a stall is detected by the SPC.

▼ : Trim Clacker

Trim Clacker	⇒ No response is required. The trim clacker sounds when the pitch trim is in motion.

▼ : Vertical Track Alert

Vertical Track Alert FMS VNAV Alert	⇒ Monitor the FMS vertical flight plan. The pilot may manually change any of the AFCS mode selections or associated reference settings made by the FMS. Such pilot action suspends the current FMS VNAV control of the AFCS, but does not preclude reassertion of control by the FMS. Pilot intervention may adversely affect the FMS ability to keep the aircraft on the FMS vertical flight plan. The vertical track alert is provided prior to a VNAV change.

VISUAL ANNUNCIATIONS

PFD – FLAGS



TPN7703_01

▼ PFD: Altitude Flag

ALT (red)	⇒ Select the cross-side air data source. A red, boxed ALT flag shows in place of the barometric altitude pointer when altitude data from the Air Data Computer is missing or invalid. The barometric altitude scale, digital readout, and metric altitude are removed when the ALT flag is in view. The flag flashes for 5 seconds and then shows steady.

▼ PFD: AOA Flag

AOA (red)	⇒ No response required. A red, boxed AOA flag shows in place of the AOA scale when AOA data is missing or invalid. The AOA scale, pointer and digital readout are removed when the ALT flag is in view. The flag flashes for 5 seconds and then shows steady.

▼ PFD: Autopilot Engage Failure

AP
(red)

⇒ No response required.

When the Autopilot is disconnected manually by the pilot, AP changes color to red and flashes for approximately 2 seconds. When the Autopilot is disconnected automatically by the system, AP changes color to red and flashes for approximately 5 seconds. If the system disconnects automatically, the crew must take action to clear the aural. Action is to re-engage or acknowledge the disconnect by pressing a disconnect switch. The transfer arrow flashes in red with the AP message. The AP disengage warn is accompanied by a Cavalry Charge aural.

▼ PFD: Flight Director Flag

FD
(red)

⇒ Select the cross-side flight guidance channel.

A red, boxed FD flag shows in the upper left side of the attitude ball when pitch or roll guidance is invalid. The FD command bars are removed when the flag is in view. The active mode messages flash for 5 seconds, then are removed. All other AFCS messages also flash for 5 seconds and then are removed.

▼ PFD: Glideslope Flag

GS (red)	⇒ Select a different NAV source. A red, boxed GS flag shows at the lower end of the vertical deviation scale when a LOC is the active NAV source and vertical deviation from the active NAV source is missing or invalid. The diamond pointer is removed when the flag is in view. The flag flashes for 5 seconds and then shows steady.

▼ PFD: Heading Flag

HDG (red)	⇒ Select the cross-side attitude/heading source. A red, boxed HDG flag shows in place of the heading digital readout when heading data from the Attitude Heading Computer is missing or invalid. The compass card is positioned north-up and the lubber line is removed when the HDG flag is in view. The flag flashes for 5 seconds and then shows steady.

▼ PFD: Lateral Deviation Flag

LOC or FMS (red)	⇒ Select the cross-side receiver (for example, if LOC1 select LOC2 with same frequency tuned).

	A red, boxed LOC or FMS flag shows when the LOC or FMS lateral deviation is missing or invalid and the FMS flag is in view. The flag shows to the left of the attitude display lateral deviation pointer. The flag flashes for 5 seconds and then shows steady.

▼ PFD: Airspeed Flag

IAS or MACH (red)	⇒ Select the alternate air data source with the AIR DATA knob on the Reversion Switch Panel. The red, boxed IAS flag shows instead of the airspeed pointer when the airspeed tape is displaying IAS data and the IAS data from the ADC is missing or invalid. The red, boxed MACH flag shows instead of the airspeed pointer when the airspeed tape is displaying Mach data and the Mach data from the ADC is missing or invalid. The MACH flag also shows when the airspeed reaches or exceeds .99 Mach. The IAS/Mach digital readout, IAS/Mach scale, trend vector, overspeed marker, LSC markers, IAS/Mach bug and all VSPEEDS are removed when the IAS or MACH flag is in view. The flag flashes for 5 seconds, then remains steady.

▼ PFD: Attitude Flag

ATT (red)	⇒ Select the cross-side or 3rd IRS with the IRS Reversion switch.

	The red, boxed, ATT flag shows when attitude data from the selected IRS is missing or invalid. The flag flashes for 5 seconds, then remains steady. The attitude ball, pitch tape, and roll pointer are removed when the ATT flag is in view.

▼ PFD: Map Flag

MAP (yellow)	⇒ No response required. The flag MAP shows in yellow in the top-center part of the display if a fault is detected in any data required to display the map. The MAP symbology is removed when the MAP flag is in view.

▼ PFD: NAV Source Flag

VOR, LOC, or FMS (red)	⇒ Select a different NAV source. A red, boxed NAV flag shows in place of the active NAV source message when data from the active NAV source is missing or invalid. The NAV source flag shows as VOR1, VOR2, LOC1, LOC2, FMS1, or FMS2. The flag flashes for 5 seconds and then shows steady. A red, boxed LOC1, LOC2, FMS1, or FMS2 flag, depending on the active NAV source, also shows to the left of the lateral deviation scale in the attitude display.
VOR, LOC, or FMS	⇒ Select a different NAV source.

The active NAV source message flashes for 15 seconds, then shows steady, when MAG or DG heading is selected and the active NAV source is FMS and the FMS supplied magnetic variation is missing or failed.

▼ PFD: Preselect Altitude Flag

Preselect Altitude Flag
(yellow)

⇒ No response required.

Preselect altitude is synchronized between the pilot and copilot-side displays. When a split is detected between the on-side and cross-side preselect altitude, a flashing yellow underline shows below the preselect altitude readout for 5 seconds. The preselect reference bug on the altitude scale is removed when the preselect altitude readout underline is flashing yellow.

▼ PFD: Preset NAV Source Fail

LOC
(cyan with strike-through)

⇒ No response required.

After the preselect lateral deviation bar shows during NAV to NAV approach operation, if it is removed because it goes invalid, the preselect NAV source annunciation LOC shows with a horizontal cyan strike-through line.

▼ PFD: Radio Altitude Flag

RA
(red)

⇒ No response required.

A red, boxed RA flag shows to the left of the Radio Altitude digital readout when radio altitude data is missing or invalid. The flag flashes for 5 seconds and then shows steady. The RA flag shows in the same location as the RA miscompare flag and has priority over the RA miscompare flag.

▼ PFD: TCAS Flag

TCAS FAIL
(yellow)

⇒ No response required.

The flag TCAS FAIL shows in yellow if the TCAS system is failed. Failures related exclusively to the traffic display cause the message when TFC is selected.

▼ PFD: Resolution Advisory Flag

NO VSI RA
(yellow)

⇒ No response required.

The flag NO VSI RA shows in yellow if the TCAS vertical resolution advisory word can’t provide advisory information or if vertical speed is not in view on the PFD.

▼ PFD: Terrain Fail Flag

TERRAIN FAIL
(yellow)

⇒ No response required.

The flag TERRAIN FAIL shows in yellow on the PFD and MFD when the Terrain Awareness and Warning System (TAWS) is reported as failed. The flag flashes for 5 seconds and then shows steady. The message is removed if the terrain overlay is deselected. The message does not show if a format incompatible with Terrain is active. When the flag is in view, the Terrain overlay is not available.

▼ PFD: Vertical Speed Flag

VS
(red)

⇒ Select the cross-side Air Data Computer.

A red, boxed VS flag shows in place of the vertical speed scale when vertical speed data from the Air Data Computer is missing or invalid. The vertical speed digital readout, vertical speed scale, and pointer are removed when the VS flag is in view. The flag flashes for 5 seconds and then shows steady.

▼ PFD: Vertical Navigation Flag

VNV
(red)

⇒ No response required.

A red, boxed VNV flag shows at the upper end of the vertical deviation scale when VNAV data from the FMS is missing or invalid. The VNAV deviation pointer is removed when the flag is in view. The flag flashes for 5 seconds and then shows steady.

▼ PFD: Weather Radar Flag

WX FAIL
(yellow)

⇒ No response required.

The flag WX FAIL shows when Weather Radar data is not available to the display and the Weather Radar overlay is active. The flag flashes for 5 seconds and then shows steady. The Weather Radar symbology is removed from the display when the flag is in view.

▼ PFD/MFD: Weather Radar Flag

WX FAULT
(yellow)

⇒ No response required.

The flag WX FAULT shows in yellow when the Weather Radar reports itself failed. The flag is displayed at the bottom of the weather radar status block. The flag flashes for 5 seconds and then shows steady. When WX FAULT is in view, the Weather Radar display may be degraded in a manner that is not obvious to the pilot. Some faults may cause the Weather Radar to cease providing reflectivity information to the display.

LX FAULT (yellow)	⇒ No response required. The flag LX FAULT shows in yellow when the lightning detection system data is failed or reports itself failed. The flag is displayed at the bottom of the weather radar status block. The flag flashes for 5 seconds and then shows steady.
USTB (RTA-854 Only) (yellow)	⇒ No response required. The stabilization inputs to the weather radar exceed limits and weather may not be depicted accurately while this annunciation is displayed.
AUTO FAULT (RTA-4114 Only) (yellow)	⇒ No response required. Indicates to the flight crew that the Multiscan function of the Weather Radar is not available.
CTRL FAULT (RTA-4114 Only) (yellow)	⇒ No response required. Indicates to the flight crew that a Control Fault was detected. When this fault is annunciated, the cross side radar controls will be used by the weather radar to provide the commands for the affected side displays.
TURB FAULT (RTA-4114 Only) (yellow)	⇒ No response required. Indicates to the flight crew that the turbulence function is not available.

▼ PFD: Crosstalk Flag

XTLK
(yellow)

⇒ No response required.

A yellow, boxed XTLK flag shows above the AOA scale on the PFD when crosstalk is lost between the PFD and either of the MFDs or the other PFD. The XTLK flag flashes for 5 seconds and then shows steady. The flag is removed when communication is restored. The message does not show for PFD or MFD that have been deselected by way of display reversion.

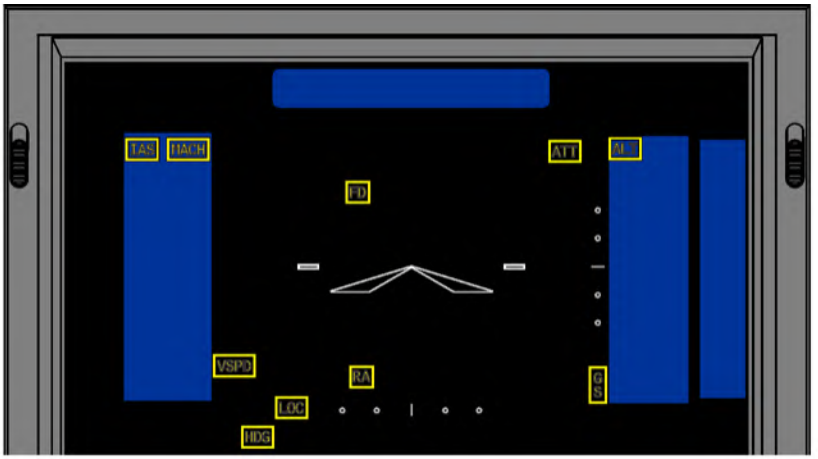
▼ PFD: Yaw Damper Disconnect

YD
(yellow)

⇒ No response required.

The message YD shows in yellow when the Yaw Damper is disconnected for any reason. After 10 seconds of operation, the YD disconnect caution is automatically cancelled.

PFD – MISCOMPARE FLAGS



TPO4650_27

▼ PFD: Altitude Miscompare Flag

ALT (yellow)	⇒ The miscompare flag is a caution to the flight crew that the associated data may be in error. When the difference between the pilot-side and copilot-side pressure altitude values is not within the limit, the altitude miscompare flag shows on the PFD.

▼ PFD: Airspeed Miscompare Flag

IAS (yellow)	⇒ The miscompare flag is a caution to the flight crew that the associated data may be in error.

When the difference between the pilot-side and copilot-side airspeed values is not within the limit, the indicated airspeed miscompare flag shows on the PFD.

▼ PFD: Attitude Miscompare Flag

PIT, ROL, or ATT
(yellow)

⇒ The miscompare flag is a caution to the flight crew that the associated data may be in error.

When the difference between the pilot-side and copilot-side attitude data is not within the limit, the attitude miscompare flag shows on the PFD. The flag is PIT when the pitch data is out of limits. The flag is ROL when roll data is out of limits. The flag is ATT when both the pitch and roll data is out of limits.

▼ PFD: Mach Miscompare Flag

MACH
(yellow)

⇒ The miscompare flag is a caution to the flight crew that the associated data may be in error.

When the difference between the pilot-side and copilot-side Mach values is greater than the limit, the Mach miscompare flag shows on the PFD.

▼ PFD: Heading Miscompare Flag

HDG
(yellow)

⇒ The miscompare flag is a caution to the flight crew that the associated data may be in error.

When the heading compare function is active and the difference between the pilot-side and copilot-side heading values is not within the limit, the heading data miscompare flag shows on the PFD.

▼ PFD: Localizer Miscompare Flag

LOC
(yellow)

⇒ The miscompare flag is a caution to the flight crew that the associated data may be in error.

When the localizer compare function is active and the difference between the pilot-side and copilot-side localizer deviation is not within the limit, the localizer deviation miscompare flag shows on the PFD.

▼ PFD: Flight Director Miscompare Flag

FD
(yellow)

⇒ The miscompare flag is a caution to the flight crew that the associated data may be in error.

When the difference between the pilot-side and copilot-side flight director steering commands is not within the limit, the flight director miscompare flag shows on the PFD.

▼ PFD: Glideslope Miscompare Flag

GS
(yellow)

⇒ The miscompare flag is a caution to the flight crew that the associated data may be in error.

When the glideslope compare function is active and the difference between the pilot-side and copilot-side glideslope deviation is not within the limit, the glideslope deviation miscompare flag GS shows on the PFD.

▼ PFD: Radio Altitude Miscompare Flag

RA
(yellow)

⇒ The miscompare flag is a caution to the flight crew that the associated data may be in error.

When the radio altitude compare function is active and the difference between the pilot-side and copilot-side radio altitude is not within the limit, the radio altitude miscompare flag shows on the PFD.

▼ PFD: Vspeed Miscompare Flag

VSPD
(yellow)

⇒ Check the values posted against those on the CDU Vspeed pages. If the values match, send the Vspeeds again to attempt to clear the flag. If unable to clear the flag, set the VSPEEDS manually using the DCP.

When the V_{SPEED} compare function is active and the difference between the pilot-side and copilot-side radio V_{SPEED} data is not within the limit, the V_{SPEED} miscompare flag shows on the PFD. The flag is also displayed if the PFD received VSPEED data from only one of the two active FMS.

▼ PFD: FMS COMPARATOR FLAG

FMS
(yellow)

⇒ The LPV approach cannot be continued. Execute a Go Around. A different type of approach may need to be selected.

During an LPV approach, each AFD compares the FMS computed and displayed lateral deviaton. When the displayed lateral deviations from each AFD do not match within the comparator limits, a yellow FMS comparator flag is displayed to the left of the lateral deviation scale in the ADI.

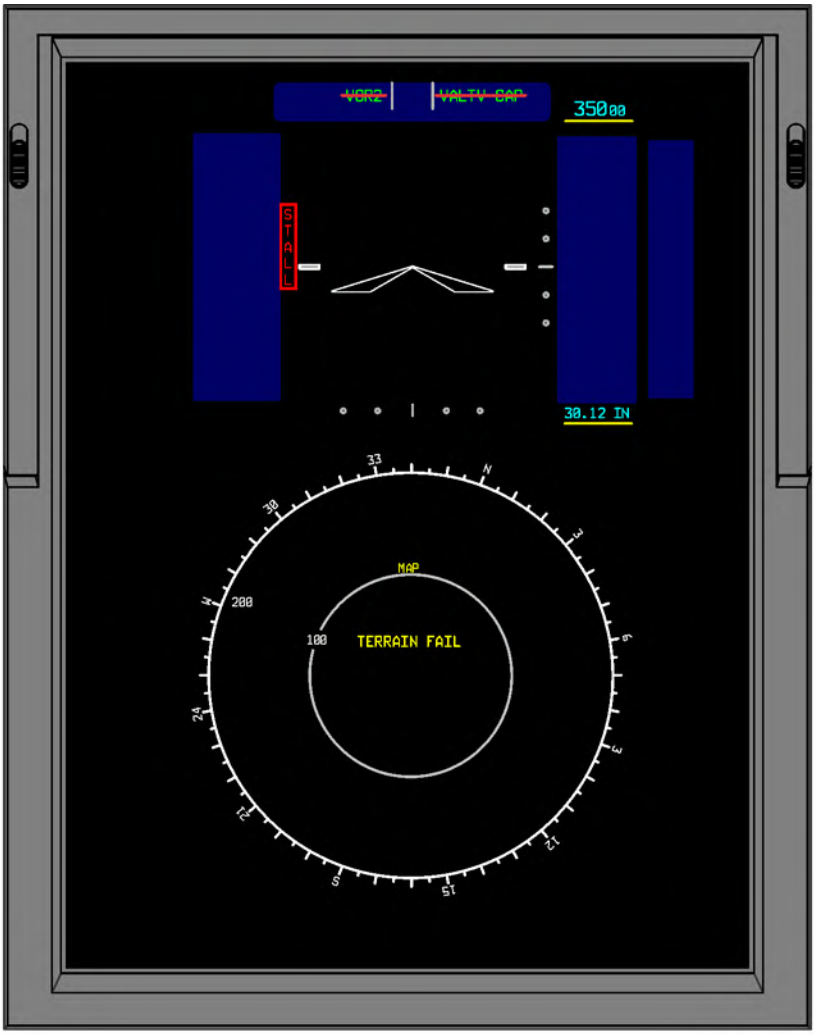
▼ PFD: VNAV COMPARATOR FLAG

VNV
(yellow)

⇒ The LPV approach cannot be continued. Execute a Go Around. A different type of approach may need to be selected.

During an LPV approach, each AFD compares the FMS computed and displayed vertical deviatons. When the displayed lateral deviations from each AFD do not match within the comparator limits, a yellow VNAV comparator flag is displayed at the top of the vertical deviation scale in the ADI.

PFD – CAUTION FLAGS



TPO4650_31

▼ PFD: AFCS Outer Loop Fault

AFCS Outer Loop Fault (Red line is displayed through the affected captured lateral or vertical mode)	⇒ A different navigation source, sensor or flight director mode must be selected to continue to use the flight director on the affected side. If the data required for the active lateral or vertical flight director mode(s) is failed or missing (for example, no valid VOR signal when NAV mode is active and VOR is active NAV source), the failure is annunciated by display of the red FD flag, removal of the flight director cue(s) and by display of a red line through the affected active mode.

▼ PFD/MFD: Auto Tilt State Comparator

Auto Tilt State Comparator	⇒ The comparator flag is a caution to the flight crew that the associated data may be in error. When Auto Tilt is active, the A suffix at the end of the tilt value changes to yellow whenever the Auto Tilt state (on or off) commanded does not match the Auto Tilt state received by the Weather Radar.

▼ PFD: Barometric Pressure Crosscheck

Barometric Pressure Setting Crosscheck	⇒ Set the barometric pressure setting on both sides to the correct value.

If the barometric pressure settings for the pilot and copilot-side are different by more than 0.02 inches of mercury, a yellow line is drawn beneath the barometric pressure digits/legend display.

▼ PFD: Preselect Altitude Flag

Preselect Altitude Flag	⇒ No response required. If a PFD detects the Preselect Altitude from the cross-side PFD is not received, the Preselect Altitude readout is underlined in yellow. If one PFD detects its Preselect Altitude is changing, and it is not receiving a Preselect Altitude change from the cross-side PFD, the Preselect Altitude readout shows with a flashing yellow underline. The yellow underline flashes for 5 seconds and then shows steady. The Preselect altitude bug on the altitude scale is removed when the Preselect Altitude Readout underline is flashing yellow.

▼ PFD: STALL Warning

Stall Protection System STALL Warning Message	⇒ No response required. The Stall Protection System STALL annunciation shows in red to the left of the attitude ball when a stall condition is detected by the Stall Protection Computer. The flight deck aural stall horn also sounds and the stick shaker activates.

▼ PFD/MFD: Terrain Not Available Flag

TERRAIN NOT AVAILABLE	⇒ No response required. The message TERRAIN NOT AVAILABLE shows in yellow on the PFD and MFD when the TAWS is reporting Terrain Awareness not available and not failed. A TAWS TERR NOT AVAIL CAS Message (advisory) shows when terrain awareness not available is asserted. The graphical terrain display is removed when TERRAIN NOT AVAILABLE is in view.

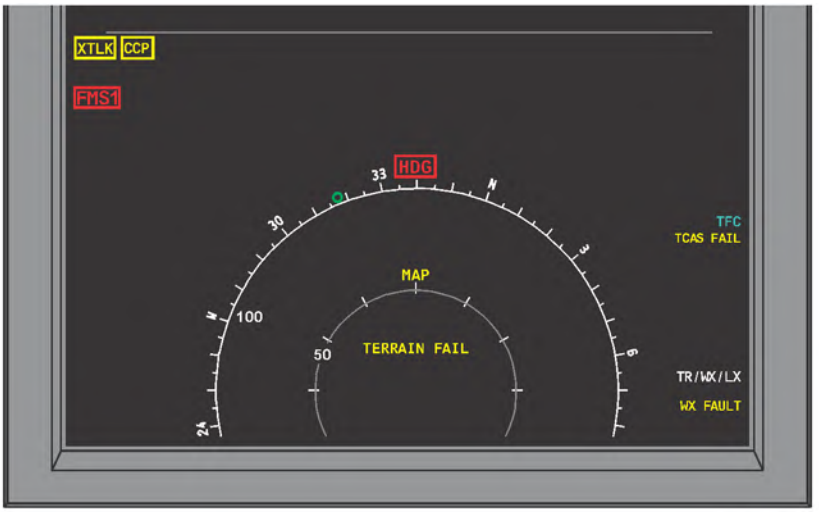
▲

▼ PFD: LOC and G/S Excessive Deviation Indication

LOC and G/S Excessive Deviation Indications	⇒ No response is required. A LOC excessive deviation is indicated by flashing the LOC deviation pointer and turning the pointer yellow for the duration of the excessive deviation. A Glideslope (GS) excessive deviation is indicated by flashing the GS deviation pointer and turning the pointer yellow for the duration of the excessive duration. These indications are only available during a Category 2 (CAT2) approach when the on-side GS is captured and the radio altitude is between 600 feet and RA MIN or 100 feet whichever is higher. The requirements for radio tuning, active NAV source selection, and AFCS mode selection that apply to establish the monitored approach are as follows: • Left PFD active NAV source is LOC1, right PFD active NAV source is LOC2, and both sides LOC frequencies are identical and valid.

- RA MIN Alert is set and RA MIN is set to less than 200 feet.
- Back Course mode not active.
- Go-Around mode not active.

MFD – FLAGS



TPO4650_30

▼ MFD: Cursor Control Panel Flag

CCP (yellow)	⇒ No response required. The cursor control panel flag shows on the MFD when the CCP is reported failed. The flag flashes for 5 seconds and then shows steady.

▼ MFD: Crosstalk Flag

XTLK
(yellow)

⇒ No response required.

The cross-talk flag shows on the MFD when the MFD loses communication with any other PFD or MFD. The flag flashes for 5 seconds and then shows steady. The message does not show for MFD that have been deselected by way of pilot reversion.

▼ MFD: Heading Flag

HDG
(red)

⇒ Select the cross-side attitude/heading source.

The heading flag shows on the MFD when heading data is missing or invalid. The compass card is positioned north-up and the lubber line is removed when the HDG flag is in view. The flag flashes for 5 seconds and then shows steady.

▼ MFD: TCAS Flag

TCAS FAIL
(yellow)

⇒ No response required.

The TCAS flag shows on the PFD when the TCAS system is reported as failed. The message also shows for failures related exclusively to the traffic display when TFC is selected.

▼ MFD: Resolution Advisory Flag

NO VSI RA
(yellow)

⇒ No response required.

The vertical speed resolution advisory flag shows on the PFD when the TCAS vertical resolution advisory information is failed. The flag also shows when vertical speed is not in view on the PFD.

▼ MFD: Weather Radar Flag

WX FAULT
(yellow)

⇒ No response required.

The Weather Radar fault flag shows on the PFD when the Weather Radar reports itself failed. The flag replaces the Tilt/USTB message. The flag flashes for 5 seconds and then shows steady. When the flag is in view, the Weather Radar display may be degraded in a manner that is not obvious to the pilot. Some faults may cause the Weather Radar to cease providing reflectivity information to the display.

MFD – MISCOMPARE FLAGS



TPO4650_13

▼ MFD: N1 Miscompare Flag

N1
(yellow)

⇒ The miscompare flag is a caution to the flight crew that the associated data may be in error.

When the difference between the pilot-side and copilot-side N1 value is not within the limit, the N1 miscompare flag shows in place of the N1 legend. An arrow on the left or right side of the flag points to the engine with the miscompare condition. The flag flashes for 5 seconds and then shows steady.

▼ MFD: ITT Miscompare Flag

ITT
(yellow)

⇒ The miscompare flag is a caution to the flight crew that the associated data may be in error.

When the difference between the pilot-side and copilot-side ITT value is not within the limit, the ITT miscompare flag shows in place of the ITT legend. An arrow on the left or right side of the flag points to the engine with the miscompare condition. The flag flashes for 5 seconds and then shows steady.

▼ MFD: N2 Miscompare Flag

N2
(yellow)

⇒ The miscompare flag is a caution to the flight crew that the associated data may be in error.

When the difference between the pilot-side and copilot-side N2 value is not within the limit, the N2 miscompare flag shows in place of the N2 legend. An arrow on the left or right side of the flag points to the engine with the miscompare condition. The flag flashes for 5 seconds and then shows steady.

MFD – FORMAT QUICK ACCESS MESSAGES

▼ MFD: Message Field

CANNOT STORE	Shows when a fault is detected that prevents the system from storing the current MFD values/states when a CCP MEM1, MEM2, or MEM3 quick access button is pushed.
STORE	⇒ Release the button to complete the STORE action. Shows when a CCP MEM1, MEM2, or MEM3 quick access button is pushed for more than 3 seconds.
STORE COMPLETE	Shows when a CCP MEM1, MEM2, or MEM3 quick access button is released and the current MFD display parameters have successfully been stored.
RECALL COMPLETE	Shows when a recall action is complete.
RECALL FAULT	Shows when a recall action cannot be achieved due to a fault.

MFD – DISPLAY MESSAGE WINDOW

▼ MFD: Display Message Window

FSU INOP (Yellow)	Shows when a fault is detected in the FSU.
CONFIG UPDATED	Shows when the FSU configuration is changed. This message is cancelled when the FILE SERVER CONFIGURATION page is in view.
CONFIG ERROR (Yellow)	Shows when the FSU reports that an EAK within the system cannot be recognized in the FSU configuration table, or the FSU Node software part number does not match the one expected by the FSA.

MFD – E-CHART CYCLE MESSAGES

▼ MFD: Chart Status Field

OUT OF DATE	⇒ Chart may still be usable, but data may no longer be valid. Use with caution. Shows when the selected E-Chart is out of date.
CHECK DATE	⇒ If satellites are not in view (for example, while the aircraft is in a hangar) move the aircraft to another location. Shows when the system date or time is invalid. The message may appear for a short period of time while the GNSS tries to acquire satellites. If loss of system time or date is temporary, the message will clear once the system time/date has been regained.

MFD – E-CHART FAULT MESSAGES

▼ MFD: Message Field

CHART FAULT	Shows when the chart data is corrupted or the data cannot be retrieved.
NO CHART AVAILABLE OR SELECTED	Shows when no chart has yet to be selected or the previously selected chart was associated with an airport that has been removed from the Chart Main Index menu, when the display chart state is entered. Shows when departure for the destination airport is selected and the Shortcut for the procedure is populated in the Origin Airport Departure field instead of in the Destination Airport area. The Origin Airport Departure is overwritten.

MFD – E-CHART MESSAGES

▼ MFD: Message Window

PROCESSING REQUEST	Shows when the requested chart is being retrieved.

▼ MFD: Chart Main Index Page: Charts

NO CHART COVERAGE AVAILABLE	Shows when the origin, destination, alternate, or other airport is not covered by the regions contained within the current chart subscription.

▼ MFD: NOTAM Page:

PROCESSING REQUEST	Shows when the requested NOTAM data is being retrieved.
NOTAM DATA FAULT	Shows when the requested NOTAM data is corrupted of the data cannot be retrieved.

MFD – DATALINK GRAPHICAL WEATHER MESSAGES (GWX-5000 KEY)

▼ MFD: GWX Message Field

GWX FAULT	Shows when the GWX data is corrupted or the data cannot be retrieved.
NO IMAGE AVAILABLE	Shows when the GWX only mode is selected and: • No GWX image has been selected yet • The previously selected GWX image is no longer available • There are no images for display.
PROCESSING REQUEST	Shows when the system is retrieving a requested image.

MFD – XM GRAPHICAL WEATHER MESSAGES (GWX-3002 KEY)

▼ MFD: GWX Message Field

ORIGIN OR DESTINATION UNAVAILABLE	Shows when the origin or destination airports are not populated in the flight plan and A/C FLIGHT INFO selection has been made
NOT RECEIVING DATA	Shows when there is no satellite coverage or the information received from the satellite is invalid.
GWX FAULT	Shows when the GWX data is corrupted or the data cannot be retrieved.
SUBSCRIPTION EXPIRED	⇒ Move the aircraft out of the hangar. Shows when one of the items that follow occurs: • Shows when the XM receiver indicates that the aircraft owner's subscription to XM weather products has expired. • Shows when the XM Receiver does not receive a valid satellite signal at power-up (for example, while the aircraft is in a hangar). Without satellite reception, the XM Receiver is unable to authorize an XM Weather subscription. The message clears when the XM Receiver receives a valid satellite signal and completes subscription authorization processing.

SIGMET OVERLAY INCOMPLETE CHECK TEXT	Shows when the SIGMET overlay is selected and the graphical plot data for a specific SIGMET is not received by the datalink. (NO GRAPHICAL DATA) shows in the Weather Report sub-heading for the associated SIGMET.

▼ MFD: SIGMET Report Page

NO GRAPHICAL SIGMET	Shows in the upper right corner of the SIGMET Report page when no graphical SIGMET data is available.

▼ MFD: IMAGES & OVERLAYS MENU ACQUIRING IMAGES

ACQUIRING IMAGES	Shows next to any image (US ANIMATED NEXRAD, CAN ANIMATED NEXRAD, SATELLITE, US WINDS ALOFT, CAN WINDS ALOFT, ICING, TURBULENCE) when not enough images are available to display the item. The menu cursor indicator skips over any image item when there are not enough images to display.
AVAILABLE	Shows next to each item which has enough images available to display the item. For items that provide images at different altitudes, images for some altitudes will not be available.
AVAILABLE (ALL ALTITUDES)	Shows next to items that allow selection of different altitudes for display of the image (SATELLITE, US WINDS ALOFT, CAN WINDS ALOFT, ICING and TURBULENCE) when images are available at all altitudes.

MFD – E-MAP MESSAGES

▼ MFD: E-Map Overlay Fault Messages

AIRSPACE FAULT	Shows when the E-Map data is corrupted or the data cannot be retrieved.
AIRWAYS FAULT	Shows when the E-Map data is corrupted or the data cannot be retrieved.
GEO-POL FAULT	Shows when the E-Map data is corrupted or the data cannot be retrieved.
NO AIRSPACE AVAILABLE	Shows when a condition exists that prevents the proper display of the data.
NO AIRWAYS AVAILABLE	Shows when a condition exists that prevents the proper display of the data.
NO GEO-POL AVAILABLE	Shows when a condition exists that prevents the proper display of the data.
PROCESSING REQUEST	Shows when the requested E-Map overlay is being retrieved.
TFR FAULT	Shows when graphical TFR data is corrupted or the data cannot be retrieved.
NO TFRS AVAILABLE	Shows when a condition exists that prevents proper display of the data.

▼ MFD: GWX Status Messages

NEXRAD AGE HH:MM (White)	Shows when the requested USA or CAN NEXRAD image is available and the image has a valid image time. The HH:MM readout is the age of the NEXRAD image in hours and minutes.

NEXRAD AGE HH:MM (Yellow)	Shows with the age readout in yellow within a yellow box when the product is old. Images are considered old when the current time exceeds the product's next available time by more than 30 minutes.
NO USA NEXRAD AVAILABLE	Shows when the GWX overlay is selected on, US NEXRAD overlay is selected, and there is no USA NEXRAD image stored (no data available).
NO CAN NEXRAD AVAILABLE	Shows when the GWX overlay is selected on, CAN NEXRAD overlay is selected, and there is no CAN NEXRAD image stored (no data available).

▼ MFD: GWX Fault Messages

— — — — (four dashes)	Shows in place of the image age readout when the current time is missing or invalid or the image valid time is not available.
GWX FAULT	Shows when the GWX data is corrupted or the data cannot be retrieved.

MFD – DATABASE MESSAGES

▼ MFD: DATABASE EFFECTIVITY Page: Database Messages

SEE SUBSCRIPTIONS PAGE	Shows in place of the last region listed in the database details window when the number of coverage regions exceeds 10.
NO REGIONS AVAILABLE FOR CURRENT SUBSCRIPTION	Shows in the coverage region portion of the database details window when there are no coverage regions.
CHECK DATABASE STATUS	⇒ Go to the DATABASE EFFECTIVITY page to acknowledge the message. Shows in yellow near the bottom of the MFD when a database has been determined to be expired or a database effectivity cannot be determined (due to the loss of system time). This message will show on every MFD format until it is cleared or acknowledged by viewing the DATABASE EFFECTIVITY page. The message shows only when the aircraft is on the ground.
PROCESSING REQUEST	Shows in the list field when the system is retrieving data for display.
DATABASE EFFECTIVITY FAULT	Shows in the list field instead of list data for IFIS related databases, when IFIS database effectivity data is not available or is invalid.

▼ MFD: DATABASE EFFECTIVITY Page: Database Status Messages

CURRENT	Shows when the associated installed database is current.

NOT CURRENT	Shows when the associated installed database is not current.

MFD – EVS MESSAGES

▼ MFD: EVS CAS Caution Messages: Enhanced Vision System Caution Messages

EVS FAIL (Yellow)	⇒ Turn EVS off. This message displays when an Infrared Sensor Unit (ISU) failure occurs.
EVS HEAT FAIL (Yellow)	⇒ Switch EVS off if image is degraded or unusable. If the system is left on, avoid icing conditions. This message displays when the ISU determines that heat is not being applied to the unit when window heat is turned on.
EVS HEAT OVHT (Yellow)	⇒ Switch EVS off if image is degraded or unusable. If the system is left on, avoid icing conditions. EVS HEAT OVHT displays when the ISU determines that too much heat is being applied.
EVS IR SENSOR OVHT (Yellow)	⇒ Switch EVS off to allow sensor to cool. This message displays when and ISU is in thermal shutdown state to prevent damage to components.

▼ MFD EVS CAS Advisory Messages: Enhanced Vision System Advisory Messages

EVS CAL (Green)	EVS CAL displays when an ISU is being calibrated.

EVS ON (Green)	EVS ON displays when power is applied to the ISU.

▼ MFD EVS CAS Status Messages: Enhanced Vision System Status Messages

EVS DEFOG FAULT (White)	EVS DEFOG displays when the ISU determines that the Forward Looking Infrared (FLIR) window heat fails.

MFD – SUBSCRIPTION MESSAGES

▼ MFD: CHART SUBSCRIPTION Page: Regions Enabled Messages

NO REGIONS AVAILABLE FOR CURRENT SUBSCRIPTION	Shows when no chart regions are associated with the subscription serial number.
PROCESSING REQUEST	Shows when the system is retrieving data for display.
SUBSCRIPTION FAULT	Shows when a regions enabled data fault is reported.

▼ MFD: CHART SUBSCRIPTION Page: Chart Subscription Messages

ACCEPTED KEY	Shows next to the associated access code data entry field when a valid access code is entered.
INVALID ENTRY	Shows in yellow next to the associated access code data entry field when an invalid access code is entered. This message is removed when a valid access code is entered.
PROCESSING	Shows in white when the system is processing access code message data.
CODE FAULT	Shows in white when access code message data is not received.

MFD – FSU CONFIGURATION MESSAGES

▼ MFD: FILE SERVER CONFIGURATION Page: Priced Option List Field Messages

NO PURCHASED OPTIONS	Shows centered in place of the priced option list when no priced option information is available.
PROCESSING REQUEST	Shows when the system is retrieving data for display.
FILE SERVER CONFIGURATION FAULT	Shows when a file server configuration data fault is reported.

▼ MFD: FILE SERVER CONFIGURATION Page: EAK Data Entry Field Messages

ENTRY DISABLED IN AIR	Shows while the system is airborne. The EAK Data entry is disabled when this message is in view.
KEY ACCEPTED	Shows when the entered EAK is valid. Alphanumeric data entry mode is exited upon display of this message.
INVALID KEY	Shows when the entered Software Key is invalid. Alphanumeric data entry mode is exited upon display of this message. The message is cleared when a valid EAK is entered.
PROCESSING	Shows in white when the system is processing software key data.
CONFIG FAULT	Shows in white when data Entry Status Field data is not received.

▼ MFD: FILE SERVER CONFIGURATION Page: Software Update Status Field Messages

CYCLE FSU POWER TO COMPLETE SOFTWARE UP-DATES	Shows when a valid EAK has been entered, but the associated FSU function has not yet been enabled. The software mismatch message has priority over this message.
FSA / NODE SOFT-WARE ARE INCOM-PATIBLE	Shows when there is a mismatch between the Node S/W part number that is contained in the FSA configuration file (part of FSA load set) and the actual Node S/W part number. This message has priority over the cycle FSU power message.

CDU – ABNORMAL OPERATION MESSAGES

▼ CDU: NO INPUT DATA <XXXXX>

NO INPUT DATA <XXXXX> where XXXXXX may be FMS 1, FMS 2, FMS 3, DL, DBU, HGS, SATCOM, or it may be blank between the carets.

The NO INPUT DATA <XXXXX> message is posted by the CDU when the function, application or LRU that was driving the CDU page is no longer available. With the exception of transient conditions or multiple failures, the "string" value will be the same as the corresponding MCDU INDEX prompt label (FMS 1/2/3, DL, DBU, HGS, SATCOM) that accesses to the page that was displayed at the time of the failure.

In every case that the NO INPUT DATA <XXXXX> message is shown, the appropriate flight crew action is to select the IDX key, which will result in the display of either the FMS INDEX page if the FMC is still available, or the MCDU INDEX page if the FMC is not available. From these pages, the flight crew would determine which functions are still available from that particular CDU.

▲

PFD/MFD – ABNORMAL OPERATION MESSAGES

▼ PFD: RADAR ON

RADAR ON	⇒ Select Weather Radar OFF. The message RADAR ON shows at the bottom center of the display when the Weather Radar is transmitting with weight-on-wheels. When any Weather Radar mode other than STBY or TEST is selected and it is more than 65 seconds after an air-to-ground transition (with weight-on-wheels), the Weather Radar turns off. If the Weather Radar is turned on, with weight-on-wheels and the Weather Radar overlay is not selected, the RADAR ON message shows. The message flashes for 5 seconds and then shows steady. The message also shows when any Weather Radar mode other than STBY or TEST is selected, it is more than 3 seconds after power-up, and there is weight-on-wheels. The RADAR ON message does not show on the MFD.

▲

▼ PFD: SVS INOP

SVS INOP	⇒ Select the opposite side as the transfer side using the FCP XFR switch to see if condition clears Shows below the left tape when the SVS is selected on and any of the following occurs: • On initial power up while the system does a self-test. • There is an AFD/SVC miscompare.

- When there is a SVC terrain database failure.
- When the SVC in missing required input data. in this case, a change of XFR side may clear the condition.

▼ PFD: SVS RWY INOP

SVS RWY INOP

Shows below the left tape when there is a SVC airport/runway database failure and the SVS is on. This shows in the same place as SVS INOP. If both messages activate, only SVS INOP shows.

▼ PFD/MFD: RADAR RANGE XXXNM

RADAR RANGE XXXNM

⇒ No response required.

The message RADAR RANGE XXXNM shows in white at the bottom center of the display if the display range and Weather Radar range do not agree. The XXXNM portion of the message is the Weather Radar range currently reported from the Weather Radar. The graphical Weather Radar returns are removed if the display and Weather Radar range do not agree.

NOTE: In most failure conditions, the XXXNM range will be sourced from the cross side cockpit displays, and setting the ranges equal will remove the message and restore the overlay to the display.

▼ PFD/MFD: TERRAIN RANGE XXXNM

TERRAIN RANGE XXXNM	⇒ No response required. The message TERRAIN RANGE XXXNM shows in white at the bottom center of the display if the AFD range and TAWS range do not agree. The XXXNM portion of the message is the TAWS range currently reported from the TAWS Computer. The graphical Terrain display is removed if the AFD and TAWS range do not agree. NOTE: In most failure conditions, the XXXNM range will be sourced from the cross side cockpit displays, and setting the ranges equal will remove the message and restore the overlay to the display.

▼ PFD/MFD: MAP RANGE XXXNM

MAP RANGE XXXNM	⇒ No response required. The message MAP RANGE XXXNM shows in white at the bottom center of the display if FMS Map range does not agree with the range on the display. The XXX portion of the message represents the actual range the FMS Map is set to. The MAP symbology is removed when MAP RANGE XXXNM shows. As long as the condition that created the range disagreement persists the MAP symbology generated by the FMS will continue to be removed regardless of the range selection.

NOTE: When the weather radar or terrain overlays are selected on, the RADAR RANGE XXXNM or TERRAIN RANGE XXXNM annunciations have a higher priority than the MAP RANGE XXXNM annunciation.

NOTE: When the map range is set to the cross cockpit value, weather radar or terrain overlay data will be displayed if selected on. TCAS traffic overlay data will remain displayed regardless of range selection.

▼ PFD/MFD: MAP INCOMPLETE

MAP INCOMPLETE

⇒ No response required.

The message MAP INCOMPLETE shows in white at the bottom center of the display if the FMS cannot provide all the requested map data, or if the PFD cannot display all the map data sent by the FMS.

▼ PFD/MFD: NO FLIGHT PLAN

NO FLIGHT PLAN

⇒ Make sure an active flight plan is loaded into the FMS.

The message NO FLIGHT PLAN shows in white above the aircraft symbol if the active NAV source is FMS and no flight plan exists. The message flashes for 5 seconds and then shows steady.

▼ PFD/MFD: DISCONTINUITY

DISCONTINUITY

⇒ Check the FMS flight plan/active legs to delete the discontinuity in the flight plan.

The message DISCONTINUITY shows in white above the aircraft symbol if the FMS is the active NAV source and the FMS flight plan has reached a discontinuity. The message flashes for 5 seconds and then shows steady.

▼ PFD/MFD: Tilt Compare Flag

Tilt Compare Flag

⇒ The tilt compare flag is a caution to the flight crew that the associated data may be in error.

If the tilt value commanded and the tilt value returned from the Weather Radar disagree for more than 5 seconds, the Tilt value changes to yellow if the Weather Radar overlay is in view.

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APPENDIX A

FANS AND LINK 2000+ INTRODUCTION

The Future Air Navigation System (FANS) and Link 2000+ applications are Controller/Pilot Datalink Communication (CPDLC) systems for communicating between the aircraft and an Air Traffic Control (ATC) and is available on all aircraft equipped with a CPDLC interface.

NOTE

This FANS and Link 2000+ operators guide appendix is not a training manual. It is an overview covering operations specific to the FANS and Link 2000+ on the Challenger 605/650. General operation/use of FANS and Link2000+ is covered in OG 523-0822435 ATN and FANS Interlocked CPDLC Applications.

NOTE

FANS refers to the oceanic and non-European CPDLC remote areas portion of the datalink system. Link 2000+ refers to European CPDLC regulatory requirements. In this appendix, the term CPDLC implies both FANS and Link 2000+ unless specified.

CAUTION

Some aircraft operators can have special procedures that are different from those covered in this generic CPDLC appendix. Refer to the applicable Aircraft Flight Manual (AFM) for instructions specific to the aircraft.

CAUTION

Systems and components related to the CPDLC exhibit a very high degree of functional integrity. However, users must recognize that it is not practical to provide monitoring for all conceivable system functions and failures. Erroneous operation could occur without a fault indication. The operator has the responsibility to detect such an occurrence by means of cross-checks with redundant or correlated information available in the flight deck.

CAUTION

Content, display illustrations, and values used in this appendix may be generic and used for illustrative purposes only. Specific aircraft operating parameters may not be represented in this chapter.

OVERVIEW

The Controller-Pilot Data Link Communication (CPDLC) application allows digital communication between the aircraft and an Air Traffic Control (ATC) center through either the Aeronautical Telecommunications Network (ATN) or the Aircraft Communications Addressing and Reporting System (ACARS). Routine air traffic instructions and requests are transmitted through text messages between the aircraft and the ATC Center, precluding the need for voice communications.

The Link 2000+ function is used in European airspace only. The European system provides VHF messaging capability via the ATN for aircraft that operate above 28 500 feet MSL.

The FANS function is primarily used for oceanic and remote regions using Satellite Communication SATCOM, and VHF methods of datalink via ACARS. It contains over 200 messages similar to the Link 2000+ function in format and usage.

NOTE

For aircraft specific information refer to the applicable operator's manuals, e.g., FCOM or AFM.

COMPONENTS

System configurations for CPDLC functions include: Communication Management Unit (CMU), Control Display Unit (CDU), Flight Management Computer (FMC), Adaptive Fight Display (AFD), Data Concentrator Unit (DCU), External Compensation Unit (ECU), SATCOM, VHF Data Link (VDL) Capable VHF radio and optional printer.

CONTROLS AND INDICATORS (C&I)

All controls and associated indicators used by CPDLC are generic to the specific aircraft.

The CDU is the primary CPDLC interface for the flight crew. Function keys shown on the graphics in this appendix are visually similar to most CDU displays and are generic in nature. Display pages used in the graphics in this appendix are visually similar between each CDU and are generic in nature. On installations with more than one CDU, data is automatically synced with other units.

NOTE

Descriptions of CDU operations found in this guide are generic.

NOTE

Pages shown in this appendix are typical of what may be shown in a specific aircraft. They may not be exactly what is shown for every individual instance. Refer to the specific aircraft Flight Management System (FMS) operator guide and Aircraft Flight Manual (AFM) for specific system function keys/pages information.

The Link 2000+/DATALINK menu page can be shown by accomplishing the following:

1. INDEX function key – push (shows the INDEX page).

2. DATALINK LSK – push (shows the DL APPLICATION MENU page).
3. ATN CPDLC LSK – push (shows the ATN —CPLDC MENU page).

The FANS MAIN MENU can be shown by accomplishing the following:

1. INDEX function key – push (shows the INDEX page).
2. FANS LSK – push (shows the FANS MAIN MENU page).

NOTE

Graphics used to show the following Controls and Indicators (C&I) are not shown under the following subheadings. Only generic text overviews are supplied.

CDU LINE SELECT AND DISPLAY PAGE SELECTION KEYS

All LSKs are accessed on the main CDU page. The following callout applies.

LSKs:

- Are used to make selections.
- Each LSK is numbered to facilitate location and ease of operation. L1 (top left) through L6 (bottom left) are on the left side of the CDU. R1 (top right) through R6 (bottom right) are on the right side of the CDU. LSKs are used to:
  - Transfer data from the scratchpad to the data line.
  - Copy data from a data line to the scratchpad.
  - Select an option/mode from a menu of available options.

CDU DISPLAY PAGES

When entering or changing information on a CDU datalink page, the change is automatically applied, and shown, the other CDU datalink pages. This happens whether or not the datalink page is currently shown on another CDU.

All display pages can show up to 15 lines of text and 24 characters per line. The following callout applies:

Font:

- Two font sizes may appear on any data line. Data supplied by any aircraft system shows as a smaller font size.
- Data entered by an operator shows as a larger font size.

CDU SYMBOLS

Symbols appear next to a LSK. Selecting the LSK associated with the symbol advances the CDU display to that page. The following callouts apply:

SYMBOL	MEANING
< or >	Appears next to a LSK to indicate pushing the associated LSK changes the display to that display page.
* (asterisk)	Appears next to a LSK to indicate pushing the associated LSK activates an internal function. The page may change as a result of the selection.
↓ Arrow (pointed down)	Appears next to a LSK to indicate pushing the associated LSK steps through a selection list.
□ Square Box (Prompt)	Generally shown in multiples that indicate the field size, this symbol indicates the data field requires an entry to perform a specific function.
[ ] (Brackets)	Generally shown in multiples that indicate the field size, this symbol indicates the data field content is optional.

CDU COLORS

The CL605/650 is equipped with a color CDU. The following callouts apply:

White:

- Page and field titles
- Comments
- Page change text
- Scratchpad entry and error messages
- Inactive advisories

- Mandatory user entry (boxes).

Cyan:

- Optional user entry
- Function calls
- Active advisories
- Select lists.

Green:

- Information supplied by System
- No user interface.

CDU DATA FORMAT

The following applies to CDU data formats as they are used in this appendix:

DD/MM/YY or dd/mm/yy:

- Date: day, month, year
- Example: 31/12/10.

HH:MM or hh:mm:

- Amount of time: hours, minutes (HH = 0 - 23, and MM = 00 - 59)
- Example: 12:15.

NNN or nnn:

- Numbers (0 - 9)
- Example: 555.

SNNN:

- Signed number (plus + or minus −)
- Example: +555.

AAA or aaa:

- Letters (A - Z)
- Example: ABC.

XXX or xxx:

- Alphanumeric characters (0 - 9, A - Z)

- Example: A9B.

CDU DISPLAY PAGE GENERAL OPERATIONS

The following refers to general operation of the display pages:

- Enter data manually into data fields by using the keypad to type the desired data into the scratchpad. Then, transfer the information to the appropriate data field by pushing the LSK for that field.
- To select prefilled data (selection list), push the associated LSK as many times as needed to toggle or scroll through available options. A down or slew arrow indicates a selection list is available.
- The <RETURN LSK causes the CDU to show either the previous menu page or the previous function page, even if an active advisory activated the page that is currently shown.

Some display pages include a *PRINT select that allows the crew to print various information and messages. This selection is only shown if a printer is available and has a status of READY. When selected, the *PRINT prompt disappears for approximately five seconds to indicate the data has been sent to the printer. If the printer is not in READY status, the *PRINT select will be replaced by one of the following status indicators:

- PRN BUSY: Any of the following are true:
  - The printer is initializing.
  - The printer is running a functional test.
  - The printer is busy.
- PRN ERR: The printer buffer has been overrun or there is an incomplete message.
- PRN PAPR: The printer is out of paper.
- PRN DOOR: The printer door is open.
- PRN INOP: The printer is not broadcasting a signal to the system.

MANUAL DATA ENTRY RANGES AND SELECTIONS

The following displays the range or selection list choices for manual entry of data into each CPDLC data field. The following callouts apply:

ALT (Level):

- Altitude expressed in either feet or flight level. Feet: 25000, Flight Level: FL250.
- Single or Block Altitude.
- Shown as feet from 600 – 70 000, resolution of 10.
- Shown as FL from 30 – 700, resolution of 1.

ETA (Time):

- Estimated time of arrival
- Expressed as HHMM.

TIME:

- Time value
- Expressed as HHMM from, 0 — 2359.

FIX (Position):

- Navigation fix expressed in alphanumeric characters
- Shows from 1 – 5 characters.

AIRPORT:

- 4 letter ICAO Airport
- ID expressed in capital letters.

LAT/LON (Position):

- Latitude and longitude in whole degrees, degrees, minutes and seconds, or degrees and minutes
- Written as {N or S} 0–90 {E or W} 0–180 or {N or S} 0–89 00–59.99 {E or W} 0–180 00–59.99 or {N or S} 0–89 [0–59] {E or W} 0–180 [0–59]. (For example, N40E110 or N4004.76E11040.55).

NAVAID:

- NAVAID identifier expressed in alphanumeric characters
- Shows from 1 – 4 characters.

BEARING (Position):

- Magnetic bearing in degrees. Magnetic (default) or True
- Shows from 001 – 360 in one degree increments.

FAC DESIGN (Unit Name):

- Ground facility designation

- Shows from 4 – 8 alphanumeric characters.

FACILITY:

- Ground facility function
- Includes CENTER, CLEARANCE DELIVERY, APPROACH, DEPARTURE, TOWER, CONTROL, FINAL, RADIO, or GROUND CONTROL.

FAC ID (Unit Name):

- Ground facility identifier (name)
- 3 – 18 characters.

VHF (Frequency):

- Radio frequency
- Displays from 118.00 – 136.990
- Resolution of .005.

IAS (Speed):

- Indicated airspeed
- Displays from 0 – 400
- Resolution of 1.

MACH (Speed):

- Mach indicated
- Displays from (.61 – .92)
- Resolution of .001.

DISTANCE:

- IAS (FANS): Indicated airspeed from 70–380
- TAS/GS (FANS): True airspeed or groundspeed from 70–700. Add T for TAS or G for GS
- Distance in nautical miles
- Displays from 1 – 999.9 nautical miles or 1 — 1024 kilometers.

DIRECTION:

- LEFT, RIGHT, or EITHER
- Displays any or all.

FLT ID:

- Flight identification
- Displays from 2 – 8 alphanumeric characters.

LOGON TO:

- Data authority/ATC facility
- Displays from 4 – 8 alphanumeric characters.

ORIG STA:

- Origin airport/station
- Expressed as ICAO identifier.

DEST STA:

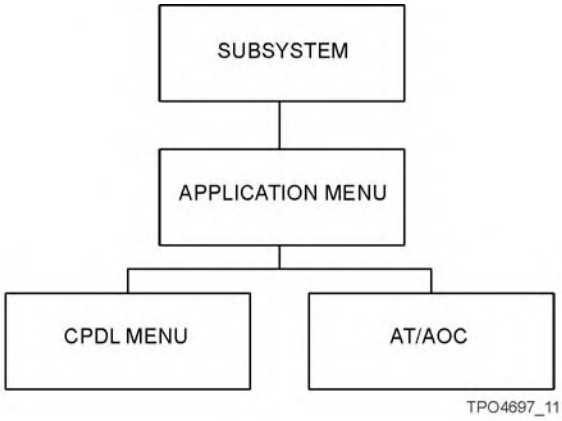
- Destination airport/station
- Expressed as ICAO identifier.

ETD (Time):

- Estimated time of departure
- Expressed as HHMM.
- Transition Entry: Written in 1 — 5 alphanumeric characters.
- Procedure Entry: Written in 1 — 6 alphanumeric characters.

MENUS AND DISPLAYS

Figure A-1 Link 2000+ Introduction



INTRODUCTION

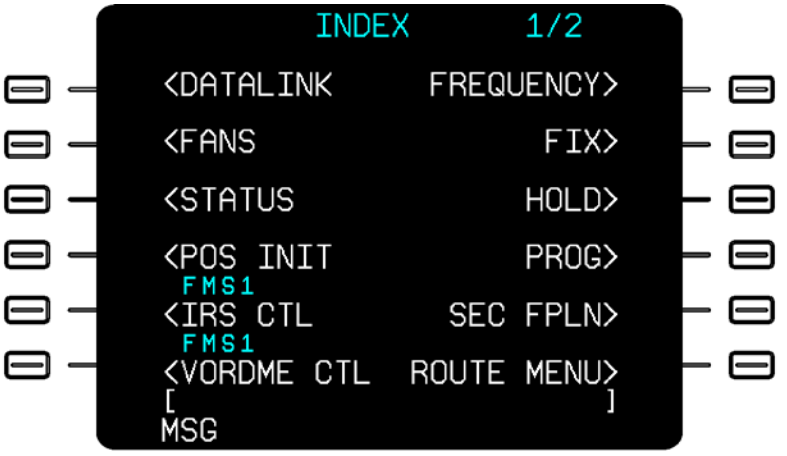
The Link 2000+ application pages follow the hierarchy of the system tree shown above. Selection of an active advisory may show a display page outside of the normal hierarchy, and the <RETURN prompt must be selected to remove that page before the normal hierarchy may be restored. The Air Operations Center (AOC) function is accessed through the DL Application menu. The FANS application pages start from the FANS MAIN MENU page, not the DL APPLICATION MENU page.

NOTE

Only major menus are shown.

INDEX PAGE

Figure A-2 CDU INDEX Page



The CDU INDEX page:

- Shows multiple selections
- Push the LSK to show that page.

DL–APPLICATION MENU PAGE

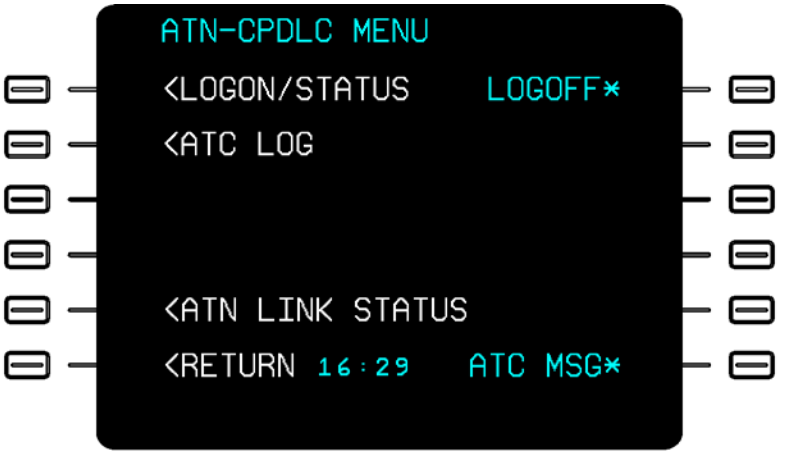
Figure A-3 APPLICATION MENU



To access ATN CPDLC push the ATN CPDLC LSK. The ATN CPDLC page shows. If an ATC MSG is pending, the operator can navigate to the message by selecting RLK6 (ATC MSG).

LINK 2000+ ATN-CPDLC MENU PAGE

Figure A-4 Link 2000+ ATN-CPDLC MENU

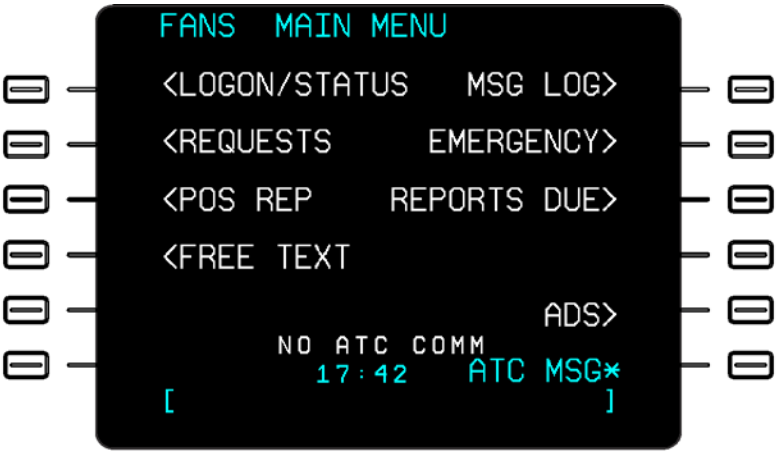


Push the ATN CPDLC LSK on the DL APPLICATION MENU to show the ATN CPDLC MENU page. ATN CPDLC MENU available options include:

- <LOGON / STATUS shows the current connection status, FLIGHT ID, current and next data authority, and attempted connection with the ground.
- <ATC LOG shows when the received message is shown.
- <ATN LINK STATUS shows the current ATN status.
- <REPORTS/REQUESTS LSK (only available after the LOGON is accomplished) shows available options.
- If an ATC MSG is pending, the operator can navigate to the message by selecting RLK6 (ATC MSG).
- If the LOGOFF is displayed, the operator can logoff ATN network by selecting RLSK1 (LOGOFF). The LOGOFF prompt will only be shown if the operator has previously completed ATN logon.

FANS MAIN MENU PAGE

Figure A-5 FANS MAIN MENU

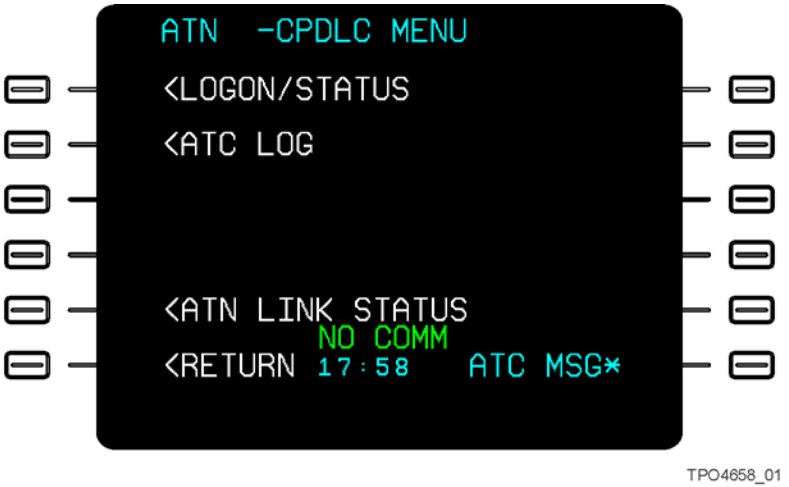


Push the FANS LSK on the INDEX page to show the FANS MAIN MENU page. The FANS MAIN MENU options include:

- <LOGON / STATUS shows the current connection status, FLIGHT ID, current and next data authority, and attempted connection with the ground.
- <REQUESTS shows the available requests that the pilot can send to ATC.
- <POS REP shows the position reporting page.
- <FREE TEXT shows a free text format that allows the pilot to send ATC information that is not available in the current messages.
- <MSG LOG shows a list of all messages to and from the aircraft and the responses to the messages.
- <EMERGENCY shows a list of emergency actions that the pilots can use to request from or to inform ATC.
- <REPORTS DUE shows a list of ATC-requested reports and options.
- <ADS shows whether the aircraft is connected to CPDLC.
- If an ATC MSG is pending, the operator can navigate to the message by selecting RLK6 (ATC MSG).

CPDLC – ATC MSG ANNUNCIATION

Figure A-6 Message Review



Receipt of a new message triggers the ATC MSG advisory. The Message (MSG) advisory alerts the crew that a new message is available that requires a response.

ATC LOG

The following tables are applicable to both the REPORTS/REQUESTS pages and on the ATC LOG page.

Uplink Message Status Annunciations

UPLINK MESSAGE STATUS	DESCRIPTION
NEW	An uplink message has not yet been viewed.
OPEN	An uplink message has been viewed but no response has been sent.

Downlink Message Status Annunciations

DOWNLINK MESSAGE STATUS	DESCRIPTION
PENDING	A downlink message was sent to the ground facility, but the ground facility has not returned Logical Acknowledgment (LACK) of the receipt of the message.
OPEN	A downlink message has an acknowledgement response from the ground but no direct response has been received.

MESSAGES

Active advisories alert the crew that:

- A new message has been received.
- An active advisory causes the initiation of a specific function.
- The new message requires attention.
- When more than one advisory is shown, the highest priority is activated first.

Active advisories are shown in the lower right corner of the display.

CPDLC MESSAGES, TYPICAL

CPDLC additional messages:

New Messages

NEW MESSAGES
ATN – NO COMM, ATN – IN COMM
LOGON – DATA CHANGE
LOGON – IN PROGRESS
LOGON – LOGGED OFF

CPDLC – CREW ALERTING MESSAGES

The following Cyan Crew Alerting System (CAS) messages associated with CPDLC operation are displayed in the CAS section of the MFD or PFD:

- DATALINK ATN CPDLC N indicates receipt of Link 2000+ ATC message. Accompanied by the datalink chime aural. Select the ATC MSG RLSK6 on any CDU ATN or Datalink page to review the message and clear the annunciation. If the CDU is not currently displaying an ATN or Datalink page, access a Datalink page as follows:
  - INDEX function key — Push.
  - DATALINK LSK — Push.
- DATALINK FANS CPDLC indicates receipt of FANS ATC message. Accompanied by the datalink chime aural. Select the ATC MSG LSK on the CDU FANS to review any message and clear the annunciation. If the CDU is not currently displaying a FANS page, access a FANS page as follows:
  - INDEX function key - Push
  - FANS LSK - Push
  - A FANS message that is received that cannot be displayed on a single page (longer than 5 lines on a CDU) must be completely viewed before the operator is given the option to ACPT (accept), STBY (standby), or REJ (reject). When viewing the message, the CDU NEXT and PREV keys cannot be used. The operator must use the soft keys NEXT and PREV to traverse through the entire message.

- DATALINK FAIL indicates total loss of datalink capability or loss of datalink (ATN\FANS\ACARS) via VHF. Revert to voice communications if unable to restore a LOGON.
- DATALINK FANS 1/A FAIL indicates loss of FANS capability. Revert to voice communications if unable to restore a LOGON.

NOTE

Refer to Rockwell Collins publication ATN and FANS Interlocked CPDLC Applications (523-0822435) for information on CPDLC operations.

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TAWS Colors B-1

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APPENDIX B

TAWS COLORS

The Terrain Awareness Warning System (TAWS) display colors represent various terrain elevations with respect to aircraft altitude. The terrain elevations associated with the different terrain display colors are as follows:

TAWS Colors

Color	Meaning
Solid Red	Terrain Threat Area - Warning
Solid Yellow	Terrain Threat Area - Caution
50% Red Dots	Terrain that is more than 2000 feet above aircraft altitude.
50% Yellow Dots	Terrain that is between 1000 and 2000 feet above aircraft altitude.
25% Yellow Dots	Terrain that is 500 feet (250 feet with gear down) below to 1000 feet above aircraft altitude (Peaks Only).
Solid Green	Shows only when no red or yellow terrain areas are within range on the display. Highest terrain not within 500 feet (250 feet with gear down) of aircraft altitude (Peaks Only).
50% Green Dots	Terrain that is 500 feet (250 feet with gear down) below to 1000 below aircraft altitude.
	Terrain that is the middle elevation band when there are no red or yellow terrain areas within range on the display (Peaks Only).

Color	Meaning
16% Green Dots	Terrain that is 1000 to 2000 feet below aircraft altitude.
	Terrain that is the lower elevation band when there are no red or yellow terrain areas within range on the display (Peaks Only).
Black	No significant terrain.
16% Cyan	Terrain Elevation equal to 0 feet MSL. (Peaks Only requires compatible display.)
Magenta Dots	Unknown terrain.

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IFIS Menu Navigation C-1

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APPENDIX C

IFIS MENU NAVIGATION

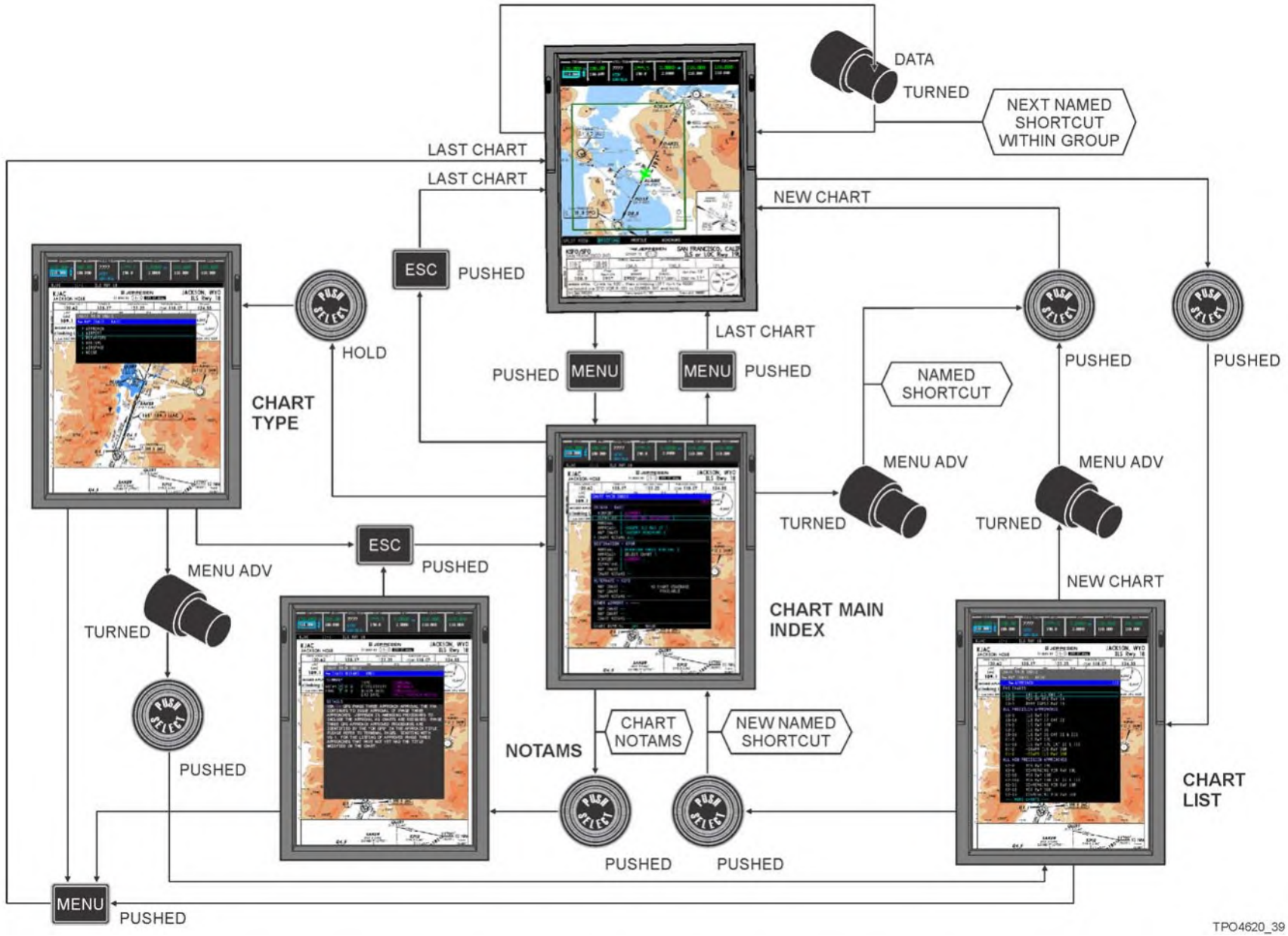
The diagrams that follow depict IFIS menu navigation. The diagrams are designed to assist the operator with understanding how the system interacts with CCP control inputs and onscreen menu selections.

NOTE

The IFIS illustrations in this operator's guide are for illustrative purposes only.

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Figure C-1 IFIS – Menu Architecture



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Figure C-2 IFIS – Chart Display to Chart Main Index

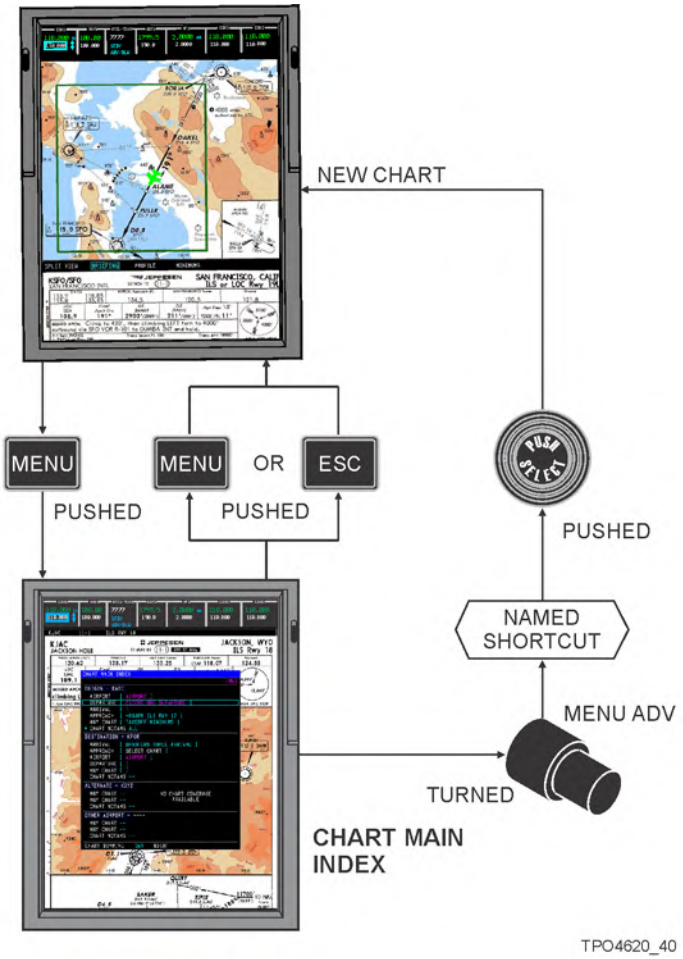


Figure C-3 IFIS – Chart Display to Named Shortcut

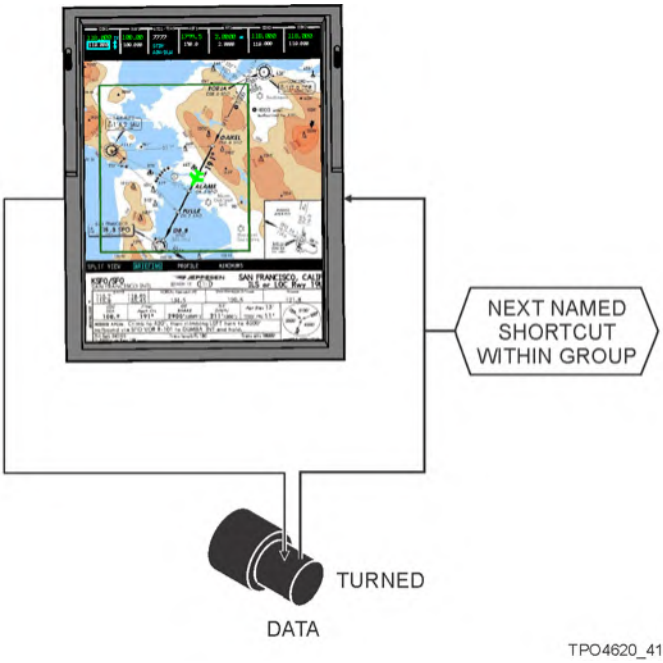
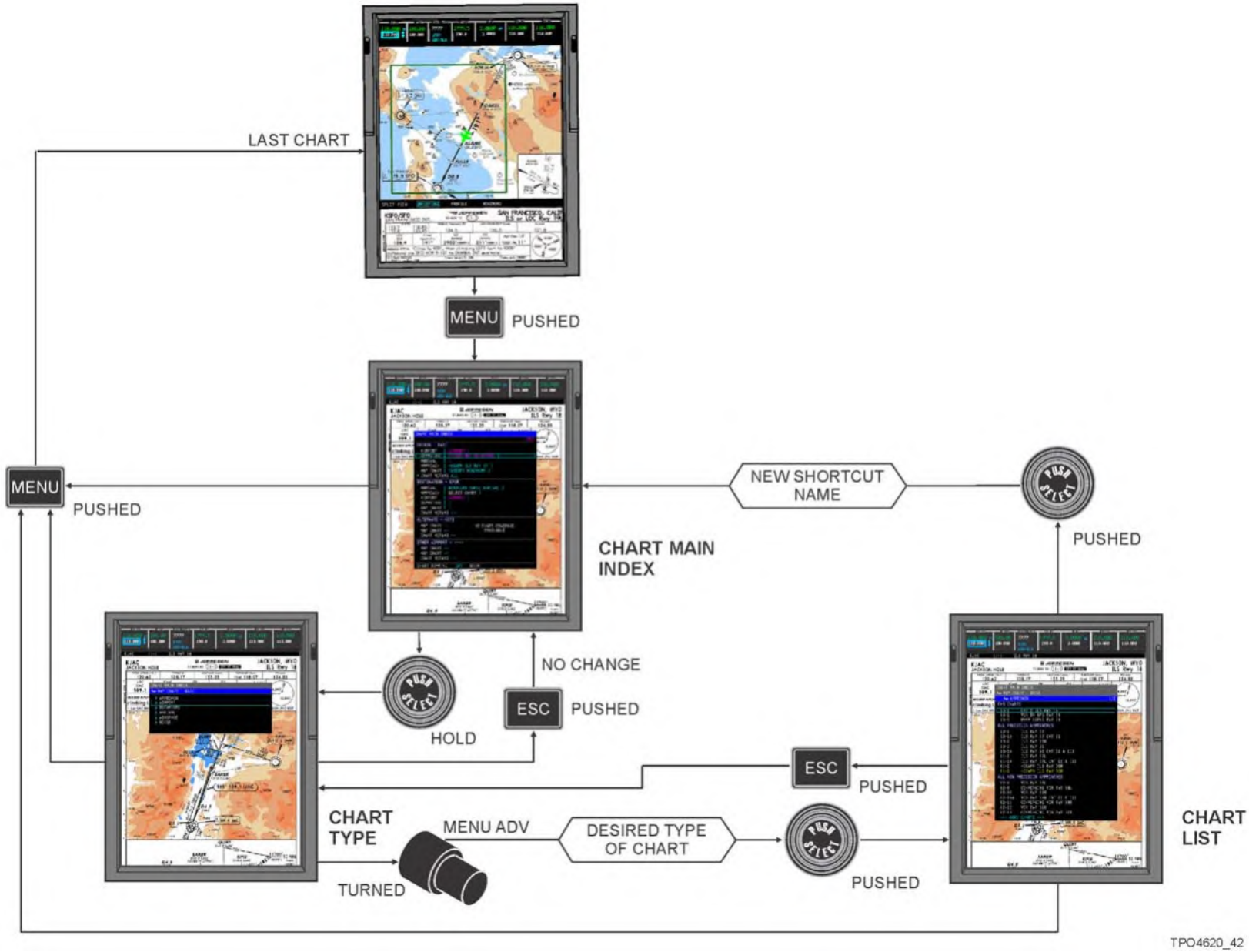


Figure C-4 IFIS – Chart Display to Chart Type Menu



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Figure C-5 IFIS – Chart Display to Chart List

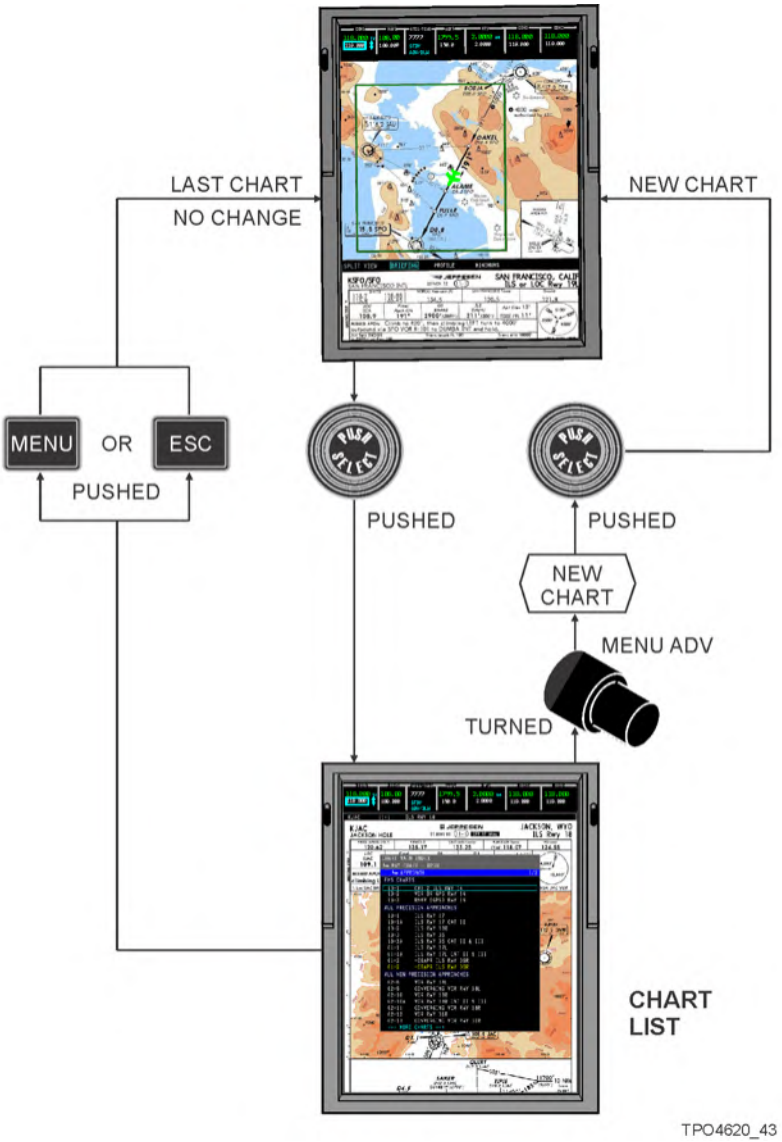


Figure C-6 IFIS – Chart Display to NOTAM

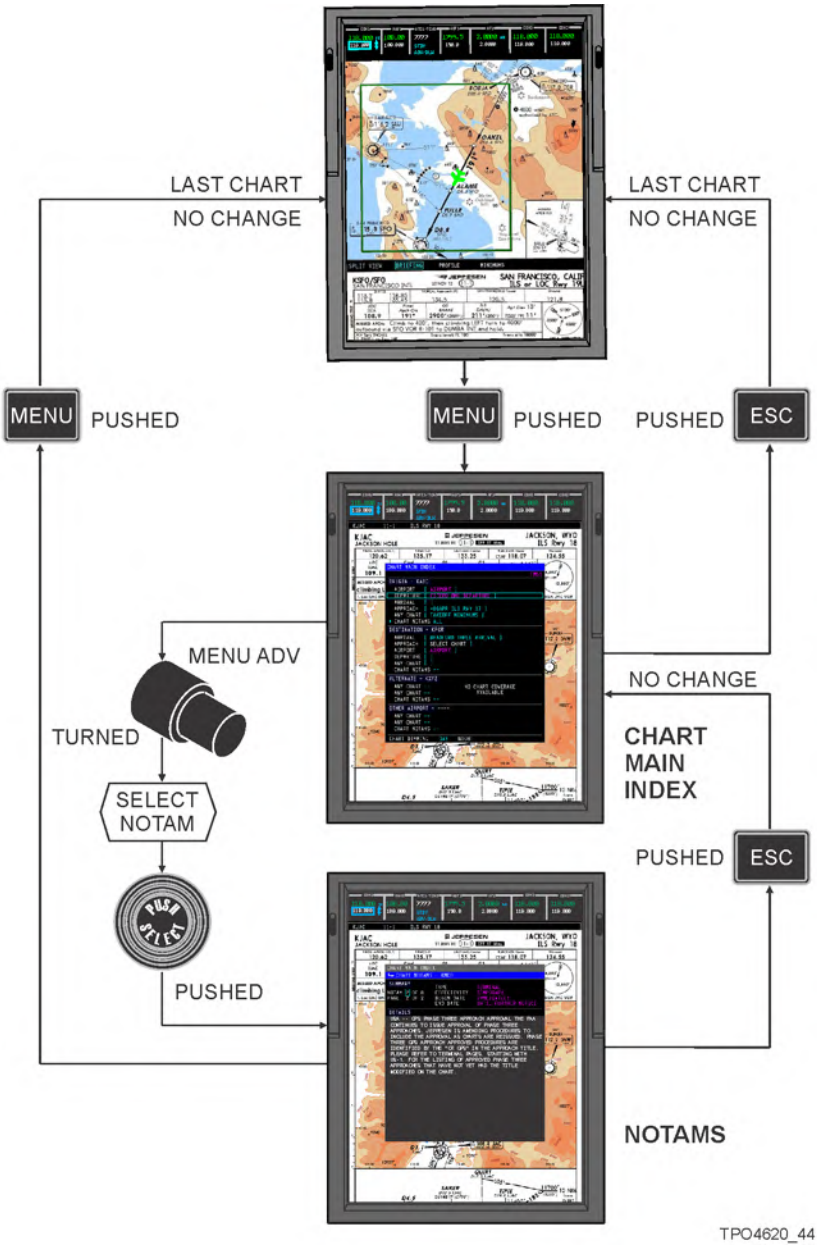
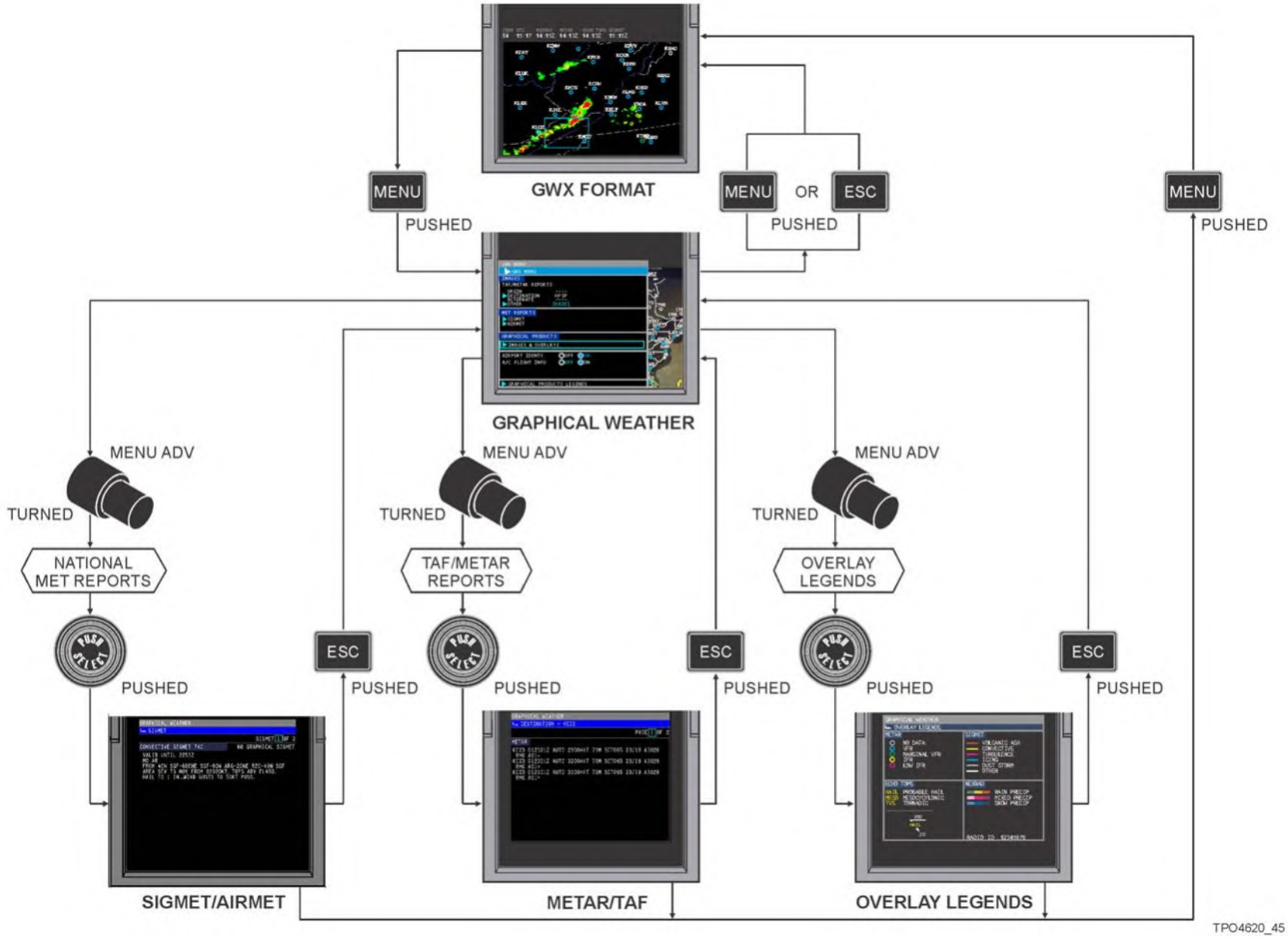


Figure C-7 IFIS – XM Weather Menu Architecture



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TCAS Aurals D-1

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APPENDIX D

TCAS AURALS

The TCAS-4000 system gives an aural RA over the aircraft flight deck audio system or an independent speaker system in addition to the RA that shows on the AFD. If the TCAS senses that the other aircraft is not responding in the planned manner, it issues opposite corrective commands. For example, if the TCAS issues a corrective climb command. The following table lists the TCAS aurals and gives a brief description of each.

TCAS Aurals

Advisory	Aural	Response
Clear	CLEAR OF CONFLICT	Resume normal flight, apparent conflict of airspace has been resolved.
Traffic	TRAFFIC, TRAFFIC	Gain visual contact of traffic. Check TCAS display for traffic bearing and range, if necessary. Assess the threat and prepare to execute the evasive maneuver TCAS issues.

Advisory	Aural	Response
Preventive RA	MONITOR VERTICAL SPEED	Be alert for approaching traffic. Ensure that the aircraft pitch attitude or vertical speed does not enter the red zone on the attitude display or vertical speed scale.
	MAINTAIN VERTICAL SPEED MAINTAIN	Maintain present pitch or vertical speed and direction. Ensure that the aircraft pitch or vertical speed does not enter the red zone on the attitude display or vertical speed scale.
	MAINTAIN VERTICAL SPEED CROSSING MAINTAIN	A flight path crossing is predicted, but being monitored by TCAS. Maintain present pitch or vertical speed and direction. Ensure that the aircraft pitch or vertical speed does not enter the red zone on the attitude display or vertical speed scale.
	LEVEL OFF, LEVEL OFF	Reduce vertical speed to zero feet per minute. A green arc is illuminated beginning at zero feet per minute.

Advisory	Aural	Response
Corrective RA	CLIMB, CLIMB	Change vertical speed to 1500 ft/min climbing or as indicated on attitude display or vertical speed scale.
	CLIMB, CROSSING CLIMB CLIMB, CROSSING CLIMB	Same as previous except that it further indicates that flight paths will cross at some altitude.
	DESCEND, DESCEND	Same as CLIMB except that a descending vertical speed is indicated.
	DESCEND, CROSSING DESCEND DESCEND, CROSSING DESCEND	Same as previous except that it further indicates that flight paths will cross at some altitude.
	LEVEL OFF, LEVEL OFF	Reduce vertical speed to zero feet per minute. A green arc is illuminated beginning at zero feet per minute.

Advisory	Aural	Response
Corrective RA (Cont.)	INCREASE CLIMB INCREASE CLIMB	This follows a CLIMB advisory. The climb pitch or vertical speed should be increased, typically to 2500 ft/min, as shown on the attitude display or vertical speed scale.
	INCREASE DESCENT INCREASE DESCENT	This follows a DESCEND advisory. The descent pitch or vertical speed should be increased, typically to 2500 ft/min, as shown on the attitude display or vertical speed scale.
	CLIMB, CLIMB NOW CLIMB, CLIMB NOW	This follows a DESCEND advisory. This advisory indicates that a reversal of vertical speed from descent to climb is needed to provide adequate separation.
	DESCEND, DESCEND NOW DESCEND, DESCEND NOW	This follows a CLIMB advisory. This advisory indicates that a reversal of vertical speed from climb to descent is needed to provide adequate separation.

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APPENDIX E

WEATHER RADAR TECHNIQUES (RTA-854 AND RTA-4114)

The primary function of the Weather Radar System is to aid the crew in the detection and avoidance of areas of precipitation. The Turbulence Weather Radar System aids in the detection of precipitation-related turbulence in and around thunderstorms. The Turbulence Weather Radar System cannot detect clear air turbulence, and the Weather Radar System cannot detect windshear, clouds, or lightning. However, rain, wet hail, moderate to heavy wet snow, and in some instances, possible icing conditions can be detected by the system. The Weather Radar System may also be used to map the terrain.

Each operator normally develops specific techniques and procedures for using Weather Radar. Basic operational techniques for the Weather Radar are no different from the techniques used with earlier generation radars equipped with flat plate antennas. The Weather Radar includes many features to aid the crew in interpreting the weather. However, as with most tools, operational experience is a valuable teacher. This experience enables the crew to properly analyze the various types of storm displays.

The Weather Radar should be used on every flight. This allows the crew to make sure of proper operation and gain proficiency in operating the system on a regular basis. If there is no significant weather in the immediate vicinity, or if operating in Visual Meteorological Conditions (VMC), the MAP (Ground Mapping) mode may be selected and the Weather Radar display can be compared with terrain features. Practice ground mapping on routes frequently flown and in familiar terminal areas.

Select Standby mode when on the ground unless the Weather Radar is being used to check the terminal area weather prior to departure.

The RTA-854 and RTA-4114 provide similar features as described below. The RTA-4114 offers the following additional features:

- The MultiScan™ Radar is designed to work in the fully automatic mode at all times. Operators select only the desired range and operating mode. Tilt and gain inputs are not required.

- Weather data from multiple scans, at varying tilt angles, is stored in memory. When a range is selected, the display presentation represents an optimized weather picture regardless of the aircraft altitude.
- The MultiScan™ Radar provides true weather information out to a 320 NM range.
- Overflight™ Protection reduces the possibility of inadvertent thunderstorm top penetration.

PATH ATTENUATION COMPENSATION

The WXR uses the Path Attenuation Compensation (PAC) feature to maintain the correct display of storm intensity. The PAC feature compensates for Weather Radar energy that is absorbed when the Weather Radar beam penetrates a precipitation cell.

Non-compensated radars have a tendency to underestimate the true level of a precipitation cell simply because of Weather Radar energy absorption.

For the RTA-854:

The PAC feature is automatically enabled whenever the WX (weather Only), WX+TURB (Weather Plus Turbulence), or TURB (Turbulence Only) modes is selected. The PAC feature is active out to a range of approximately 65 NM. Therefore, any cells beyond this range are uncompensated and should be avoided. The PAC circuits are disabled when any of the conditions that follow occur:

- When the 5 or 10 NM ranges are selected (in any mode)
- When the MAP mode is selected
- When the GAIN setting is any value except NORM.

For the RTA-4114:

The PAC feature is automatically enabled whenever the WX or WX + TURB modes are selected and is active out to a maximum range of 80 NM. Therefore, any cells beyond this range are uncompensated and should be avoided. The PAC circuits are disabled for any of the conditions that follow:

- MAP or TURB Only mode is selected
- GAIN setting is any value except NORM
- Manual mode of operation is selected.

When the Weather Radar beam encounters a weather target of sufficient intensity and depth that requires the full-range of attenuation correction, a yellow PAC alert arc shows at the edge of the outer heading-range arc. The PAC alert arc informs the crew that the Weather Radar beam, in the direction of the PAC alert arc, is being severely attenuated, and that the area or shadow behind the intervening rainfall may contain hidden (and possibly significant) precipitation. Returns that display a PAC alert arc should be treated as potentially dangerous. Always avoid the area between the weather that shows and the PAC alert arc.

ANTENNA STABILIZATION

The STAB (Antenna Stabilization) feature uses inputs from the non-Rockwell Collins Inertial Reference System (IRS) to automatically stabilize the Weather Radar antenna during changes in aircraft attitude. These inputs eliminate the effects of aircraft pitch and roll on the antenna to maintain the desired antenna tilt-to-range ratio. The Antenna Stabilization feature is always set to ON. When the Antenna Stabilization feature is reported off by the RTA-854 radar, the advisory message USTB shows in place of the antenna tilt angle in the Weather Radar data field. The USTB message is not implemented for the RTA-4114.

SECTOR SCAN (RTA-854 ONLY)

The SEC (Sector Scan) feature is selected with the SEC SCAN selection on the RADAR menu. The Sector Scan feature reduces the antenna scan/sweep angle from ±60 degrees (120 degrees total) to ±30 degrees (60 degrees total). By reducing the antenna scan angle, the amount of time needed for the antenna to complete its sweep is also reduced. This effectively increases the Weather Radar update rate. Selection of the Sector Scan feature on either PFD reduces the scan arc on all displays.

RECEIVER GAIN

CAUTION

The crew should always return the gain setting to NORM (zero) when finished analyzing the display. Failure to do so may result in missing significant targets when operating in one of the minus gain settings.

The DATA knob on the DCP is used to set the level of receiver gain. The range is -3, -2, -1, NORM, +1, +2, and +3.

Normally, receiver gain should be at normal (NORM) position. Six other positions, including three of increased gain for maximum sensitivity, are available for analyzing the details of the returns. The higher settings are useful in identifying the lightest levels of precipitation while the lower settings are useful for more in-depth studies of the most intense weather targets. Each of the minus settings (-1, -2, and -3) reduces the sensitivity of the Weather Radar System by approximately one color level. This results in a total reduction of three color levels when operating with the GAIN set to the -3 position. Therefore, if a target was magenta (RTA-854) or red (RTA-4114) in NORM, the same target at -3 is almost totally removed from the display. From this example, it should be clear that operating the Weather Radar in one of the reduced gain settings for an extended period of time is not recommended.

NOTE

When the GAIN value is set to any value other than NORM, the Weather Radar returns are uncalibrated, so display colors may no longer accurately indicate rainfall rates.

ANTENNA TILT

The antenna tilt control is perhaps the most used and most useful control of the RTA–854 radar system. By changing the angle of the antenna relative to the horizon, the crew can point the Weather Radar energy beam at close-range weather or ground returns, or direct the beam farther out to paint distant weather.

When operated in the preferred AUTO mode, The RTA-4114 radar transmits at two tilt angles sequentially. The data from the two beams is stored in memory for later extraction. The data from the two pages is combined to separate ground from weather returns and to display weather to the flight crew at altitudes relevant to the aircraft flight altitude.

The basic MultiScan process is described in Section 5 of the RTA-4100 Series MutliScan Automatic Weather Radar Operator's Guide. Similar information is provided in Sections 3 and 4 of the RTA-4100 Series MutliScan Fully Automatic Weather Radar Installation manual. In cruise, the lower beam is pointed near the horizon and is automatically adjusted with flight altitude and local terrain. The upper beam operates at a fixed tilt offset from the lower beam.

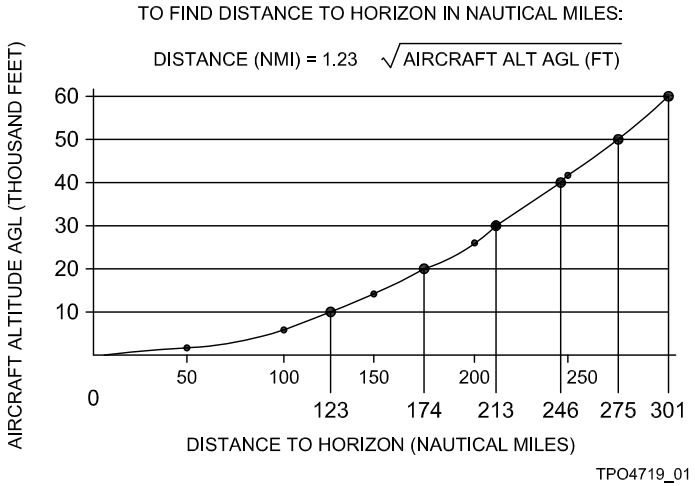
At lower altitudes, the beams are raised above the horizon to minimize ground clutter contamination. The radar beam management in AUTO mode is automatic and transparent to the flight crew. Since the displayed weather data is based on stored memory based information, the crew does not see the alternating upper and lower beam sweeps. The displayed tilt angle is the average of the upper and lower beam tilts.

Although AUTO mode is the preferred mode of operation for the RTA-4114, manual operation may be used as a backup or in the event of an automatic operation failure, annunciated with the AUTO FAULT message. During manual operation, the radar can be operated as conventional manually operated radar.

The TILT knob on the DCP is used to manually adjust the antenna tilt angle. The tilt range is ±15 degrees. Turning the TILT knob clockwise selects a positive (upward) tilt angle and turning the TILT knob counterclockwise selects a negative (downward) tilt angle. The TILT knob remains functional while the Auto Tilt (RTA-854 only) feature is active, enabling the operator to change the antenna tilt/range ratio that is being maintained. Proper use of the TILT knob allows the operator to achieve the best picture of storm cell size, height, and relative direction of movement. Procedures for adjusting the TILT knob vary depending upon user requirements. Proper use can only be achieved through experience and regular practice.

When operating over land, the best general guideline is to select a range that is within the line-of-sight distance to the horizon (refer to the illustration that follows), then adjust the antenna tilt until a small amount of ground clutter appears at about the outer third of the display.

Figure E-1 Distance to Horizon

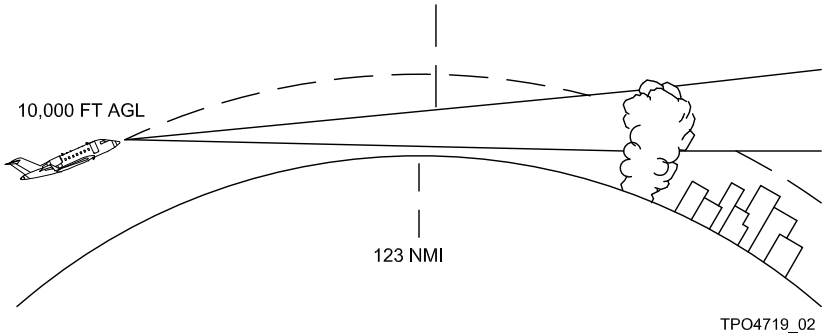


TIP

Maximum rainfall rates in a thunderstorm usually occur about mid-level in the storm. This is normally the area that paints the strongest returns. When the aircraft is above or below that altitude, some antenna tilt is needed. The amount of TILT needed varies with the estimated distance to the storm – the closer the storm, the more TILT required. In either instance, it is good practice to periodically operate the TILT knob throughout its range to reduce the possibility of missing close-in targets.

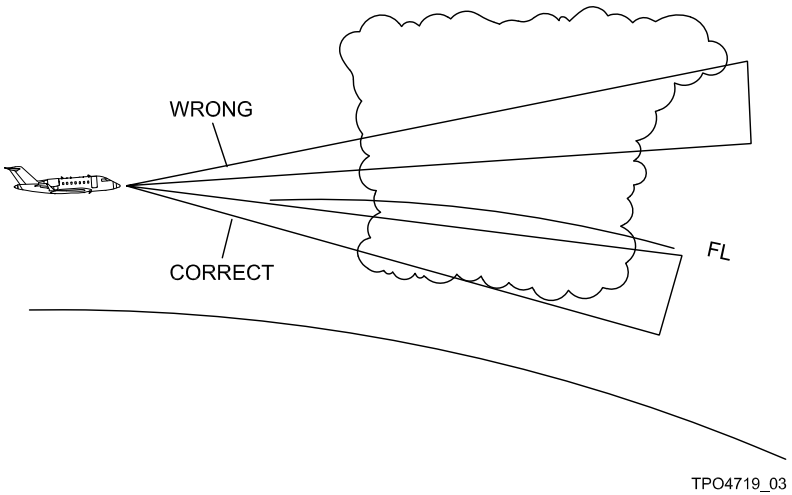
As the illustration that follows shows, an aircraft at 10 000 feet AGL with the 300-NM range selected is not able to paint ground clutter much beyond 123 NM. A better range selection at this altitude may be the 100-NM setting.

Figure E-2 Line-of-Sight Range



Once weather activity is identified, it is important to keep the Weather Radar beam pointed at the liquid portion of the cell. Ice crystals reflect less energy than liquid precipitation. Tilting the beam above the freezing level may result in underestimating the intensity of the cell. Move the TILT control up and down to determine the most reflective portion of the cell.

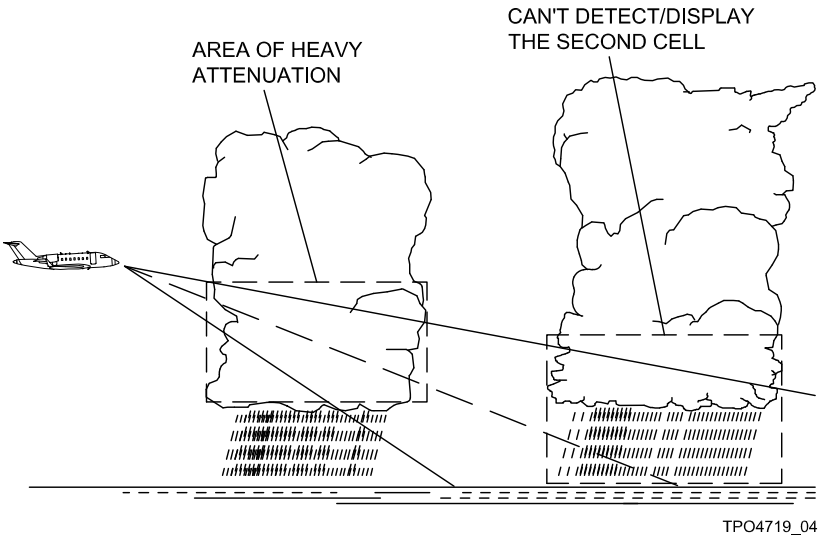
Figure E-3 Reflective Parts of a Cell



Proper tilt adjustment allows the crew to detect Weather Radar shadows. Once weather is detected, the crew can continue to decrease the tilt setting until more ground clutter shows around the precipitation

cell. Significant precipitation cells can attenuate the Weather Radar beam so much that Weather Radar returns which are behind the intervening cell (ground and/or weather returns) are greatly reduced in intensity. In some instances, the attenuation may also be great enough to cause a PAC alert arc to appear.

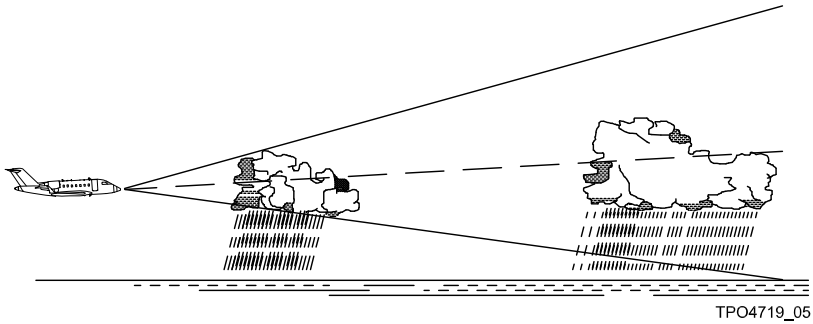
Figure E-4 Down Tilt for Weather Radar Shadows



TILT CONTROL AT MIDDLE ALTITUDES

While scanning for storm targets and with one of the weather detection modes selected, the optimum tilt angle of the antenna depends upon the altitude of the aircraft and the selected range. Antenna tilt for an aircraft flying at 20 000 feet (middle altitudes) is typically set near 0 (zero) degrees or slightly down.

Figure E-5 Tilt Control at Middle Altitudes

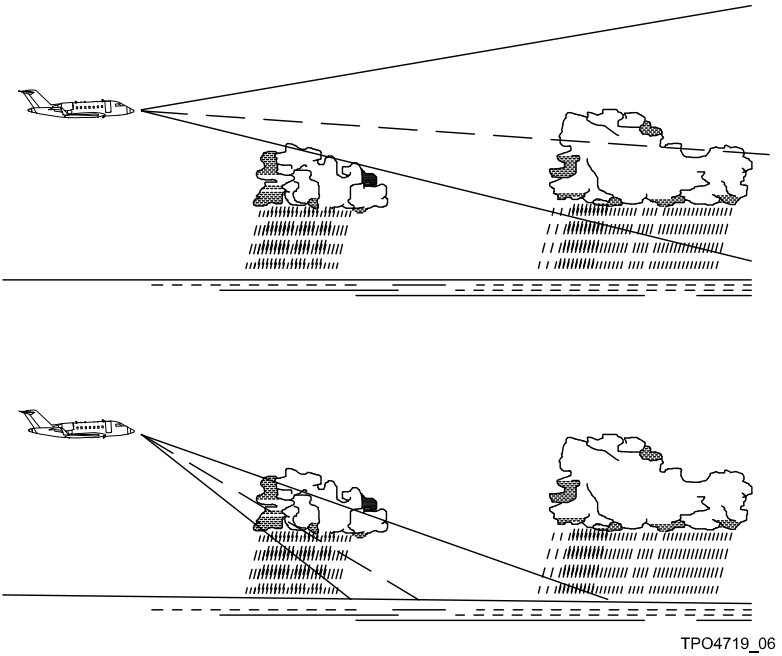


TILT CONTROL AT HIGHER ALTITUDES (ABOVE FL 350)

The TILT control guidelines just discussed for middle altitudes may not be effective for flights above 35 000 feet. Typically, at high altitudes a longer range is selected and the tilt is adjusted slightly down. When selecting the operating range, keep in mind the line-of-sight distance to the horizon.

As the illustration that follows shows, when operating at the higher altitudes, it is particularly easy to scan over the top of significant storm cells.

Figure E-6 Tilt Control at Higher Altitudes



Do not attempt to overfly targets. It is possible that dry hail (which generally cannot be detected) and severe turbulence may be present far above the top of any areas of detected precipitation activity. The crew should always remember that the Weather Radar is an avoidance tool. It is strongly recommended that pilots never attempt to overfly, underfly, or penetrate storm cells or squall lines. For safest operation, the crew should plan ahead to establish a flight path that avoids all returns by the distance established by the flight operations department.

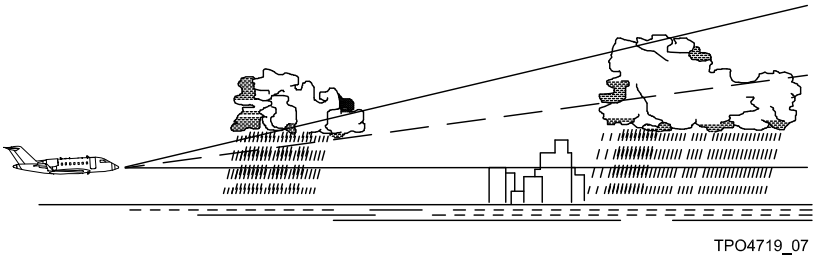
To view targets inside the 40-NM range, large down tilt settings may be necessary. The crew should recognize that a large down-tilt setting may prevent more distant storms from being detected, and may cause excessive ground clutter to appear when flying over land.

TILT CONTROL BELOW 10 000 FEET

Flight operations below 10 000 feet, for example, takeoffs and landings, typically require a shorter range selection (for example, the 50-NM

range) with 3 or 4 degrees of uptilt. These settings typically reduce ground clutter and provide the necessary target information. As altitude and speed increase, tilt settings typically need to be lowered to maintain an acceptable display, and the range needs to be increased.

Figure E-7 Uptilt at Low Altitudes



AUTO VERSUS MANUAL (RTA-4114)

To discuss intended function of the AUTO mode, it is useful to compare manual and automatic operation.

When performing the weather detection function in MAN Operation, the operator must adjust the tilt to suppress ground clutter. Higher beam tilt reduces the amount of ground clutter but also reduces the strength of the weather returns since the most reflective parts of weather cells are low to the ground where large amounts of liquid water are present. Higher in the cell, where liquid water has frozen to ice crystals, reflectivity is greatly reduced.

For optimal weather detection and ground clutter rejection, the operator typically selects a tilt where the beam is just skimming the ground. The specific tilt setting is dependent on the altitude of the aircraft above the ground and the operator must continually adjust the tilt to account for altitude variation and local ground terrain variation. A skilled radar operator observes weather signal strength as the beam is raised above the ground in order to assess the height of weather relative to the aircraft.

When the crew selects AUTO operation, the automatic tilt function is engaged and the beam-tilt, in conjunction with signal processing techniques, is automatically managed to provide the crew with either a ground map view or weather detection view, regardless of the altitude

or ground returns for ground mapping purposes, or to suppress ground returns so that only weather is displayed. When only weather display is desired, the radar not only manages beam tilt, but also invokes additional signal processing to further suppress ground returns and low lying weather which in not relevant to the flight altitude.

The specific Methodology for ground clutter suppression in AUTO Operation involves collecting return data with sweeps at multiple tilt angles. All the data is stored in a memory system relative to ground coordinates. In essence, the radar builds an image of the weather in its memory. The data is filtered and the radar determines the expected return from ground and automatically calculates a threshold to determine whether the measured returns are from ground or weather. Thresholds are designed to display weather relevant to the flight altitude and to reject low flying weather. Finally, an output is generated to the display by extracting the stored data from memory.

These advanced threshold techniques are key to the radar's ability to suppress ground clutter and low flying weather effectively. In principal, however, the intended function is no different than would be expected of traditional manual mode radar where the operator adjusts beam tilt to manually suppress clutter and to identify weather relevant to the aircraft's altitude.

While the radar is in AUTO Operation mode, the automatic nature of the tilt control and the status of the ground clutter suppression mode are clearly annunciated to the crew. AUTO operation can be disabled simply by selecting the MAN operation where traditional tilt control is available. Regardless of whether the radar is operating in AUTO operation or MAN operation, all performance requirements specified by DO-220 are met and the fact that the radar is performing scans at multiple tilt angles during AUTO operation is transparent to the crew.

AUTO TILT (RTA-854)

This RTA–854 feature is designed to reduce the number of times it is necessary to adjust the antenna tilt control while changing aircraft altitude or range settings. The Auto Tilt feature automatically adjusts the antenna tilt angle to attempt to maintain the picture in the same area of the display. The Auto Tilt feature is continuously enabled as the system always uses the current manual tilt setting as the starting point.

Figure E-8 Auto Tilt Antenna Movement with Altitude Changes

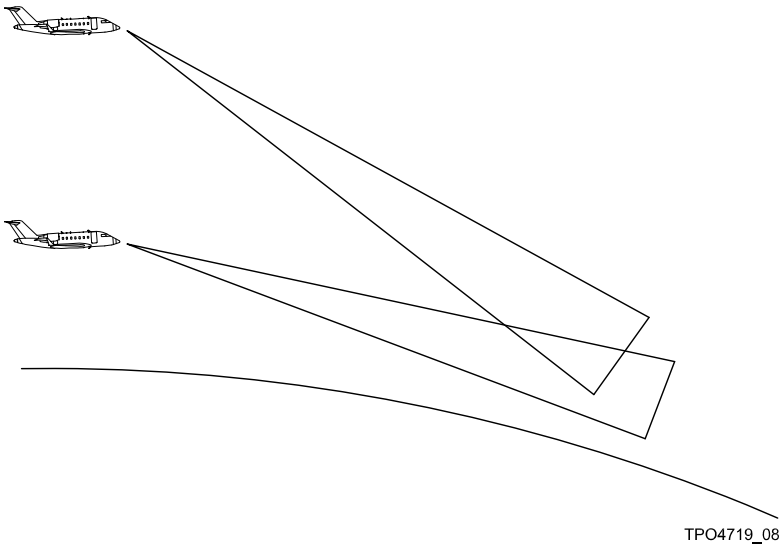
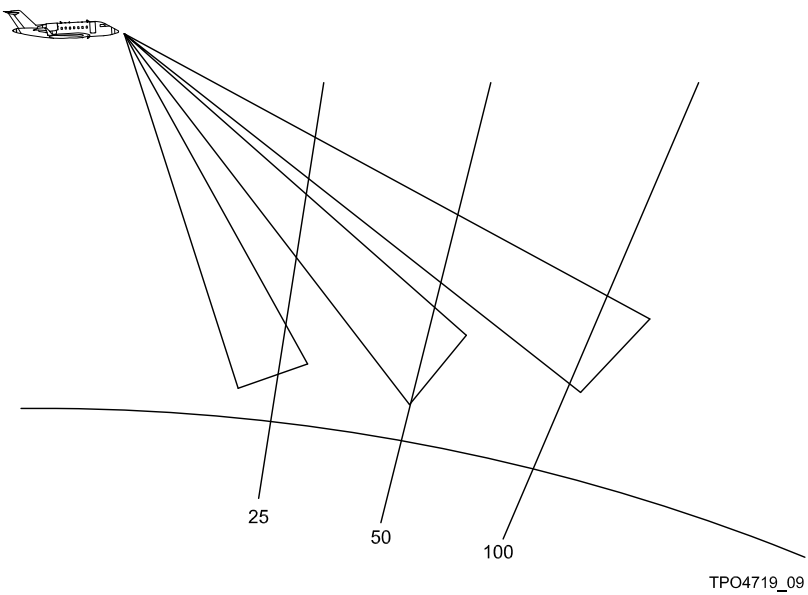


Figure E-9 Auto Tilt Antenna Movement with Range Changes



TARGET ALERT (RTA-854)

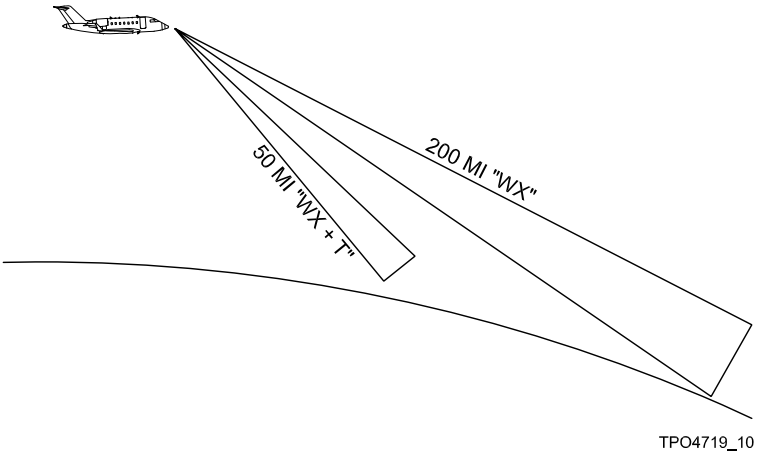
The Target alert feature permits detectable weather targets (precipitation or precipitation-related turbulence) to be detected and annunciated without the targets themselves appearing on the Weather Radar display. This mode is normally used as a background mode when other data (for example, checklists, remote data, etc.) are active on the MFD.

When target mode is selected, the Weather Radar transmitter is energized, the antenna begins to scan, and the TWR automatically selects the range and tilt required (the crew has no control of range and tilt in TGT mode). The legend TGT shows on the AFD when target mode is enabled.

TGT mode automatically evaluates two target alert windows (refer to the illustration that follows) to provide search coverage from 7 to 200-NM range (50-NM maximum range for turbulence detection) and ±15 degrees of aircraft heading regardless of heading and regardless of the range selected. Target alert notifies the crew of the two weather phenomena that follow.

- When red or magenta weather targets are detected within the entire range. (Rainfall rates greater than 0.2 in/hr or 5.1 mm/hr.)
- When areas of precipitation related turbulence are detected within the 50-NM window. (Wind velocity shifts in excess of 16.4 feet/second or 5 meters/second.)

Figure E-10 Auto Tilt and Range Setting in Target Alert Mode



GROUND CLUTTER SUPPRESSION (RTA-854)

The Ground Clutter Suppression (GCS) button on the DCP is used to temporarily select the GCS feature. GCS reduces the intensity of ground returns (clutter) in the WX, WX+TURB and TURB modes, making precipitation returns easier to see. GCS removes some very light rainfall, especially rainfall that is not associated with convective cells. This type of unagitated rain (rain with no associated turbulence) appears ground-like to the Doppler processing circuitry. GCS is most effective when tilt is set so the Weather Radar beam just skims the horizon. It is less effective when the antenna tilt is set to scan below the horizon and the resulting ground returns are strong. Ground Clutter Suppression automatically times out after 30 seconds.

GROUND CLUTTER SUPPRESSION (WXR-4114)

The Ground Clutter Suppression (GCS) control is used during automatic operation to deselect the GCS feature. GCS is only available in the AUTO mode of operation. When active, GCS removes approximately 98 percent of ground clutter. The GCS feature can also remove some very light rainfall that is not associated with convective cells. This type of non-agitated rain (rain with no associated turbulence) appears like a ground target to the GCS processing circuitry.

The GCS feature is available in the Weather (WX), Weather Plus Turbulence (WX+TURB), and Turbulence (TURB) modes of operation. The default setting is ON in the AUTO mode. GCS can be turned off, but will revert to ON after approximately 30 seconds.

When operated in AUTO mode with ground clutter suppression enabled, the detection threshold in RTA-4114 MultiScan Weather radar is designed to display weather at altitudes 6000 – 10 000 feet below the aircraft. The radar senses weather far below this altitude but low flying weather is removed by design to provide the flight crew with a view of weather relevant to their flight altitude.

Weather far below the aircraft flight altitude does not result in a safety impact to the flight crew. If this low flying weather is displayed, it may result in unnecessary diversions and maneuvers.

WEATHER RECOGNITION AND AVOIDANCE

Experience soon enables the crew to properly analyze various types of storm displays. The key to avoiding detected weather is to first determine the heading change needed to bypass a storm safely. Establish the aircraft on the appropriate heading, then recheck the Weather Radar display to see if further heading changes are required. The crew should remember that the Weather Radar was designed as a weather avoidance tool.

Most operators establish guidelines for storm cell avoidance distances. However, the crew has the sole responsibility to decide how close to approach various types of storm displays. Most convective weather systems in the continental United States travel from south/southwest to north/northeast. The areas ahead of these storms (north/northeast) can be expected to contain gust fronts, turbulence, heavy rain, and possibly hail. It is suggested that these areas be avoided by no less than the minimum distance established by the flight operations department.



TIP

A good operating technique is for one pilot to set his Weather Radar display at a shorter range with some uptilt (depending on altitude) for low altitude maneuvering. The other pilot sets his Weather Radar display at a longer range with some down tilt (depending on altitude) to see the big picture for cruise altitude maneuvering. In this manner, the shorter range allows the crew to monitor any close-in targets, while the longer range simultaneously helps the crew in plotting any required course changes.

TERRAIN MAPPING

Interpreting the terrain maps that show on the display is largely a matter of experience and understanding the factors involved when the Weather Radar beam strikes a ground target. Again, regular use of the Weather Radar will build proficiency in interpreting Weather Radar returns.

Terrain mapping should be done with the MAP (Ground Mapping) mode selected under the RADAR menu, one of the shorter ranges selected, and the antenna tilted down. The receiver gain may have to be adjusted to change the apparent intensity of ground returns for the highest contrast and ease of interpretation.

The extent to which ground targets show depends upon the selected range, antenna beam width, aircraft altitude, aircraft attitude, the angle at which the Weather Radar beam strikes the ground target (incidence angle), and the reflective properties of the ground target. As experience is gained, the crew should be able to interpret displays that indicate lakes, rivers, coastlines, mountains, cities, and larger structures.

The receiver gain can be adjusted so coastlines become apparent and cities are well defined. The use of increased gain settings (+1, +2, and +3) should be avoided, as increased gain may cause the Weather Radar returns to paint between targets, which can obscure some landmarks.

BODIES OF WATER

Bodies of water, (for example, lakes, rivers, and oceans) usually do not provide a strong return and show up as dark areas on the display. This is because the Weather Radar beam is reflected away from the antenna and very little, or none, of the signal is returned. Rough or choppy water, however, provides a return proportional to the activity of the water surface.

Figure E-11 Calm Water

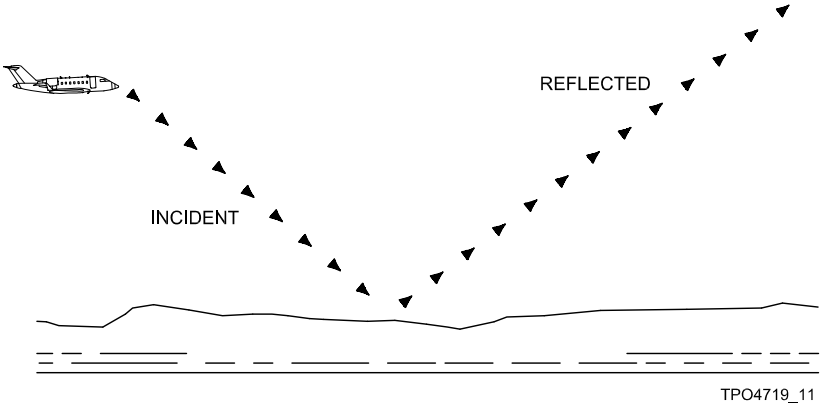
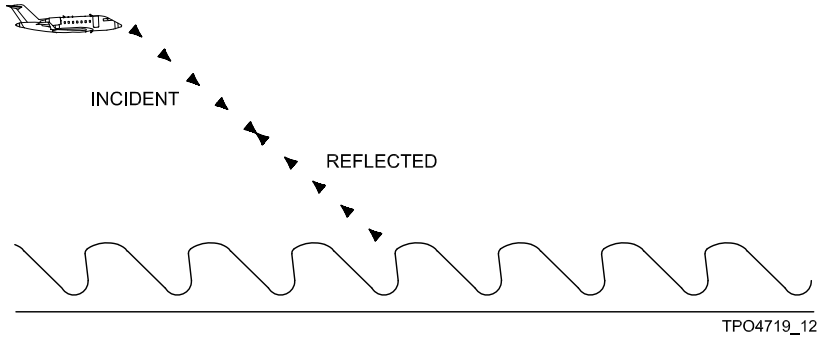


Figure E-12 Choppy Water



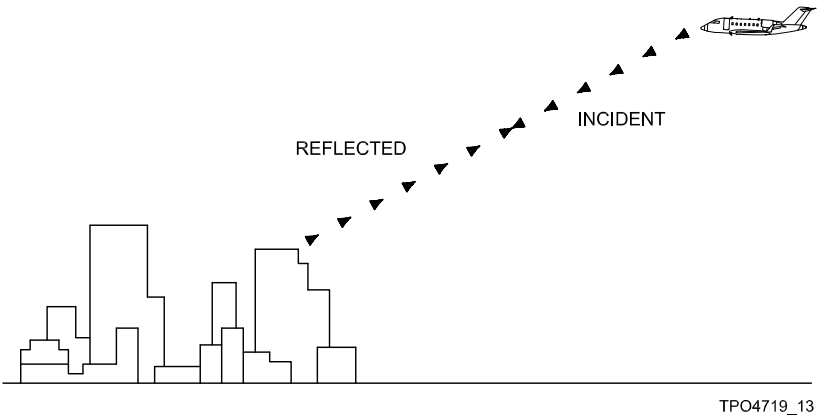
CITIES

Cities usually provide a good return signal, although their intensity depends on selected range, gain setting, aircraft attitude and altitude, and antenna tilt if one of the longer ranges is selected.

Typically, large buildings and structures will provide a return while small buildings are shadowed from the Weather Radar beam by taller buildings. As the aircraft approaches closer to the city and a shorter range is selected, details become more noticeable as the regular lines and edges of the city appear.

In areas where several cities are grouped together, the area can be identified by the pattern of the return signal. For example, cities located along the shore of a large lake such as Lake Michigan may appear in yellow. The lake itself does not provide a good return signal, and therefore appears black.

Figure E-13 Cities



WEATHER RADAR OPERATIONAL CONSIDERATIONS

DETECTION AND ATTENUATION

The Weather Radar, like all other X-band airborne weather radars, has limitations that must be understood and recognized by the crew. The first of these has to do with exactly what the systems can detect and display. The Weather Radar has been designed to accurately detect and display rainfall and precipitation-related turbulence. However, what are the rainfall rates for the green, yellow, red, and magenta display levels, and what is the minimum amount of precipitation-related turbulence (also shown in magenta) that can be detected? The reflectivity level chart shown below shows the relationship between reflectively levels, precipitation rates, and display colors.

Weather Radar Reflectivity Levels and Associated Precipitation Rates — RTA-4114

COLORS	COLORS REFLECTIVITY LEVELS IN MM AND IN./HR.		
Black	<20 dB	0 – 0.77 mm/hr.	0 – 0.03 in./hr.
Green	20 to 30 dB	0.77 – 3.81 mm/hr.	0.03 – 0.15 in./hr.
Yellow	30 to 40 dB	3.81 – 12.7 mm/hr.	0.15 – 0.50 in./hr.
Red	> 40 dB	>12.7 mm/hr.	>0.50 in./hr.
Magenta	Precipitation related Turbulence		

Weather Radar Reflectivity Levels and Associated Precipitation Rates — RTA-854

REFLECTIVITY LEVEL	PRECIPITATION RATE		COLOR
	in/hr	mm/hr	
Z1	Less than 0.03	0.52 and greater	No Display
Z2	0.03 to 0.07	0.52 and greater	Green
Z3	0.07 to 0.20	0.8 to 1.8	Yellow
Z4	0.20 to 0.52	1.8 to 5.1	Red
Z5	0.52 and greater	5.1 to 13.2	Magenta
Turbulence	Greater than 0.02	Greater than 0.5	Magenta

NOTE

Proper use of the GAIN setting allows the operator with Weather Radar operating experience to estimate rainfall rates greater than a red or magenta return. Targets that show in a reduced gain condition imply that severe turbulence, hail, and very heavy rainfall is likely.

The second limitation of the Weather Radar that must be understood by the operator is attenuation of the airborne Weather Radar signal. As a Weather Radar pulse travels through the atmosphere, it progressively loses its ability to return to the receiver with enough energy to produce

any given display level. That loss of signal strength is called attenuation, and is due primarily to distance and precipitation.

Attenuation of the Weather Radar signal due to distance is a function of basic physics. As a rule of thumb, doubling the distance to a target reduces the strength of the returning signals by at least a factor of 4, and in some instances, by a factor of 16. Any return (green, yellow, red, or magenta display) from a distant storm usually indicates extremely high rates of rainfall, and moderate to severe turbulence can be expected. These areas of rainfall should be avoided whenever possible.

The second type of attenuation that can reduce the performance of the airborne Weather Radar is attenuation due to precipitation. Moisture, especially rainfall, attenuates Weather Radar signals. The amount of attenuation due to precipitation depends entirely upon the rainfall rate.

Figure E-14 RTA-854 Weather Radar Color Levels vs. NWS VIP Levels

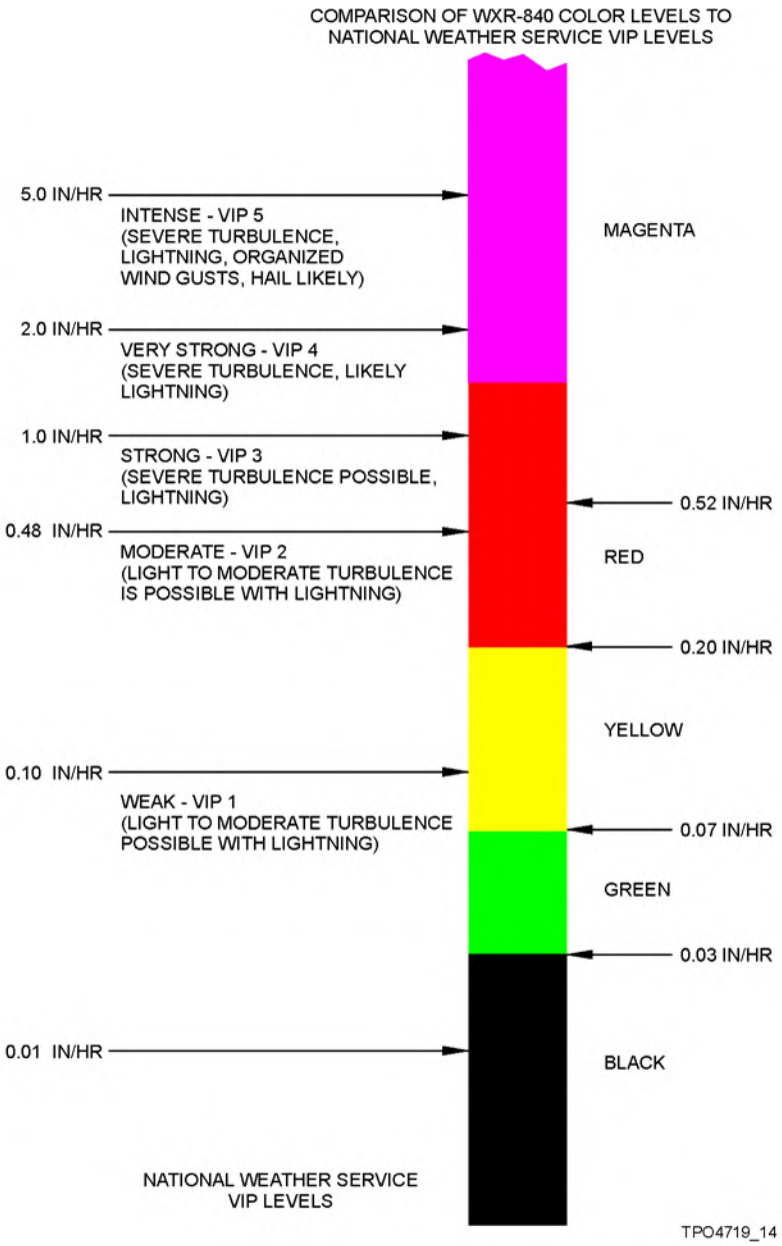
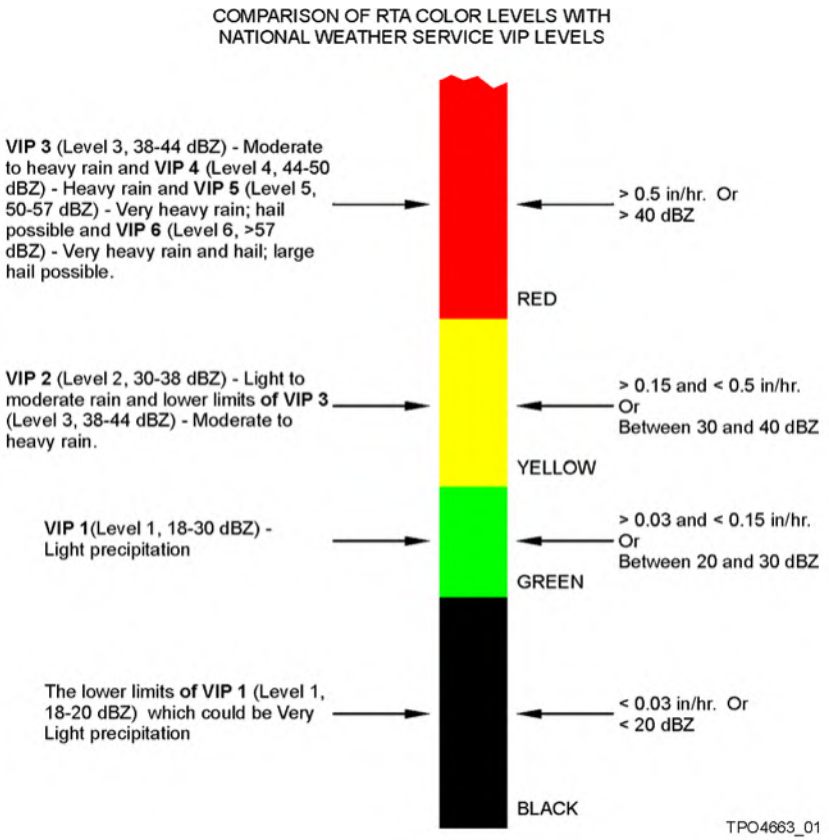


Figure E-15 RTA-4114 Weather Radar Color Levels vs. NWS VIP Levels



There are instances where extremely heavy rainfall within just a few miles of the aircraft can attenuate the transmitted pulse of any Weather Radar to useless levels. This attenuation could be so great that the operator could interpret the return as an indication of no rainfall ahead. In fact, the aircraft may be entering an area of serious and destructive thunderstorm activity. These areas of no returns (or Weather Radar shadows) are caused by the inability of the Weather Radar pulses to penetrate the storm. The reason the pulses cannot get through is that the line of storms contains extremely heavy rain, possibly large hail, and almost certainly severe turbulence. (Refer to the Use of Tilt

Control paragraphs for information on tilt techniques required to identify Weather Radar shadows.)

RANGE COMPENSATION

As previously discussed, attenuation due to distance reduces the strength of Weather Radar signals. As the aircraft approaches a storm, this type of attenuation rapidly decreases, thus increasing signal strength and giving the appearance on the displays of rapid intensification of the target. At ranges of approximately 65 NM (with a 14-inch antenna), the Weather Radar begins to compensate for this problem with its Sensitivity Time Control (STC) circuit. As the aircraft approaches a storm, the targets being painted grow in intensity (due to lessened attenuation) right up to the 65-NM STC boundary. From the STC boundary inward, the display paints an image that allows accurate assessment of the targets (assuming attenuation due to precipitation is taken into account). For most radars, targets beyond the STC boundary do not show properly according to the standard relationship between return level and rainfall intensity. However, the Weather Radar provides range-adjusted thresholds to help compensate for this usual deficiency at ranges beyond approximately 70 NM.

ANTENNA SIZE

The Weather Radar installed in the Challenger 605/650 uses a 14-inch flat-plate, phased-array antenna. The illustration that follows shows the beam-width for the 14-inch antenna. Note that the beam of energy radiating from the antenna is cone-shaped rather than a pencil-shaped beam as commonly thought. This illustration shows that the 14-inch antenna has a relatively narrow beam width. A narrower beam concentrates the radiated energy into a smaller volume. The result is that more energy is available to illuminate the relatively small target. The concentrated energy improves the ability of the Weather Radar System to see targets at the longer ranges. In addition to focusing the transmitted energy, it also increases the ability of the receiver to receive returns, specifically from the areas of interest. The Weather Radar System performance at lower altitudes is also improved. The narrower beam may be pointed more directly at lower and midsections of a storm without painting too much ground clutter.

NOTE

The beam of energy radiating from the antenna is cone-shaped rather than a pencil-shaped beam as commonly thought.

Since the airborne Weather Radar measures the intensity of a storm by the amount of signal reflected back from the target, distortions occur as the beam gets broader with distance. Note the beam width at 50 NM for the 14-inch antenna in the illustration that follows. At 50 NM, the 14-inch antenna has a beam approximately 6.4 NM in diameter. If a storm that is 3 miles wide by 3 miles high shows on the Weather Radar at 50 NM, at that distance more than 50 percent of the transmitted energy will pass over, under, and to the sides of the storm due to the beam width. This results in the Weather Radar measuring reflectivity levels that are less than true value. The low reflectivity levels occur because the storm is significantly less than beam filling and therefore, shows as a weak storm that in fact could be a strong and hazardous one, even though it is relatively small.

Figure E-16 Antenna Size vs. Beam Width

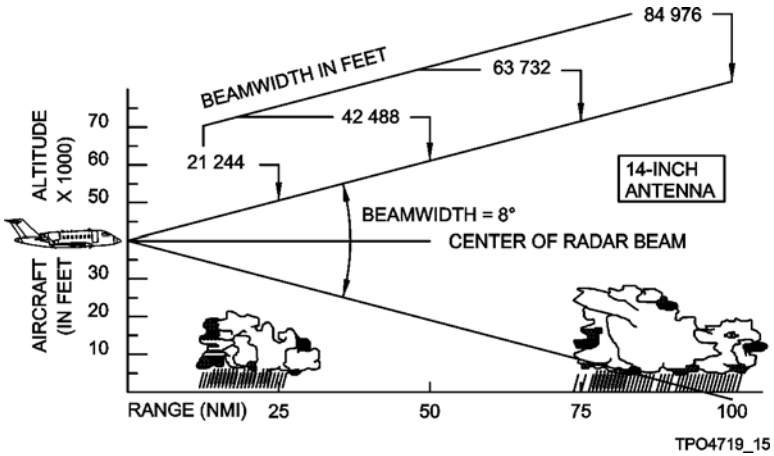
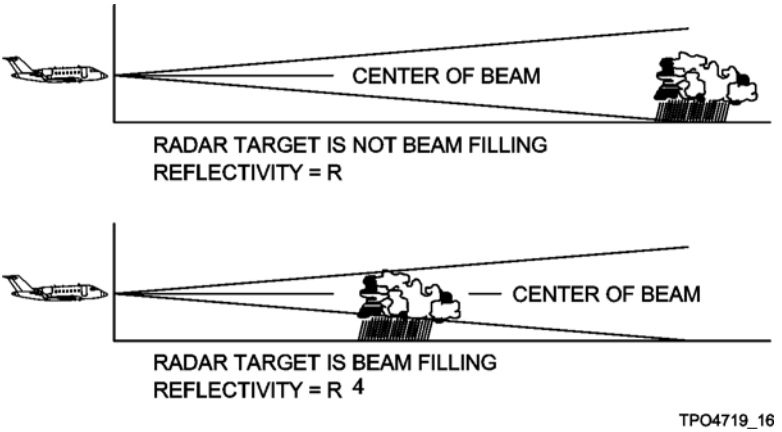


Figure E-17 Weather Radar Target, Beam Filling vs. Non-Beam Filling



The crew should be aware of two other kinds of beam width distortion that apply to any airborne Weather Radar. First, if the same 3 mile high by 3 mile wide storm is being analyzed with a beam that is over 6 NM in diameter, it is obvious that the storm cannot be investigated at various levels. The Weather Radar System would be painting the storm from top to bottom and the displays would show a composite of storm reflectivity at all levels. Secondly, if a storm 3 miles wide is being painted with a beam that is greater than 6 NM wide, the target may show smearing from side to side, making the storm appear much wider than it is deep. In some cases, the 3 mile wide storm could appear as though it is 6 to 10 miles across. This smearing effect is not critical where a single storm is present, but if several 3-mile-wide storms spaced 5 to 10 miles apart are ahead of the aircraft, they could smear into a solid line.

NOTE

For any radar, the larger the antenna, the less this smearing distortion.

AIRPLANE RADOMES

Although not part of the Weather Radar, the radome contributes significantly to overall system performance. In effect, the radome is a window for the Weather Radar to the targets of interest. A poorly designed, constructed, repaired, or maintained radome can result in poor transmission and reception of signals. There are many instances

where the crew reports the Weather Radar as being inoperable or weak when the problem is actually the radome.

A properly designed and maintained aircraft radome should allow a minimum transmissivity of 90 percent. (Transmissivity is the ratio of the amount of power transmitted through the radome to the amount of power that would be transmitted with the radome removed.) Any transmissivity factor of less than 100 percent causes reduced performance.

When properly applied, radome paint causes a loss of about 3.5 to 5 percent in transmissivity. Rubber boots, when properly installed, present a loss of about 5 percent. If a boot is used, it must be adequately sealed to prevent moisture accumulation between the boot and the radome surface. Plastic boots usually present a loss of 20 to 50 percent, which makes them totally unacceptable.

If there is reason to question the transmissivity of the radome, it should be checked by a reputable radome authority, for example, Cair Radomes, Plastics and Synthetics Division of Norton Company, Akron, Ohio. To further assist the crew in selecting or evaluating a radome, refer to FAA Advisory Circular 43-14.

WEATHER RADAR INTERPRETATION

Flight hazards due to weather conditions are primarily the result of turbulence and hail. Wet hail can be detected by Weather Radar, but turbulent air by itself (for example clear-air turbulence) does not provide a Weather Radar echo. Areas having high rainfall rates are ordinarily associated with turbulence, and it is from this rainfall that Weather Radar echoes are reflected and the accompanying turbulence associated with the rainfall is implied.

Some clouds, often of the stratus type, do not contain sufficient moisture to reflect a detectable echo, however, these clouds are usually not a hazard to flight.

THUNDERSTORMS

Updrafts and downdrafts in thunderstorms carry water throughout the cloud. The more severe the drafts, the greater the amount of water contained in the cloud. From the intensity of Weather Radar echoes from this moisture, assumptions can be made about the turbulence involved. When the intensity of the target that shows is high (red or

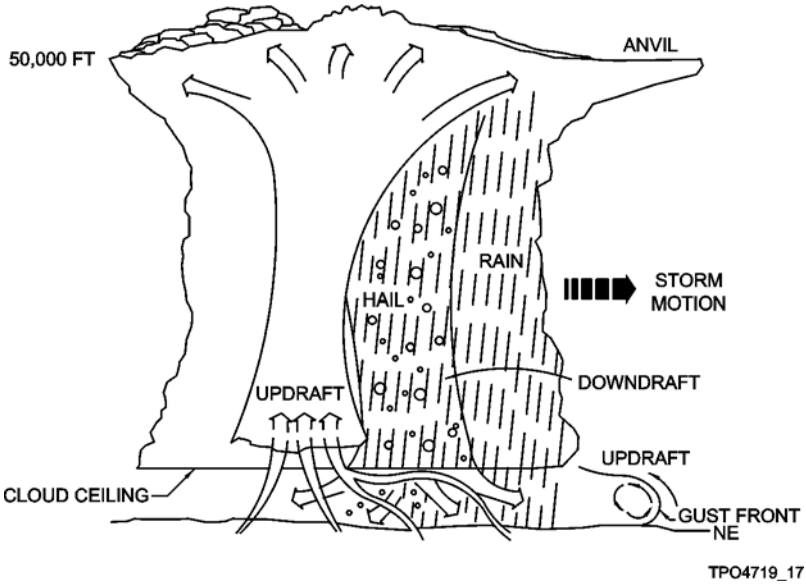
magenta) due to large amounts of water, expect the turbulence to be more severe. Regardless of the operating mode (WX or WX+TURB), areas that show a solid red or magenta display should be avoided by a wide margin.

As a general rule, the steeper the rainfall gradient of a target, the stronger and more severe the turbulence will be. A steep rainfall gradient shows as areas of very narrow, closely spaced green and yellow, or yellow and red, or red and magenta returns. This indicates a large change in the rate of rainfall in a small distance within the storm cell.

Along squall lines, individual cells are in different stages of development. Areas between closely spaced, intense echoes may contain developing clouds that do not have enough moisture to produce an echo. The lightest precipitation may or may not show. However, these areas may have strong updrafts, downdrafts, or turbulence. If approaching a squall line, fly as far from building cells as possible. A good rule of thumb for the safest operation is to avoid red or magenta areas (areas of greatest turbulence) by at least the minimum distance established by the flight operations department. Targets that show broad areas of green are generally precipitation without turbulence, but keep in mind attenuation due to distance and precipitation.

Thunderstorm development is rapid. A flight course that appears clear may contain active and growing cells a few minutes later. When viewing the shorter ranges, periodically select one of the longer ranges to observe distant conditions. This permits early planning of necessary avoidance maneuvers.

Figure E-18 Thunderstorms



Studies have shown that thunderstorms tend to travel in the direction of the winds at approximately 10 000 feet. New cells generally form on the side of a cloud in the direction toward which it is moving, usually an easterly direction. Newly developing cells often do not contain sufficient water to reflect an echo, yet they can cause severe turbulence. In general, detour to the diminishing side of thunderstorms, keeping in mind the minimum distances established by the flight operations department.

TORNADOES

An extreme case of severe turbulence is a tornado. The vast majority of tornadoes are produced by severe thunderstorms and super cells. Weather Radar displays of clouds from which tornadoes were confirmed have occasionally shown the formation of a hook pattern in connection with the tornado. A narrow, finger-like portion extends from the cloud display, and in a short time, curls into a hook and closes on itself. Other echoes associated with tornadoes are V-shaped notches and doughnut shapes. These shapes do not always indicate tornadoes, nor are tornado echoes limited to these characteristic patterns. Of the confirmed Weather Radar observation of tornadoes from target

thunderstorms, most displays have not shown shapes different from those of a normal thunderstorm display.

Conditions conducive to tornado formation produce severe updrafts and downdrafts that carry large amounts of water to great heights. Red or magenta areas with steep rainfall gradients that produce echoes at high altitudes (TILT control up more than usual) generally indicate conditions favorable for producing tornadoes. In no case should these areas be approached or penetrated. Avoid them by at least the minimum distance established by the flight operations department, since turbulence may extend outward from the echo-producing area for large distances.

HAIL

Hail results from updrafts carrying water high enough to freeze. Consequently, the greater the height of a thunderstorm echo, the greater the probability that it contains hail. An estimate of the height can be made by the amount of antenna up tilt required to view the upper part of the target echo. In the upper regions of a cloud where ice particles are dry (no liquid coating on the particles) echoes are less intense. Liquid (water) reflects about five times more Weather Radar energy than solid ice particles of the same mass. Since hailstones are considerably larger than water drops and are usually coated with a thin layer of water, the echo intensity from wet hail is greater than that from rainfall. Thunderstorm targets having an intensity greater than that associated with maximum rainfall will most likely contain hail.

It is not always possible to determine from the display whether the echo is from hail or from rain. Instances have been reported of hail targets producing finger-like protrusions up to 5 miles long, and blunt protuberances up to 3 miles from the edge of thunderstorm echoes. In parts of the country where hail occurs often, extensions from thunderstorms, shown in either red or magenta, generally indicate the presence of hail. This same type of display is also associated with new convective cells that may not yet contain hail.

As with tornadoes, there are no uniquely distinctive displays that are in all cases associated with hail. Protruding fingers, hooks, scalloped edges, and U-shapes are display shapes that have been associated with hail, yet hail echoes are not limited to these shapes. These displays, however, do indicate areas of severe turbulence and must be avoided by a wide margin.

Echoes from hail can appear quickly and along any edge of a storm cell. These echoes can also change in shape and intensity in a very short period of time. For this reason, close and careful monitoring of the display is required. If the Turbulence Weather Radar System is installed, the WX+TURB mode should be used to provide continuous monitoring for turbulence.

WEATHER AVOIDANCE

The paragraphs that follow are included in this section to reinforce the Weather Recognition and Avoidance section presented earlier. Remember, the key to avoiding detected weather is to first determine the heading change needed to safely bypass a storm. Although there is no definite minimum distance rule in avoiding storms, most flight operation departments have established guidelines for storm cell avoidance distances. However, the crew has the sole responsibility to decide how close to approach various types of storm displays.

Monitor the weather at the longer ranges whenever possible. Periodically switch to a lower range and tilt setting for a quick look in front of the aircraft. This procedure reduces the chance of overflying a target, yet still allows time to evaluate weather development and plan any heading changes. When using the shorter ranges, it is a good idea to periodically switch to one of the longer ranges to determine the extent of the weather activity. Operate in the normal mode with one display monitoring a short range for maneuvering, and the other display on a long range to aid in longer term planning.

Proper management of the TILT control for the RTA-854 and for the RTA-4114 operating in Manual mode is paramount in obtaining the full benefits of an airborne Weather Radar. Use the TILT control to look at a target at various levels. If at all possible, avoid targets that are painting areas of red or magenta by at least the minimum distance established by the flight operations department. Do not fly toward an area when a PAC alert arc is present at the perimeter of the last range arc.

It is also a considered good idea to have some amount of ground return in view in the outer regions of the display. This serves two purposes — it enables confirmation that the system is transmitting and receiving properly, and it enables Weather Radar shadows to be seen. A target that shows a Weather Radar shadow may also cause a PAC alert. **Do not fly toward these areas!**

NOTE

While in the vicinity of detectable weather, the crew should temporarily select a shorter range setting either during or immediately after significant heading changes.

Experience gained with the use of the Weather Radar and the guidelines discussed in this guide soon allows the crew to properly analyze the various types of storms and determine the distance necessary to avoid them. Remember that the Weather Radar can only detect precipitation, and the turbulence Weather Radar can only detect precipitation and precipitation-based turbulence. Neither system detects clouds, lightning, or clear-air turbulence hazards.

NOTE

FAA Advisory Circular 20-68B provides criteria for determining the safe distance for human exposure to Weather Radar radiation. This criteria applied to the Weather Radar produces a safe distance of 0.65 meters (2 feet). Users should take necessary and reasonable precautions to make sure that personnel and equipment sensitive to microwave radiation remain safely beyond this distance while within the illumination pattern of an operating Weather Radar.

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